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Effects of Low-salt Therapy on Chronic Kidney Disease Outcomes: A meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050843
Article Type:	Original research
Date Submitted by the Author:	05-Mar-2021
Complete List of Authors:	Shi, Honghong; Shanxi Medical University Second Affiliated Hospital, Renal Division Su, Xiaole; Shanxi Medical University Second Affiliated Hospital Li, Chunfang; Shanxi Cardiovascular Hospital Guo, Wenjuan; Shanxi Medical University Second Affiliated Hospital Wang, Lihua; Shanxi Medical University Second Affiliated Hospital
Keywords:	Nephrology < INTERNAL MEDICINE, Chronic renal failure < NEPHROLOGY, HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effects of Low-salt Therapy on Chronic Kidney Disease Outcomes: A meta-analysis

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ABSTRACT

Objective The benefits of low-salt diet therapy for patients with chronic kidney disease (CKD) are still controversial. We performed a systematic review and meta-analysis to investigate the effects of low-salt diet therapy on major clinical outcomes.

Methods We systematically searched MEDLINE by Ovid (from 1946 to February 2020), EMBASE (from 1966 to February 2020), and the Cochrane Library database (no date restriction) without language restriction. We included all eligible clinical studiesthat involved adults with CKD and compared low-salt diet therapy with controls.

Results A total of 30 eligible studies, including 13 randomized controlled trials (RCTs) and 17 cohort studies reported in 32 publications with 99011 participants were included. Low-salt diet therapy produced a 34% reduction in renal failure events (relative risk (RR): 0.66; 95% confidence interval (95%CI): 0.54-0.82) compared with control groups. But no significant effects were observed on change of proteinuria (standardized mean difference: -0.71; 95% CI: -1.66 to 0.24), decline rate of estimated glomerular filtration rate (eGFR) (mean difference: 0.55; 95% CI: -2.68 to 3.77), the risk of

all-cause mortality (RR: 0.89; 95% CI: 0.49-1.63), and cardiovascular (CV) events (RR: 1.01; 95% CI: 0.46-2.22).

Conclusion Low-salt diet seems to reduce the risk of kidney failure events in CKD patients . However, there was no compelling evidence indicating that low-salt diet would reduce rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. More definitive studies are needed.

PROSPERO registration number CRD42017072395.

Strengths and limitations of this study

This study is the first meta-analysis to evaluate the effect of low-salt diet treatment on hard clinical outcomes, such as renal failure, all-cause mortality and CV events for CKD patients.

No study design or language restriction, therefore allowing inclusion of the maximum number of studies in this review.

The existence of significant heterogeneity might limit the interpretation and clinical application of the results.

Confounding factors in the observational studies cannot be neglected, which might impact our results.

INTRODUCTION

Chronic kidney disease (CKD) has become a global public health burden. Dietary interventions that aim to delay the progression of CKD and decrease complications have a pivotal role in the management of CKD. Restriction of dietary sodium intake can augment the antiproteinuric effects of angiotensin converting enzymes inhibitors/angiotensin receptor blockers (ACEIs/ARBs) [1] and

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decrease blood pressure (BP) [2]. In the 2020 KDIGO guidelines for glomerulonephritis, dietary restriction of sodium to < 2 g/d is a primary tenet for controlling BP and edema and improving urinary protein excretion independently of medications. However, there remains an open question of whether these effects can translate into a worthwhile absolute risk reduction in patients with CKD, including kidney failure events, cardiovascular (CV) events and all-cause mortality. In the Chronic Renal Insufficiency Cohort (CRIC) Study, recruiting patients eGFR between 20 and 70 ml/min/1.73 m^2 depending on age, compared with the lowest quartile of urinary sodium excretion (<116.8 mmol/24 h), high urinary sodium excretion (>194.6 mmol/24 h) was associated with a 54% risk increase in end-stage renal disease (ESRD) or halving of estimated glomerular filtration rate (eGFR) and with a 43% risk increase in all-cause mortality [3]. Contrasting results have been observed in other cohort studies, in which sodium intake does not appear to impact the progression of CKD to ESRD and a decline in eGFR [4,5]. Furthermore, a low salt intake of <6 g/day was associated with an increased risk of all-cause mortality and CV mortality in hemodialysis patients, resulting in an L-shaped association curve [6]. According to data from the NHANE II study, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans [7]. Randomized controlled trials (RCTs) evidence that discussed the effect of a low-salt diet in CKD patients on kidney failure events, CV events and all-cause mortality are controversial. It is difficult to base clinical decisions on findings from these studies, given their small sample sizes, highly selected participants, short-term follow-up and lack of assessment of hard clinical outcomes.

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Thus, the association between dietary salt restriction and clinical outcomes in patients with CKD remains unclear. We performed this systematic review to summarize all eligible clinical studies data and evaluate the effects of low-salt diet therapy on kidney disease outcomes, all-cause mortality and CV events in CKD patients.

MATERIALS AND METHODSData sources and search strategy

We performed this systematic review according to a pre-specified protocol registered at the International Prospective Register of Systematic Reviews (CRD42017072395) [8], and reporting was done in line with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines [9]. A comprehensive search was conducted using the following database: MEDLINE by Ovid (from 1946 to February 2020), EMBASE (from 1966 to February 2020), and Cochrane Central Register of Controlled Trials (no date restriction), with relevant keywords and medical subject headings that included all spellings of “CKD”, “RCT”, “Cohort Studies”, “Sodium Chloride”, “Sodium”, “Sodium-Restricted” (see Supplementary item 1 for full search terms). Studies were considered without any language restriction. To ensure a comprehensive literature search, we also screened reference lists from included articles. The ClinicalTrials.gov website was also searched for studies the ongoing, but unpublished in this field.

Study selection and outcome estimation

We included data from RCTs and cohort studies in which low-alt diet was given to adults with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management

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of CKD) compared with usual therapy or different salt intake.

Pre-defined outcomes that contained analyzable data were listed as follows: First, renal failure events, defined as a more than 25% or 50% decline in eGFR from baseline during follow-up [10], doubling of serum creatinine or ESRD. Second, the rate of change in eGFR per year. Third, changes in urinary protein or urinary albumin during follow-up, including urinary protein excretion, urinary albumin excretion and urinary albumin/creatinine ratio. Fourth, all-cause mortality events. Fifth, CV events, defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, CV disease and CV death. Fifth, change in systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Data extraction and quality assessment

Two independent reviewers (H.S. and X.S.) extracted data, and assessed quality, according to the pre-specified protocol. Disagreements were resolved through a third reviewer (L.W.). Data of all eligible studies was extracted into a spreadsheet. The data sought included characteristics of studies (study type, randomization method, follow-up time, withdrawals/dropouts), baseline patient characteristics (age, gender, baseline eGFR, baseline BP), intake of salt, and outcome events.

We used Cochrane Collaboration risk-of-bias tool [11] to assess all potential sources of bias for the included RCTs. Trails were assessed as being at low, other, some concerns, or high, and the overall risk of bias generally corresponds to the worst risk of bias in any of the domains. However, if a study was judged to have some concerns about risk of bias for multiple domains, it might be judged as at high risk of bias overall. In addition, the quality of RCTs were assessed using the Jadad scale [12]. We

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used the Newcastle-Ottawa Scale (NOS) to assess quality of cohort studies from 3 aspects, including selection of cohorts, comparability of cohorts, and assessment of outcome [13].

Data synthesis and statistical analyses

When dichotomous outcome data from individual trials were analyzed, relative risks (RRs) measure and its 95% confidence intervals (CIs) were used. If individual study RR was unavailable in the original article, RR and 95% CI were calculated from event numbers extracted from each study before data pooling. In calculating the RR values, we used the total number of patients randomized in each group as the corresponding denominator. Continuous outcome data from individual trials were analyzed using the difference in means (MD) with 95% CI to pool eGFR, and the standardized mean differences (SMD) with 95% CI were used to pool the proteinuria or albuminuria data. When continuous outcome data were analyzed, the difference in mean change between baseline and end of treatment value were used. If data for change between baseline and end of treatment were not available in the studies, we calculated it using correlations that were estimated from other included studies that had a similar follow-up period and reported in considerable detail according to the imputed formulation and its related interpretations in Cochrane Handbook [14].

In consideration of poor stability of the Der Simonian-Laird procedure for small numbers of studies [15], we used the empirical Bayes procedure [16] toestimate all outcomes. We also used Der Simonian-Laird random effects model [17] and restricted maximum likelihood [18] to assess the summary effects as a part of the sensitivity analyses. Considering the inevitable heterogeneity between studies, subgroup analyses, meta-regression, and sensitivity analyses were performed. Subgroup analyses were performed according to pre-specified protocol, including study type, baseline eGFR,

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and baseline BP. In addition, we performed sensitivity analyses by using different random-effects estimation methods as above mentioned, excluding studies with sample size <50, studies with follow-up time <12 months, studies with low quality (Jadad score <3, NOS <5★), and studies with extreme outliers. Heterogeneity between studies was evaluated by using I^2 statistics or τ^2 statistics. Stata version 15.0 (StataCorp LP, College Station, TX, USA) was used for statistical analyses , and a two-sided *P*-value < 0.05 was considered statistically significant.

RESULTS

Overview of included trials

We identified 9002 potentially relevant records. After removal of duplicates, we screened the titles and abstracts, and 218 publications were reviewed in full text. As shown in figure 1, Supplementary table 1 and Supplementary table 2, a total of 30 eligible studies [3-6,19-46] including 13 RCTs [19,21,26-28,32,33,36-41,46] and 17 cohort studies [3-6,20,22-25,29,31,34,35,42-45] reported in 32 publications with 99011 participants, were included in this review. The median follow-up duration was 6 months (interquartile range 2.8 to 19.5). Participants were enrolled at an average age of 54 years, and male participants accounted for 60% of the total participants. The average eGFR of the included participants was 48.15 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and eighteen trials compared different levels of sodium intake.

Figure 1 PRISMA flow chart for the included studies.

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The risk of bias varied substantially across the RCTs. Nine trials had a Jadad scale of 3-5, and others scored less than 3 (Supplementary table 1). The results from the Cochrane Collaboration risk-of-bias tool are shown in Supplementary table 3. In all the RCTs, bias arising from the randomization process was assessed as low risk in 61.54% of the trails; risk of bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result were low risk in all the trails. In terms of overall bias, 61.54% of the trails were assessed as low risk, and 38.46% of the trails were assessed as having some concerns.

According to the NOS checklist, all included cohort studies were generally high quality and awarded 6★- 8★ (Supplementary table 1 and Supplementary table 4).

Effect of low-salt diet therapy on renal outcomes

One RCT and 7 observational studies including 7596 patients with 1842 events compared the low-salt diet with the control in preventing renal failure events. As shown in figure 2, compared with the control, low-salt diet therapy produced a 34% reduction in the risk of renal failure events (RR: 0.66; 95% CI: 0.54, 0.82), with substantial heterogeneity (I^2 =78.5%, P = 0.04). Subgroup analyses and meta-regression were conducted to discuss the source of heterogeneity. No significant difference between RCTs and observational studies was found, while the effect sizes were greater in studies that enrolled patients aged < 56 years (P = 0.04) (table 1). On univariate meta-regression, no clear relationship between 24-hour urine sodium (P = 0.88) or baseline eGFR (P = 0.57) and the reduction in renal failure events was observed (Supplementary figure 1, Supplementary figure 2).

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Figure 2 Forest plot for renal failure events. Renal failure was defined as a more than 25% or 50% decline in eGFR from baseline during follow-up, doubling of serum creatinine or ESRD. CI, confidence interval; RR, relative risk.

Table 1 Subgroup Analysis of Renal Failure Events.

Subgroup	No. of Trials	n	RR (95% CI)	P for RR	I^2	P for Heterogeneity Test[a]
Study type						
RCT	1	7178	0.64 (0.43, 0.98)	0.04	-	0.10
Cohort study	7	418	0.66 (0.53, 0.82)	<0.001	81.6%	
Baseline eGFR (mL/min/1.73m^2)						
60-89	1	171	0.14 (0.02, 1.14)	0.07	-	0.20
30-59	6	7084	0.65 (0.53, 0.79)	<0.001	81.0%	
15-29	1	338	0.86 (0.62, 1.18)	0.34	-	
Mean age (years)						
<56	4	860	0.55 (0.46, 0.64)	<0.001	16.5%	**0.04**
≥56	4	5662	0.75 (0.64, 0.88)	<0.001	44.3%	
Baseline BP (mmHg)						
<140/90	5	5609	0.60 (0.50, 0.72)	<0.001	58.4%	0.30
SBP≥140 or DBP≥90	2	1646	0.70 (0.44, 1.11)	0.13	74.6%	

Note: a P value calculated by meta regression. BP,blood pressure; CI, confidence interval; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk.

Five RCTs and six observational studies with a total of 3203 participants reported data regarding the effects of a low-salt diet on the rate of change in eGFR. Low-salt diet therapy did not produce an

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apparent beneficial effect on the rate of change in eGFR (MD: 0.55; 95% CI: -2.68 to 3.77). There was evidence of significant heterogeneity across the included studies (I^2= 98.0%, P< 0.001) (figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to prespecified characteristics (Supplementary table 5).

Figure 3 Forest plot for rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; MD, mean difference; SD, standard deviation.

Data for the effects of low-salt diet therapy on proteinuria or albuminuria were available in 13 studies including 2922 participants. The SMD for change in proteinuria or albuminuria was not statistically significant at -0.71 (95% CI: -1.66 to 0.24) compared with that in the control groups, with significant heterogeneity (I^2 = 97.3%, P=0.009) (figure 4). No statistical heterogeneity was found in any subgroup analyses for the effects of a low-salt diet on proteinuria or albuminuria (Supplementary table 5).

Figure 4 Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; SD, standard deviation; SMD, standard mean difference.

Effect of low-salt diet therapy on all-cause mortality and CV events

Data for the effects of low-salt diet therapy on all-cause mortality and CV events were available for 7 studies (two RCTs and five observational studies) and 5 studies (one RCT and four observational studies), respectively. There were no clear effects of a low-salt diet on the risk of all-cause mortality (RR: 0.89; 95% CI: 0.49-1.63; 94015 participants and 4192 events) or CV events (RR: 1.01; 95% CI:

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0.46-2.22; 93573 participants and 1989 events) compared with those in the control groups, with significant heterogeneity across the included studies (*P* values for heterogeneity both less than 0.001) (figure 5). More significant benefits were found in studies that included patients with an eGFR of 30-59 mL/min/1.73 m^2 (*P* for subgroup heterogeneity = 0.02) in the subgroup analysis of CV events (Supplementary table 5).

Figure 5 Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death. CI, confidence interval; N, number of trials; RR, relative risk.

Effect of low-salt diet therapy on BP

Twenty studies reported the effect of a low-salt diet on BP, including 19 studies (3608 individuals) for SBP and 18 studies (3608 participants) for DBP. The MD changes in SBP and DBP were both statistically significant at -5.81 mmHg (95% CI: -9.63 to -1.99) and -2.3 mmHg (95% CI, -3.61 to -0.98), respectively (Supplementary figure 3). Significant heterogeneity across the included studies was found (*P* values for heterogeneity both less than 0.05). Subgroup analyses showed that the association between the change in SBP or DBP and a low-salt diet was modified by the baseline BP of patients. Patients with baseline SBP >140 mmHg or DBP > 90 mmHg had an average reduction in SBP of 10 mmHg and in DBP of 5 mmHg (Supplementary table 5).

Sensitivity analyses and publication bias

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All sensitivity analyses results were robust (Supplementary table 6). The most notable exception was that the effect of low-salt diet therapy on proteinuria or albuminuria became significant when the DL method was used to pool data.

DISCUSSION

Behavioral modifications can make an important contribution to delaying CKD progression to ESRD, and dietary salt restriction is one of the main concerns. Our meta-analysis showed that compared with control groups, low-salt diet therapy significantly reduced renal failure events and lowered BP (SBP and DBP) in CKD patients. No significant effects were observed regarding the decline rate of eGFR, change in proteinuria, risk of all-cause mortality or CV events. These results remained broadly consistent across major subgroups and sensitivity analyses. However, the existence of significant heterogeneity might limit the interpretation and clinical application of these results.

The previous review of 16 studies in 2006 addressing salt intake and kidney disease set out to establish whether variations in dietary sodium consumption influence kidney outcomes in patients with CKD [47]. The review suggested that increased salt intake was associated with worsening albuminuria and an increased likelihood of a reduction in GFR. In 2015 [48] and 2018 [49], the two meta-analyses of RCTs both suggested that moderate dietary salt restriction significantly reduced BP and proteinuria. However, it remains unanswered whether these benefits may translate to clinically significant reductions in kidney failure events, all-cause mortality events and CV events. Our study, including 30 studies (13 RCTs and 17 observational studies), more than 90,000 patients, and 1842 kidney failure events, first concluded that compared with the control groups, a low-salt diet produced a significant 34% reduction in kidney failure events despite marked heterogeneity. Furthermore, the significant

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effect of a low-salt diet on BP (decreases in SBP and DBP of 5.8 mmHg and 2.3 mmHg, respectively) was confirmed again in our study and was recognized as an important target to slow the progression of CKD towards ESRD. The benefit of proteinuria from low-salt diet therapy became nonsignificant in our study when the restricted maximum likelihood or empirical Bayes methods were used to pool data. The standard DerSimonian-Laird procedure that was applied in previous meta-analyses can be very unstable with small numbers of studies. However, it remains unanswered whether these benefits may translate to clinically significant reductions in kidney failure events, all-cause mortality events and CV events. Our study, including 30 studies (13 RCTs and 17 observational studies), more than 90,000 patients, and 1842 kidney failure events, first concluded that compared with the control groups, a low-salt diet produced a significant 34% reduction in kidney failure events despite marked heterogeneity. Furthermore, the significant effect of a low-salt diet on BP (decreases in SBP and DBP of 5.8 mmHg and 2.3 mmHg, respectively) was confirmed again in our study and was recognized as an important target to slow the progression of CKD towards ESRD. The benefit of proteinuria from low-salt diet therapy became nonsignificant in our study when the restricted maximum likelihood or empirical Bayes methods were used to pool data. The standard DerSimonian-Laird procedure that was applied in previous meta-analyses can be very unstable with small numbers of studies [50,51].

Of note, statistical heterogeneity was found in almost all outcomes, and clinical heterogeneity was also an inevitable issue. Low-salt diets are not palatable to many [52], therefore, it is difficult to ensure sustained adherence to a low-sodium diet over long periods of follow-up. Sodium intake is quantified in distinct ways between studies (for example, 24 h urine collection, spot urine collection or dietary questionnaire). These inherent problems were reflected in the paradoxically opposite

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findings of various studies, especially observational studies with more confounding factors. Only three studies (1 RCT and 2 observational studies) simultaneously reported the outcomes of eGFR and kidney failure events. Furthermore, the different follow-up periods with 52 months of kidney failure events and 12 months of eGFR should be noted. The inconsistent results of eGFR and kidney failure events may partly be explained by the heterogeneity mentioned above. It should be assumed that heterogeneity limited the interpretation and generalizability of our results.

In CKD, BP is usually sodium sensitive, and RAAS blockers are considered the first-line therapy for hypertension and proteinuria. It has been known for over 2 decades that the effects of RAAS blockers are blunted by high sodium intake [53-55]. Thus, the significant effect of dietary sodium restriction on lowering BP and supporting the function of RAAS blockers was speculated to play important roles in reducing the risk of kidney failure events. Interestingly, a BP-independent effect of dietary sodium on the kidney is substantiated by data in healthy volunteers, in which dietary sodium restriction reduces albuminuria within the normal range, without a detectable effect on BP [56]. Similarly, proteinuria reduction by sodium restriction remained significant after adjustment for the decrease in BP[54]. These results suggested an independent renoprotective effect of low-sodium diets, both as a single measure and in combination with RAAS blockers.

In our study, there was no convincing evidence that a low-salt diet was associated with a lower incidence of all-cause mortality and CV events. Limited data on all-cause mortality (seven studies with 2618 events) and CV events (five studies with 1989 events) for the meta-analysis might introduce a risk of false-negative results because of low statistical power. More significant benefits of CV events were found in studies that included patients with eGFR 30-59 mL/min/1.73 m^2 in the subgroup analysis,

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while the effects were not significant in patients with eGFR less than 15 mL/min/1.73 m^2. The significant subgroup heterogeneity (P =0.02) could be attributable to more CV risk factors in patients with ESRD than in those with eGFR 30-59 mL/min/1.73 m^2. The benefits of CV events from a low-salt diet were not enough to outweigh the increase in CV events caused by more risk factors (such as hyperuricemia, hyperphosphatemia, and vascular calcification). It is a hint at least for clinicians that low-salt diet education needs to be emphasized in the early stage of CKD.

Observational data from various studies showing a J-curve between sodium intake and renal and CV outcomes have raised concern regarding the safety of rigorous sodium restriction [57-59]. It is uncertain whether the presence of CKD modifies this association. In our meta-regression analysis, no clear relationships between 24-hour urine sodium (P = 0.88) or baseline eGFR (P = 0.57) and the reduction in kidney failure events were observed. A regimen of ACEIs and rigorous sodium restriction reduced blood pressure, proteinuria, and glomerular damage but unexpectedly aggravated tubule-interstitial damage in experimental renal disease [60]. This could provide a potential explanation for the association of very low sodium intake with worse renal outcome. Both high or extremely low dietary salt intake might be inappropriate for CKD patients. KDIGO recommends a daily sodium intake of < 2.0 g/d in patients with CKD without a clear lower limit. Taken together, these results should be interpreted, and the proper levels of dietary salt intake in CKD patients should be discussed. The current study might not be the final answer for the question of low-sodium diets and progression of kidney disease. Interventional studies are needed to ascertain the optimum level of dietary sodium intake for renal protection in patients with different CKD stages.

Our study has several potential limitations. First, as has been noted, the existence of statistical

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heterogeneity and inevitable clinical heterogeneity might raise critical concerns regarding validity, although we tried to resolve these concerns through subgroup analysis and sensitivity analysis. Heterogeneity in the study population as included in the umbrella term "CKD patients" remains a major concern for meaningful interpretation of the results. Second, many of the included studies with low quality had small sample sizes and a low incidence of events. Consequently, the uncertainty in the analyses was increased, resulting in wide CIs of effect measures. Third, confounding factors in the observational studies cannot be neglected, which might impact our results. Therefore, the findings of this study should be viewed as hypothesis-generating and need to be confirmed by further studies.

CONCLUSION

In conclusion, our meta-analyses demonstrated that a low-salt diet may reduce the risk of kidney failure events in patients with CKD. However, there was no compelling evidence indicating that a low-salt diet could reduce the rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. The optimal dietary salt intake for patients with different CKD stages remains unanswered. Further well-designed RCTs targeting patients with different CKDs are required to assess and integrate these results.

Acknowledgments We thank American Journal Experts Corporation for helping in proofreading and editing the English of the final manuscrip.

Contributors All authors take responsibility for the integrity of the data and the accuracy of data analysis. Study concept and design: L.W., X.S., S.H. and C.L.. Extraction, analysis and interpretation of data: H.S. and X.S.. Drafting of the manuscript: H.S. and L.W.. Critical revision of the manuscript:

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L.W. and X.S. Statistical analysis: H.S., C.L. and W.G.. Technical support: L.W., X.S. and W.G.. All authors have read and agreed to the submission to this journal of the manuscript.

Funding Support was provided by the National Natural Science Foundation of China (no. 82000655) and the Science and Technology Project of Shanxi Province (no. 201801D121218).The funders had no role in decision to publish and preparation of the manuscript.

Competing interests None declared.

Patient and Public Involvement No patient involved.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data used to support the findings of this study are included within the article and supplementary information.

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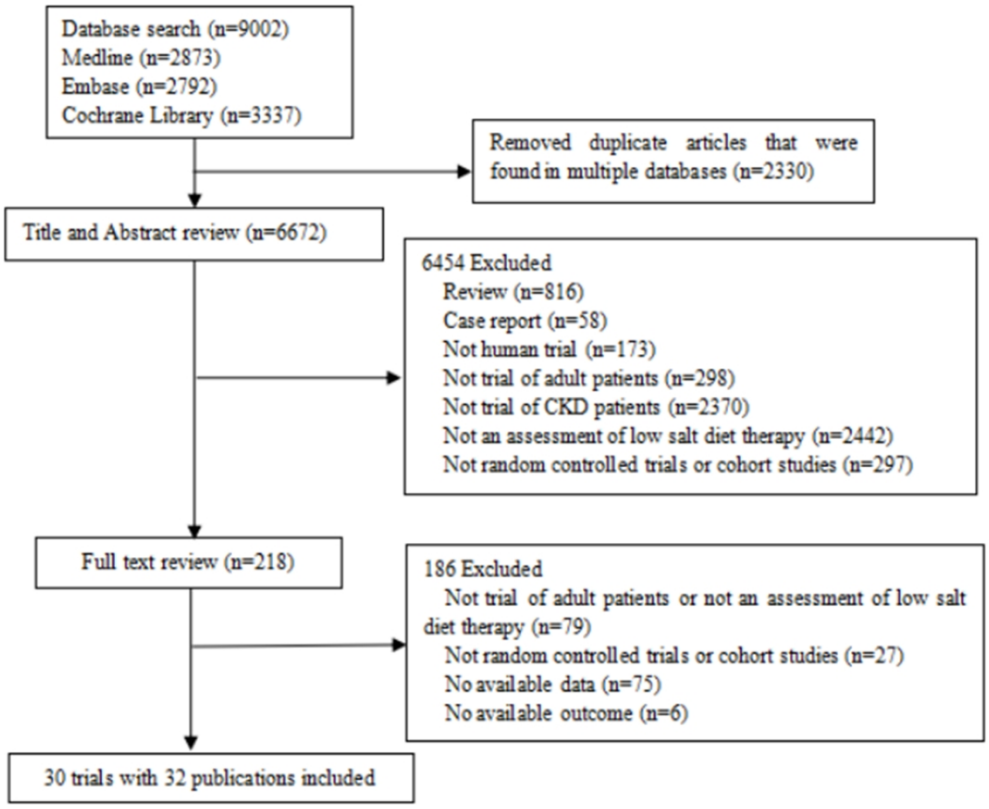
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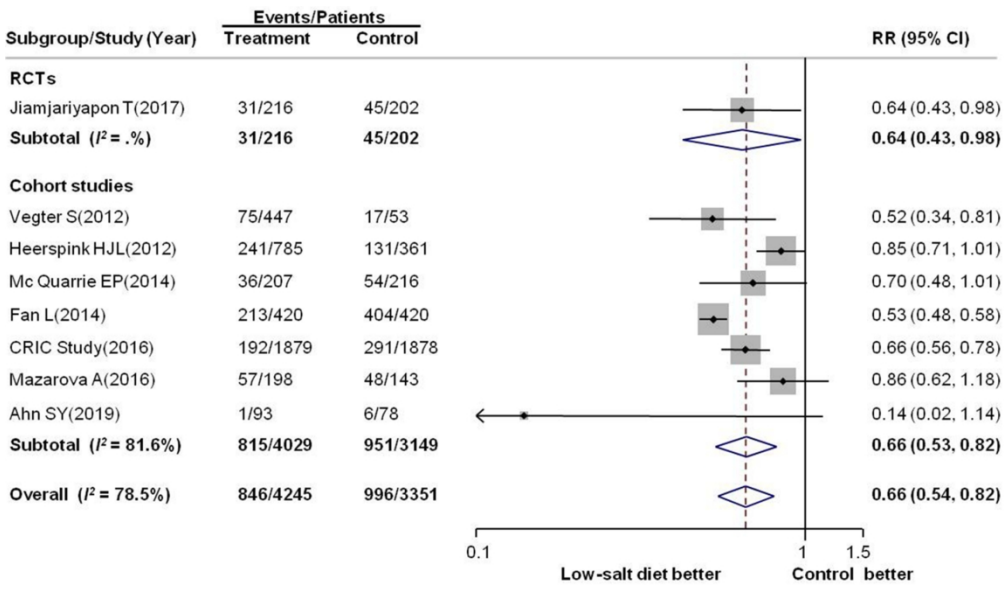
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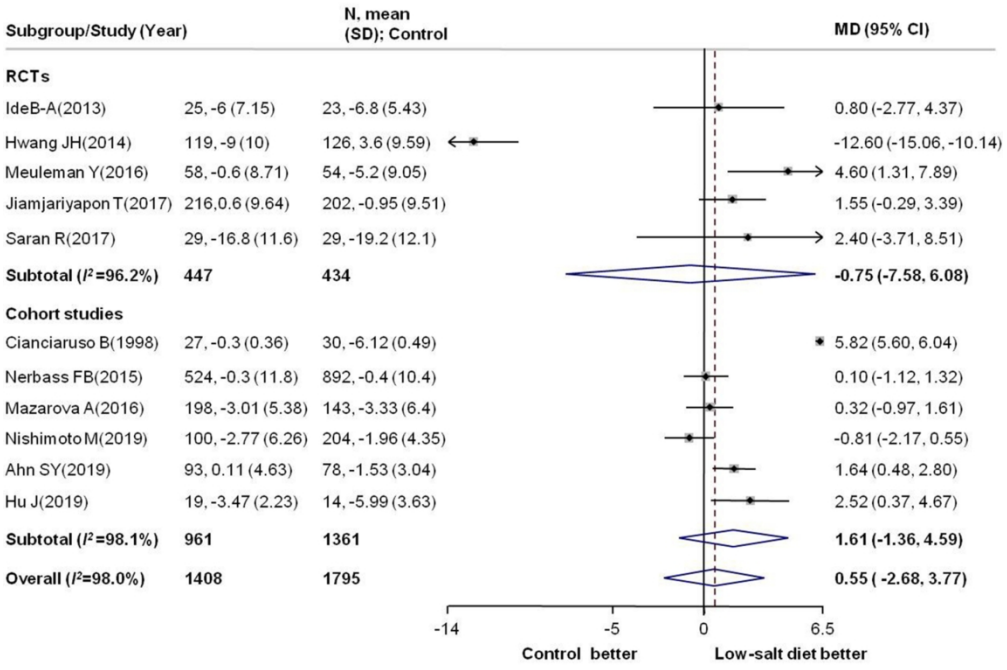
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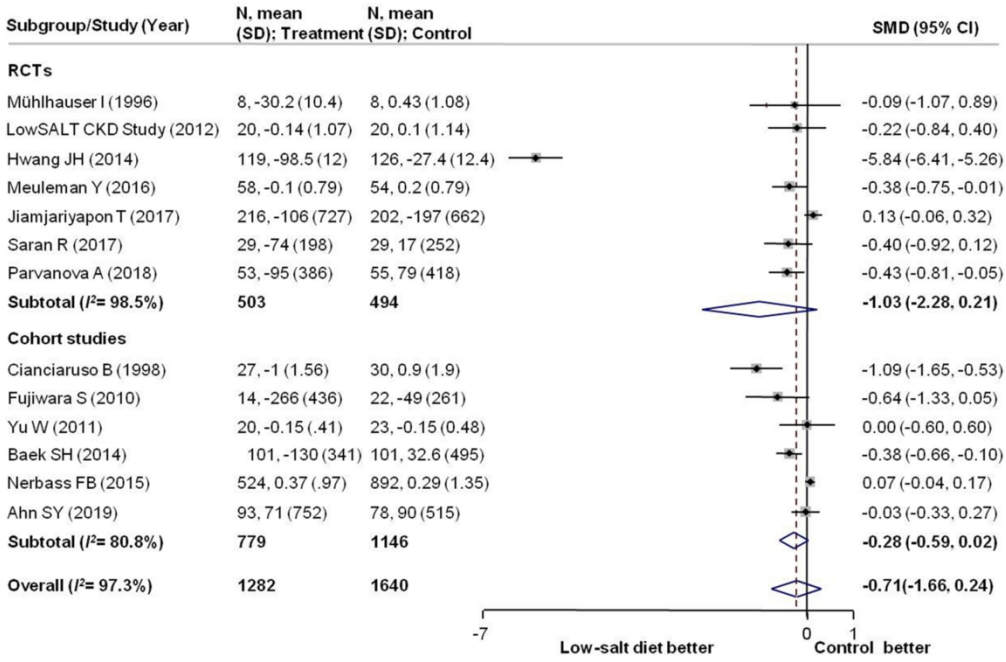
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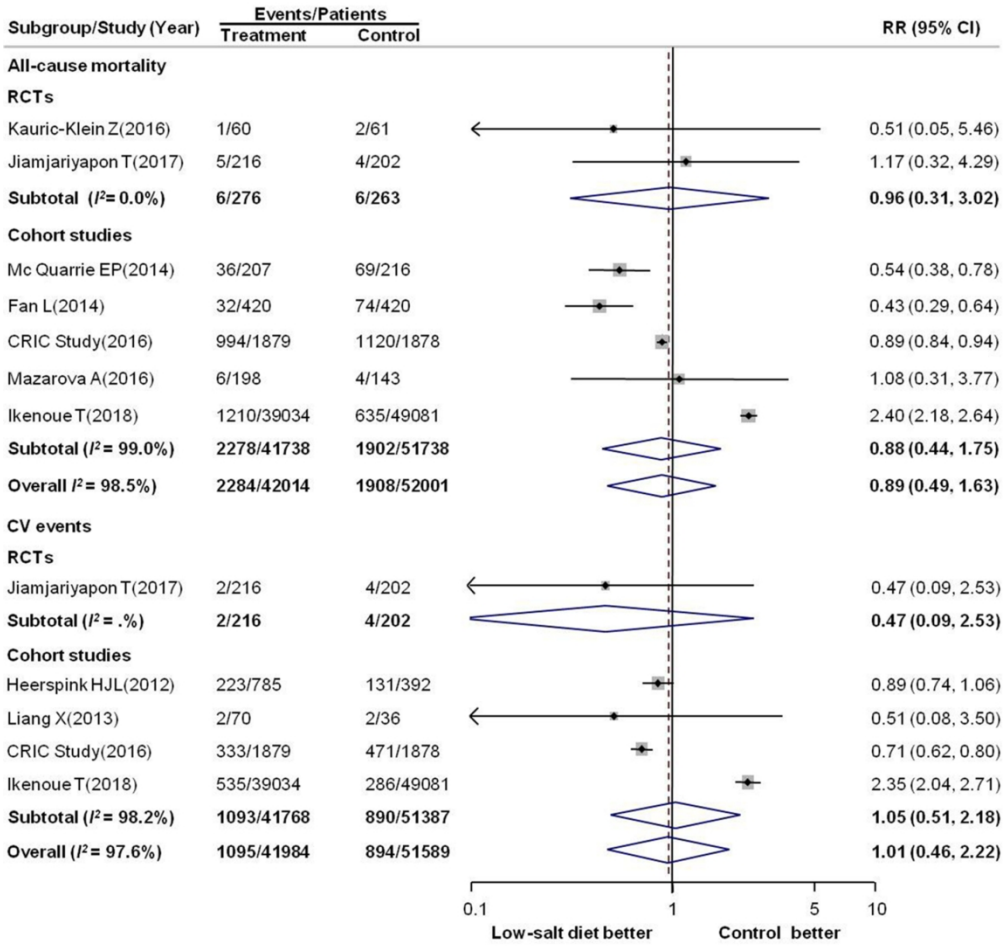
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Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang

Citation

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang. Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease. PROSPERO 2017 CRD42017072395 Available from: http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42017072395

Review question

This meta-analysis aims to evaluate the effects of different levels of dietary sodium intake on renal outcome in patients with chronic kidney disease (CKD).

Searches

All relevant randomized controlled trails (RCTs) and cohort studies will be identified and searched in following electronic databases without language restriction: Ovid MEDLINE SP (from 1946 to July 2017), EMBASE (from 1966 to July 2017), and the Cochrane Central Register of Controlled Trails(no date restriction). A search in www.clinicaltrials.gov will be used to identify the ongoing but unpublished studies about this field. We combined this strategy with a manual search of reference lists from identified trials and review articles.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/72395_STRATEGY_20170618.pdf

Types of study to be included

We will include RCTs and cohort studies that compared different levels of sodium intake for adult patients with CKD as well as studies in broader populations in which outcome data for subgroups with CKD could be disaggregated.

Condition or domain being studied

Chronic kidney disease

Participants/population

Inclusion criteria: Adults (no less than 18 years old) with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) at all disease stages will included without restrictions on gender or race. Data from subgroups of participants with CKD within studies with broader inclusion criteria (e.g. people from the general population, people with diabetes, people with cardiovascular disease) were also included.
Exclusion criteria: Studies which do not include any people with chronic kidney disease.

Intervention(s), exposure(s)

Low-sodium diet

Comparator(s)/control

No intervention or different levels of dietary sodium intake

Context

None

Primary outcome(s)

If the data allow:

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1) Kidney failure events: It will include more than 25% or 50% decrease in estimated glomerular filtration rate (eGFR), doubling of serum creatinine, end stage renal disease (ESRD);
2) Rate of change in the eGFR: The difference from baseline in eGFR divided by the number of years between creatinine measurements (ml/min/1.73m2 per year);
3) Change in proteinuria or albuminuria, including urinary protein excretion (UPE), urinary albumin excretion (UAE), protein to creatinine (PCR) and albumin to creatinine ratio (ACR);
4) Major cardiovascular events: It will include a composite including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death.

Secondary outcome(s)

1) 24-hour urinary sodium excretion;
2) Change in blood pressure;
3) All-cause mortality;
4) Change in markers of fluid overload (brain natriuretic peptide, N-terminal pro-brain natriuretic peptide, weight, bio-impedance analysis);
5) Adverse events: hypotensive episodes, undesirable change in blood lipids (low density lipoprotein, high-density lipoprotein);
6) Hospitalization.

Data extraction (selection and coding)

Data extraction will be carried out independently by two reviewers using standard data extraction forms. Disagreement will be resolved by consensus or by discussion with a third author. When detailed information that was needed for the analysis was not available, the original authors will be contacted to obtain the missing information through e-mail. Where multiple publications of one study exists, data will be extracted and grouped together as a single study with the most complete data.
The extracted data will include: study characteristics (study design, method of randomization, and withdrawals / dropouts); baseline patient characteristics (age, sex, race, country, BMI, sodium intake and its measurement, 24-hour urinary sodium excretion, mean proteinuria or albuminuria, eGFR, serum creatinine concentrations, blood lipids, blood pressure, diabetic status, cardiovascular events, smoking status, glycosylated hemoglobin, hemoglobin, antihypertensive drugs, antidiabetic drugs, different kidney disease diagnosis); change in eGFR; follow-up duration; outcomes: as listed in Primary outcomes and Secondary outcomes. When the required quantitative data is only provided in published figures, we will use g3 data software (http://www.frantz.fi/software/g3data.php) to extract exact numbers from figures.

Risk of bias (quality) assessment

For the RCTs, we will assess risk of bias for sequence generation, allocation concealment, blinding, selective reporting, incomplete outcome data and other sources of bias, and determined overall risk of bias based on predefined rules, utilizing the Cochrane Collaboration risk of bias tool. (reference: Higgins JP, Altman DG, et al. Cochrane Statistical Methods G. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials. BMJ 2011;343:d5928.). Furthermore, the quality of RCT study will be quantified with Jadad scale. (reference: Jadad AR, Moore RA, Carroll D, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Control Clin Trials 1996;17:1– 12.). The quality of cohort study will be independently evaluated by each investigator using Newcastle-Ottawa quality assessment scale. (reference: Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010; 25:603–5.).

Strategy for data synthesis

The results of the data extraction and quality assessment for each study will be presented in structured tables. For dichotomous outcomes, relative risks (RR) and 95% Cis will be calculated from the numbers of events and participants at risk for events. For continuous outcomes, the mean difference (MD) will be used, or the standardized mean difference (SMD) if different scales were used. Data will be pooled using the random-effects model but the fixed-effect model will also be used to ensure robustness of the model chosen and susceptibility to outliers.
Heterogeneity will be analyzed using Cochrane's Q and the I-squared statistic. We will explore potential

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heterogeneity by subgroup analysis and meta-regression.
Sensitivity analyses will be conducted to check the robustness of the effect of sodium-restricted treatment by excluding of trials with small sample size and low Jadad scores.
A two-sided p-value less than 0.05 was considered statistically significant and statistical analyses were performed using RevMan version 5.3 and STATA version 12.0.

Analysis of subgroups or subsets

If the data allow, subgroup analysis will be conducted by study design, follow-up duration, measures of sodium intake, different levels of sodium intake, difference in sodium intake between two groups, and baseline measures, including age, sex, BMI, eGFR, 24-hour urinary sodium excretion,mean serum creatinine, blood pressure, mean proteinuria or albuminuria, stage of CKD, presence of comorbidities (hypertension, diabetes and CVD), different kidney disease diagnosis (e.g. GN or Kidney Transplant Recipients) and whether RAAS blocker or other antihypertensive drug treatments.

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Anticipated or actual start date

15 July 2017

Anticipated completion date

01 February 2018

Funding sources/sponsors

None

Conflicts of interest

None known

Language

English

Country

China

Stage of review

Review_Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Acids, Noncarboxylic; Humans; Kidney; Renal Insufficiency, Chronic; Sodium; Sodium Chloride, Dietary; Sodium, Dietary

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Date of registration in PROSPERO
18 July 2017

Date of publication of this version
18 July 2017

Details of any existing review of the same topic by the same authors
None

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions
18 July 2017

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Item S1 Search Strategy

Ovid MEDLINE

1. exp Sodium Chloride/ or sodium chloride.mp.

2. exp Sodium, Dietary/

3. exp Diet, Sodium-Restricted/

4. ((sodium or salt) adj5 (low or high or alter* or reduce* or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)).mp.

5. (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free).mp.

6. 1 or 2 or 3 or 4 or 5

7. exp Kidney/ or kidney.mp.

8. exp Nephrosis/ or nephrosis*.mp.

9. (renal or nephropath*).mp.

10. (chronic kidney or chronic renal).mp.

11. exp Kidney Diseases/

12. ((kidney or renal) adj (disease* or failure)).mp.

13. exp Kidney Function Tests/ or kidney function test*.mp.

14. (CKF or CKD or CRF or CRD).mp.

15. (end-stage renal or end-stage kidney or endstage renal or endstage kidney).mp.

16. (ESRF or ESKF or ESRD or ESKD).mp.

17. exp Uremia/ or ur?emi*.mp.

18. exp Renal Replacement Therapy/ or renal replacement therap*.mp.

19. exp Renal Insufficiency/ or renal insufficien*.mp.

20. exp Dialysis/ or dialysis.mp.

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21. exp Renal Dialysis/ or renal dialys*.mp.

22. ("pre dialysis" or pre-dialysis or predialysis).mp.

23. (h*emodialysis or h*emofiltration or h*emodiafiltration).mp.

24. exp Peritoneal Dialysis/ or peritoneal dialysis.mp.

25. (PD or CAPD or CCPD or APD or IPD).mp.

26. exp Kidney Transplantation/

27. ((kidney or renal) adj (transplant* or graft* or allograft*)).mp.

28. 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27

29. randomized controlled trial.pt.

30. controlled clinical trial.pt.

31. randomized.ab.

32. placebo.ab.

33. randomly.ab.

34. clinical trials as topic.sh.

35. trial.ti.

36. 29 or 30 or 31 or 32 or 33 or34 or 35

37. (animals not (humans and animals)).sh.

38. 36 not 37

39. exp Cohort Studies/

40. exp Longitudinal Studies/

41. exp Follow-Up Studies/

42. exp Prospective Studies/

43. exp Retrospective Studies/

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44. cohort.ti,ab.

45. longitudinal.ti,ab.

46. prospective.ti,ab.

47. retrospective.ti,ab.

48.38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47

49. 6 and 28 and 48

Embase

#1. 'sodium chloride'/exp OR 'sodium chloride'

#2. 'sodium intake'/exp

#3. 'salt intake'/exp

#4. 'sodium restriction'/exp

#5. 'high sodium intake'/exp

#6. 'high salt diet'/exp

#7. 'high sodium' OR 'low sodium' OR 'sodium alter*' OR 'sodium reduce*' OR 'sodium reduction' OR 'sodium restrict*' OR 'sodium intake*' OR 'sodium diet*' OR 'sodium food' OR 'sodium increas*' OR 'sodium decreas*' OR 'sodium change*' OR 'sodium changing' OR 'sodium free'

#8. 'high salt' OR 'low salt' OR 'salt alter*' OR 'salt reduce*' OR 'salt reduction' OR 'salt restrict*' OR 'salt intake*' OR 'salt diet*' OR 'salt food' OR 'salt increas*' OR 'salt decreas*' OR 'salt change*' OR 'salt changing' OR 'salt free'

#9. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8

#10. 'kidney'/exp OR 'nephrosis'/exp

#11. kidney OR renal OR nephropath* OR nephrosis

#12. 'chronic kidney' OR 'chronic renal'

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#13. 'kidney disease'/exp OR 'kidney failure'/exp

#14. 'kidney disease*' OR 'renal disease*' OR 'kidney failure' OR 'renal failure'

#15. 'kidney function test'/exp OR 'kidney function test*'

#16. ckf OR ckd OR crf OR crd

#17. 'end-stage kidney' OR 'end-stage renal' OR 'endstage kidney' OR 'endstage renal'

#18. esrf OR eskf OR esrd OR eskd

#19. 'uremia'/exp OR ur?emi*

#20. 'renal replacement therapy'/exp OR 'renal replacement therap*'

#21. 'renal insufficien*'

#22. 'dialysis'/exp OR dialys*

#23. 'pre dialysis' OR predialysis

#24. 'hemodialysis'/exp OR 'hemofiltration'/exp OR 'hemodiafiltration'/exp

#25. hemodialysis OR haemodialysis OR hemofiltration OR haemofiltration OR hemodiafiltration OR haemodiafiltration

#26. 'peritoneal dialysis'/exp OR 'peritoneal dialysis'

#27. pd OR capd OR ccpd OR apd OR ipd

#28. 'kidney transplantation'/exp OR 'kidney graft'/exp OR 'kidney allograft'/exp

#29. 'kidney transplant*' OR 'renal transplant*' OR 'kidney graft*' OR 'renal graft*' OR 'kidney allograft*' OR 'renal allograft*'

#30. #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29

#31. 'randomized controlled trial'/exp OR 'randomized controlled trial' OR 'controlled clinical trial'/exp OR 'controlled clinical trial' OR random*OR 'clinical trials'/exp OR 'clinical trials' OR trial:ti

#32. 'human'

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#33. animal$ OR nonhuman

#34. #32 AND #33

#35. #33 NOT #34

#36. #31 NOT #35

#37.'cohort analysis'/exp OR 'longitudinal study'/exp OR 'prospective study'/exp OR 'retrospective study'/exp

#38. (cohort OR concurrent OR incidence OR longitudinal OR followup OR 'follow up' OR prospective OR retrospective) NEXT/1 (analys* OR design* OR evaluation* OR research OR stud* OR survey* OR trial*)

#39. #36 OR #37 OR #38

#40. #9 AND #30 AND #39

CENTRAL

#1 MeSH descriptor: [Sodium Chloride] explode all trees

#2 "sodium chloride":ti,ab,kw (Word variations have been searched)

#3 MeSH descriptor: [Sodium, Dietary] explode all trees

#4 MeSH descriptor: [Diet, Sodium-Restricted] explode all trees

#5 ((sodium or salt) near/5 (low or high or alter* or reduce* or reducing or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)):ti,ab,kw

#6 (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free):ti,ab,kw

#7 (#1 or #2 or #3 or #4 or #5 or #6)

#8 MeSH descriptor: [Kidney] explode all trees

#9 MeSH descriptor: [Nephrosis] explode all trees

#10 (kidney or renal or nephropath* or nephrosis):ti,ab,kw

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#11 (chronic next kidney or chronic next renal):ti,ab,kw

#12 MeSH descriptor: [Kidney Diseases] explode all trees

#13 (kidney or renal) next (disease* or failure):ti,ab,kw

#14 MeSH descriptor: [Kidney Function Tests] explode all trees

#15 “kidney function test*”:ti,ab,kw

#16 (CKF or CKD or CRF or CRD):ti,ab,kw

#17 ("end-stage kidney" or "end-stage renal" or "endstage kidney" or "endstage renal"):ti,ab,kw

#18 (ESRF or ESKF or ESRD or ESKD):ti,ab,kw

#19 MeSH descriptor: [Uremia] explode all trees

#20 ur?emi*:ti,ab,kw

#21 MeSH descriptor: [Renal Replacement Therapy] explode all trees

#22 "renal replacement therap*":ti,ab,kw

#23 MeSH descriptor: [Renal Insufficiency] explode all trees

#24 "renal insufficien*":ti,ab,kw

#25 MeSH descriptor: [Dialysis] explode all trees

#26 dialys*:ti,ab,kw

#27 MeSH descriptor: [Renal Dialysis] explode all trees

#28 "renal dialys*":ti,ab,kw

#29 "pre dialysis" or pre-dialysis or predialysis:ti,ab,kw

#30 (h*emodialysis or h*emofiltration or h*emodiafiltration):ti,ab,kw

#31 MeSH descriptor: [Peritoneal Dialysis] explode all trees

#32 "peritoneal dialys*":ti,ab,kw

#33 (PD or CAPD or CCPD or APD or IPD):ti,ab,kw

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#34 MeSH descriptor: [Kidney Transplantation] explode all trees

#35 (kidney or renal) near (transplant* or graft* or allograft*):ti,ab,kw

#36 (#8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35)

#37 (#7 and #36)

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Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up, months	Outcome	Treatment group	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source	Quality assessment*
Mühlhauser I[19] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Sodium intake about 190 mmol/d	87.9[a]	132/101	Non- industry	4
Cianciaruso B[20] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium > 200 mmol/d	38	NA	NA	7★
Keven K[21] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Usual care	1.39c	143/88	NA	2
Fujiwara S[22] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	Usual care	65.6	140/67	Non- industry	7★
Yu W[23] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Usual diet	70.2	118/75	NA	7★
Vegter S[24] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine ≥ 200 mmol/g	42.5[b]	144/95	Non- industry	7★

Heerspink HJL[25] (2012)	Type 2 diabetic nephropathy	1177	65	59	Cohort study	30	Renal failure, CV events	Urinary sodium/creatinine < 153mmol/g	Urinary sodium/creatinine ≥ 153mmol/g	44	155/84	Industry and non- industry	7★
LowSALT CKD Study[26,27] (2012)	Hypertensive patients with stage 3–4 CKD	20	75	69	Randomised crossover trial	1	UACR, SBP, DBP	Sodium intake < 100 mmol/d	Sodium intake ≥ 220 mmol/d	32	151/82	Non- industry	5
de Brito-Ashurst I[28] (2013)	eGFR < 60 mL/min and mean BP > 130/80 mm Hg	48	58	58	Randomised parallel-group controlled trial	6	Change in eGFR	Intensive low-salt diet education	Usual care	41.5	153/85	Non- industry	3
Liang X[29] (2013)	PD	106	54	33	Cohort study	1	CV events, SBP, DBP	Health education to restrict water-sodium intake strictly	Usual care	NA	149/88	NA	7★
Mc Quarrie EP[30] (2014)	Patients with CKD and had at least one 24-hour urinary sodium measurement	423	50	52	Cohort study	102	Renal failure, all- cause mortality	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine > 16	48	139/79	Non- industry	8★
Fan L[4] (2014)	GFR 13–55 mL/min/1.73m^2	840	60	52	Cohort study	72	Renal failure, all- cause mortality	Urinary sodium < 3 g/d	Urinary sodium ≥ 3 g/d	32.5	132/81	Non- industry	6★
Baek SH[31] (2014)	Hypertensive patients with albuminuria	202	51	50	Cohort study	2	UAE, SBP, DBP	Good compliance with an low salt diet	Poor compliance with an low salt diet	77.1[c]	121/73	Industry and non- industry	6★

Hwang JH[32] (2014)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	245	50	50	Randomised parallel-group controlled trial	2	eGFR, UAE, SBP, DBP	Sodium intake < 100mmol/d	Usual care	67.3	131/79	Industry	3
Rodrigues Telini LS[33] (2014)	HD	39	38	58	Randomised parallel-group controlled trial	4	SBP, DBP	Sodium intake 2g/d	Usual diet	NA	146/86	Non- industry	2
Nerbass FB[34] (2015)	eGFR 59–30 ml/min/1·73 m^2	1416	39	73	Cohort study	12	eGFR, UACR, SBP, DBP	Sodium intake ≤ 100 mmol/d	Sodium intake > 100 mmol/d	52.7	134/73	Industry and non- industry	7★
CRIC Study[3,35] (2016)	eGFR20-70mL/min/1.73 m^2	3757	56	58	Cohort study	78	ESRD, all- cause mortality, CV events	Urinary sodium < 153.6mmol/d	Urinary sodium ≥153.7 mmol/d	44.5	128/71	Non- industry	8★
Mazarova A[5] (2016)	eGFR < 30 mL/min/1.73 m^2	341	62	64	Cohort study	18	Change in eGFR, renal failure, all-cause mortality	Urinary sodium ≤ 150mmol/d	Urinary sodium> 150 mmol/d	17.1	NA	No funding supported	7★
Kauric-Klein Z[36] (2017)	HD	121	50	60	Randomised parallel-group controlled trial	3	All- cause mortality, SBP, DBP	Sodium intake < 2g/d	Usual care	NA	165/87	Non- industry	2
Meuleman Y[37] (2017)	eGFR ≥ 20mL/min/1.73m^2 and UPE > 0.2 g/L	112	82	55	Randomised parallel-group controlled trial	6	eGFR, UPE, SBP, DBP	Usual care and low salt diet education	Usual care	49.7	129/76	Non- industry	3

Jiamjariyapon T[38] (2017)	Stage 3-4 CKD with diabetes or hypertension	418	27	62	Cluster randomized controlled trial	24	Renal failure, all-cause mortality, eGFR, CV events, UACR, SBP, DBP	Usual care and intensive low-salt diet education	Usual care	41.5	125/76	Non- industry	2
Saran R[39] (2017)	eGFR 15–60 ml/min/1.73 m^2	58	60	61	Randomized crossover trial	1	eGFR, UACR, SBP, DBP	Sodium intake < 2g/d	Usual care	38.7	130.3/NA	Non- industry	3
Huang B[40] (2017)	HD	90	43	55	Cluster randomized controlled trial	6	SBP, DBP	Self-management support, restrict salt intake	Usual care	NA	154/86	Non- industry	4
Ikenoue T[6] (2018)	HD	88115	60	65	Cohort study	12	All-cause mortality, CV events	Salt intake < 6 g/d	Salt intake ≥ 6 g/d	NA	NA	No funding supported	7★
Parvanova A[41] (2018)	Patients with type 2 diabetes and macroalbuminuria	108	89	64	Randomised parallel-group controlled trial	3	UACR	Sodium intake ≤ 100mmol/d	Sodium intake ≥ 200mmol/d	87.36	146/80	Industry	3
Martinez MG[42] (2019)	Pre-dialysis CKD with UPE > 500 mg/d	84	52	56	Cohort study	6	Change in SBP	Urinary sodium reduced	Urinary sodium increased	49[b]	142/NA	NA	8★
Nishimoto M[43] (2019)	UACR 30–599 mg/g and eGFR ≥ 50 ml/min/1.73 m^2	304	70	59	Cohort study	12	Change in eGFR, SBP, and DBP	Urinary sodium ≤ 189 mmol/d	Urinary sodium > 189 mmol/d	64.93	140/83	Non- industry	7★

Ahn SY[44] (2019)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	171	49	50	Cohort study	12	Rate of change in eGFR, renal failure, change in UAE, SBP, DBP	Intensive low-salt diet education	Usual care	64.2	124/75	Non- industry	7★
Hu J[45] (2019)	PD	33	58	42	Cohort study	12	Residual renal Ccr	Sodium intake ≤ 3.0 g/d	Sodium intake > 3.0 g/d	8.3	146/95	Non- industry	7★
Humalda JK[46] (2020)	CKD stages 1 to 4 or kidney transplants recipients, and eGFR was ≥ 25 mL/min/ 1.73 m2	99	84	57	RCT	6	SBP, DBP	Self-management to restrict salt intake	Usual care	55	139/83	Non- industry	3

Note: Ccr, clearance of creatinine; CKD, chronic kidney disease; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; ESRD, end stage renal disease; HD, hemodialysis; NA, no available; PD, peritoneal dialysis; SBP, systolic blood pressure; UACR, urinary albumin to creatinine ratio; UAE, urine albumin excretion; UPE, urinary protein excretion; the units for eGFR: mL/min/1.73m². * Quality was assessed by using the Jadad scale for randomized clinical trial and the Newcastle-Ottawa Scale (NOS) for cohort study; [a] The units for eGFR: mL/min, [b] Only Ccr obtained, the units for Ccr: mL/min; [c] Only serum creatinine obtained, the units for serum creatinine: μmol/L.

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Table S2 Summary of characteristics of included trials and patients.

Outcomes	No. of Studies/ No. of Patients	Sample Size	Follow-up, months	Age, years	Male (%)	Lost follow-up rate
Renal failure events	8/7596	462 (360-1093)	44 (26-77)	55 (52-62)	59.5%	3.6%
Rate of change in eGFR	11/3203	171 (57-341)	12 (6-18)	55(48-61)	56.1%	7.6%
Proteinuria	13/2922	108 (40-224)	3 (1-12)	50 (43-66)	57.2%	8,8%
All-cause death	7/94015	423 (341-3757)	24 (12-78)	60 (52-64)	60.6%	0.03%
CV events	5/93573	1177 (262-45936)	24 (6.5-54)	59(45-63)	60.8%	0.02%
SBP	19/3608	103 (38-213)	3 (1-6)	55 (41-60)	57.1%	7.5%
DBP	18/3524	36 (20-90)	12 (2-54)	55 (40-60)	57.3%	7.7%

Note:Values are given as median (quartile). CV, cardiovascular; eGFR, estimated glomerular filtration rate.

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Table S3 Assessment of risk of bias for RCTs.

Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
1. Bias arising from the randomization process																
1.1 Was the allocation sequence random?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias arising from the randomisation process?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
2. Risk of bias due to deviations from intended interventions																
2.1 Were participants aware of their assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	Other	High	High	High	High	Low	High	High
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	High	High	High	High	High	High	High	High
2.3 If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?	N/PN	Y/PY	NA/NI		Low		Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.4 If Y/PY/NI to 2.3: Were these deviations likely to have affected the outcome?	N/PN	Y/PY	NA/NI													
2.5 If Y/PY to 2.4: Were these deviations from intended intervention balanced between groups?	Y/PY	N/PN	NA/NI													

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomised?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to deviations from intended interventions?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
3. Bias due to missing outcome data																
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	High	High	High	High	Low	High	High	Low
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?	Y/PY	N/PN	NA	Low	Low				Low	Low	Low	Low		Low	Low	
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	N/PN	Y/PY	NA/NI													
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to missing outcome data?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
4. Bias in measurement of the outcome																
4.1 Was the method of measuring the outcome inappropriate?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?	N/PN	Y/PY	NI	Low	High	Low	High	High	Low	High	High	High	Low	High	Low	High
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI		Low		Low	Low		Low	Low	Low		Low		Low
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias in measurement of the outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
5. Bias in selection of the reported result																
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:																
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
5.3 ... multiple eligible analyses of the data?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction bias due to selection of the reported results?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
Overall bias																
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the overall predicted direction of bias for this outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null

Note: Y, yes; PY, probably yes; PN, probably no; N, no; NA, not applicable; NI, no information; RCT, randomized controlled trial.

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Table S4 Quality assessment for cohort studies.

Study (Year)	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Scores
Cianciaruso B[20] 1998	★	★	★	★	—	★	★	★	7★
Fujiwara S[22] 2010	★	★	★	★	★	★	—	★	7★
Yu W[23] 2011	★	★	★	★	★	★	—	★	7★
Vegter S[24] 2012	★	★	★	★	—	★	★	★	7★
Heerspink HJL[25] 2012	★	★	★	★	★	★	—	★	7★
Liang X[29] 2013	★	★	★	★	★	★	—	★	7★
Mc Quarrie EP[30] 2014	★	★	★	★	★	★	★	★	8★
Fan L[4] 2014	★	★	★	★	—	★	★	—	6★
Baek SH[31] 2014	★	★	★	★	—	★	—	★	6★
Nerbass FB[34] 2015	★	★	★	★	★	★	—	★	7★
CRIC Study[3,35] 2016	★	★	★	★	★	★	★	★	8★
Mazarova A[5] 2016	★	★	★	★	★	★	—	★	7★
Ikenoue T[6] 2018	★	★	★	★	★	★	—	★	7★
Martinez MG[42] 2019	★	★	★	★	★	★	★	★	8★

Nishimoto M[43] 2019	★	★	★	★	★	★	—	★	7★
Ahn SY[44] 2019	★	★	★	★	★	★	—	★	7★
Hu J[45] 2019	★	★	★	★	★	★	—	★	7★

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability. —: no score; It is considered that the queue follow-up is appropriate when the follow-up rate is more than 80%.

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Table S5 Subgroup analysis of outcomes except renal failure events.

Subgroup	No. of trials	n	RR/MD/SMD (95%CI)	*P* for RR/MD/SMD	I^2	P for heterogeneity test[a]
Change in eGFR（mL/min/1.73m²）						
Study type						
RCT	5	881	-0.75 (-7.58, 6.08)	0.83	96.2%	0.42
Cohort studies	6	2322	1.61 (-1.36, 4.59)	0.29	98.1%	
Baseline eGFR (mL/min/1.73m²)						
60-89	3	720	-3.82 (-10.29, 2.65)	0.25	98.1%	0.18
30-59	6	2109	2.58 (-0.44, 5.61)	0.10	95.4%	
<15	2	374	1.24 (-0.88, 3.37)	0.25	66.2%	
Mean age (years)						
<56	5	106	0.44 (-4.40, 5.27)	0.86	98.5%	0.95
≥56	6	93467	0.19 (-0.49, 0.86)	0.59	0.0%	
Baseline BP (mmHg)						
<140/90	6	2420	-0.52 (-4.36, 3.32)	0.79	95.8%	0.73
SBP≥140 或 DBP≥90	3	385	0.72 (-1.62, 3.06)	0.55	70.1%	
Proteinuria or Albuminuria						
Study type						
RCT	7	997	-1.03 (-2.28, 0.21)	0.11	98.4%	0.45
Cohort studies	6	1925	-0.28 (-0.59, 0.02)	0.07	80.8%	
Baseline eGFR (mL/min/1.73m²)						
60-89	7	821	-1.06 (-2.32, 0.20)	0.10	98.2%	0.42
30-59	6	1201	-0.23 (-0.50, 0.04)	0.10	79.6%	
Mean age (years)						
<56	7	846	-1.12 (-2.35, 0.12)	0.08	98.2%	0.34
≥56	6	2076	-0.13 (-0.34, 0.09)	0.24	64.2%	
Baseline BP (mmHg)						
<140/90	8	200	-0.82 (-1.55, -0.10)	0.01	98.3%	0.65
SBP≥140 或 DBP≥90	4	2665	-0.39 (-0.68, -0.11)	0.03	0.0%	
All-cause mortality						
Study type						
RCT	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.98
Cohort studies	5	93476	0.88 (0.44, 1.75)	0.72	99.0%	
Baseline eGFR (mL/min/1.73m²)						
30-59	4	5438	0.65 (0.41, 1.02)	0.06	85.2%	0.02
15-29	1	341	1.08 (0.31, 3.77)	0.90	-	
<15	2	88236	1.77 (0.53, 5.90)	0.35	38.9%	
Mean age (years)						
<56	2	704	0.49 (0.38, 0.64)	<0.001	0.0%	0.07
≥56	5	92752	1.23 (0.55, 2.73)	0.62	98.9%	
Baseline BP (mmHg)						
<140/90	4	5438	0.65 (0.41, 1.02)	0.06	85.2%	0.86
SBP≥140 或 DBP≥90	1	121	0.51 (0.05, 5.46)	0.58	-	

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CV events						
Study type						
RCT	1	418	0.47 (0.09, 2.53)	0.88	-	0.51
Cohort studies	4	93155	1.05 (0.51, 2.18)	0.89	98.2%	
Baseline eGFR (mL/min/1.73m^2)						
30-59	3	5352	0.78 (0.63, 0.95)	0.02	57.1%	**0.02**
<15	2	88221	1.50 (0.39, 5.85)	0.56	58.3%	
Mean age (years)						
<56	1	106	0.51 (0.08, 3.50)	0.50	-	0.60
≥56	4	93467	1.03 (0.50, 2.12)	0.94	98.2%	
Baseline BP (mmHg)						
<140/90	2	4175	0.71 (0.62, 0.80)	<0.001	0.0%	0.62
SBP≥140 或 DBP≥90	2	1283	0.88 (0.74, 1.05)	0.17	0.0%	
SBP（mmHg）						
Study type						
RCT	11	1246	-3.54 (-6.40, -0.68)	0.02	65.6%	0.43
Cohort studies	8	2362	-7.22 (-11.69, -2.76)	0.002	93.6%	
Baseline eGFR (mL/min/1.73m^2)						
60-89	7	1017	-3.30 (-5.26, -1.34)	0.001	31.7%	0.64
30-59	7	2210	-6.10 (-11.62, -0.59)	0.03	92.8%	
< 15	4	349	-5.66 (-11.62, 0.30)	0.06	76.0%	
Mean age (years)						
< 56	9	1010	-4.79 (-8.00, -1.59)	0.003	70.3%	0.82
≥ 56	10	2598	-5.85 (-9.78, -1.93)	0.003	91.7%	
Baseline BP (mmHg)						
< 140/90	10	2786	-0.97 (-2.75, 0.81)	0.28	46.9%	**0.02**
SBP ≥ 140 or DBP ≥ 90	9	822	-10.24 (-15.37, -5.12)	< 0.001	88.4%	
DBP（mmHg）						
Study type						
RCT	7	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.85
Cohort studies	11	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	
Baseline eGFR (mL/min/1.73m^2)						
30-59	7	1017	-1.74 (-3.40, -0.08)	0.04	56.7%	0.13
15-29	6	2126	-0.53 (-1.25, 0.20)	0.16	11.9%	
< 15	4	349	-4.94 (-7.08, -2.80)	<0.001	23.2%	
Mean age (years)						
< 56	9	1010	-3.11 (-4.81, -1.40)	<0.001	56.4%	0.19
≥ 56	9	2514	-0.81 (-1.65, 0.03)	<0.001	52.3%	
Baseline BP (mmHg)						
< 140/90	9	2747	-0.94 (-1.90, 0.02)	0.056	5.6%	**0.04**
SBP ≥ 140 或 DBP ≥ 90	9	777	-4.05 (-6.61, -1.49)	0.002	0.2%	

Note: [a] *P* value calculated by meta regression. BP, blood pressure; CI, confidence interval; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

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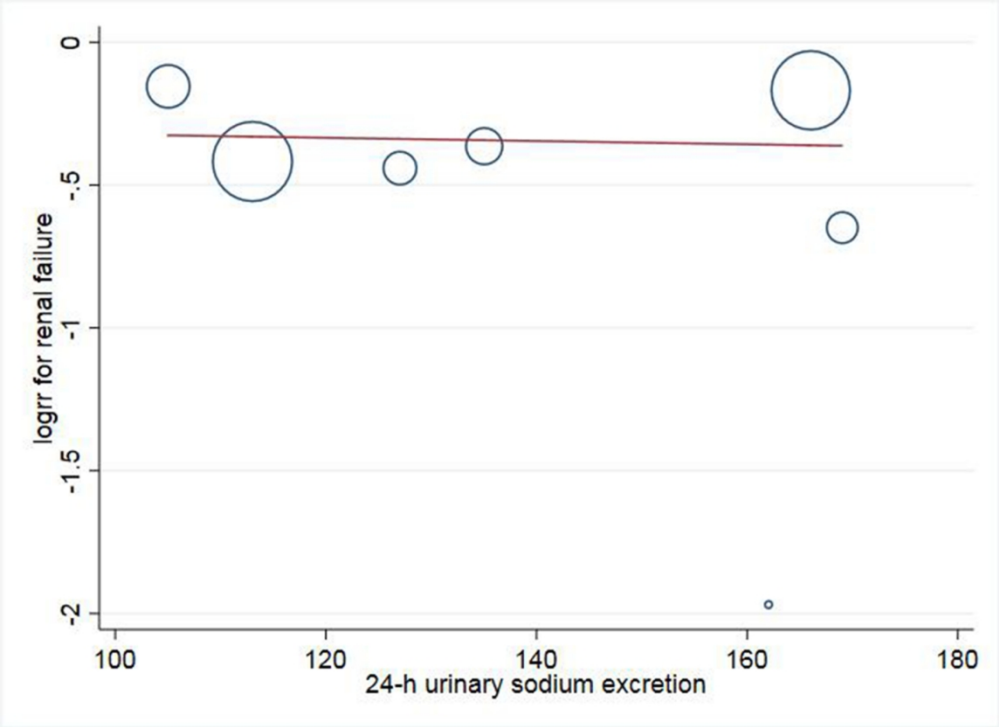
Table S6 Sensitivity analysis of outcomes.

Sensitivity analysis	Renal failure events			The rate of change in eGFR			Proteinuria or albuminuria			All-cause mortaity		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD/SMD (95%CI)	τ^2	No./n	RR (95% CI)	τ^2
Omit studies of follow-up time <12 months	-	-	-	7/2740	1.60 (-1.10, 4.31)	12.78	-	-	-	6/93894	0.91 (0.48, 1.73)	0.54
Omit studies of sample size <50	-	-	-	9/3122	0.30 (-2.82, 3.42)	21.19	9/2787	-0.89 (-1.56, -0.23)	0.99	-	-	-
Omit studies of Jadad score <3 or NOS<5★	7/7209	0.66 (0.53, 0.83)	0.06	10/2785	0.44 (-2.51, 3.40)	21.01	12/2504	-0.78 (-1.42, -0.15)	1.18	5/93476	0.88 (0.44, 1.75)	0.55
Omit studies with extreme outliers	-	-	-	10/2958	1.86 (-0.44, 4.16)	12.20	12/2677	-0.24 (-0.43, -0.05)	0.06	10/5900	0.67 (0.45, 1.00)	0.13
Different statistical estimators												
DL	8/7596	0.66 (0.54, 0.80)	0.05	11/3203	0.55 (-2.19, 3.29)	19.82	13/2922	-0.70 (-1.24, -0.1)	0.89	7/94015	0.88 (0.47, 1.65)	0.54
EB	8/7596	0.66 (0.54, 0.82)	0.03	11/3203	0.55 (-2.68, 3.77)	21.36	13/2922	**-0.71(-1.66, 0.24)**	2.39	7/94015	0.89 (0.49, 1.63)	0.29
REML	8/7596	0.66 (0.54, 0.82)	0.03	11/3203	0.55 (-2.69, 3.79)	21.58	13/2922	**-0.71(-1.66, 0.24)**	2.38	7/94015	0.89 (0.45, 1.76)	0.39

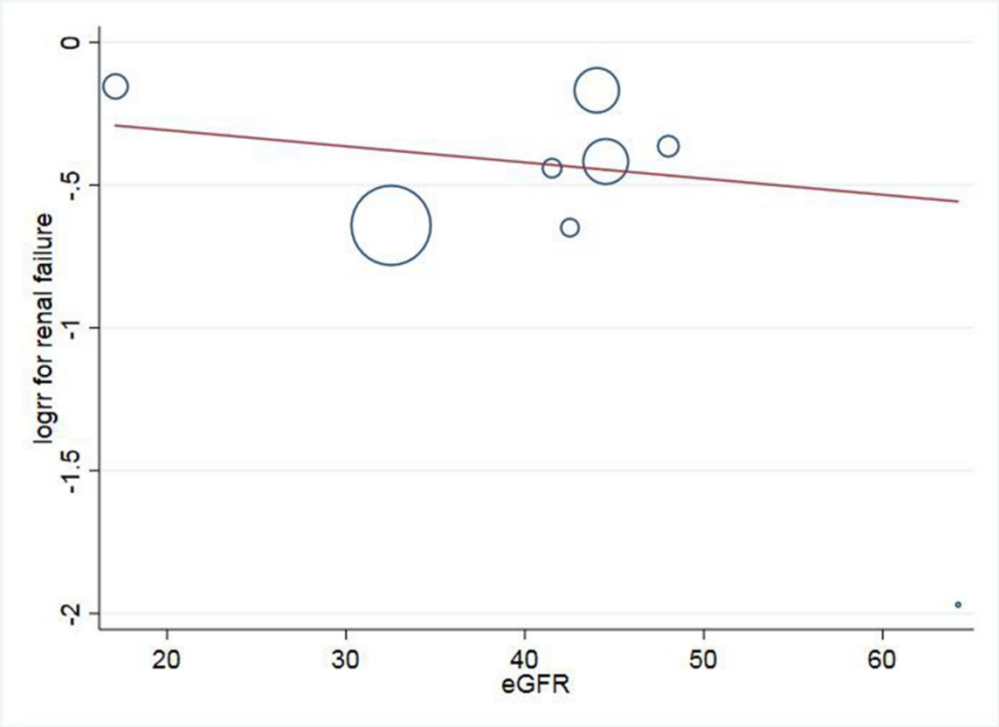
Sensitivity analysis	CV events			SBP			DBP		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD (95%CI)	τ^2
Omit studies of follow-up time <12 months	4/93467	1.03 (0.50, 2.12)	0.45	-	-	-	-	-	-
Omit studies of sample size <50	-	-	-	13/3402	-4.05 (-6.93, -1.17)	22.57	12/3318	-1.45 (-2.34, -0.56)	1.05
Omit studies of Jadad score <3 or NOS<5★	4/93155	1.05 (0.51, 2.18)	0.45	15/2998	-5.64 (-8.49, -2.79)	23.26	14/2914	-2.16 (-3.33, -0.99)	2.33
Different statistical estimators									
DL	5/93573	0.97 (0.49, 1.93)	0.45	19/3608	-5.34 (-7.93, -2.75)	22.80	15/3524	-2.00(-2.93, -1.06)	1.56
EB	5/93573	1.01 (0.46, 2.22)	0.28	19/3608	-5.81(-9.63, -1.99)	14.19	15/3524	-2.30(-3.61, -0.98)	3.78
REML	5/93573	0.99 (0.41, 2.37)	0.35	19/3608	-5.78(-9.59, -1.97)	52.37	15/3524	-2.27(-3.57, -0.96)	3.41

Note: CI, confidence interval; CV, cardiovascular; DL, DerSimonian-Laird; EB, empirical Bayes; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; No., number of trials; NOS, Newcastle-Ottawa Scale; REML, restricted maximum likelihood; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

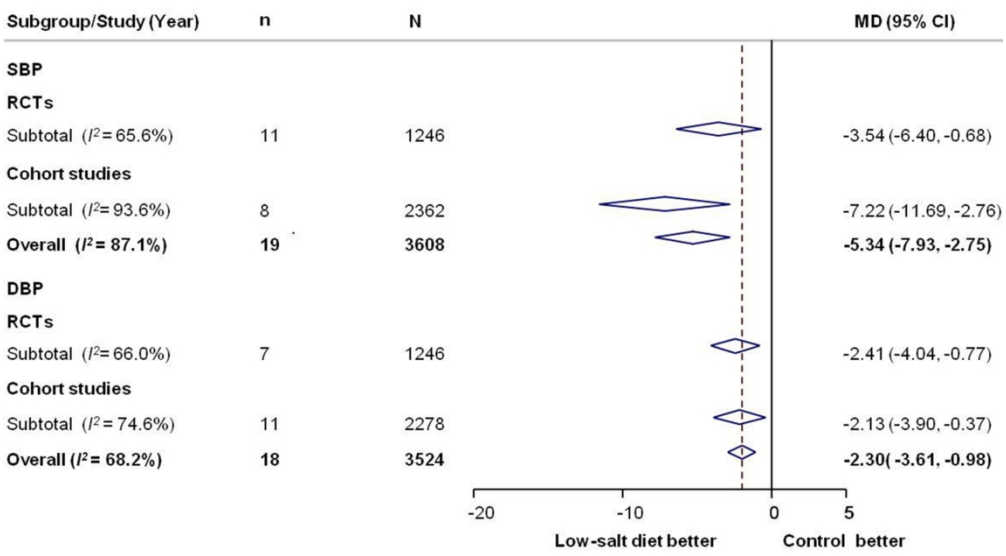
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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2-3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5-6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5-6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5-6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5-6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6-7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7-10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7-10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14-15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	15

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Effects of Low-salt Diet Intervention on Chronic Kidney Disease Outcomes: A meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050843.R1
Article Type:	Original research
Date Submitted by the Author:	12-Aug-2021
Complete List of Authors:	Shi, Honghong; Shanxi Medical University Second Affiliated Hospital, Renal Division Su, Xiaole; Shanxi Medical University Second Affiliated Hospital Li, Chunfang; Shanxi Cardiovascular Hospital Guo, Wenjuan; Shanxi Medical University Second Affiliated Hospital Wang, Lihua; Shanxi Medical University Second Affiliated Hospital
Primary Subject Heading:	Renal medicine
Secondary Subject Heading:	Health services research
Keywords:	Nephrology < INTERNAL MEDICINE, Chronic renal failure < NEPHROLOGY, HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effects of Low-salt Diet Intervention on Chronic Kidney Disease Outcomes: A meta-analysis

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ABSTRACT

Objective The benefits of low-salt diet intervention for CKD patients are still controversial. We performed a meta-analysis to investigate the effects of low-salt diet on major clinical outcomes.

Design Systematic review and meta-analysis.

Data sources MEDLINE by Ovid, EMBASE, and the Cochrane Library database.

Eligibility criteria for selecting studies We included RCTs and cohort studies that evaluated the effects of low-salt diet on the renal composite outcomes (more than 25% or 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD), decline rate in eGFR, changes in proteinuria, all-cause mortality events, cardiovascular (CV) events, change in SBP and DBP.

Data extraction and synthesis Two independent reviewers extracted data, and assessed quality. RR with 95% CI were used for dichotomous data . Difference in means (MD) or standardized mean differences (SMD) with 95% CI were used to pool continuous outcome data. We used Cochrane Collaboration risk-of-bias tool to assess quality of RCTs, and NOS to assess quality of cohort studies.

Results We identified 9948 potentially records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. Thirty-three studies with 101077

participants were included. Low-salt diet produced a 28% reduction in renal composite outcomes events (RR: 0.72; 95%CI: 0.58-0.89) compared with control groups. But no significant effects were observed on change of proteinuria (SMD: -0.71; 95% CI: -1.66 to 0.24), decline rate of eGFR (MD: 1.16; 95% CI: -2.02 to 4.33), the risk of all-cause mortality (RR: 0.92; 95% CI: 0.58-1.46), and CV events (RR: 1.01; 95% CI: 0.46-2.22).

Conclusion Low-salt diet seems to reduce the risk of renal composite outcomes events in CKD patients. However, no compelling evidence indicated that low-salt diet would reduce rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. More definitive studies are needed.

PROSPERO registration number CRD42017072395.

Strengths and limitations of this study

This study is the first meta-analysis to evaluate the effect of low-salt diet intervention on hard clinical outcomes, such as renal composite outcomes, all-cause mortality and CV events for CKD patients.

No study design or language restriction, therefore allowing inclusion of the maximum number of studies in this review.

The existence of significant heterogeneity might limit the interpretation and clinical application of the results.

Confounding factors in the observational studies cannot be neglected, which might impact our results.

INTRODUCTION

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Chronic kidney disease (CKD) has become a global public health burden. Dietary interventions that aim to delay the progression of CKD and decrease complications have a pivotal role in the management of CKD. Restriction of dietary sodium intake can augment the antiproteinuric effects of angiotensin converting enzymes inhibitors/angiotensin receptor blockers (ACEIs/ARBs) [1] and decrease blood pressure (BP) [2]. In the 2020 KDIGO guidelines for glomerulonephritis, dietary restriction of sodium to < 2 g/d is a primary tenet for controlling BP and edema and improving urinary protein excretion independently of medications. However, there remains an open question of whether these effects can translate into a worthwhile absolute risk reduction in patients with CKD, including kidney failure events, cardiovascular (CV) events and all-cause mortality. In the Chronic Renal Insufficiency Cohort (CRIC) Study, recruiting patients eGFR between 20 and 70 ml/min/1.73 m^2 depending on age, compared with the lowest quartile of urinary sodium excretion (<116.8 mmol/24 h), high urinary sodium excretion (>194.6 mmol/24 h) was associated with a 54% risk increase in end-stage renal disease (ESRD) or halving of estimated glomerular filtration rate (eGFR) and with a 43% risk increase in all-cause mortality [3]. Contrasting results have been observed in other cohort studies, in which sodium intake does not appear to impact the progression of CKD to ESRD and a decline in eGFR [4,5]. Furthermore, a low salt intake of <6 g/day was associated with an increased risk of all-cause mortality and CV mortality in hemodialysis patients, resulting in an L-shaped association curve [6]. According to data from the NHANE II study, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans [7]. Randomized controlled trials (RCTs) evidence that discussed the effect of a low-salt diet in CKD patients on kidney failure events, CV events and all-cause mortality are controversial. It is difficult to

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base clinical decisions on findings from these studies, given their small sample sizes, highly selected participants, short-term follow-up and lack of assessment of hard clinical outcomes.

Thus, the association between dietary salt restriction and clinical outcomes in patients with CKD remains unclear. We performed this systematic review to summarize all eligible clinical studies data and evaluate the effects of low-salt diet intervention on kidney disease outcomes, all-cause mortality and CV events in CKD patients.

MATERIALS AND METHODS

Data sources and search strategy

We performed this systematic review according to a pre-specified protocol registered at the International Prospective Register of Systematic Reviews (CRD42017072395) (Supplementary file 1) [8], and reporting was done in line with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines [9]. A comprehensive search was conducted using the following database: MEDLINE by Ovid (from 1946 to July 2021), EMBASE (from 1966 to July 2021), and Cochrane Central Register of Controlled Trials (no date restriction), with relevant keywords and medical subject headings that included all spellings of “CKD”, “RCT”, “Cohort Studies”, “Sodium Chloride”, “Sodium”, “Sodium-Restricted” (see Supplementary file 2 for full search terms). Studies were considered without any language restriction. To ensure a comprehensive literature search, we also screened reference lists from included articles. The ClinicalTrials.gov website was also searched for studies the ongoing, but unpublished in this field.

Study selection and outcome estimation

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We included data from RCTs and cohort studies in which low-alt diet was given to adults with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) compared with usual therapy or different salt intake. Salt reduction was defined as that recommended by guidelines sodium <2.3 g/day (<100 mmol of sodium, <6g of salt), or as defined by the authors of each study.

Pre-defined outcomes that contained analyzable data were listed as follows: First, renal composite outcomes events, defined as a more than 25% or 50% decline in eGFR from baseline during follow-up [10], doubling of serum creatinine or ESRD. Second, the rate of change in eGFR per year. Third, changes in urinary protein or urinary albumin during follow-up, including urinary protein excretion, urinary albumin excretion and urinary albumin/creatinine ratio. Fourth, all-cause mortality events. Fifth, CV events, defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, CV disease and CV death. Fifth, change in systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Data extraction and quality assessment

Two independent reviewers (H.S. and X.S.) extracted data, and assessed quality, according to the pre-specified protocol. Disagreements were resolved through a third reviewer (L.W.). Data of all eligible studies was extracted into a spreadsheet. The data sought included characteristics of studies (study type, randomization method, follow-up time, withdrawals/dropouts), baseline patient characteristics (age, gender, baseline eGFR, baseline BP), intake of salt, and outcome events.

We used Cochrane Collaboration risk-of-bias tool [11] to assess all potential sources of bias for the included RCTs. Trails were assessed as being at low, other, some concerns, or high, and the overall risk of bias generally corresponds to the worst risk of bias in any of the domains. However, if

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a study was judged to have some concerns about risk of bias for multiple domains, it might be judged as at high risk of bias overall. We used the Newcastle-Ottawa Scale (NOS) to assess quality of cohort studies. The NOS contains eight items, categorized into three dimensions, including selection of cohorts, comparability of cohorts, and assessment of outcome[12]. A star system is used to allow a semi-quantitative assessment of study quality, such that the highest quality studies are awarded a maximum of one star for each item with the exception of the item related to comparability that allows the assignment of two stars. The NOS ranges between zero up to nine stars.

Data synthesis and statistical analyses

When dichotomous outcome data from individual trials were analyzed, relative risks (RRs) measure and its 95% confidence intervals (CIs) were used. If individual study RR was unavailable in the original article, RR and 95% CI were calculated from event numbers extracted from each study before data pooling. In calculating the RR values, we used the total number of patients randomized in each group as the corresponding denominator. Continuous outcome data from individual trials were analyzed using the difference in means (MD) with 95% CI to pool eGFR, and the standardized mean differences (SMD) with 95% CI were used to pool the proteinuria or albuminuria data. When continuous outcome data were analyzed, the difference in mean change between baseline and end of treatment value were used. If data for change between baseline and end of treatment were not available in the studies, we calculated it using correlations that were estimated from other included studies that had a similar follow-up period and reported in considerable detail according to the imputed formulation and its related interpretations in Cochrane Handbook [13].

In consideration of poor stability of the Der Simonian-Laird procedure for small numbers of studies [14], we used the empirical Bayes procedure [15] toestimate all outcomes. We also used Der

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Simonian-Laird random effects model [16] and restricted maximum likelihood [17] to assess the summary effects as a part of the sensitivity analyses. Considering the inevitable heterogeneity between studies, subgroup analyses, meta-regression, and sensitivity analyses were performed. Subgroup analyses were performed according to pre-specified protocol, including study type, baseline eGFR, and baseline BP, and comparator of control group. In addition, we performed sensitivity analyses by using different random-effects estimation methods as above mentioned, excluding studies with sample size <50, studies with follow-up time <12 months, studies with low quality (high risk of overall bias for RCTs, NOS <5★ for cohort studies), and studies with extreme outliers. Heterogeneity between studies was evaluated by using I^2 statistics or τ^2 statistics. The I^2 value of 25%, 50%, 75% respectively represent low, moderate, and high heterogeneity. Publication bias was assessed using Egger's test and Begg's test, and *P*-value < 0.05 indicated obvious publication bias. Stata version 15.0 (StataCorp LP, College Station, TX, USA) was used for statistical analyses , and a two-sided *P*-value < 0.05 was considered statistically significant.

Patient and Public Involvement statement

No patients and the public were involved in the development of the research question, selection of the outcome measures, design and implementation of the study, or interpretation of the results.

RESULTS

Overview of included trials

We identified 9948 potentially relevant records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. As shown in figure 1, Supplementary file 3 and Supplementary file 4, a total of 33 eligible studies [3-6, 18-48] including

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13 RCTs [18,20,25-27,31,32,35-40,45] and 19 cohort studies [3-6,19,21-24,28,30,33,34,41-44,46-48] reported in 35 publications with 101077 participants, were included in this review. The median follow-up duration was 6 months (interquartile range 3 to 21). Participants were enrolled at an average age of 56 years, and male participants accounted for 60% of the total participants. The average eGFR of the included participants was 47.45 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and twenty-one trials compared different levels of sodium intake.

Figure 1 PRISMA flow chart for the included studies.

The risk of bias varied substantially across the RCTs. The results from the Cochrane Collaboration risk-of-bias tool are shown in Supplementary file 5. In all the RCTs, bias arising from the randomization process was assessed as low risk in 61.54% of the trails; risk of bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result were low risk in all the trails. In terms of overall bias, 61.54% of the trails were assessed as low risk, and 38.46% of the trails were assessed as having some concerns.

According to the NOS checklist, all included cohort studies were generally high quality and awarded 6★- 8★ (Supplementary file 6).

Effect of low-salt diet intervention on renal outcomes

One RCT and 9 observational studies including 9578 patients with 2329 events compared the low-salt diet with the control in preventing renal composite outcomes events. As shown in figure 2, compared with the control, low-salt diet intervention produced a 34% reduction in the risk of renal composite outcomes events (RR: 0.72; 95% CI: 0.58, 0.89), with substantial heterogeneity (I^2

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=87.2%, P <0.001). Subgroup analyses and meta-regression were conducted to discuss the source of heterogeneity. No significant difference between RCTs and observational studies was found, while the effect sizes were greater in studies that enrolled patients aged < 58 years (P = 0.03) (table 1). On univariate meta-regression, no clear relationship between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcomes events was observed (Supplementary file 7, Supplementary file 8).

Figure 2 Forest plot for renal composite outcomes events. Renal failure was defined as a more than 25% or 50% decline in eGFR from baseline during follow-up, doubling of serum creatinine or ESRD. CI, confidence interval; RR, relative risk.

Table 1 Subgroup Analysis of Renal Composite Outcomes Events.

Subgroup	No. of Trials	n	RR (95% CI)	*P* for RR	I^2	P for Heterogeneity Test[a]
Study type						
RCT	1	418	0.64 (0.43, 0.98)	0.04	-	0.76
Cohort study	9	9160	0.72 (0.57, 0.90)	0.005	88.6%	
Baseline eGFR (mL/min/1.73m²)						
60-89	1	171	0.14 (0.02, 1.14)	0.07	-	0.48
30-59	7	8869	0.71 (0.55, 0.92)	0.009	90.8%	
15-29	2	538	0.77 (0.63, 0.94)	0.01	0.0%	
Mean age (years)						
<58	5	5691	0.58 (0.49, 0.70)	<0.001	56.4%	**0.03**
≥58	5	3887	0.85 (0.69, 1.05)	0.14	70.1%	
Baseline BP (mmHg)						
<140/90	6	5806	0.62 (0.52, 0.73)	<0.001	58.5%	0.32
SBP≥140 or DBP≥90	2	1646	0.70 (0.44, 1.11)	0.13	74.6%	
Comparator of control group						
Usual care or usual diet	2	589	0.43 (0.11, 1.64)	0.21	50.2%	0.12
Different levels of sodium intake	8	8989	0.73 (0.58, 0.92)	0.008	89.8%	

Note: a *P* value calculated by meta regression. BP,blood pressure; CI, confidence interval; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk.

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Five RCTs and 7 observational studies with a total of 3287 participants reported data regarding the effects of a low-salt diet on the rate of change in eGFR. Low-salt diet therapy did not produce an apparent beneficial effect on the rate of change in eGFR (MD:1.16; 95% CI: -2.02 to 4.33). There was evidence of significant heterogeneity across the included studies (I^2= 98.0%, P< 0.001) (figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to prespecified characteristics (Supplementary file 9).

Figure 3 Forest plot for rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; MD, mean difference; SD, standard deviation.

Data for the effects of low-salt diet intervention on proteinuria or albuminuria were available in 13 studies including 2922 participants. The SMD for change in proteinuria or albuminuria was not statistically significant at -0.71 (95% CI: -1.66 to 0.24) compared with that in the control groups, with significant heterogeneity (I^2 = 97.3%, P=0.009) (figure 4). No statistical heterogeneity was found in any subgroup analyses for the effects of a low-salt diet on proteinuria or albuminuria (Supplementary file 9).

Figure 4 Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; SD, standard deviation; SMD, standard mean difference.

Effect of low-salt diet intervention on all-cause mortality and CV events

Data for the effects of low-salt diet intervention on all-cause mortality and CV events were available for 9 studies (two RCTs and seven observational studies) and 5 studies (one RCT and four observational studies), respectively. There were no clear effects of a low-salt diet on the risk of all-cause mortality (RR: 0.92; 95% CI: 0.53-1.46; 95997 participants and 4455 events) or CV events

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(RR: 1.01; 95% CI: 0.46-2.22; 93573 participants and 1989 events) compared with those in the control groups, with significant heterogeneity across the included studies (I^2 = 98.0%, 97.6% for all-cause mortality and CV events respectively, and *P* values for heterogeneity both less than 0.001) (figure 5). More significant benefits were found in studies that included patients with an eGFR of 30-59 mL/min/1.73 m^2 (*P* for subgroup heterogeneity = 0.02) in the subgroup analysis of CV events (Supplementary file 9).

Figure 5 Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death. CI, confidence interval; N, number of trials; RR, relative risk.

Effect of low-salt diet intervention on BP

Twenty studies reported the effect of a low-salt diet on BP, including 19 studies (3608 individuals) for SBP and 18 studies (3608 participants) for DBP. The MD changes in SBP and DBP were both statistically significant at -5.81 mmHg (95% CI: -9.63 to -1.99) and -2.3 mmHg (95% CI, -3.61 to -0.98), respectively (Supplementary file 10). Significant heterogeneity across the included studies was found (I^2 = 87.1%, 68.2% for SBP and DBP respectively, and *P* values for heterogeneity both less than 0.05). Subgroup analyses showed that the association between the change in SBP or DBP and a low-salt diet was modified by the baseline BP of patients. Patients with baseline SBP >140 mmHg or DBP > 90 mmHg had an average reduction in SBP of 10 mmHg and in DBP of 5 mmHg (Supplementary file 9).

Sensitivity analyses and publication bias

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All sensitivity analyses results were robust (Supplementary file 11). The most notable exception was that the effect of low-salt diet therapy on the rate of change in eGFR became significant when studies with extreme outliers were omitted, and proteinuria or albuminuria became significant when the DL method was used to pool data. The risk of publication bias was found to be significant for the rate of change in eGFR and BP (Supplementary file 12).

DISCUSSION

Behavioral modifications can make an important contribution to delaying CKD progression to ESRD, and dietary salt restriction is one of the main concerns. Our meta-analysis showed that compared with control groups, low-salt diet intervention seems to reduced renal composite outcomes events and lowered BP (SBP and DBP) in CKD patients. No significant effects were observed regarding the decline rate of eGFR, change in proteinuria, risk of all-cause mortality or CV events. These results remained broadly consistent across major subgroups and sensitivity analyses. However, the existence of significant heterogeneity might limit the interpretation and clinical application of these results.

The previous review of 16 studies in 2006 addressing salt intake and kidney disease set out to establish whether variations in dietary sodium consumption influence kidney outcomes in patients with CKD [49]. The review suggested that increased salt intake was associated with worsening albuminuria and an increased likelihood of a reduction in GFR. In 2015 [50] and 2018 [51], the two meta-analyses of RCTs both suggested that moderate dietary salt restriction significantly reduced BP and proteinuria. However, it remains unanswered whether these benefits may translate to clinically significant reductions in renal composite outcomes events, all-cause mortality events and CV events.

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Our study, including 33 studies (13 RCTs and 20 observational studies), more than 90,000 patients, and 2329 renal composite outcomes events, first concluded that compared with the control groups, a low-salt diet produced a significant 28% reduction in renal composite outcomes events despite marked heterogeneity. Furthermore, the significant effect of a low-salt diet on BP (decreases in SBP and DBP of 5.8 mmHg and 2.3 mmHg, respectively) was confirmed again in our study and was recognized as an important target to slow the progression of CKD towards ESRD. The benefit of proteinuria from low-salt diet intervention became nonsignificant in our study when the restricted maximum likelihood or empirical Bayes methods were used to pool data. The standard DerSimonian-Laird procedure that was applied in previous meta-analyses can be very unstable with small numbers of studies [52,53].

Of note, statistical heterogeneity was found in all outcomes, and clinical heterogeneity was also an inevitable issue. Low-salt diets are not palatable to many [54], therefore, it is difficult to ensure sustained adherence to a low-sodium diet over long periods of follow-up. Sodium intake is quantified in distinct ways between studies (for example, 24 h urine collection, spot urine collection or dietary questionnaire). Given the use of diuretic may influnence the urine sodium excretion, and eGFR, and it would influence the accuracy of quantified salt intake. We extracted data about use of diuretic. Ten studies have the data about diuretic use in the study we included, but only about the percentages of patients using diuretics in the intervention and control group. The percentage of patients using diuretics was no difference in most studies. There were no data about type and dose of diuretics, or the associations with the eGFR. These inherent problems were reflected in the paradoxically opposite findings of various studies, especially observational studies with more confounding factors. Only three studies (1 RCT and 2 observational studies) simultaneously reported the outcomes of eGFR and

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renal composite outcomes events. Furthermore, the different follow-up periods with 52 months of renal composite outcomes events and 12 months of eGFR should be noted. The inconsistent results of eGFR and renal composite outcomes events may partly be explained by the heterogeneity mentioned above. It should be assumed that heterogeneity limited the interpretation and generalizability of our results.

In CKD, BP is usually sodium sensitive, and RAAS blockers are considered the first-line therapy for hypertension and proteinuria. It has been known for over 2 decades that the effects of RAAS blockers are blunted by high sodium intake [55-57]. Thus, the significant effect of dietary sodium restriction on lowering BP and supporting the function of RAAS blockers was speculated to play important roles in reducing the risk of kidney failure events. Interestingly, a BP-independent effect of dietary sodium on the kidney is substantiated by data in healthy volunteers, in which dietary sodium restriction reduces albuminuria within the normal range, without a detectable effect on BP [58]. Similarly, proteinuria reduction by sodium restriction remained significant after adjustment for the decrease in BP [57]. These results suggested an independent renoprotective effect of low-sodium diets, both as a single measure and in combination with RAAS blockers.

In our study, there was no convincing evidence that a low-salt diet was associated with a lower incidence of all-cause mortality and CV events. Limited data on all-cause mortality (nine studies with 4455 events) and CV events (five studies with 1989 events) for the meta-analysis might introduce a risk of false-negative results because of low statistical power. More significant benefits of CV events were found in studies that included patients with eGFR 30-59 mL/min/1.73 m^2 in the subgroup analysis, while the effects were not significant in patients with eGFR less than 15 mL/min/1.73 m^2. The significant subgroup heterogeneity (P =0.02) could be attributable to more CV

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risk factors in patients with ESRD than in those with eGFR 30-59 mL/min/1.73 m^2. The benefits of CV events from a low-salt diet were not enough to outweigh the increase in CV events caused by more risk factors (such as hyperuricemia, hyperphosphatemia, and vascular calcification). It is a hint at least for clinicians that low-salt diet education needs to be emphasized in the early stage of CKD.

Observational data from various studies showing a J-curve between sodium intake and renal and CV outcomes have raised concern regarding the safety of rigorous sodium restriction [59-61]. It is uncertain whether the presence of CKD modifies this association. In our meta-regression analysis, no clear relationships between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in kidney failure events were observed. A regimen of ACEIs and rigorous sodium restriction reduced blood pressure, proteinuria, and glomerular damage but unexpectedly aggravated tubule-interstitial damage in experimental renal disease [62]. This could provide a potential explanation for the association of very low sodium intake with worse renal outcome. Both high or extremely low dietary salt intake might be inappropriate for CKD patients. KDIGO recommends a daily sodium intake of < 2.0 g/d in patients with CKD without a clear lower limit. Taken together, these results should be interpreted, and the proper levels of dietary salt intake in CKD patients should be discussed. The current study might not be the final answer for the question of low-sodium diets and progression of kidney disease. Interventional studies are needed to ascertain the optimum level of dietary sodium intake for renal protection in patients with different CKD stages.

Our study has several potential limitations. First, as has been noted, the existence of statistical heterogeneity and inevitable clinical heterogeneity might raise critical concerns regarding validity, although we tried to resolve these concerns through subgroup analysis and sensitivity analysis. Heterogeneity in the study population as included in the umbrella term “CKD patients” remains a

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major concern for meaningful interpretation of the results. Because of the stages of CKD patients in the studies we included varies a lot, some are CKD 1-5, some are CKD 1-3, some are CKD 3-4, some are CKD 2-4, some are CKD 4-5, some are CKD 3-4. So it is difficult to stratify our analysis our outcomes by stage of CKD. We can only perform subgroup analyses according to the baseline eGFR of CKD patients. Second, many of the included studies with low quality had small sample sizes and a low incidence of events. Consequently, the uncertainty in the analyses was increased, resulting in wide CIs of effect measures. Third, confounding factors in the observational studies cannot be neglected, which might impact our results. Therefore, the findings of this study should be viewed as hypothesis-generating and need to be confirmed by further studies.

CONCLUSION

In conclusion, our meta-analyses suggested that a low-salt diet may reduce the risk of renal composite outcomes events in patients with CKD. However, there was no compelling evidence indicating that a low-salt diet could reduce the rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. The optimal dietary salt intake for patients with different CKD stages remains unanswered. Further well-designed RCTs targeting patients with different CKDs are required to assess and integrate these results.

Acknowledgments We thank American Journal Experts Corporation for helping in proofreading and editing the English of the final manuscript.

Contributors All authors take responsibility for the integrity of the data and the accuracy of data analysis. Study concept and design: L.W., X.S., H.S. and C.L.. Extraction, analysis and interpretation of data: H.S. and X.S.. Drafting of the manuscript: H.S. and L.W.. Critical revision of the manuscript:

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L.W. and X.S. Statistical analysis: H.S., C.L. and W.G.. Technical support: L.W., X.S. and W.G.. All authors have read and agreed to the submission to this journal of the manuscript.

Funding Support was provided by the National Natural Science Foundation of China (no. 82000655) and the Science and Technology Project of Shanxi Province (no. 201801D121218).The funders had no role in decision to publish and preparation of the manuscript.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data used to support the findings of this study are included within the article and supplementary information.

Ethics statement Our research include published studies, and does not require ethical committee approval.

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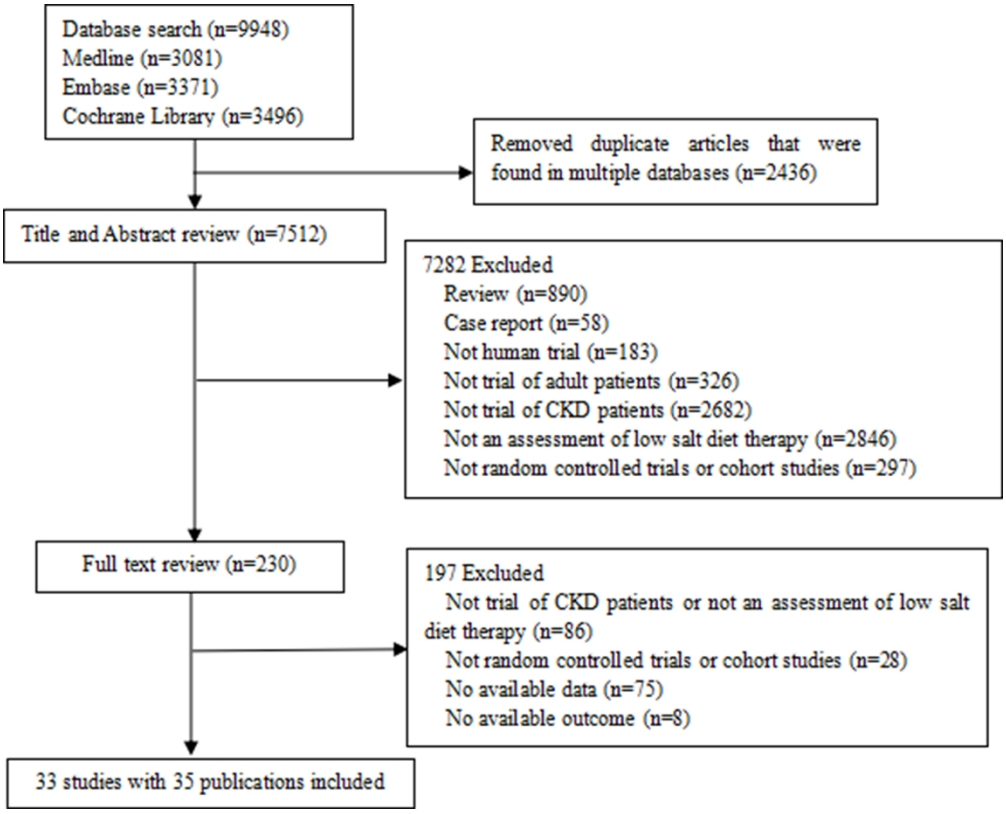

Database search (n=9948)
Medline (n=3081)
Embase (n=3371)
Cochrane Library (n=3496)
Removed duplicate articles that were found in multiple databases (n=2436)
Title and Abstract review (n=7512)
7282 Excluded
Review (n=890)
Case report (n=58)
Not human trial (n=183)
Not trial of adult patients (n=326)
Not trial of CKD patients (n=2682)
Not an assessment of low salt diet therapy (n=2846)
Not random controlled trials or cohort studies (n=297)
Full text review (n=230)
197 Excluded
Not trial of CKD patients or not an assessment of low salt diet therapy (n=86)
Not random controlled trials or cohort studies (n=28)
No available data (n=75)
No available outcome (n=8)
33 studies with 35 publications included

Figure 1

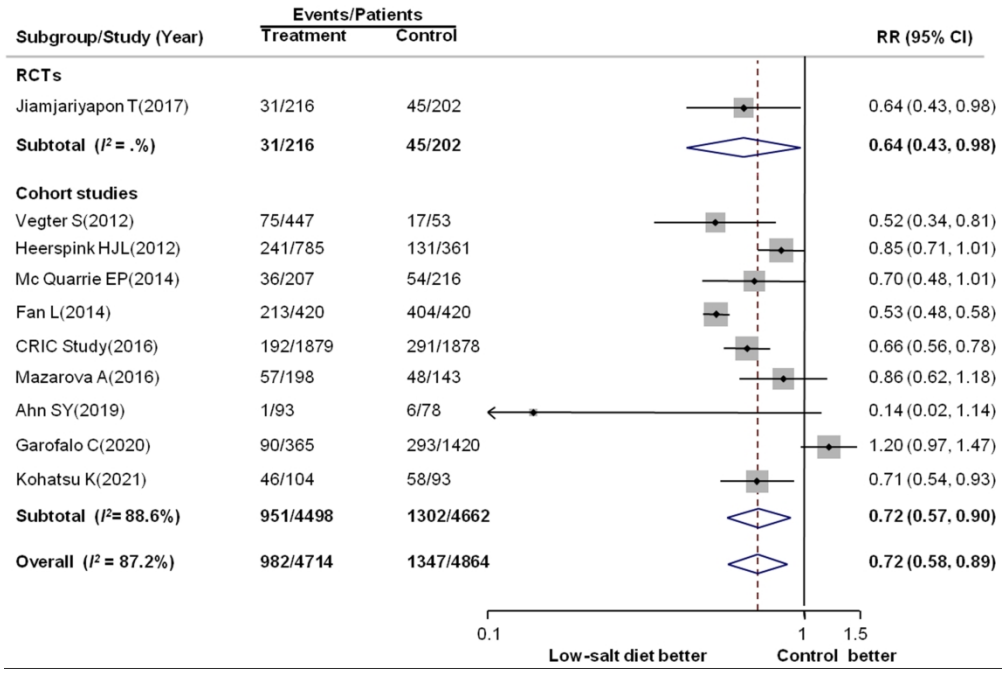


Figure 2

203x136mm (300 x 300 DPI)

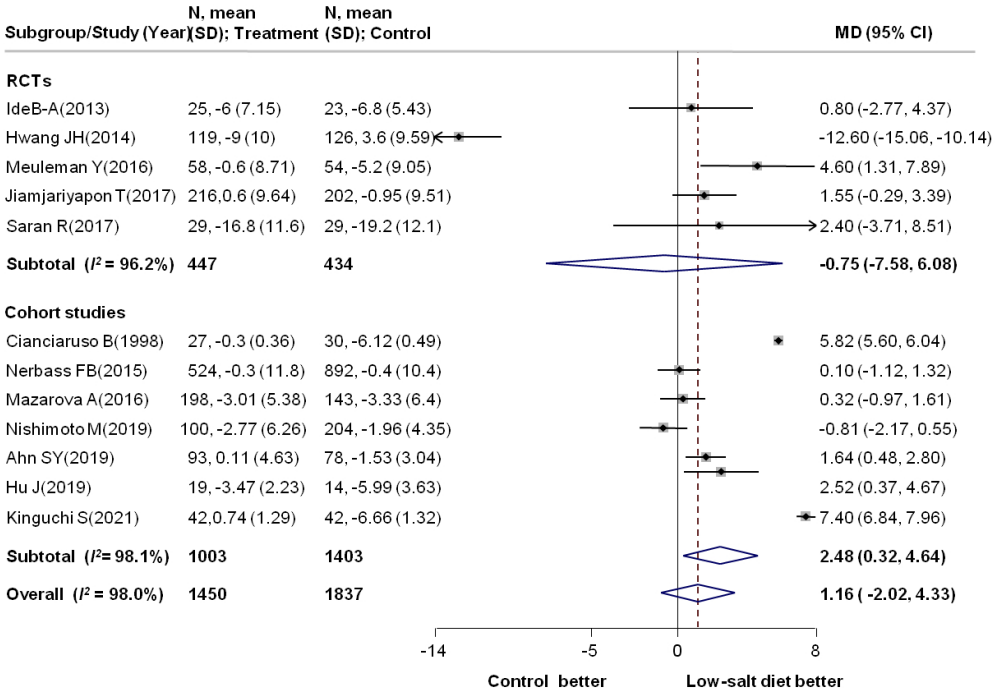

Subgroup/Study (Year)
N, mean (SD); Treatment
N, mean (SD); Control
MD (95% CI)
RCTs
IdeB-A(2013) 25, -6 (7.15) 23, -6.8 (5.43) 0.80 (-2.77, 4.37)
Hwang JH(2014) 119, -9 (10) 126, 3.6 (9.59) -12.60 (-15.06, -10.14)
Meuleman Y(2016) 58, -0.6 (8.71) 54, -5.2 (9.05) 4.60 (1.31, 7.89)
Jiamjariyapon T(2017) 216,0.6 (9.64) 202, -0.95 (9.51) 1.55 (-0.29, 3.39)
Saran R(2017) 29, -16.8 (11.6) 29, -19.2 (12.1) 2.40 (-3.71, 8.51)
Subtotal (I^2 = 96.2%) 447 434 -0.75 (-7.58, 6.08)
Cohort studies
Cianciaruso B(1998) 27, -0.3 (0.36) 30, -6.12 (0.49) 5.82 (5.60, 6.04)
Nerbass FB(2015) 524, -0.3 (11.8) 892, -0.4 (10.4) 0.10 (-1.12, 1.32)
Mazarova A(2016) 198, -3.01 (5.38) 143, -3.33 (6.4) 0.32 (-0.97, 1.61)
Nishimoto M(2019) 100, -2.77 (6.26) 204, -1.96 (4.35) -0.81 (-2.17, 0.55)
Ahn SY(2019) 93, 0.11 (4.63) 78, -1.53 (3.04) 1.64 (0.48, 2.80)
Hu J(2019) 19, -3.47 (2.23) 14, -5.99 (3.63) 2.52 (0.37, 4.67)
Kinguchi S(2021) 42,0.74 (1.29) 42, -6.66 (1.32) 7.40 (6.84, 7.96)
Subtotal (I^2= 98.1%) 1003 1403 2.48 (0.32, 4.64)
Overall (I^2 = 98.0%) 1450 1837 1.16 (-2.02, 4.33)
-14
-5
0
8
Control better
Low-salt diet better

Figure 3

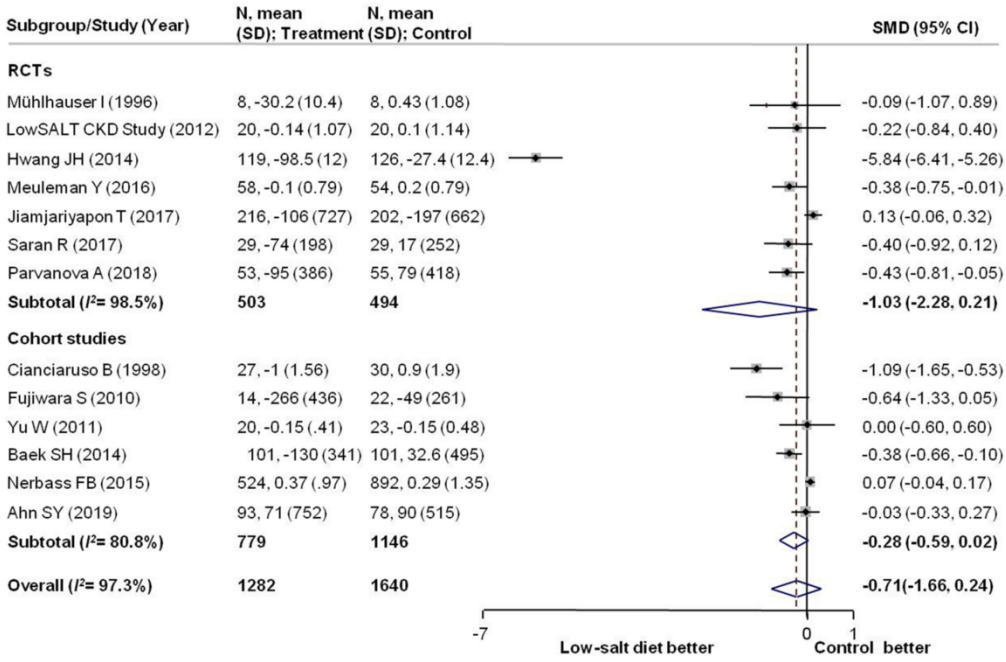


Figure 4

171x113mm (300 x 300 DPI)

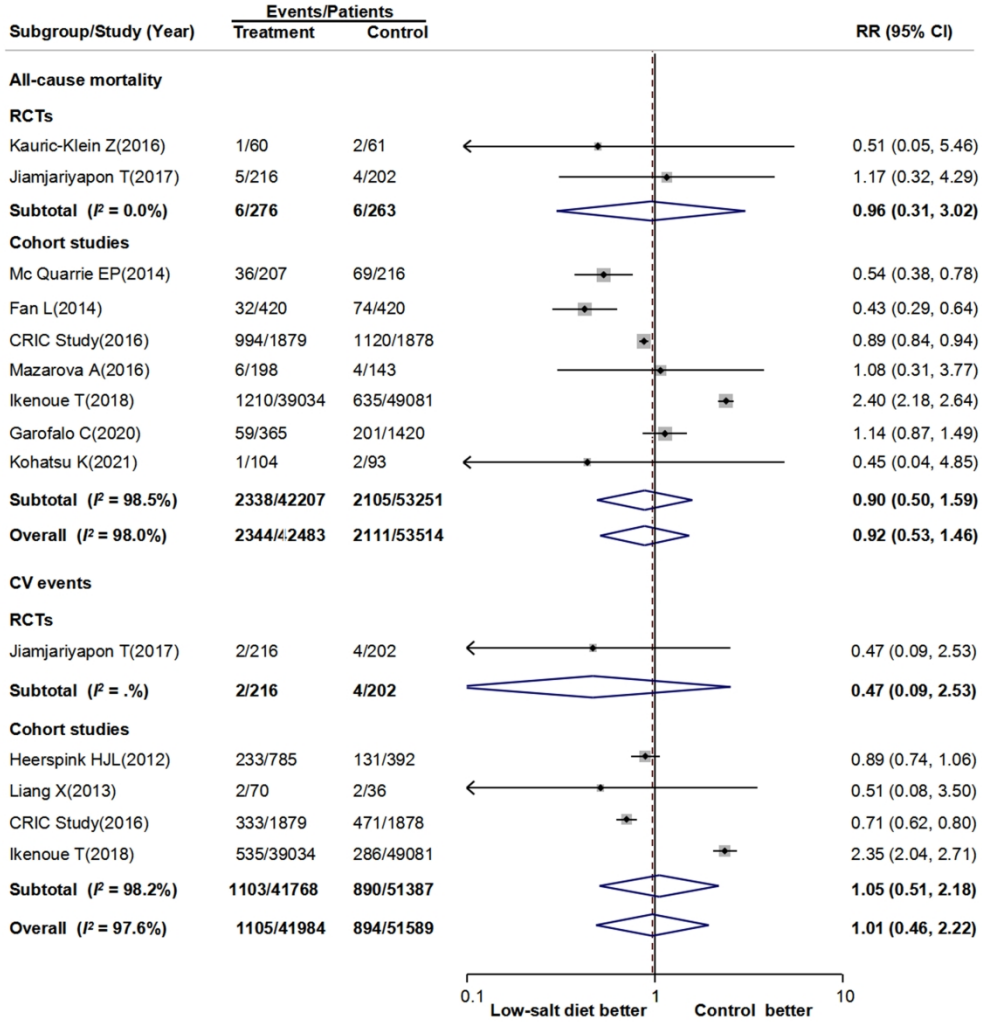


Figure 5

206x216mm (300 x 300 DPI)

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Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang

Citation

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang. Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease. PROSPERO 2017 CRD42017072395 Available from: http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42017072395

Review question

This meta-analysis aims to evaluate the effects of different levels of dietary sodium intake on renal outcome in patients with chronic kidney disease (CKD).

Searches

All relevant randomized controlled trails (RCTs) and cohort studies will be identified and searched in following electronic databases without language restriction: Ovid MEDLINE SP (from 1946 to July 2017), EMBASE (from 1966 to July 2017), and the Cochrane Central Register of Controlled Trails(no date restriction). A search in www.clinicaltrials.gov will be used to identify the ongoing but unpublished studies about this field. We combined this strategy with a manual search of reference lists from identified trials and review articles.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/72395_STRATEGY_20170618.pdf

Types of study to be included

We will include RCTs and cohort studies that compared different levels of sodium intake for adult patients with CKD as well as studies in broader populations in which outcome data for subgroups with CKD could be disaggregated.

Condition or domain being studied

Chronic kidney disease

Participants/population

Inclusion criteria: Adults (no less than 18 years old) with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) at all disease stages will included without restrictions on gender or race. Data from subgroups of participants with CKD within studies with broader inclusion criteria (e.g. people from the general population, people with diabetes, people with cardiovascular disease) were also included.
Exclusion criteria: Studies which do not include any people with chronic kidney disease.

Intervention(s), exposure(s)

Low-sodium diet

Comparator(s)/control

No intervention or different levels of dietary sodium intake

Context

None

Primary outcome(s)

If the data allow:

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1) Kidney failure events: It will include more than 25% or 50% decrease in estimated glomerular filtration rate (eGFR), doubling of serum creatinine, end stage renal disease (ESRD);
2) Rate of change in the eGFR: The difference from baseline in eGFR divided by the number of years between creatinine measurements (ml/min/1.73m2 per year);
3) Change in proteinuria or albuminuria, including urinary protein excretion (UPE), urinary albumin excretion (UAE), protein to creatinine (PCR) and albumin to creatinine ratio (ACR);
4) Major cardiovascular events: It will include a composite including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death.

Secondary outcome(s)

1) 24-hour urinary sodium excretion;
2) Change in blood pressure;
3) All-cause mortality;
4) Change in markers of fluid overload (brain natriuretic peptide, N-terminal pro-brain natriuretic peptide, weight, bio-impedance analysis);
5) Adverse events: hypotensive episodes, undesirable change in blood lipids (low density lipoprotein, high-density lipoprotein);
6) Hospitalization.

Data extraction (selection and coding)

Data extraction will be carried out independently by two reviewers using standard data extraction forms. Disagreement will be resolved by consensus or by discussion with a third author. When detailed information that was needed for the analysis was not available, the original authors will be contacted to obtain the missing information through e-mail. Where multiple publications of one study exists, data will be extracted and grouped together as a single study with the most complete data.
The extracted data will include: study characteristics (study design, method of randomization, and withdrawals / dropouts); baseline patient characteristics (age, sex, race, country, BMI, sodium intake and its measurement, 24-hour urinary sodium excretion, mean proteinuria or albuminuria, eGFR, serum creatinine concentrations, blood lipids, blood pressure, diabetic status, cardiovascular events, smoking status, glycosylated hemoglobin, hemoglobin, antihypertensive drugs, antidiabetic drugs, different kidney disease diagnosis); change in eGFR; follow-up duration; outcomes: as listed in Primary outcomes and Secondary outcomes. When the required quantitative data is only provided in published figures, we will use g3 data software (http://www.frantz.fi/software/g3data.php) to extract exact numbers from figures.

Risk of bias (quality) assessment

For the RCTs, we will assess risk of bias for sequence generation, allocation concealment, blinding, selective reporting, incomplete outcome data and other sources of bias, and determined overall risk of bias based on predefined rules, utilizing the Cochrane Collaboration risk of bias tool. (reference: Higgins JP, Altman DG, et al. Cochrane Statistical Methods G. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials. BMJ 2011;343:d5928.). Furthermore, the quality of RCT study will be quantified with Jadad scale. (reference: Jadad AR, Moore RA, Carroll D, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Control Clin Trials 1996;17:1– 12.). The quality of cohort study will be independently evaluated by each investigator using Newcastle-Ottawa quality assessment scale. (reference: Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010; 25:603–5.).

Strategy for data synthesis

The results of the data extraction and quality assessment for each study will be presented in structured tables. For dichotomous outcomes, relative risks (RR) and 95% Cis will be calculated from the numbers of events and participants at risk for events. For continuous outcomes, the mean difference (MD) will be used, or the standardized mean difference (SMD) if different scales were used. Data will be pooled using the random-effects model but the fixed-effect model will also be used to ensure robustness of the model chosen and susceptibility to outliers.
Heterogeneity will be analyzed using Cochrane's Q and the I-squared statistic. We will explore potential

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heterogeneity by subgroup analysis and meta-regression.
Sensitivity analyses will be conducted to check the robustness of the effect of sodium-restricted treatment by excluding of trials with small sample size and low Jadad scores.
A two-sided p-value less than 0.05 was considered statistically significant and statistical analyses were performed using RevMan version 5.3 and STATA version 12.0.

Analysis of subgroups or subsets

If the data allow, subgroup analysis will be conducted by study design, follow-up duration, measures of sodium intake, different levels of sodium intake, difference in sodium intake between two groups, and baseline measures, including age, sex, BMI, eGFR, 24-hour urinary sodium excretion,mean serum creatinine, blood pressure, mean proteinuria or albuminuria, stage of CKD, presence of comorbidities (hypertension, diabetes and CVD), different kidney disease diagnosis (e.g. GN or Kidney Transplant Recipients) and whether RAAS blocker or other antihypertensive drug treatments.

Contact details for further information

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Organisational affiliation of the review

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http://www.sydey.com/

Review team members and their organisational affiliations

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Dr Lihua Wang. Shanxi Medical University Second Hospital
Dr Xiaole Su. Shanxi Medical University Second Hospital
Dr Xuejiao Wang. Shanxi Medical University Second Hospital

Anticipated or actual start date

15 July 2017

Anticipated completion date

01 February 2018

Funding sources/sponsors

None

Conflicts of interest

None known

Language

English

Country

China

Stage of review

Review_Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Acids, Noncarboxylic; Humans; Kidney; Renal Insufficiency, Chronic; Sodium; Sodium Chloride, Dietary; Sodium, Dietary

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NHS National Institute for Health Research

Date of registration in PROSPERO
18 July 2017

Date of publication of this version
18 July 2017

Details of any existing review of the same topic by the same authors
None

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions
18 July 2017

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Supplementary file 2 Search Strategy

Ovid MEDLINE

1. exp Sodium Chloride/ or sodium chloride.mp.

2. exp Sodium, Dietary/

3. exp Diet, Sodium-Restricted/

4. ((sodium or salt) adj5 (low or high or alter* or reduce* or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)).mp.

5. (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free).mp.

6. 1 or 2 or 3 or 4 or 5

7. exp Kidney/ or kidney.mp.

8. exp Nephrosis/ or nephrosis*.mp.

9. (renal or nephropath*).mp.

10. (chronic kidney or chronic renal).mp.

11. exp Kidney Diseases/

12. ((kidney or renal) adj (disease* or failure)).mp.

13. exp Kidney Function Tests/ or kidney function test*.mp.

14. (CKF or CKD or CRF or CRD).mp.

15. (end-stage renal or end-stage kidney or endstage renal or endstage kidney).mp.

16. (ESRF or ESKF or ESRD or ESKD).mp.

17. exp Uremia/ or ur?emi*.mp.

18. exp Renal Replacement Therapy/ or renal replacement therap*.mp.

19. exp Renal Insufficiency/ or renal insufficien*.mp.

20. exp Dialysis/ or dialysis.mp.

21. exp Renal Dialysis/ or renal dialys*.mp.

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22. ("pre dialysis" or pre-dialysis or predialysis).mp.

23. (h*emodialysis or h*emofiltration or h*emodiafiltration).mp.

24. exp Peritoneal Dialysis/ or peritoneal dialysis.mp.

25. (PD or CAPD or CCPD or APD or IPD).mp.

26. exp Kidney Transplantation/

27. ((kidney or renal) adj (transplant* or graft* or allograft*)).mp.

28. 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27

29. randomized controlled trial.pt.

30. controlled clinical trial.pt.

31. randomized.ab.

32. placebo.ab.

33. randomly.ab.

34. clinical trials as topic.sh.

35. trial.ti.

36. 29 or 30 or 31 or 32 or 33 or34 or 35

37. (animals not (humans and animals)).sh.

38. 36 not 37

39. exp Cohort Studies/

40. exp Longitudinal Studies/

41. exp Follow-Up Studies/

42. exp Prospective Studies/

43. exp Retrospective Studies/

44. cohort.ti,ab.

45. longitudinal.ti,ab.

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46. prospective.ti,ab.

47. retrospective.ti,ab.

48.38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47

49. 6 and 28 and 48

Embase

#1. 'sodium chloride'/exp OR 'sodium chloride'

#2. 'sodium intake'/exp

#3. 'salt intake'/exp

#4. 'sodium restriction'/exp

#5. 'high sodium intake'/exp

#6. 'high salt diet'/exp

#7. 'high sodium' OR 'low sodium' OR 'sodium alter*' OR 'sodium reduce*' OR 'sodium reduction' OR 'sodium restrict*' OR 'sodium intake*' OR 'sodium diet*' OR 'sodium food' OR 'sodium increas*' OR 'sodium decreas*' OR 'sodium change*' OR 'sodium changing' OR 'sodium free'

#8. 'high salt' OR 'low salt' OR 'salt alter*' OR 'salt reduce*' OR 'salt reduction' OR 'salt restrict*' OR 'salt intake*' OR 'salt diet*' OR 'salt food' OR 'salt increas*' OR 'salt decreas*' OR 'salt change*' OR 'salt changing' OR 'salt free'

#9. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8

#10. 'kidney'/exp OR 'nephrosis'/exp

#11. kidney OR renal OR nephropath* OR nephrosis

#12. 'chronic kidney' OR 'chronic renal'

#13. 'kidney disease'/exp OR 'kidney failure'/exp

#14. 'kidney disease*' OR 'renal disease*' OR 'kidney failure' OR 'renal failure'

#15. 'kidney function test'/exp OR 'kidney function test*'

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#16. ckf OR ckd OR crf OR crd

#17. 'end-stage kidney' OR 'end-stage renal' OR 'endstage kidney' OR 'endstage renal'

#18. esrf OR eskf OR esrd OR eskd

#19. 'uremia'/exp OR ur?emi*

#20. 'renal replacement therapy'/exp OR 'renal replacement therap*'

#21. 'renal insufficien*'

#22. 'dialysis'/exp OR dialys*

#23. 'pre dialysis' OR predialysis

#24. 'hemodialysis'/exp OR 'hemofiltration'/exp OR 'hemodiafiltration'/exp

#25. hemodialysis OR haemodialysis OR hemofiltration OR haemofiltration OR hemodiafiltration OR haemodiafiltration

#26. 'peritoneal dialysis'/exp OR 'peritoneal dialysis'

#27. pd OR capd OR ccpd OR apd OR ipd

#28. 'kidney transplantation'/exp OR 'kidney graft'/exp OR 'kidney allograft'/exp

#29. 'kidney transplant*' OR 'renal transplant*' OR 'kidney graft*' OR 'renal graft*' OR 'kidney allograft*' OR 'renal allograft*'

#30. #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29

#31. 'randomized controlled trial'/exp OR 'randomized controlled trial' OR 'controlled clinical trial'/exp OR 'controlled clinical trial' OR random*OR 'clinical trials'/exp OR 'clinical trials' OR trial:ti

#32. 'human'

#33. animal$ OR nonhuman

#34. #32 AND #33

#35. #33 NOT #34

#36. #31 NOT #35

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#37.'cohort analysis'/exp OR 'longitudinal study'/exp OR 'prospective study'/exp OR 'retrospective study'/exp

#38. (cohort OR concurrent OR incidence OR longitudinal OR followup OR 'follow up' OR prospective OR retrospective) NEXT/1 (analys* OR design* OR evaluation* OR research OR stud* OR survey* OR trial*)

#39. #36 OR #37 OR #38

#40. #9 AND #30 AND #39

CENTRAL

#1 MeSH descriptor: [Sodium Chloride] explode all trees

#2 "sodium chloride":ti,ab,kw (Word variations have been searched)

#3 MeSH descriptor: [Sodium, Dietary] explode all trees

#4 MeSH descriptor: [Diet, Sodium-Restricted] explode all trees

#5 ((sodium or salt) near/5 (low or high or alter* or reduce* or reducing or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)):ti,ab,kw

#6 (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free):ti,ab,kw

#7 (#1 or #2 or #3 or #4 or #5 or #6)

#8 MeSH descriptor: [Kidney] explode all trees

#9 MeSH descriptor: [Nephrosis] explode all trees

#10 (kidney or renal or nephropath* or nephrosis):ti,ab,kw

#11 (chronic next kidney or chronic next renal):ti,ab,kw

#12 MeSH descriptor: [Kidney Diseases] explode all trees

#13 (kidney or renal) next (disease* or failure):ti,ab,kw

#14 MeSH descriptor: [Kidney Function Tests] explode all trees

#15 “kidney function test*”:ti,ab,kw

#16 (CKF or CKD or CRF or CRD):ti,ab,kw

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#17 ("end-stage kidney" or "end-stage renal" or "endstage kidney" or "endstage renal"):ti,ab,kw

#18 (ESRF or ESKF or ESRD or ESKD):ti,ab,kw

#19 MeSH descriptor: [Uremia] explode all trees

#20 ur?emi*:ti,ab,kw

#21 MeSH descriptor: [Renal Replacement Therapy] explode all trees

#22 "renal replacement therap*":ti,ab,kw

#23 MeSH descriptor: [Renal Insufficiency] explode all trees

#24 "renal insufficien*":ti,ab,kw

#25 MeSH descriptor: [Dialysis] explode all trees

#26 dialys*:ti,ab,kw

#27 MeSH descriptor: [Renal Dialysis] explode all trees

#28 "renal dialys*":ti,ab,kw

#29 "pre dialysis" or pre-dialysis or predialysis:ti,ab,kw

#30 (h*emodialysis or h*emofiltration or h*emodiafiltration):ti,ab,kw

#31 MeSH descriptor: [Peritoneal Dialysis] explode all trees

#32 "peritoneal dialys*":ti,ab,kw

#33 (PD or CAPD or CCPD or APD or IPD):ti,ab,kw

#34 MeSH descriptor: [Kidney Transplantation] explode all trees

#35 (kidney or renal) near (transplant* or graft* or allograft*):ti,ab,kw

#36 (#8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35)

#37 (#7 and #36)

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Supplementary file 3 Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up , months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9[a]	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Sodium intake 80-100 mmol/d	24-h urinary sodium	Usual care	1.39c	143/88	NA
Fujiwara S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPAKUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5[b]	144/95	Non- industry

Heerspink HJL[24] (2012)	Type 2 diabetic nephropathy	1177	65	59	Cohort study	30	Renal composite outcomes, CV events	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 153mmol/g	44	155/84	Industry and non- industry
LowSALT CKD Study[25,26] (2012)	Hypertensive patients with stage 3–4 CKD	20	75	69	Randomised crossover trial	1	UACR, SBP, DBP	Sodium intake < 100 mmol/d	Sodium intake < 100 mmol/d	24-h urinary sodium	Sodium intake ≥ 220 mmol/d	32	151/82	Non- industry
de Brito-Ashurst I[27] (2013)	eGFR < 60 mL/min and mean BP > 130/80 mm Hg	48	58	58	Randomised parallel-group controlled trial	6	Change in eGFR	Intensive low-salt diet education	Traditional meals had salt reduced by 50%	24-h urinary sodium	Usual care	41.5	153/85	Non- industry
Liang X[28] (2013)	PD	106	54	33	Cohort study	1	CV events, SBP, DBP	Health education to restrict water-sodium intake strictly	Salt intake not more than 3 g/d	The 3-day diet diary was collected once a month by the special nurse. Compared with the Food Composition, the daily salt intake were estimated	Usual care	NA	149/88	NA
Mc Quarrie EP[29] (2014)	Patients with CKD and had at least one 24-hour urinary sodium measurement	423	50	52	Cohort study	102	Renal composite outcomes, all- cause mortality	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine	Urinary sodium/creatinine > 16	48	139/79	Non- industry
Fan L[4] (2014)	GFR 13–55 mL/min/1.73m^2	840	60	52	Cohort study	72	Renal composite outcomes, all- cause mortality	Urinary sodium < 3 g/d	Urinary sodium < 3 g/d	24-h urinary sodium	Urinary sodium ≥ 3 g/d	32.5	132/81	Non- industry
Baek SH[30] (2014)	Hypertensive patients with albuminuria	202	51	50	Cohort study	2	UAE, SBP, DBP	Good compliance with an low salt diet	Good compliance with an low salt diet (24-hurinary sodium reduction ratio < 50%)	24-hurinary sodium	Poor compliance with an low salt diet	77.1[c]	121/73	Industry and non- industry

Hwang JH[31] (2014)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	245	50	50	Randomised parallel-group controlled trial	2	eGFR, UAE, SBP, DBP	Sodium intake < 100mmol/d	Sodium intake < 100mmol/d	24-hurinary sodium	Usual care	67.3	131/79	Industry
Rodrigues Telini LS[32] (2014)	HD	39	38	58	Randomised parallel-group controlled trial	4	SBP, DBP	2 g of sodium reduction in their habitual diet	2 g of sodium reduction in their habitual diet	Salt intake was based on a 72-halimentary registry	Usual diet	NA	146/86	Non- industry
Nerbass FB[33] (2015)	eGFR 59–30 ml/min/1·73 m^2	1416	39	73	Cohort study	12	eGFR, UACR, SBP, DBP	Sodium intake decrease	Sodium intake decrease	Estimated 24-hurinary sodium	Sodium intake unchanged or increase	52.7	134/73	Industry and non- industry
CRIC Study[3,34] (2016)	eGFR20-70mL/min/1.73 m^2	3757	56	58	Cohort study	78	Renal composite outcomes, all- cause mortality, CV events	Urinary sodium < 153.6mmol/d	Urinary sodium < 153.6mmol/d	24-hurinary sodium	Urinary sodium ≥153.7 mmol/d	44.5	128/71	Non- industry
Mazarova A[5] (2016)	eGFR < 30 mL/min/1.73 m^2	341	62	64	Cohort study	18	Change in eGFR, renal failure, all-cause mortality	Urinary sodium ≤ 150mmol/d	Urinary sodium ≤ 150mmol/d	24-hurinary sodium	Urinary sodium> 150 mmol/d	17.1	NA	No funding supported
Kauric-Klein Z[35] (2017)	HD	121	50	60	Randomised parallel-group controlled trial	3	All- cause mortality, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	Sodium intake checklist developed by Millar and Beard	Usual care	NA	165/87	Non- industry
Meuleman Y[36] (2017)	eGFR ≥ 20mL/min/1.73m^2 and UPE > 0.2 g/L	112	82	55	Randomised parallel-group controlled trial	6	eGFR, UPE, SBP, DBP	Usual care and low salt diet education	low salt diet education	24-hurinary sodium	Usual care	49.7	129/76	Non- industry

Jiamjariyapon T[37] (2017)	Stage 3-4 CKD with diabetes or hypertension	418	27	62	Cluster randomized controlled trial	24	Renal composite outcomes, all-cause mortality, eGFR, CV events, UACR, SBP, DBP	Usual care and intensive low-salt diet education	Intensive low-salt diet educatio (Sodium intake less than 2,400 mg/d)	24-hurinary sodium	Usual care	41.5	125/76	Non- industry
Saran R[38] (2017)	eGFR 15–60 ml/min/1.73 m^2	58	60	61	Randomized crossover trial	1	eGFR, UACR, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	24-hurinary sodium	Usual care	38.7	130.3/NA	Non- industry
Huang B[39] (2017)	HD	90	43	55	Cluster randomized controlled trial	6	SBP, DBP	Self-management support, restrict salt intake	Self-management support, restrict salt intake	Dietary salt intake was measured using slightly modified balance formulas: (1) Sodium intake (mol)=[Na+1(mmol/L)×ECW(L)－Na+ 2(mmol/L)×(ECW– IDWG)(L)]/1000; (2) Salt intake (g/day)=sodium intake(mol)×58.4 (g/mol)/2(day)	Usual care	NA	154/86	Non- industry
Ikenoue T[6] (2018)	HD	88115	60	65	Cohort study	12	All-cause mortality, CV events	Salt intake < 6 g/d	Salt intake < 6 g/d	Intra-dialytic weight loss was used to calculate salt intake	Salt intake ≥ 6 g/d	NA	NA	No funding supported
Parvanova A[40] (2018)	Patients with type 2 diabetes and macroalbuminuria	108	89	64	Randomised parallel-group controlled trial	3	UACR	Sodium intake ≤ 100mmol/d	Sodium intake ≤ 100mmol/d	24-hurinary sodium	Sodium intake ≥ 200mmol/d	87.36	146/80	Industry
Martinez MG[41] (2019)	Pre-dialysis CKD with UPE > 500 mg/d	84	52	56	Cohort study	6	Change in SBP	Urinary sodium reduced	Urinary sodium reduced	24-hurinary sodium	Urinary sodium increased	49[b]	142/NA	NA
Nishimoto M[42] (2019)	UACR 30–599 mg/g and eGFR ≥ 50 ml/min/1.73 m^2	304	70	59	Cohort study	12	Change in eGFR, SBP, and DBP	Urinary sodium ≤ 189 mmol/d	Urinary sodium ≤ 189 mmol/d	24-hurinary sodium	Urinary sodium > 189 mmol/d	64.93	140/83	Non- industry

Ahn SY[43] (2019)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	171	49	50	Cohort study	12	Rate of change in eGFR, renal composite outcomes, change in UAE, SBP, DBP	Intensive low-salt diet education	Intensive low-salt diet education to achieve urinary sodium ≤ 100mmol/d	24-hurinary sodium	Usual care	64.2	124/75	Non- industry
Hu J[44] (2019)	PD	33	58	42	Cohort study	12	Residual renal Ccr	Sodium intake ≤ 3.0 g/d	Sodium intake ≤ 3.0 g/d	Daily sodium intake of the patients was calculated based on the 3-day dietary record	Sodium intake > 3.0 g/d	8.3	146/95	Non- industry
Humalda JK[45] (2020)	CKD stages 1 to 4 or kidney transplants recipients, and eGFR was ≥ 25 mL/min/ 1.73 m2	99	84	57	RCT	6	SBP, DBP	Self-management to restrict salt intake	Self-management to restrict salt intake	A detailed interactive food diary	Usual care	55	139/83	Non- industry
Garofalo C[46] (2020)	CKD stages 1–5	1785	58	64	Cohort study	49	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	37	NA	NA
Kinguchi S[47] (2021)	Diabetic kidney disease patients with HbA1c 7.0–10.0%, eGFR ≥ 45 ml/min/1.73 m2, and UACR ≥ 30 mg/g	84	76	65	Cohort study	6	UACR	urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios	urinary sodium-to-creatinine ratios ≥ 1.14	67	142/79	Non- industry
Kohatsu K[48] (2021)	CKD stages 3–5	197	76	71	Cohort study	17	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	24.2	133/74	Non- industry

Note: Ccr, clearance of creatinine; CKD, chronic kidney disease; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; ESRD, end stage renal disease; ECW, extracellular water; HD, hemodialysis; IDWG, interdialytic weight gain; NA, no available; PD, peritoneal dialysis; SBP, systolic blood pressure; UACR, urinary albumin to creatinine ratio; UAE, urine albumin excretion; UPE, urinary protein excretion; the units for eGFR: mL/min/1.73m². [a] The units for eGFR: mL/min, [b] Only Ccr obtained, the units for Ccr: mL/min; [c] Only serum creatinine obtained, the units for serum creatinine: μmol/L.

Supplementary file 4 Table S2 Summary of characteristics of included trials and patients.

Outcomes	No. of Studies/ No. of Patients	Sample Size	Follow-up, months	Age, years	Male (%)	Lost follow-up rate
Renal composite outcomes	10/9578	462 (305-1329)	43 (23-74)	58 (52-64)	59.2%	3.2%
Rate of change in eGFR	12/3287	142 (57-332)	12 (6-17)	55(48-61)	56.6%	7.5%
Proteinuria	13/2922	108 (40-224)	3 (1-12)	50 (43-66)	57.2%	8.8%
All-cause death	9/95997	423 (231-2771)	24 (15-75)	62 (55-65)	60.6%	0.03%
CV events	5/93573	1177 (262-45936)	24 (6.5-54)	59(45-63)	60.8%	0.02%
SBP	19/3608	103 (38-213)	3 (1-6)	55 (41-60)	57.1%	7.5%
DBP	18/3524	36 (20-90)	12 (2-54)	55 (40-60)	57.3%	7.7%

Note:Values are given as median (quartile). CV, cardiovascular; eGFR, estimated glomerular filtration rate.

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Supplementary file 5 Table S3 Assessment of risk of bias for RCTs.

Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
1. Bias arising from the randomization process																
1.1 Was the allocation sequence random?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias arising from the randomisation process?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
2. Risk of bias due to deviations from intended interventions																
2.1 Were participants aware of their assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	Other	High	High	High	High	Low	High	High
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	High	High	High	High	High	High	High	High
2.3 If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?	N/PN	Y/PY	NA/NI		Low		Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.4 If Y/PY/NI to 2.3: Were these deviations likely to have affected the outcome?	N/PN	Y/PY	NA/NI													
2.5 If Y/PY to 2.4: Were these deviations from intended intervention balanced between groups?	Y/PY	N/PN	NA/NI													

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomised?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to deviations from intended interventions?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
3. Bias due to missing outcome data																
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	High	High	High	High	Low	High	High	Low
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?	Y/PY	N/PN	NA	Low	Low				Low	Low	Low	Low		Low	Low	
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	N/PN	Y/PY	NA/NI													
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to missing outcome data?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
4. Bias in measurement of the outcome																
4.1 Was the method of measuring the outcome inappropriate?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?	N/PN	Y/PY	NI	Low	High	Low	High	High	Low	High	High	High	Low	High	Low	High
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI		Low		Low	Low		Low	Low	Low		Low		Low
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias in measurement of the outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
5. Bias in selection of the reported result																
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:																
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
5.3 ... multiple eligible analyses of the data?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction bias due to selection of the reported results?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
Overall bias																
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the overall predicted direction of bias for this outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null

Note: Y, yes; PY, probably yes; PN, probably no; N, no; NA, not applicable; NI, no information; RCT, randomized controlled trial.

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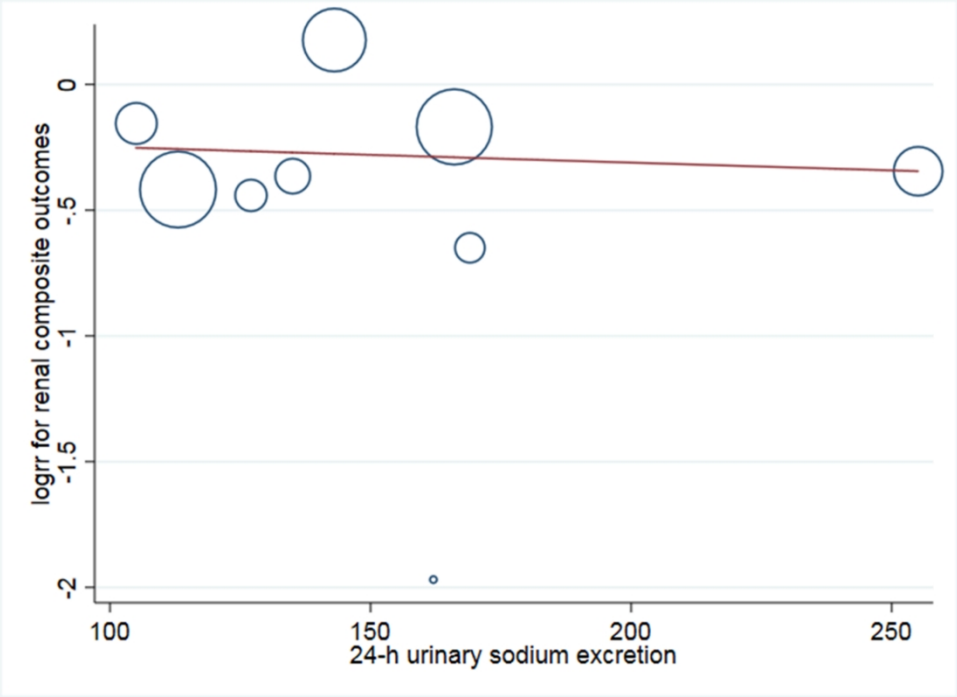
Supplementary file 6 Table S4 Quality assessment for cohort studies.

Study (Year)	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Scores
Cianciaruso B[19] 1998	★	★	★	★	—	★	★	★	7★
Fujiwara S[21] 2010	★	★	★	★	★	★	—	★	7★
Yu W[22] 2011	★	★	★	★	★	★	—	★	7★
Vegter S[23] 2012	★	★	★	★	—	★	★	★	7★
Heerspink HJL[24] 2012	★	★	★	★	★	★	—	★	7★
Liang X[28] 2013	★	★	★	★	★	★	—	★	7★
Mc Quarrie EP[29] 2014	★	★	★	★	★	★	★	★	8★
Fan L[3] 2014	★	★	★	★	—	★	★	—	6★
Baek SH[30] 2014	★	★	★	★	—	★	—	★	6★
Nerbass FB[33] 2015	★	★	★	★	★	★	—	★	7★
CRIC Study[3,34] 2016	★	★	★	★	★	★	★	★	8★
Mazarova A[5] 2016	★	★	★	★	★	★	—	★	7★
Ikenoue T[6] 2018	★	★	★	★	★	★	—	★	7★
Martinez MG[41] 2019	★	★	★	★	★	★	★	★	8★

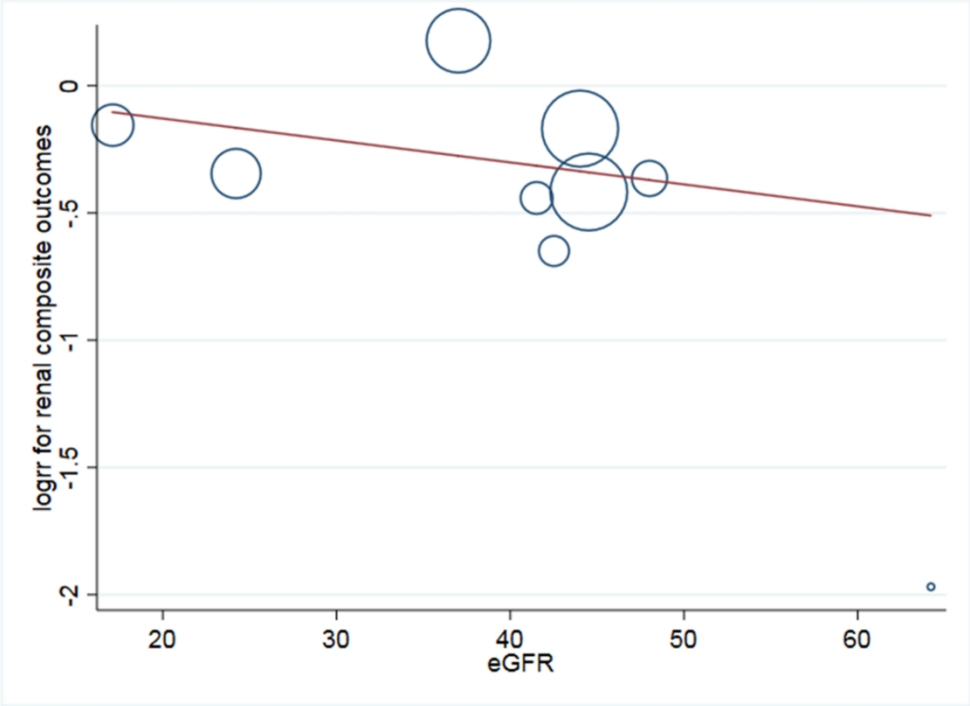
Nishimoto M[42] 2019	★	★	★	★	★	★	—	★	7★
Ahn SY[43] 2019	★	★	★	★	★	★	—	★	7★
Hu J[44] 2019	★	★	★	★	★	★	—	★	7★
Garofalo C[46] 2020	★	★	★	★	★	★	★	★	8★
Kinguchi S[47] 2021	★	★	★	★	★	★	—	★	7★
Kohatsu K[48] 2021	★	★	★	★	★	★	—	★	7★

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability. —: no score; It is considered that the queue follow-up is appropriate when the follow-up rate is more than 80%.

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Supplementary file 7



Supplementary file 8

Supplementary file 9 Table S5 Subgroup analysis of outcomes except renal failure events.

Subgroup	No. of trials	n	RR/MD/SMD (95%CI)	*P* for RR/MD/SMD	I^2	*P* for heterogeneity test[a]
Change in eGFR（mL/min/1.73m²）						
Study type						
RCT	5	881	-0.75 (-7.58, 6.08)	0.83	96.2%	0.28
Cohort studies	7	2406	2.48 (0.32, 4.64)	0.02	98.1%	
Baseline eGFR (mL/min/1.73m²)						
60-89	4	804	-1.00 (-7.41, 5.42)	0.76	99.2%	0.48
30-59	6	2109	2.58 (-0.44, 5.61)	0.10	95.4%	
<15	2	374	1.24 (-0.88, 3.37)	0.25	66.2%	
Mean age (years)						
<58	5	106	0.44 (-4.40, 5.27)	0.86	98.5%	0.70
≥58	6	2669	1.68 (-1.83, 5.18)	0.35	97.8%	
Baseline BP (mmHg)						
<140/90	6	2420	-0.52 (-4.36, 3.32)	0.32	97.8%	0.41
SBP≥140 或 DBP≥90	4	469	2.54 (-2.47, 7.55)	0.55	70.1%	
Comparator of control group						
Usual care or usual diet	6	1052	-0.38 (-5.14, 4.39)	0.88	95.8%	0.85
Different levels of sodium intake	6	2235	2.62 (0.31, 4.94)	0.03	98.2%	
Proteinuria or Albuminuria						
Study type						
RCT	7	997	-1.03 (-2.28, 0.21)	0.11	98.4%	0.45
Cohort studies	6	1925	-0.28 (-0.59, 0.02)	0.07	80.8%	
Baseline eGFR (mL/min/1.73m²)						
60-89	7	821	-1.06 (-2.32, 0.20)	0.10	98.2%	0.42
30-59	6	1201	-0.23 (-0.50, 0.04)	0.10	79.6%	
Mean age (years)						
<58	7	846	-1.12 (-2.35, 0.12)	0.08	98.2%	0.34
≥58	6	2076	-0.13 (-0.34, 0.09)	0.24	64.2%	
Baseline BP (mmHg)						
<140/90	8	200	-0.82 (-1.55, -0.10)	0.01	98.3%	0.65
SBP≥140 或 DBP≥90	4	2665	-0.39 (-0.68, -0.11)	0.03	0.0%	
Comparator of control group						
Usual care or usual diet	7	1083	-1.01 (-2.18, 0.15)	0.09	98.4%	0.12
Different levels of sodium intake	6	1839	-0.34 (-0.69, 0.00)	0.05	81.7%	
All-cause mortality						
Study type						
RCT	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.99
Cohort studies	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
Baseline eGFR (mL/min/1.73m²)						
30-59	4	5438	0.65 (0.41, 1.02)	0.06	85.2%	0.008
15-29	2	2126	1.14 (0.88, 1.48)	0.33	0.0%	
<15	3	88433	1.34 (0.41, 4.39)	0.62	43.4%	

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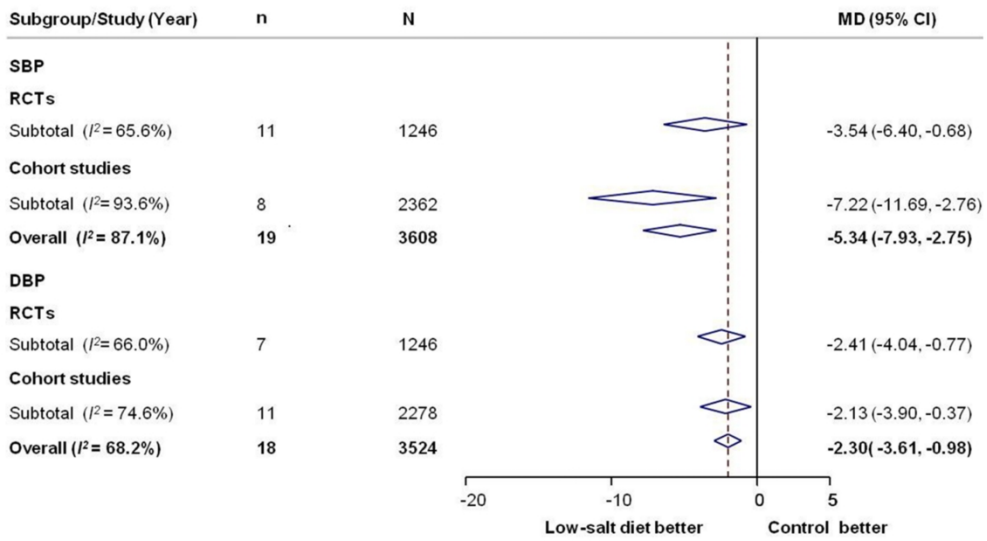
Mean age (years)						
<58	3	704	0.61 (0.37, 1.00)	0.05	90.1%	0.08
≥58	6	90977	1.34 (0.76, 2.37)	0.32	84.2%	
Baseline BP (mmHg)						
<140/90	5	5438	0.64 (0.41, 0.99)	0.05	80.6%	0.86
SBP≥140 或 DBP≥90	1	121	0.51 (0.05, 5.46)	0.58	-	
Comparator of control group						
Usual care or usual diet	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.90
Different levels of sodium intake	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
CV events						
Study type						
RCT	1	418	0.47 (0.09, 2.53)	0.88	-	0.51
Cohort studies	4	93155	1.05 (0.51, 2.18)	0.89	98.2%	
Baseline eGFR (mL/min/1.73m^2)						
30-59	3	5352	0.78 (0.63, 0.95)	0.02	57.1%	**0.02**
<15	2	88221	1.50 (0.39, 5.85)	0.56	58.3%	
Mean age (years)						
<58	2	3863	0.71 (0.62, 0.80)	<0.001	0.0%	0.40
≥58	3	89710	1.20 (0.49, 2.91)	0.69	97.3%	
Baseline BP (mmHg)						
<140/90	2	4175	0.71 (0.62, 0.80)	<0.001	0.0%	0.62
SBP≥140 或 DBP≥90	2	1283	0.88 (0.74, 1.05)	0.17	0.0%	
Comparator of control group						
Usual care or usual diet	2	524	0.49 (0.14, 1.73)	0.27	0.0%	0.52
Different levels of sodium intake	3	93049	1.14 (0.53, 2.46)	0.74	98.8%	
SBP（mmHg）						
Study type						
RCT	11	1246	-3.54 (-6.40, -0.68)	0.02	65.6%	0.43
Cohort studies	8	2362	-7.22 (-11.69, -2.76)	0.002	93.6%	
Baseline eGFR (mL/min/1.73m^2)						
60-89	7	1017	-3.30 (-5.26, -1.34)	0.001	31.7%	0.64
30-59	7	2210	-6.10 (-11.62, -0.59)	0.03	92.8%	
< 15	4	349	-5.66 (-11.62, 0.30)	0.06	76.0%	
Mean age (years)						
< 58	12	1215	-7.45 (-11.76, -3.14)	0.001	85.6%	0.25
≥ 58	7	2393	-2.39 (-5.65, 0.87)	0.15	87.3%	
Baseline BP (mmHg)						
< 140/90	10	2786	-0.97 (-2.75, 0.81)	0.28	46.9%	**0.02**
SBP ≥ 140 or DBP ≥ 90	9	822	-10.24 (-15.37, -5.12)	< 0.001	88.4%	
Comparator of control group						
Usual care or usual diet	13	1546	-4.28 (-7.20, -1.36)	0.004	74.3%	0.05
Different levels of sodium intake	6	2062	-7.55 (-12.83, -2.28)	0.005	94.6%	
DBP（mmHg）						
Study type						
RCT	7	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.85
Cohort studies	11	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

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Baseline eGFR (mL/min/1.73m^2)						
30-59	7	1017	-1.74 (-3.40, -0.08)	0.04	56.7%	0.13
15-29	6	2126	-0.53 (-1.25, 0.20)	0.16	11.9%	
< 15	4	349	-4.94 (-7.08, -2.80)	<0.001	23.2%	
Mean age (years)						
< 58	11	1131	-2.78 (-4.35, -1.22)	<0.001	50.8%	0.34
≥ 58	7	2393	-0.86 (-1.77, 0.04)	0.06	62.2%	
Baseline BP (mmHg)						
< 140/90	9	2747	-0.94 (-1.90, 0.02)	0.056	5.6%	**0.04**
SBP ≥ 140 或 DBP ≥ 90	9	777	-4.05 (-6.61, -1.49)	0.002	0.2%	
Comparator of control group						
Usual care or usual diet	11	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.12
Different levels of sodium intake	7	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

Note: [a] *P* value calculated by meta regression. BP, blood pressure; CI, confidence interval; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

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Supplementary file 10

Supplementary file 11 Table S6 Sensitivity analysis of outcomes.

Sensitivity analysis	Renal composite outcomes			The rate of change in eGFR			Proteinuria or albuminuria			All-cause mortaity		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD/SMD (95%CI)	τ^2	No./n	RR (95% CI)	τ^2
Omit studies of follow-up time <12 months	-	-	-	7/2740	1.60 (-1.10, 4.31)	12.78	-	-	-	8/95876	0.92 (0.53, 1.59)	0.48
Omit studies of sample size <50	-	-	-	10/3206	1.07 (-1.26, 3.41)	12.86	9/2787	-0.89 (-1.56, -0.23)	0.99	-	-	-
Omit studies with low quality	-	-	-	-	-	-	-	-	-	-		
Omit studies with extreme outliers	-	-	-	11/3042	**2.43 (0.63, 4.23)**	8.01	12/2677	-0.24 (-0.43, -0.05)	0.06	8/7882	0.75 (0.56, 1.00)	0.08
Different statistical estimators												
DL	10/9578	0.71 (0.57, 0.88)	0.09	12/3287	1.18 (-0.94, 3.29)	12.57	13/2922	-0.70 (-1.24, -0.1)	0.89	9/95997	0.90 (0.52, 1.53)	0.48
EB	10/9578	0.72 (0.58, 0.89)	0.06	12/3287	1.16 (-2.02, 4.33)	23.4	13/2922	**-0.71(-1.66, 0.24)**	2.39	9/95997	0.92 (0.53, 1.46)	0.23
REML	10/9578	0.72 (0.58, 0.89)	0.03	12/3287	1.16 (-2.04, 4.35)	23.73	13/2922	**-0.71(-1.66, 0.24)**	2.38	9/95997	0.91 (0.53, 1.55)	0.33

Sensitivity analysis	CV events			SBP			DBP		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD (95%CI)	τ^2
Omit studies of follow-up time <12 months	4/93467	1.03 (0.50, 2.12)	0.45	-	-	-	-	-	-
Omit studies of sample size <50	-	-	-	13/3402	-4.05 (-6.93, -1.17)	22.57	12/3318	-1.45 (-2.34, -0.56)	1.05
Omit studies with low quality	-	-	-	-	-	-	-	-	
Different statistical estimators									
DL	5/93573	0.97 (0.49, 1.93)	0.45	19/3608	-5.34 (-7.93, -2.75)	22.80	15/3524	-2.00(-2.93, -1.06)	1.56
EB	5/93573	1.01 (0.46, 2.22)	0.28	19/3608	-5.81(-9.63, -1.99)	14.19	15/3524	-2.30(-3.61, -0.98)	3.78
REML	5/93573	0.99 (0.41, 2.37)	0.35	19/3608	-5.78(-9.59, -1.97)	52.37	15/3524	-2.27(-3.57, -0.96)	3.41

Note: CI, confidence interval; CV, cardiovascular; DL, DerSimonian-Laird; EB, empirical Bayes; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; No., number of trials; NOS, Newcastle-Ottawa Scale; REML, restricted maximum likelihood; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

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Supplementary file 12 Table S7 The risk of publication bias of outcomes.

	N	Egger's test(*P*-value)	Bgge's test(*P*-value)
Renal composite outcomes events	10	0.47	0.28
The rate of change in eGFR	12	0.01	0.95
Proteinuria or albuminuria	13	0.10	0.30
All-cause mortality	9	0.09	1.00
CV events	5	0.91	0.46
BP	19	＜0.001	0.005

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5-6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-7

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8-12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15-16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Effects of Low-salt Diet Intervention on Chronic Kidney Disease Outcomes: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050843.R2
Article Type:	Original research
Date Submitted by the Author:	11-Oct-2021
Complete List of Authors:	Shi, Honghong; Shanxi Medical University Second Affiliated Hospital, Renal Division Su, Xiaole; Shanxi Medical University Second Affiliated Hospital Li, Chunfang; Shanxi Cardiovascular Hospital Guo, Wenjuan; Shanxi Medical University Second Affiliated Hospital Wang, Lihua; Shanxi Medical University Second Affiliated Hospital
Primary Subject Heading:	Renal medicine
Secondary Subject Heading:	Health services research
Keywords:	Nephrology < INTERNAL MEDICINE, Chronic renal failure < NEPHROLOGY, HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effects of Low-salt Diet Intervention on Chronic Kidney Disease Outcomes: A Systematic Review and Meta-analysis

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ABSTRACT

Objective The benefits of low-salt diet intervention for CKD patients are still controversial. We performed a systematic review and meta-analysis to investigate the effects of low-salt diet on major clinical outcomes.

Design Systematic review and meta-analysis.

Data sources MEDLINE by Ovid, EMBASE, and the Cochrane Library database.

Eligibility criteria for selecting studies We included RCTs and cohort studies that evaluated the effects of low-salt diet on the renal composite outcomes (more than 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD), decline rate in eGFR, changes in proteinuria, all-cause mortality events, cardiovascular (CV) events, change in SBP and DBP.

Data extraction and synthesis Two independent reviewers extracted data, and assessed quality. RR with 95% CI were used for dichotomous data. Difference in means (MD) or standardized mean differences (SMD) with 95% CI were used to pool continuous data. We used Cochrane Collaboration risk-of-bias tool to assess quality of RCTs, and NOS to assess quality of cohort studies.

Results We identified 9948 potentially records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. Thirty-three studies with 101077

participants were included. Low-salt diet produced a 28% reduction in renal composite outcomes events (RR: 0.72; 95%CI: 0.58-0.89) compared with control groups. But no significant effects were observed on change of proteinuria (SMD: -0.71; 95% CI: -1.66 to 0.24), decline rate of eGFR (MD: 1.16; 95% CI: -2.02 to 4.33), the risk of all-cause mortality (RR: 0.92; 95% CI: 0.58-1.46), and CV events (RR: 1.01; 95% CI: 0.46-2.22).

Conclusion Low-salt diet seems to reduce the risk of renal composite outcomes events in CKD patients. However, no compelling evidence indicated that low-salt diet would reduce rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. More definitive studies are needed.

PROSPERO registration number CRD42017072395.

Strengths and limitations of this study

This study is the first meta-analysis to evaluate the effect of low-salt diet intervention on hard clinical outcomes, such as renal composite outcomes, all-cause mortality and CV events for CKD patients.

We included cohort studies and RCTs without language restriction, and screened reference lists from included articles. Therefore allowing inclusion of the maximum number of studies in this review.

The existence of significant heterogeneity might limit the interpretation and clinical application of the results.

Confounding factors in the observational studies cannot be neglected, which might impact our results.

INTRODUCTION

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Chronic kidney disease (CKD) has become a global public health burden. Dietary interventions that aim to delay the progression of CKD and decrease complications have a pivotal role in the management of CKD. Restriction of dietary sodium intake can augment the antiproteinuric effects of angiotensin converting enzymes inhibitors/angiotensin receptor blockers (ACEIs/ARBs) [1] and decrease blood pressure (BP) [2]. In the 2020 KDIGO guidelines for glomerulonephritis, dietary restriction of sodium to < 2 g/d is a primary tenet for controlling BP and edema and improving urinary protein excretion independently of medications. However, there remains an open question of whether these effects can translate into a worthwhile absolute risk reduction in patients with CKD, including kidney failure events, cardiovascular (CV) events and all-cause mortality. In the Chronic Renal Insufficiency Cohort (CRIC) Study, recruiting patients eGFR between 20 and 70 ml/min/1.73 m^2 depending on age, compared with the lowest quartile of urinary sodium excretion (<116.8 mmol/24 h), high urinary sodium excretion (>194.6 mmol/24 h) was associated with a 54% risk increase in end-stage renal disease (ESRD) or halving of estimated glomerular filtration rate (eGFR) and with a 43% risk increase in all-cause mortality [3]. Contrasting results have been observed in other cohort studies, in which sodium intake does not appear to impact the progression of CKD to ESRD and a decline in eGFR [4,5]. Furthermore, a low salt intake of <6 g/day was associated with an increased risk of all-cause mortality and CV mortality in hemodialysis patients, resulting in an L-shaped association curve [6]. According to data from the NHANE II study, compared with those whose sodium intake exceed 2.3g daily, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans [7]. Randomized controlled trials (RCTs) evidence that discussed the effect of a low-salt diet in CKD patients on kidney failure events, CV events and

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all-cause mortality are controversial. It is difficult to base clinical decisions on findings from these studies, given their small sample sizes, highly selected participants, short-term follow-up and lack of assessment of hard clinical outcomes.

Thus, the association between dietary salt restriction and clinical outcomes in patients with CKD remains unclear. We performed this systematic review to summarize all eligible clinical studies data and evaluate the effects of low-salt diet intervention on kidney disease outcomes, all-cause mortality and CV events in CKD patients.

MATERIALS AND METHODS

Data sources and search strategy

We performed this systematic review according to a pre-specified protocol registered at the International Prospective Register of Systematic Reviews (CRD42017072395) (Supplementary file 1) [8], and reporting was done in line with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines [9]. A comprehensive search was conducted using the following database: MEDLINE by Ovid (from 1946 to July 2021), EMBASE (from 1966 to July 2021), and Cochrane Central Register of Controlled Trials (no date restriction), with relevant keywords and medical subject headings that included all spellings of “CKD”, “RCT”, “Cohort Studies”, “Sodium Chloride”, “Sodium”, “Sodium-Restricted” (see Supplementary file 2 for full search terms). Studies were considered without any language restriction. To ensure a comprehensive literature search, we also screened reference lists from included articles. The ClinicalTrials.gov website was also searched for studies the ongoing, but unpublished in this field.

Study selection and outcome estimation

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We included data from RCTs and cohort studies in which low-salt diet was given to adults with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) compared with usual therapy or different salt intake. Salt reduction was defined as that recommended by guidelines sodium <2.3 g/day (<100 mmol of sodium, <6g of salt), or as defined by the authors of each study.

Pre-defined outcomes that contained analyzable data were listed as follows: First, renal composite outcomes events, defined as a more than 50% decline in eGFR from baseline during follow-up [10], doubling of serum creatinine or ESRD. Second, the rate of change in eGFR per year. Third, changes in urinary protein or urinary albumin during follow-up, including urinary protein excretion, urinary albumin excretion and urinary albumin/creatinine ratio. Fourth, all-cause mortality events. Fifth, CV events, defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, CV disease and CV death. Fifth, change in systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Data extraction and quality assessment

Two independent reviewers (H.S. and X.S.) extracted data, and assessed quality, according to the pre-specified protocol. Disagreements were resolved through a third reviewer (L.W.). Data of all eligible studies was extracted into a spreadsheet. The data sought included characteristics of studies (study type, randomization method, follow-up time, withdrawals/dropouts), baseline patient characteristics (age, gender, baseline eGFR, baseline BP), intake of salt, and outcome events.

We used Cochrane Collaboration risk-of-bias tool [11] to assess all potential sources of bias for the included RCTs. Trails were assessed as being at low, other, some concerns, or high, and the overall risk of bias generally corresponds to the worst risk of bias in any of the domains. However, if

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a study was judged to have some concerns about risk of bias for multiple domains, it might be judged as at high risk of bias overall. We used the Newcastle-Ottawa Scale (NOS) to assess quality of cohort studies. The NOS contains eight items, categorized into three dimensions, including selection of cohorts, comparability of cohorts, and assessment of outcome[12]. A star system is used to allow a semi-quantitative assessment of study quality, such that the highest quality studies are awarded a maximum of one star for each item with the exception of the item related to comparability that allows the assignment of two stars. The NOS ranges between zero up to nine stars.

Data synthesis and statistical analyses

When dichotomous outcome data from individual trials were analyzed, relative risks (RRs) measure and its 95% confidence intervals (CIs) were used. If individual study RR was unavailable in the original article, RR and 95% CI were calculated from event numbers extracted from each study before data pooling. In calculating the RR values, we used the total number of patients randomized in each group as the corresponding denominator. Continuous outcome data from individual trials were analyzed using the difference in means (MD) with 95% CI to pool eGFR, and the standardized mean differences (SMD) with 95% CI were used to pool the proteinuria or albuminuria data. When continuous outcome data were analyzed, the difference in mean change between baseline and end of treatment value were used. If data for change between baseline and end of treatment were not available in the studies, we calculated it using correlations that were estimated from other included studies that had a similar follow-up period and reported in considerable detail according to the imputed formulation and its related interpretations in Cochrane Handbook [13].

In consideration of poor stability of the Der Simonian-Laird procedure for small numbers of studies [14], we used the empirical Bayes procedure [15] to estimate all outcomes. We also used Der

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Simonian-Laird random effects model [16] and restricted maximum likelihood [17] to assess the summary effects as a part of the sensitivity analyses. Considering the inevitable heterogeneity between studies, subgroup analyses, meta-regression, and sensitivity analyses were performed. Subgroup analyses were performed according to pre-specified protocol, including study type, baseline eGFR, and baseline BP, and comparator of control group. In addition, we performed sensitivity analyses by using different random-effects estimation methods as above mentioned, excluding studies with sample size <50, studies with follow-up time <12 months, studies with low quality (high risk of overall bias for RCTs, NOS <5★ for cohort studies), and studies with extreme outliers. Heterogeneity between studies was evaluated by using I^2 statistics or τ^2 statistics. The I^2 value of 25%, 50%, 75% respectively represent low, moderate, and high heterogeneity. Publication bias was assessed using Funnel plot, Egger's test and Begg's test, and P-value < 0.05 indicated obvious publication bias. Stata version 15.0 (StataCorp LP, College Station, TX, USA) was used for statistical analyses , and a two-sided P-value < 0.05 was considered statistically significant.

Patient and Public Involvement statement

No patients and the public were involved in the development of the research question, selection of the outcome measures, design and implementation of the study, or interpretation of the results.

RESULTS

Overview of included trials

We identified 9948 potentially relevant records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. As shown in figure 1, Supplementary file 3 and Supplementary file 4, a total of 33 eligible studies [3-6, 18-48] including

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13 RCTs [18,20,25-27,31,32,35-40,45] and 19 cohort studies [3-6,19,21-24,28,30,33,34,41-44,46-48] reported in 35 publications with 101077 participants, were included in this review. The median follow-up duration was 6 months (interquartile range 3 to 21). Participants were enrolled at an average age of 56 years, and male participants accounted for 60% of the total participants. The average eGFR of the included participants was 47.45 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and twenty-one trials compared different levels of sodium intake.

Figure 1 PRISMA flow chart for the included studies.

The risk of bias varied substantially across the RCTs. The results from the Cochrane Collaboration risk-of-bias tool are shown in Supplementary file 5. In all the RCTs, bias arising from the randomization process was assessed as low risk in 61.54% of the trails; risk of bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result were low risk in all the trails. In terms of overall bias, 61.54% of the trails were assessed as low risk, and 38.46% of the trails were assessed as having some concerns.

According to the NOS checklist, all included cohort studies were generally high quality and awarded 6★- 8★ (Supplementary file 6).

Effect of low-salt diet intervention on renal outcomes

One RCT and 9 observational studies including 9578 patients with 2329 events compared the low-salt diet with the control in preventing renal composite outcomes events. As shown in figure 2, compared with the control, low-salt diet intervention produced a 34% reduction in the risk of renal composite outcomes events (RR: 0.72; 95% CI: 0.58, 0.89), with substantial heterogeneity (I^2

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=87.2%, P <0.001). Subgroup analyses and meta-regression were conducted to discuss the source of heterogeneity. No significant difference between RCTs and observational studies was found, while the effect sizes were greater in studies that enrolled patients aged < 58 years (P = 0.03) (table 1). On univariate meta-regression, no clear relationship between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcomes events was observed (Supplementary file 7, Supplementary file 8).

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Figure 2 Forest plot for renal composite outcomes events. Renal composite outcomes events was defined as a more than 50% decline in eGFR from baseline during follow-up, doubling of serum creatinine or ESRD. CI, confidence interval; Cr, creatinine; Na, sodium; RR, relative risk.

Table 1 Subgroup Analysis of Renal Composite Outcomes Events.

Subgroup	No. of Trials	n	RR (95% CI)	*P* for RR	I^2	P for Heterogeneity Test[a]
Study type						
RCT	1	418	0.64 (0.43, 0.98)	0.04	-	0.76
Cohort study	9	9160	0.72 (0.57, 0.90)	0.005	88.6%	
Baseline eGFR (mL/min/1.73m²)						
60-89	1	171	0.14 (0.02, 1.14)	0.07	-	0.48
30-59	7	8869	0.71 (0.55, 0.92)	0.009	90.8%	
15-29	2	538	0.77 (0.63, 0.94)	0.01	0.0%	
Mean age (years)						
<58	5	5691	0.58 (0.49, 0.70)	<0.001	56.4%	**0.03**
≥58	5	3887	0.85 (0.69, 1.05)	0.14	70.1%	
Baseline BP (mmHg)						
<140/90	6	5806	0.62 (0.52, 0.73)	<0.001	58.5%	0.32
SBP≥140 or DBP≥90	2	1646	0.70 (0.44, 1.11)	0.13	74.6%	
Comparator of control group						
Usual care or usual diet	2	589	0.43 (0.11, 1.64)	0.21	50.2%	0.12
Different levels of sodium intake	8	8989	0.73 (0.58, 0.92)	0.008	89.8%	

Note: a *P* value calculated by meta regression. BP,blood pressure; CI, confidence interval; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk.

Five RCTs and 7 observational studies with a total of 3287 participants reported data regarding the effects of a low-salt diet on the rate of change in eGFR. Low-salt diet therapy did not produce an apparent beneficial effect on the rate of change in eGFR (MD:1.16; 95% CI: -2.02 to 4.33). There was evidence of significant heterogeneity across the included studies (I^2= 98.0%, P< 0.001) (figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to prespecified characteristics (Supplementary file 9).

Figure 3 Forest plot for rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SD, standard deviation.

Data for the effects of low-salt diet intervention on proteinuria or albuminuria were available in 13 studies including 2922 participants. The SMD for change in proteinuria or albuminuria was not statistically significant at -0.71 (95% CI: -1.66 to 0.24) compared with that in the control groups, with significant heterogeneity (I^2 = 97.3%, P=0.009) (figure 4). No statistical heterogeneity was found in any subgroup analyses for the effects of a low-salt diet on proteinuria or albuminuria (Supplementary file 9).

Figure 4 Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; Cr, creatinine; Na, sodium; SD, standard deviation; SMD, standard mean difference.

Effect of low-salt diet intervention on all-cause mortality and CV events

Data for the effects of low-salt diet intervention on all-cause mortality and CV events were available for 9 studies (two RCTs and seven observational studies) and 5 studies (one RCT and four observational studies), respectively. There were no clear effects of a low-salt diet on the risk of

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all-cause mortality (RR: 0.92; 95% CI: 0.53-1.46; 95997 participants and 4455 events) or CV events (RR: 1.01; 95% CI: 0.46-2.22; 93573 participants and 1989 events) compared with those in the control groups, with significant heterogeneity across the included studies (I^2 = 98.0%, 97.6% for all-cause mortality and CV events respectively, and *P* values for heterogeneity both less than 0.001) (figure 5). More significant benefits were found in studies that included patients with an eGFR of 30-59 mL/min/1.73 m^2 (*P* for subgroup heterogeneity = 0.02) in the subgroup analysis of CV events (Supplementary file 9).

Figure 5 Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death. CI, confidence interval; Cr, creatinine; CV, cardiovascular; Na, sodium; N, number of trials; RR, relative risk.

Effect of low-salt diet intervention on BP

Twenty studies reported the effect of a low-salt diet on BP, including 19 studies (3608 individuals) for SBP and 18 studies (3608 participants) for DBP. The MD changes in SBP and DBP were both statistically significant at -5.81 mmHg (95% CI: -9.63 to -1.99) and -2.3 mmHg (95% CI, -3.61 to -0.98), respectively (Supplementary file 10, Supplementary file 11). Significant heterogeneity across the included studies was found (I^2 = 87.1%, 68.2% for SBP and DBP respectively, and *P* values for heterogeneity both less than 0.05). Subgroup analyses showed that the association between the change in SBP or DBP and a low-salt diet was modified by the baseline BP of patients. Patients with baseline SBP >140 mmHg or DBP > 90 mmHg had an average reduction in SBP of 10 mmHg and in DBP of 5 mmHg (Supplementary file 9).

Sensitivity analyses and publication bias

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All sensitivity analyses results were robust (Supplementary file 12). The most notable exception was that the effect of low-salt diet therapy on the rate of change in eGFR became significant when studies with extreme outliers were omitted, and proteinuria or albuminuria became significant when the DL method was used to pool data. The risk of publication bias was found to be significant for the rate of change in eGFR, proteinuria or albuminuria, all-cause mortality and cardiovascular events, and BP (Supplementary file 13, Supplementary file 14).

DISCUSSION

Behavioral modifications can make an important contribution to delaying CKD progression to ESRD, and dietary salt restriction is one of the main concerns. Our meta-analysis showed that compared with control groups, low-salt diet intervention seems to reduced renal composite outcomes events and lowered BP (SBP and DBP) in CKD patients. No significant effects were observed regarding the decline rate of eGFR, change in proteinuria, risk of all-cause mortality or CV events. These results remained broadly consistent across major subgroups and sensitivity analyses. However, the existence of significant heterogeneity might limit the interpretation and clinical application of these results.

The previous review of 16 studies in 2006 addressing salt intake and kidney disease set out to establish whether variations in dietary sodium consumption influence kidney outcomes in patients with CKD [49]. The review suggested that increased salt intake was associated with worsening albuminuria and an increased likelihood of a reduction in GFR. In 2015 [50] and 2018 [51], the two meta-analyses of RCTs both suggested that moderate dietary salt restriction significantly reduced BP and proteinuria. However, it remains unanswered whether these benefits may translate to clinically

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significant reductions in renal composite outcomes events, all-cause mortality events and CV events. Our study, including 33 studies (13 RCTs and 20 observational studies), more than 90,000 patients, and 2329 renal composite outcomes events, first concluded that compared with the control groups, a low-salt diet produced a significant 28% reduction in renal composite outcomes events despite marked heterogeneity. Furthermore, the significant effect of a low-salt diet on BP (decreases in SBP and DBP of 5.8 mmHg and 2.3 mmHg, respectively) was confirmed again in our study and was recognized as an important target to slow the progression of CKD towards ESRD. The benefit of proteinuria from low-salt diet intervention became nonsignificant in our study when the restricted maximum likelihood or empirical Bayes methods were used to pool data. The standard DerSimonian-Laird procedure that was applied in previous meta-analyses can be very unstable with small numbers of studies [52,53].

Of note, statistical heterogeneity was found in all outcomes, and clinical heterogeneity was also an inevitable issue. Low-salt diets are not palatable to many [54], therefore, it is difficult to ensure sustained adherence to a low-sodium diet over long periods of follow-up. Sodium intake is quantified in distinct ways between studies (for example, 24 h urine collection, spot urine collection or dietary questionnaire). Given the use of diuretic may influence the urine sodium excretion, and eGFR, and it would influence the accuracy of quantified salt intake. We extracted data about use of diuretic. Ten studies have the data about diuretic use in the study we included, but only about the percentages of patients using diuretics in the intervention and control group. The percentage of patients using diuretics was no difference in most studies. There were no data about type and dose of diuretics, or the associations with the eGFR. These inherent problems were reflected in the paradoxically opposite findings of various studies, especially observational studies with more confounding factors. Only

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three studies (1 RCT and 2 observational studies) simultaneously reported the outcomes of eGFR and renal composite outcomes events. Furthermore, the different follow-up periods with 52 months of renal composite outcomes events and 12 months of eGFR should be noted. The inconsistent results of eGFR and renal composite outcomes events may partly be explained by the heterogeneity mentioned above. It should be assumed that heterogeneity limited the interpretation and generalizability of our results.

In CKD, BP is usually sodium sensitive, and RAAS blockers are considered the first-line therapy for hypertension and proteinuria. It has been known for over 2 decades that the effects of RAAS blockers are blunted by high sodium intake [55-57]. Thus, the significant effect of dietary sodium restriction on lowering BP and supporting the function of RAAS blockers was speculated to play important roles in reducing the risk of kidney failure events. Interestingly, a BP-independent effect of dietary sodium on the kidney is substantiated by data in healthy volunteers, in which dietary sodium restriction reduces albuminuria within the normal range, without a detectable effect on BP [58]. Similarly, proteinuria reduction by sodium restriction remained significant after adjustment for the decrease in BP [57]. These results suggested an independent renoprotective effect of low-sodium diets, both as a single measure and in combination with RAAS blockers.

In our study, there was no convincing evidence that a low-salt diet was associated with a lower incidence of all-cause mortality and CV events. Limited data on all-cause mortality (nine studies with 4455 events) and CV events (five studies with 1989 events) for the meta-analysis might introduce a risk of false-negative results because of low statistical power. More significant benefits of CV events were found in studies that included patients with eGFR 30-59 mL/min/1.73 m^2 in the subgroup analysis, while the effects were not significant in patients with eGFR less than 15

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mL/min/1.73 m^2. The significant subgroup heterogeneity (P =0.02) could be attributable to more CV risk factors in patients with ESRD than in those with eGFR 30-59 mL/min/1.73 m^2. The benefits of CV events from a low-salt diet were not enough to outweigh the increase in CV events caused by more risk factors (such as hyperuricemia, hyperphosphatemia, and vascular calcification). It is a hint at least for clinicians that low-salt diet education needs to be emphasized in the early stage of CKD.

Observational data from various studies showing a J-curve between sodium intake and renal and CV outcomes have raised concern regarding the safety of rigorous sodium restriction [59-61]. It is uncertain whether the presence of CKD modifies this association. In our meta-regression analysis, no clear relationships between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in kidney failure events were observed. A regimen of ACEIs and rigorous sodium restriction reduced blood pressure, proteinuria, and glomerular damage but unexpectedly aggravated tubule-interstitial damage in experimental renal disease [62]. This could provide a potential explanation for the association of very low sodium intake with worse renal outcome. Both high or extremely low dietary salt intake might be inappropriate for CKD patients. KDIGO recommends a daily sodium intake of < 2.0 g/d in patients with CKD without a clear lower limit. Taken together, these results should be interpreted, and the proper levels of dietary salt intake in CKD patients should be discussed. The current study might not be the final answer for the question of low-sodium diets and progression of kidney disease. Interventional studies are needed to ascertain the optimum level of dietary sodium intake for renal protection in patients with different CKD stages.

Our study has several potential limitations. First, as has been noted, the existence of statistical heterogeneity and inevitable clinical heterogeneity might raise critical concerns regarding validity, although we tried to resolve these concerns through subgroup analysis and sensitivity analysis.

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Heterogeneity in the study population as included in the umbrella term "CKD patients" remains a major concern for meaningful interpretation of the results. Because of the stages of CKD patients in the studies we included varies a lot, some are CKD 1-5, some are CKD 1-3, some are CKD 3-4, some are CKD 2-4, some are CKD 4-5, some are CKD 3-4. So it is difficult to stratify our analysis our outcomes by stage of CKD. We can only perform subgroup analyses according to the baseline eGFR of CKD patients. Second, cohort studies are prone to selection bias that is usually not found in RCT. Therefore, pooling the results from RCT and cohort studies together will bias the results of meta-analysis. So we performed subgroup analysis according to study design. There was no statistical heterogeneity according to study design for the outcomes. What is noteworthy is that low-salt diet therapy did not produce an beneficial effect on the rate of change in eGFR in RCTs (MD:-0.75; 95% CI: -7.58 to 6.08), but produce beneficial effect on the rate of change in eGFR in cohort studies (MD:2.48; 95% CI: 0.32 to 4.64). Third, many of the included studies with low quality had small sample sizes and a low incidence of events. Consequently, the uncertainty in the analyses was increased, resulting in wide CIs of effect measures. Fourth, confounding factors in the observational studies cannot be neglected, which might impact our results. Therefore, the findings of this study should be viewed as hypothesis-generating and need to be confirmed by further studies.

CONCLUSION

In conclusion, our meta-analyses suggested that a low-salt diet may reduce the risk of renal composite outcomes events in patients with CKD. However, there was no compelling evidence indicating that a low-salt diet could reduce the rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. The optimal dietary salt intake for patients with different CKD stages remains unanswered. Further well-designed RCTs targeting patients with different CKDs are

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required to assess and integrate these results.

Acknowledgments We thank American Journal Experts Corporation for helping in proofreading and editing the English of the final manuscript.

Contributors All authors take responsibility for the integrity of the data and the accuracy of data analysis. Study concept and design: L.W., X.S., H.S. and C.L.. Extraction, analysis and interpretation of data: H.S. and X.S.. Drafting of the manuscript: H.S. and L.W.. Critical revision of the manuscript: L.W. and X.S. Statistical analysis: H.S., C.L. and W.G.. Technical support: L.W., X.S. and W.G.. All authors have read and agreed to the submission to this journal of the manuscript.

Funding Support was provided by the National Natural Science Foundation of China (no. 82000655) and the Science and Technology Project of Shanxi Province (no. 201801D121218).The funders had no role in decision to publish and preparation of the manuscript.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data used to support the findings of this study are included within the article and supplementary information.

Ethics statement Our research include published studies, and does not require ethical committee approval.

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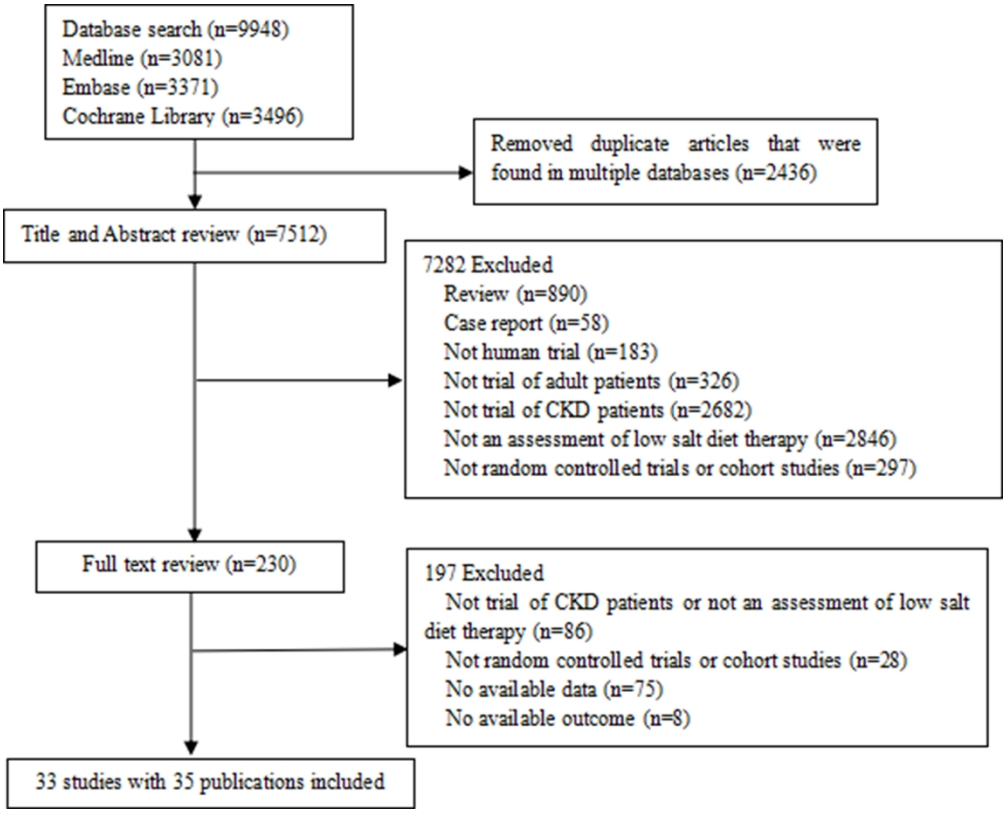
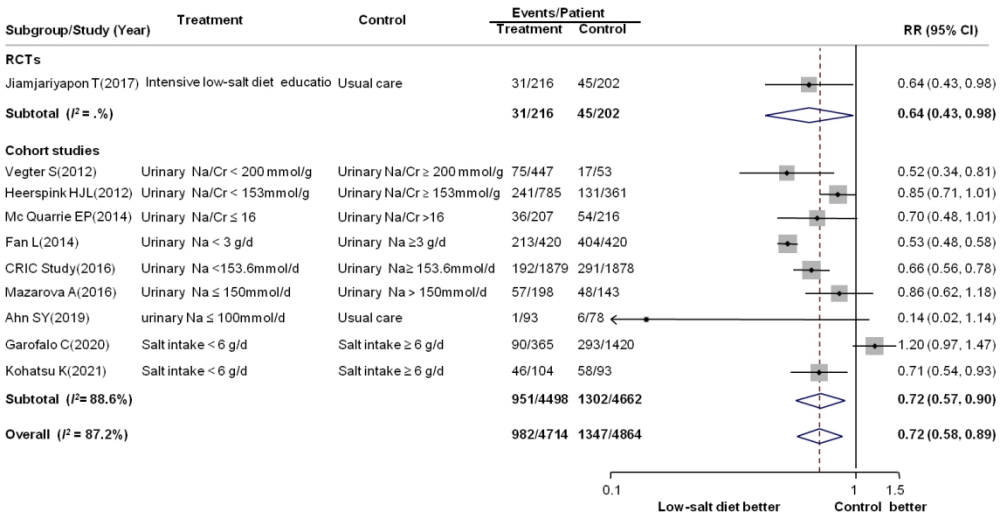
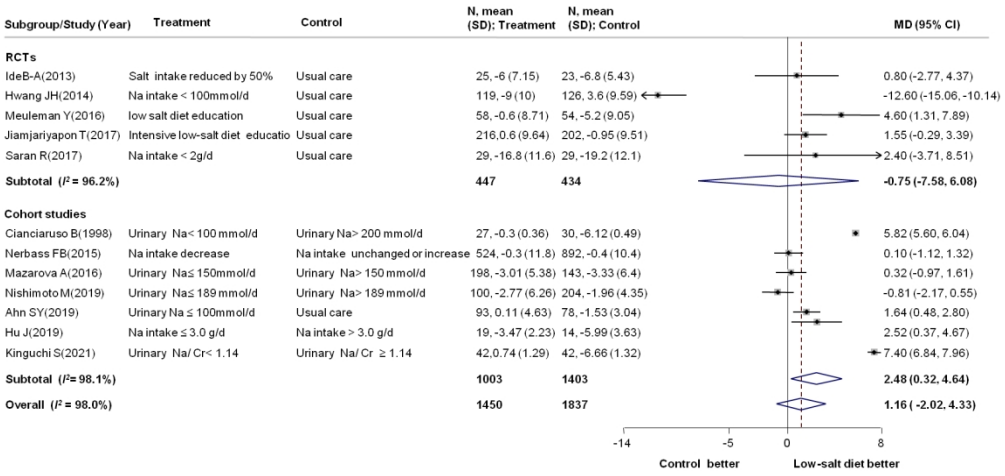

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Medline (n=3081)
Embase (n=3371)
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Removed duplicate articles that were found in multiple databases (n=2436)
Title and Abstract review (n=7512)
7282 Excluded
Review (n=890)
Case report (n=58)
Not human trial (n=183)
Not trial of adult patients (n=326)
Not trial of CKD patients (n=2682)
Not an assessment of low salt diet therapy (n=2846)
Not random controlled trials or cohort studies (n=297)
Full text review (n=230)
197 Excluded
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No available data (n=75)
No available outcome (n=8)
33 studies with 35 publications included

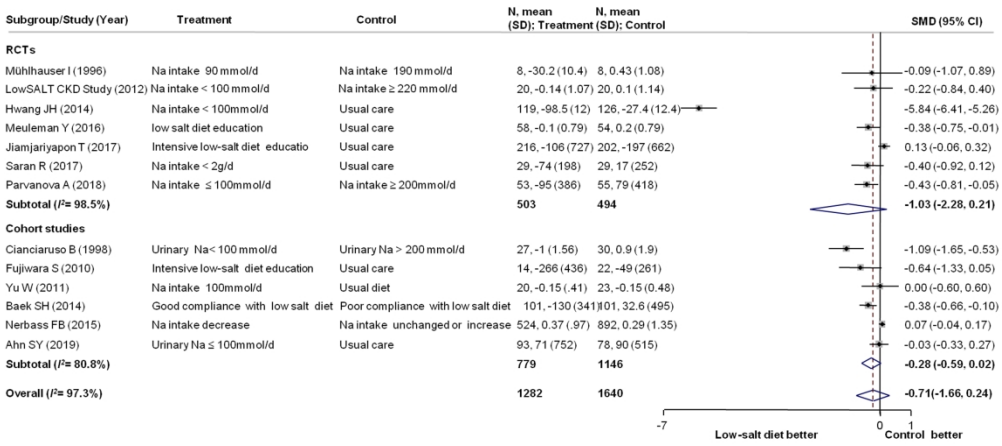
Figure 1



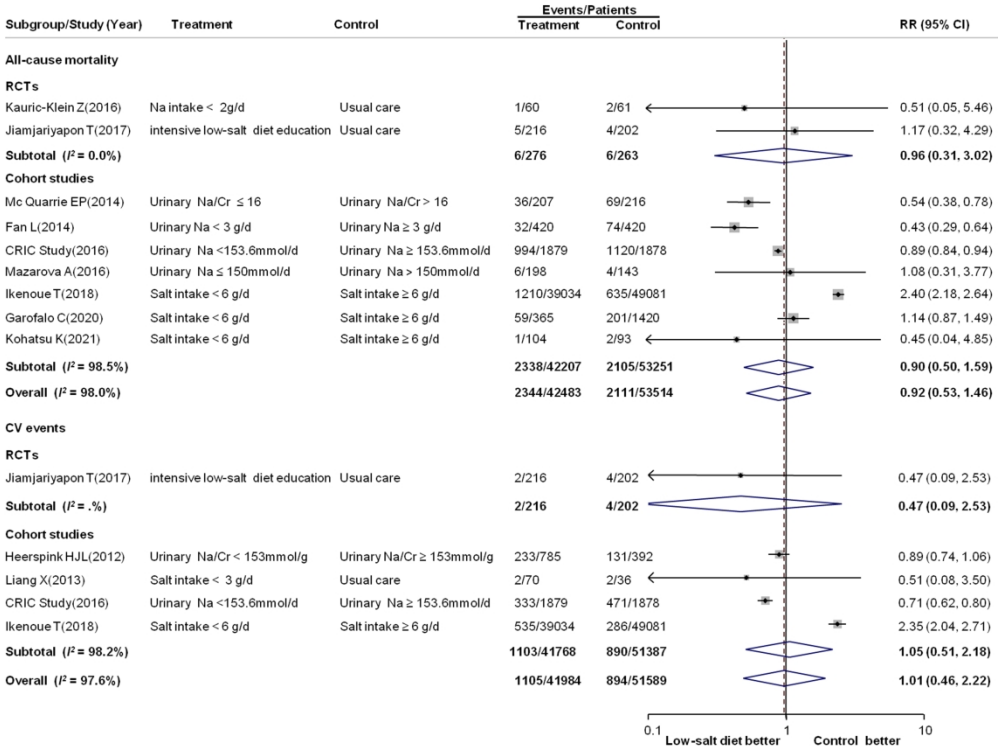
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Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang

Citation

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang. Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease. PROSPERO 2017 CRD42017072395 Available from: http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42017072395

Review question

This meta-analysis aims to evaluate the effects of different levels of dietary sodium intake on renal outcome in patients with chronic kidney disease (CKD).

Searches

All relevant randomized controlled trails (RCTs) and cohort studies will be identified and searched in following electronic databases without language restriction: Ovid MEDLINE SP (from 1946 to July 2017), EMBASE (from 1966 to July 2017), and the Cochrane Central Register of Controlled Trails(no date restriction). A search in www.clinicaltrials.gov will be used to identify the ongoing but unpublished studies about this field. We combined this strategy with a manual search of reference lists from identified trials and review articles.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/72395_STRATEGY_20170618.pdf

Types of study to be included

We will include RCTs and cohort studies that compared different levels of sodium intake for adult patients with CKD as well as studies in broader populations in which outcome data for subgroups with CKD could be disaggregated.

Condition or domain being studied

Chronic kidney disease

Participants/population

Inclusion criteria: Adults (no less than 18 years old) with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) at all disease stages will included without restrictions on gender or race. Data from subgroups of participants with CKD within studies with broader inclusion criteria (e.g. people from the general population, people with diabetes, people with cardiovascular disease) were also included.
Exclusion criteria: Studies which do not include any people with chronic kidney disease.

Intervention(s), exposure(s)

Low-sodium diet

Comparator(s)/control

No intervention or different levels of dietary sodium intake

Context

None

Primary outcome(s)

If the data allow:

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1) Kidney failure events: It will include more than 25% or 50% decrease in estimated glomerular filtration rate (eGFR), doubling of serum creatinine, end stage renal disease (ESRD);
2) Rate of change in the eGFR: The difference from baseline in eGFR divided by the number of years between creatinine measurements (ml/min/1.73m2 per year);
3) Change in proteinuria or albuminuria, including urinary protein excretion (UPE), urinary albumin excretion (UAE), protein to creatinine (PCR) and albumin to creatinine ratio (ACR);
4) Major cardiovascular events: It will include a composite including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death.

Secondary outcome(s)

1) 24-hour urinary sodium excretion;
2) Change in blood pressure;
3) All-cause mortality;
4) Change in markers of fluid overload (brain natriuretic peptide, N-terminal pro-brain natriuretic peptide, weight, bio-impedance analysis);
5) Adverse events: hypotensive episodes, undesirable change in blood lipids (low density lipoprotein, high-density lipoprotein);
6) Hospitalization.

Data extraction (selection and coding)

Data extraction will be carried out independently by two reviewers using standard data extraction forms. Disagreement will be resolved by consensus or by discussion with a third author. When detailed information that was needed for the analysis was not available, the original authors will be contacted to obtain the missing information through e-mail. Where multiple publications of one study exists, data will be extracted and grouped together as a single study with the most complete data.
The extracted data will include: study characteristics (study design, method of randomization, and withdrawals / dropouts); baseline patient characteristics (age, sex, race, country, BMI, sodium intake and its measurement, 24-hour urinary sodium excretion, mean proteinuria or albuminuria, eGFR, serum creatinine concentrations, blood lipids, blood pressure, diabetic status, cardiovascular events, smoking status, glycosylated hemoglobin, hemoglobin, antihypertensive drugs, antidiabetic drugs, different kidney disease diagnosis); change in eGFR; follow-up duration; outcomes: as listed in Primary outcomes and Secondary outcomes. When the required quantitative data is only provided in published figures, we will use g3 data software (http://www.frantz.fi/software/g3data.php) to extract exact numbers from figures.

Risk of bias (quality) assessment

For the RCTs, we will assess risk of bias for sequence generation, allocation concealment, blinding, selective reporting, incomplete outcome data and other sources of bias, and determined overall risk of bias based on predefined rules, utilizing the Cochrane Collaboration risk of bias tool. (reference: Higgins JP, Altman DG, et al. Cochrane Statistical Methods G. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials. BMJ 2011;343:d5928.). Furthermore, the quality of RCT study will be quantified with Jadad scale. (reference: Jadad AR, Moore RA, Carroll D, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Control Clin Trials 1996;17:1– 12.). The quality of cohort study will be independently evaluated by each investigator using Newcastle-Ottawa quality assessment scale. (reference: Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010; 25:603–5.).

Strategy for data synthesis

The results of the data extraction and quality assessment for each study will be presented in structured tables. For dichotomous outcomes, relative risks (RR) and 95% Cis will be calculated from the numbers of events and participants at risk for events. For continuous outcomes, the mean difference (MD) will be used, or the standardized mean difference (SMD) if different scales were used. Data will be pooled using the random-effects model but the fixed-effect model will also be used to ensure robustness of the model chosen and susceptibility to outliers.
Heterogeneity will be analyzed using Cochrane's Q and the I-squared statistic. We will explore potential

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heterogeneity by subgroup analysis and meta-regression.
Sensitivity analyses will be conducted to check the robustness of the effect of sodium-restricted treatment by excluding of trials with small sample size and low Jadad scores.
A two-sided p-value less than 0.05 was considered statistically significant and statistical analyses were performed using RevMan version 5.3 and STATA version 12.0.

Analysis of subgroups or subsets

If the data allow, subgroup analysis will be conducted by study design, follow-up duration, measures of sodium intake, different levels of sodium intake, difference in sodium intake between two groups, and baseline measures, including age, sex, BMI, eGFR, 24-hour urinary sodium excretion,mean serum creatinine, blood pressure, mean proteinuria or albuminuria, stage of CKD, presence of comorbidities (hypertension, diabetes and CVD), different kidney disease diagnosis (e.g. GN or Kidney Transplant Recipients) and whether RAAS blocker or other antihypertensive drug treatments.

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Anticipated or actual start date

15 July 2017

Anticipated completion date

01 February 2018

Funding sources/sponsors

None

Conflicts of interest

None known

Language

English

Country

China

Stage of review

Review_Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Acids, Noncarboxylic; Humans; Kidney; Renal Insufficiency, Chronic; Sodium; Sodium Chloride, Dietary; Sodium, Dietary

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NHS National Institute for Health Research

Date of registration in PROSPERO
18 July 2017

Date of publication of this version
18 July 2017

Details of any existing review of the same topic by the same authors
None

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions
18 July 2017

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Supplementary file 2 Search Strategy

Ovid MEDLINE

1. exp Sodium Chloride/ or sodium chloride.mp.

2. exp Sodium, Dietary/

3. exp Diet, Sodium-Restricted/

4. ((sodium or salt) adj5 (low or high or alter* or reduce* or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)).mp.

5. (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free).mp.

6. 1 or 2 or 3 or 4 or 5

7. exp Kidney/ or kidney.mp.

8. exp Nephrosis/ or nephrosis*.mp.

9. (renal or nephropath*).mp.

10. (chronic kidney or chronic renal).mp.

11. exp Kidney Diseases/

12. ((kidney or renal) adj (disease* or failure)).mp.

13. exp Kidney Function Tests/ or kidney function test*.mp.

14. (CKF or CKD or CRF or CRD).mp.

15. (end-stage renal or end-stage kidney or endstage renal or endstage kidney).mp.

16. (ESRF or ESKF or ESRD or ESKD).mp.

17. exp Uremia/ or ur?emi*.mp.

18. exp Renal Replacement Therapy/ or renal replacement therap*.mp.

19. exp Renal Insufficiency/ or renal insufficien*.mp.

20. exp Dialysis/ or dialysis.mp.

21. exp Renal Dialysis/ or renal dialys*.mp.

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22. ("pre dialysis" or pre-dialysis or predialysis).mp.

23. (h*emodialysis or h*emofiltration or h*emodiafiltration).mp.

24. exp Peritoneal Dialysis/ or peritoneal dialysis.mp.

25. (PD or CAPD or CCPD or APD or IPD).mp.

26. exp Kidney Transplantation/

27. ((kidney or renal) adj (transplant* or graft* or allograft*)).mp.

28. 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27

29. randomized controlled trial.pt.

30. controlled clinical trial.pt.

31. randomized.ab.

32. placebo.ab.

33. randomly.ab.

34. clinical trials as topic.sh.

35. trial.ti.

36. 29 or 30 or 31 or 32 or 33 or34 or 35

37. (animals not (humans and animals)).sh.

38. 36 not 37

39. exp Cohort Studies/

40. exp Longitudinal Studies/

41. exp Follow-Up Studies/

42. exp Prospective Studies/

43. exp Retrospective Studies/

44. cohort.ti,ab.

45. longitudinal.ti,ab.

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46. prospective.ti,ab.

47. retrospective.ti,ab.

48.38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47

49. 6 and 28 and 48

Embase

#1. 'sodium chloride'/exp OR 'sodium chloride'

#2. 'sodium intake'/exp

#3. 'salt intake'/exp

#4. 'sodium restriction'/exp

#5. 'high sodium intake'/exp

#6. 'high salt diet'/exp

#7. 'high sodium' OR 'low sodium' OR 'sodium alter*' OR 'sodium reduce*' OR 'sodium reduction' OR 'sodium restrict*' OR 'sodium intake*' OR 'sodium diet*' OR 'sodium food' OR 'sodium increas*' OR 'sodium decreas*' OR 'sodium change*' OR 'sodium changing' OR 'sodium free'

#8. 'high salt' OR 'low salt' OR 'salt alter*' OR 'salt reduce*' OR 'salt reduction' OR 'salt restrict*' OR 'salt intake*' OR 'salt diet*' OR 'salt food' OR 'salt increas*' OR 'salt decreas*' OR 'salt change*' OR 'salt changing' OR 'salt free'

#9. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8

#10. 'kidney'/exp OR 'nephrosis'/exp

#11. kidney OR renal OR nephropath* OR nephrosis

#12. 'chronic kidney' OR 'chronic renal'

#13. 'kidney disease'/exp OR 'kidney failure'/exp

#14. 'kidney disease*' OR 'renal disease*' OR 'kidney failure' OR 'renal failure'

#15. 'kidney function test'/exp OR 'kidney function test*'

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#16. ckf OR ckd OR crf OR crd

#17. 'end-stage kidney' OR 'end-stage renal' OR 'endstage kidney' OR 'endstage renal'

#18. esrf OR eskf OR esrd OR eskd

#19. 'uremia'/exp OR ur?emi*

#20. 'renal replacement therapy'/exp OR 'renal replacement therap*'

#21. 'renal insufficien*'

#22. 'dialysis'/exp OR dialys*

#23. 'pre dialysis' OR predialysis

#24. 'hemodialysis'/exp OR 'hemofiltration'/exp OR 'hemodiafiltration'/exp

#25. hemodialysis OR haemodialysis OR hemofiltration OR haemofiltration OR hemodiafiltration OR haemodiafiltration

#26. 'peritoneal dialysis'/exp OR 'peritoneal dialysis'

#27. pd OR capd OR ccpd OR apd OR ipd

#28. 'kidney transplantation'/exp OR 'kidney graft'/exp OR 'kidney allograft'/exp

#29. 'kidney transplant*' OR 'renal transplant*' OR 'kidney graft*' OR 'renal graft*' OR 'kidney allograft*' OR 'renal allograft*'

#30. #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29

#31. 'randomized controlled trial'/exp OR 'randomized controlled trial' OR 'controlled clinical trial'/exp OR 'controlled clinical trial' OR random*OR 'clinical trials'/exp OR 'clinical trials' OR trial:ti

#32. 'human'

#33. animal$ OR nonhuman

#34. #32 AND #33

#35. #33 NOT #34

#36. #31 NOT #35

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#37.'cohort analysis'/exp OR 'longitudinal study'/exp OR 'prospective study'/exp OR 'retrospective study'/exp

#38. (cohort OR concurrent OR incidence OR longitudinal OR followup OR 'follow up' OR prospective OR retrospective) NEXT/1 (analys* OR design* OR evaluation* OR research OR stud* OR survey* OR trial*)

#39. #36 OR #37 OR #38

#40. #9 AND #30 AND #39

CENTRAL

#1 MeSH descriptor: [Sodium Chloride] explode all trees

#2 "sodium chloride":ti,ab,kw (Word variations have been searched)

#3 MeSH descriptor: [Sodium, Dietary] explode all trees

#4 MeSH descriptor: [Diet, Sodium-Restricted] explode all trees

#5 ((sodium or salt) near/5 (low or high or alter* or reduce* or reducing or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)):ti,ab,kw

#6 (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free):ti,ab,kw

#7 (#1 or #2 or #3 or #4 or #5 or #6)

#8 MeSH descriptor: [Kidney] explode all trees

#9 MeSH descriptor: [Nephrosis] explode all trees

#10 (kidney or renal or nephropath* or nephrosis):ti,ab,kw

#11 (chronic next kidney or chronic next renal):ti,ab,kw

#12 MeSH descriptor: [Kidney Diseases] explode all trees

#13 (kidney or renal) next (disease* or failure):ti,ab,kw

#14 MeSH descriptor: [Kidney Function Tests] explode all trees

#15 “kidney function test*”:ti,ab,kw

#16 (CKF or CKD or CRF or CRD):ti,ab,kw

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#17 ("end-stage kidney" or "end-stage renal" or "endstage kidney" or "endstage renal"):ti,ab,kw

#18 (ESRF or ESKF or ESRD or ESKD):ti,ab,kw

#19 MeSH descriptor: [Uremia] explode all trees

#20 ur?emi*:ti,ab,kw

#21 MeSH descriptor: [Renal Replacement Therapy] explode all trees

#22 "renal replacement therap*":ti,ab,kw

#23 MeSH descriptor: [Renal Insufficiency] explode all trees

#24 "renal insufficien*":ti,ab,kw

#25 MeSH descriptor: [Dialysis] explode all trees

#26 dialys*:ti,ab,kw

#27 MeSH descriptor: [Renal Dialysis] explode all trees

#28 "renal dialys*":ti,ab,kw

#29 "pre dialysis" or pre-dialysis or predialysis:ti,ab,kw

#30 (h*emodialysis or h*emofiltration or h*emodiafiltration):ti,ab,kw

#31 MeSH descriptor: [Peritoneal Dialysis] explode all trees

#32 "peritoneal dialys*":ti,ab,kw

#33 (PD or CAPD or CCPD or APD or IPD):ti,ab,kw

#34 MeSH descriptor: [Kidney Transplantation] explode all trees

#35 (kidney or renal) near (transplant* or graft* or allograft*):ti,ab,kw

#36 (#8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35)

#37 (#7 and #36)

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Supplementary file 3 Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up, months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9[a]	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Sodium intake 80-100 mmol/d	24-h urinary sodium	Usual care	1.39c	143/88	NA
Fujiwara S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPAKUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5[b]	144/95	Non- industry

Heerspink HJL[24] (2012)	Type 2 diabetic nephropathy	1177	65	59	Cohort study	30	Renal composite outcomes, CV events	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 153mmol/g	44	155/84	Industry and non- industry
LowSALT CKD Study[25,26] (2012)	Hypertensive patients with stage 3–4 CKD	20	75	69	Randomised crossover trial	1	UACR, SBP, DBP	Sodium intake < 100 mmol/d	Sodium intake < 100 mmol/d	24-h urinary sodium	Sodium intake ≥ 220 mmol/d	32	151/82	Non- industry
de Brito-Ashurst I[27] (2013)	eGFR < 60 mL/min and mean BP > 130/80 mm Hg	48	58	58	Randomised parallel-group controlled trial	6	Change in eGFR	Intensive low-salt diet education	Traditional meals had salt reduced by 50%	24-h urinary sodium	Usual care	41.5	153/85	Non- industry
Liang X[28] (2013)	PD	106	54	33	Cohort study	1	CV events, SBP, DBP	Health education to restrict water-sodium intake strictly	Salt intake not more than 3 g/d	The 3-day diet diary was collected once a month by the special nurse. Compared with the Food Composition, the daily salt intake were estimated	Usual care	NA	149/88	NA
Mc Quarrie EP[29] (2014)	Patients with CKD and had at least one 24-hour urinary sodium measurement	423	50	52	Cohort study	102	Renal composite outcomes, all- cause mortality	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine	Urinary sodium/creatinine > 16	48	139/79	Non- industry
Fan L[4] (2014)	GFR 13–55 mL/min/1.73m^2	840	60	52	Cohort study	72	Renal composite outcomes, all- cause mortality	Urinary sodium < 3 g/d	Urinary sodium < 3 g/d	24-h urinary sodium	Urinary sodium ≥ 3 g/d	32.5	132/81	Non- industry
Baek SH[30] (2014)	Hypertensive patients with albuminuria	202	51	50	Cohort study	2	UAE, SBP, DBP	Good compliance with an low salt diet	Good compliance with an low salt diet (24-hurinary sodium reduction ratio < 50%)	24-hurinary sodium	Poor compliance with an low salt diet	77.1[c]	121/73	Industry and non- industry

Hwang JH[31] (2014)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	245	50	50	Randomised parallel-group controlled trial	2	eGFR, UAE, SBP, DBP	Sodium intake < 100mmol/d	Sodium intake < 100mmol/d	24-hurinary sodium	Usual care	67.3	131/79	Industry
Rodrigues Telini LS[32] (2014)	HD	39	38	58	Randomised parallel-group controlled trial	4	SBP, DBP	2 g of sodium reduction in their habitual diet	2 g of sodium reduction in their habitual diet	Salt intake was based on a 72-halimentary registry	Usual diet	NA	146/86	Non- industry
Nerbass FB[33] (2015)	eGFR 59–30 ml/min/1·73 m^2	1416	39	73	Cohort study	12	eGFR, UACR, SBP, DBP	Sodium intake decrease	Sodium intake decrease	Estimated 24-hurinary sodium	Sodium intake unchanged or increase	52.7	134/73	Industry and non- industry
CRIC Study[3,34] (2016)	eGFR20-70mL/min/1.73 m^2	3757	56	58	Cohort study	78	Renal composite outcomes, all- cause mortality, CV events	Urinary sodium < 153.6mmol/d	Urinary sodium < 153.6mmol/d	24-hurinary sodium	Urinary sodium ≥153.7 mmol/d	44.5	128/71	Non- industry
Mazarova A[5] (2016)	eGFR < 30 mL/min/1.73 m^2	341	62	64	Cohort study	18	Change in eGFR, renal failure, all-cause mortality	Urinary sodium ≤ 150mmol/d	Urinary sodium ≤ 150mmol/d	24-hurinary sodium	Urinary sodium> 150 mmol/d	17.1	NA	No funding supported
Kauric-Klein Z[35] (2017)	HD	121	50	60	Randomised parallel-group controlled trial	3	All- cause mortality, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	Sodium intake checklist developed by Millar and Beard	Usual care	NA	165/87	Non- industry
Meuleman Y[36] (2017)	eGFR ≥ 20mL/min/1.73m^2 and UPE > 0.2 g/L	112	82	55	Randomised parallel-group controlled trial	6	eGFR, UPE, SBP, DBP	Usual care and low salt diet education	low salt diet education	24-hurinary sodium	Usual care	49.7	129/76	Non- industry

Jiamjariyapon T[37] (2017)	Stage 3-4 CKD with diabetes or hypertension	418	27	62	Cluster randomized controlled trial	24	Renal composite outcomes, all-cause mortality, eGFR, CV events, UACR, SBP, DBP	Usual care and intensive low-salt diet education	Intensive low-salt diet educatio (Sodium intake less than 2,400 mg/d)	24-hurinary sodium	Usual care	41.5	125/76	Non- industry
Saran R[38] (2017)	eGFR 15–60 ml/min/1.73 m^2	58	60	61	Randomized crossover trial	1	eGFR, UACR, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	24-hurinary sodium	Usual care	38.7	130.3/NA	Non- industry
Huang B[39] (2017)	HD	90	43	55	Cluster randomized controlled trial	6	SBP, DBP	Self-management support, restrict salt intake	Self-management support, restrict salt intake	Dietary salt intake was measured using slightly modified balance formulas: (1) Sodium intake (mol)=[Na+1(mmol/L)×ECW(L)－Na+ 2(mmol/L)×(ECW– IDWG)(L)]/1000; (2) Salt intake (g/day)=sodium intake(mol)×58.4 (g/mol)/2(day)	Usual care	NA	154/86	Non- industry
Ikenoue T[6] (2018)	HD	88115	60	65	Cohort study	12	All-cause mortality, CV events	Salt intake < 6 g/d	Salt intake < 6 g/d	Intra-dialytic weight loss was used to calculate salt intake	Salt intake ≥ 6 g/d	NA	NA	No funding supported
Parvanova A[40] (2018)	Patients with type 2 diabetes and macroalbuminuria	108	89	64	Randomised parallel-group controlled trial	3	UACR	Sodium intake ≤ 100mmol/d	Sodium intake ≤ 100mmol/d	24-hurinary sodium	Sodium intake ≥ 200mmol/d	87.36	146/80	Industry
Martinez MG[41] (2019)	Pre-dialysis CKD with UPE > 500 mg/d	84	52	56	Cohort study	6	Change in SBP	Urinary sodium reduced	Urinary sodium reduced	24-hurinary sodium	Urinary sodium increased	49[b]	142/NA	NA
Nishimoto M[42] (2019)	UACR 30–599 mg/g and eGFR ≥ 50 ml/min/1.73 m^2	304	70	59	Cohort study	12	Change in eGFR, SBP, and DBP	Urinary sodium ≤ 189 mmol/d	Urinary sodium ≤ 189 mmol/d	24-hurinary sodium	Urinary sodium > 189 mmol/d	64.93	140/83	Non- industry

Ahn SY[43] (2019)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	171	49	50	Cohort study	12	Rate of change in eGFR, renal composite outcomes, change in UAE, SBP, DBP	Intensive low-salt diet education	Intensive low-salt diet education to achieve urinary sodium ≤ 100mmol/d	24-hurinary sodium	Usual care	64.2	124/75	Non- industry
Hu J[44] (2019)	PD	33	58	42	Cohort study	12	Residual renal Ccr	Sodium intake ≤ 3.0 g/d	Sodium intake ≤ 3.0 g/d	Daily sodium intake of the patients was calculated based on the 3-day dietary record	Sodium intake > 3.0 g/d	8.3	146/95	Non- industry
Humalda JK[45] (2020)	CKD stages 1 to 4 or kidney transplants recipients, and eGFR was ≥ 25 mL/min/ 1.73 m2	99	84	57	RCT	6	SBP, DBP	Self-management to restrict salt intake	Self-management to restrict salt intake	A detailed interactive food diary	Usual care	55	139/83	Non- industry
Garofalo C[46] (2020)	CKD stages 1–5	1785	58	64	Cohort study	49	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	37	NA	NA
Kinguchi S[47] (2021)	Diabetic kidney disease patients with HbA1c 7.0–10.0%, eGFR ≥ 45 ml/min/1.73 m2, and UACR ≥ 30 mg/g	84	76	65	Cohort study	6	UACR	urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios	urinary sodium-to-creatinine ratios ≥ 1.14	67	142/79	Non- industry
Kohatsu K[48] (2021)	CKD stages 3–5	197	76	71	Cohort study	17	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	24.2	133/74	Non- industry

Note: Ccr, clearance of creatinine; CKD, chronic kidney disease; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; ESRD, end stage renal disease; ECW, extracellular water; HD, hemodialysis; IDWG, interdialytic weight gain; NA, no available; PD, peritoneal dialysis; SBP, systolic blood pressure; UACR, urinary albumin to creatinine ratio; UAE, urine albumin excretion; UPE, urinary protein excretion; the units for eGFR: mL/min/1.73m². [a] The units for eGFR: mL/min, [b] Only Ccr obtained, the units for Ccr: mL/min; [c] Only serum creatinine obtained, the units for serum creatinine: μmol/L.

Supplementary file 4 Table S2 Summary of characteristics of included trials and patients.

Outcomes	No. of Studies/ No. of Patients	Sample Size	Follow-up, months	Age, years	Male (%)	Lost follow-up rate
Renal composite outcomes	10/9578	462 (305-1329)	43 (23-74)	58 (52-64)	59.2%	3.2%
Rate of change in eGFR	12/3287	142 (57-332)	12 (6-17)	55(48-61)	56.6%	7.5%
Proteinuria	13/2922	108 (40-224)	3 (1-12)	50 (43-66)	57.2%	8.8%
All-cause death	9/95997	423 (231-2771)	24 (15-75)	62 (55-65)	60.6%	0.03%
CV events	5/93573	1177 (262-45936)	24 (6.5-54)	59(45-63)	60.8%	0.02%
SBP	19/3608	103 (38-213)	3 (1-6)	55 (41-60)	57.1%	7.5%
DBP	18/3524	36 (20-90)	12 (2-54)	55 (40-60)	57.3%	7.7%

Note:Values are given as median (quartile). CV, cardiovascular; eGFR, estimated glomerular filtration rate.

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Supplementary file 5 Table S3 Assessment of risk of bias for RCTs.

Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
1. Bias arising from the randomization process																
1.1 Was the allocation sequence random?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias arising from the randomisation process?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
2. Risk of bias due to deviations from intended interventions																
2.1 Were participants aware of their assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	Other	High	High	High	High	Low	High	High
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	High	High	High	High	High	High	High	High
2.3 If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?	N/PN	Y/PY	NA/NI		Low		Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.4 If Y/PY/NI to 2.3: Were these deviations likely to have affected the outcome?	N/PN	Y/PY	NA/NI													
2.5 If Y/PY to 2.4: Were these deviations from intended intervention balanced between groups?	Y/PY	N/PN	NA/NI													

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomised?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to deviations from intended interventions?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
3. Bias due to missing outcome data																
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	High	High	High	High	Low	High	High	Low
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?	Y/PY	N/PN	NA	Low	Low				Low	Low	Low	Low		Low	Low	
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	N/PN	Y/PY	NA/NI													
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to missing outcome data?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
4. Bias in measurement of the outcome																
4.1 Was the method of measuring the outcome inappropriate?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?	N/PN	Y/PY	NI	Low	High	Low	High	High	Low	High	High	High	Low	High	Low	High
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI		Low		Low	Low		Low	Low	Low		Low		Low
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias in measurement of the outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
5. Bias in selection of the reported result																
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:																
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
5.3 ... multiple eligible analyses of the data?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction bias due to selection of the reported results?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
Overall bias																
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the overall predicted direction of bias for this outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null

Note: Y, yes; PY, probably yes; PN, probably no; N, no; NA, not applicable; NI, no information; RCT, randomized controlled trial.

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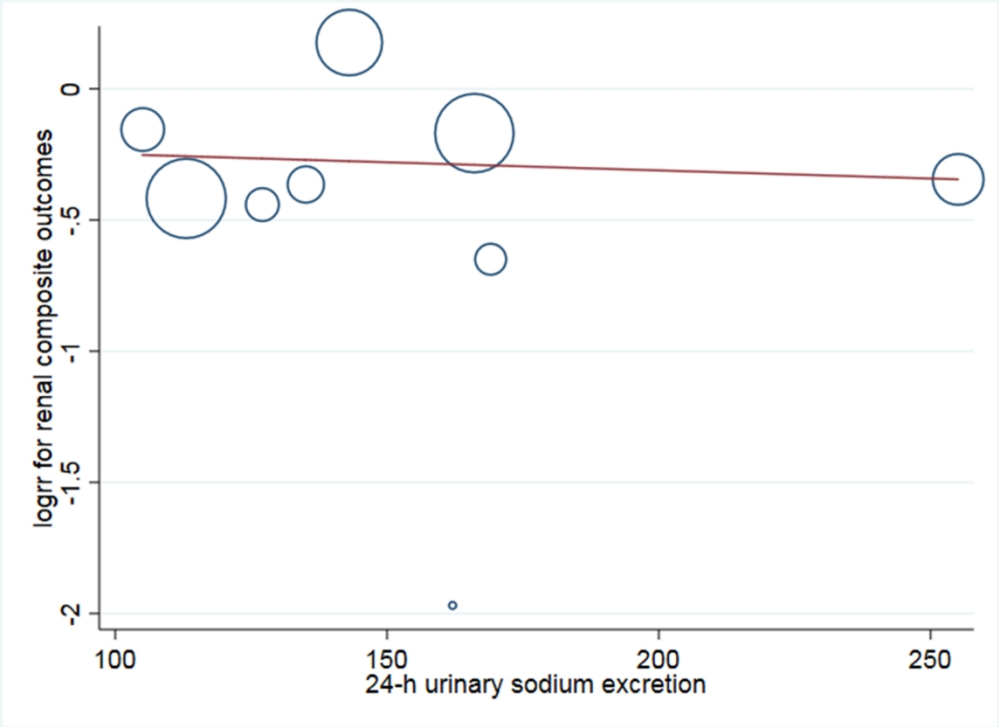
Supplementary file 6 Table S4 Quality assessment for cohort studies.

Study (Year)	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Scores
Cianciaruso B[19] 1998	★	★	★	★	—	★	★	★	7★
Fujiwara S[21] 2010	★	★	★	★	★	★	—	★	7★
Yu W[22] 2011	★	★	★	★	★	★	—	★	7★
Vegter S[23] 2012	★	★	★	★	—	★	★	★	7★
Heerspink HJL[24] 2012	★	★	★	★	★	★	—	★	7★
Liang X[28] 2013	★	★	★	★	★	★	—	★	7★
Mc Quarrie EP[29] 2014	★	★	★	★	★	★	★	★	8★
Fan L[3] 2014	★	★	★	★	—	★	★	—	6★
Baek SH[30] 2014	★	★	★	★	—	★	—	★	6★
Nerbass FB[33] 2015	★	★	★	★	★	★	—	★	7★
CRIC Study[3,34] 2016	★	★	★	★	★	★	★	★	8★
Mazarova A[5] 2016	★	★	★	★	★	★	—	★	7★
Ikenoue T[6] 2018	★	★	★	★	★	★	—	★	7★
Martinez MG[41] 2019	★	★	★	★	★	★	★	★	8★

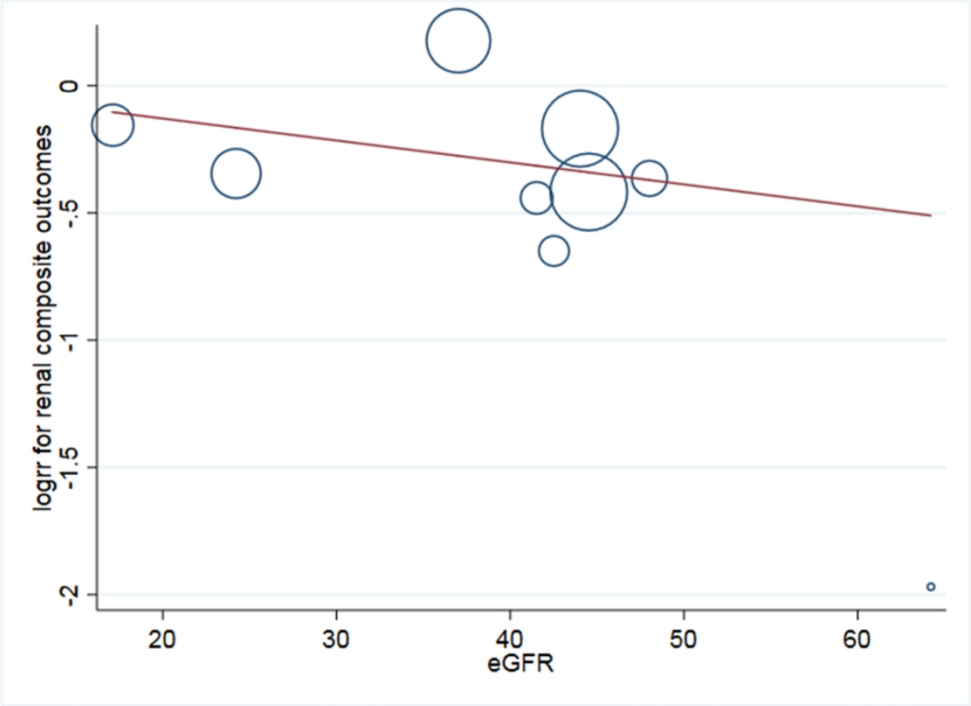
Nishimoto M[42] 2019	★	★	★	★	★	★	—	★	7★
Ahn SY[43] 2019	★	★	★	★	★	★	—	★	7★
Hu J[44] 2019	★	★	★	★	★	★	—	★	7★
Garofalo C[46] 2020	★	★	★	★	★	★	★	★	8★
Kinguchi S[47] 2021	★	★	★	★	★	★	—	★	7★
Kohatsu K[48] 2021	★	★	★	★	★	★	—	★	7★

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability. —: no score; It is considered that the queue follow-up is appropriate when the follow-up rate is more than 80%.

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Supplementary file 8

Supplementary file 9 Table S5 Subgroup analysis of outcomes except renal failure events.

Subgroup	No. of trials	n	RR/MD/SMD (95%CI)	*P* for RR/MD/SMD	I^2	*P* for heterogeneity test[a]
Change in eGFR（mL/min/1.73m²）						
Study type						
RCT	5	881	-0.75 (-7.58, 6.08)	0.83	96.2%	0.28
Cohort studies	7	2406	2.48 (0.32, 4.64)	0.02	98.1%	
Baseline eGFR (mL/min/1.73m²)						
60-89	4	804	-1.00 (-7.41, 5.42)	0.76	99.2%	0.48
30-59	6	2109	2.58 (-0.44, 5.61)	0.10	95.4%	
<15	2	374	1.24 (-0.88, 3.37)	0.25	66.2%	
Mean age (years)						
<58	5	106	0.44 (-4.40, 5.27)	0.86	98.5%	0.70
≥58	6	2669	1.68 (-1.83, 5.18)	0.35	97.8%	
Baseline BP (mmHg)						
<140/90	6	2420	-0.52 (-4.36, 3.32)	0.32	97.8%	0.41
SBP≥140 或 DBP≥90	4	469	2.54 (-2.47, 7.55)	0.55	70.1%	
Comparator of control group						
Usual care or usual diet	6	1052	-0.38 (-5.14, 4.39)	0.88	95.8%	0.85
Different levels of sodium intake	6	2235	2.62 (0.31, 4.94)	0.03	98.2%	
Proteinuria or Albuminuria						
Study type						
RCT	7	997	-1.03 (-2.28, 0.21)	0.11	98.4%	0.45
Cohort studies	6	1925	-0.28 (-0.59, 0.02)	0.07	80.8%	
Baseline eGFR (mL/min/1.73m²)						
60-89	7	821	-1.06 (-2.32, 0.20)	0.10	98.2%	0.42
30-59	6	1201	-0.23 (-0.50, 0.04)	0.10	79.6%	
Mean age (years)						
<58	7	846	-1.12 (-2.35, 0.12)	0.08	98.2%	0.34
≥58	6	2076	-0.13 (-0.34, 0.09)	0.24	64.2%	
Baseline BP (mmHg)						
<140/90	8	200	-0.82 (-1.55, -0.10)	0.01	98.3%	0.65
SBP≥140 或 DBP≥90	4	2665	-0.39 (-0.68, -0.11)	0.03	0.0%	
Comparator of control group						
Usual care or usual diet	7	1083	-1.01 (-2.18, 0.15)	0.09	98.4%	0.12
Different levels of sodium intake	6	1839	-0.34 (-0.69, 0.00)	0.05	81.7%	
All-cause mortality						
Study type						
RCT	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.99
Cohort studies	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
Baseline eGFR (mL/min/1.73m²)						
30-59	4	5438	0.65 (0.41, 1.02)	0.06	85.2%	0.008
15-29	2	2126	1.14 (0.88, 1.48)	0.33	0.0%	
<15	3	88433	1.34 (0.41, 4.39)	0.62	43.4%	

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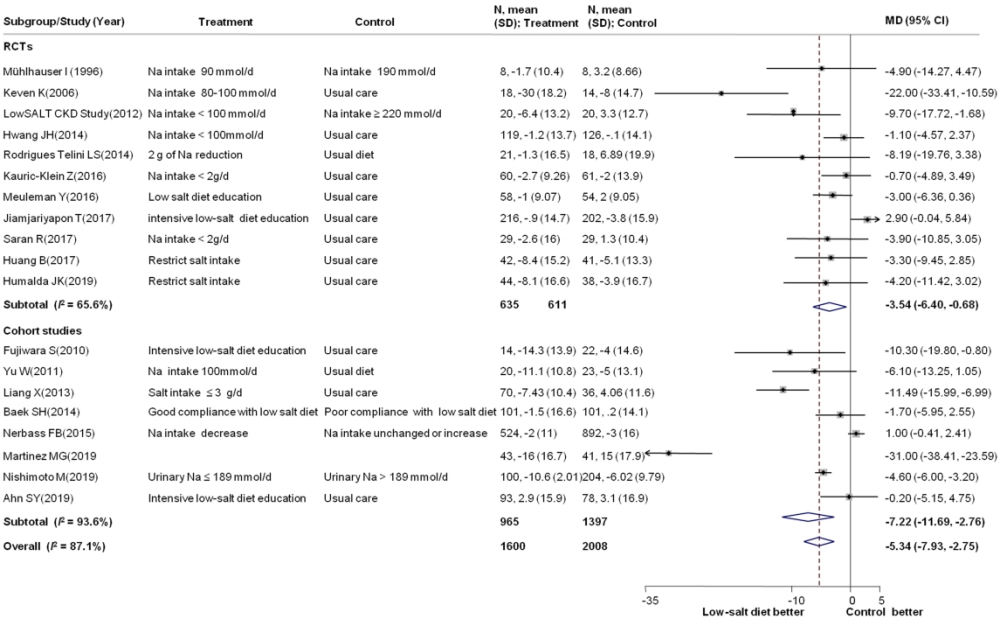
Mean age (years)						
<58	3	704	0.61 (0.37, 1.00)	0.05	90.1%	0.08
≥58	6	90977	1.34 (0.76, 2.37)	0.32	84.2%	
Baseline BP (mmHg)						
<140/90	5	5438	0.64 (0.41, 0.99)	0.05	80.6%	0.86
SBP≥140 或 DBP≥90	1	121	0.51 (0.05, 5.46)	0.58	-	
Comparator of control group						
Usual care or usual diet	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.90
Different levels of sodium intake	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
CV events						
Study type						
RCT	1	418	0.47 (0.09, 2.53)	0.88	-	0.51
Cohort studies	4	93155	1.05 (0.51, 2.18)	0.89	98.2%	
Baseline eGFR (mL/min/1.73m^2)						
30-59	3	5352	0.78 (0.63, 0.95)	0.02	57.1%	**0.02**
<15	2	88221	1.50 (0.39, 5.85)	0.56	58.3%	
Mean age (years)						
<58	2	3863	0.71 (0.62, 0.80)	<0.001	0.0%	0.40
≥58	3	89710	1.20 (0.49, 2.91)	0.69	97.3%	
Baseline BP (mmHg)						
<140/90	2	4175	0.71 (0.62, 0.80)	<0.001	0.0%	0.62
SBP≥140 或 DBP≥90	2	1283	0.88 (0.74, 1.05)	0.17	0.0%	
Comparator of control group						
Usual care or usual diet	2	524	0.49 (0.14, 1.73)	0.27	0.0%	0.52
Different levels of sodium intake	3	93049	1.14 (0.53, 2.46)	0.74	98.8%	
SBP（mmHg）						
Study type						
RCT	11	1246	-3.54 (-6.40, -0.68)	0.02	65.6%	0.43
Cohort studies	8	2362	-7.22 (-11.69, -2.76)	0.002	93.6%	
Baseline eGFR (mL/min/1.73m^2)						
60-89	7	1017	-3.30 (-5.26, -1.34)	0.001	31.7%	0.64
30-59	7	2210	-6.10 (-11.62, -0.59)	0.03	92.8%	
< 15	4	349	-5.66 (-11.62, 0.30)	0.06	76.0%	
Mean age (years)						
< 58	12	1215	-7.45 (-11.76, -3.14)	0.001	85.6%	0.25
≥ 58	7	2393	-2.39 (-5.65, 0.87)	0.15	87.3%	
Baseline BP (mmHg)						
< 140/90	10	2786	-0.97 (-2.75, 0.81)	0.28	46.9%	**0.02**
SBP ≥ 140 or DBP ≥ 90	9	822	-10.24 (-15.37, -5.12)	< 0.001	88.4%	
Comparator of control group						
Usual care or usual diet	13	1546	-4.28 (-7.20, -1.36)	0.004	74.3%	0.05
Different levels of sodium intake	6	2062	-7.55 (-12.83, -2.28)	0.005	94.6%	
DBP（mmHg）						
Study type						
RCT	7	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.85
Cohort studies	11	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

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Baseline eGFR (mL/min/1.73m^2)						
30-59	7	1017	-1.74 (-3.40, -0.08)	0.04	56.7%	0.13
15-29	6	2126	-0.53 (-1.25, 0.20)	0.16	11.9%	
< 15	4	349	-4.94 (-7.08, -2.80)	<0.001	23.2%	
Mean age (years)						
< 58	11	1131	-2.78 (-4.35, -1.22)	<0.001	50.8%	0.34
≥ 58	7	2393	-0.86 (-1.77, 0.04)	0.06	62.2%	
Baseline BP (mmHg)						
< 140/90	9	2747	-0.94 (-1.90, 0.02)	0.056	5.6%	**0.04**
SBP ≥ 140 或 DBP ≥ 90	9	777	-4.05 (-6.61, -1.49)	0.002	0.2%	
Comparator of control group						
Usual care or usual diet	11	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.12
Different levels of sodium intake	7	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

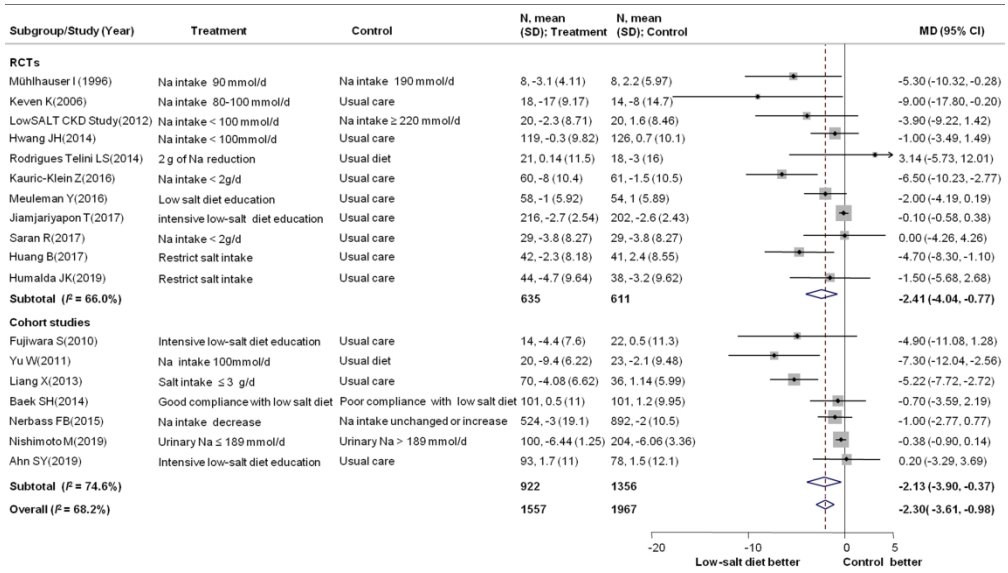
Note: [a] *P* value calculated by meta regression. BP, blood pressure; CI, confidence interval; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

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Supplementary file 10 Forest plot for SBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SBP, systolic blood pressure; SD, standard deviation.

317x210mm (300 x 300 DPI)



Supplementary file 11 Forest plot for DBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; DBP, diastolic blood pressure; SD, standard deviation.

309x186mm (300 x 300 DPI)

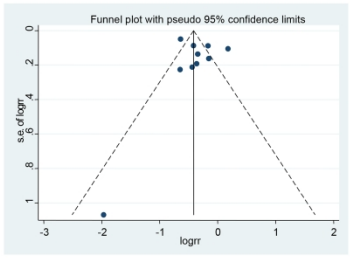
Supplementary file 12 Table S6 Sensitivity analysis of outcomes.

Sensitivity analysis	Renal composite outcomes			The rate of change in eGFR			Proteinuria or albuminuria			All-cause mortaity		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD/SMD (95%CI)	τ^2	No./n	RR (95% CI)	τ^2
Omit studies of follow-up time <12 months	-	-	-	7/2740	1.60 (-1.10, 4.31)	12.78	-	-	-	8/95876	0.92 (0.53, 1.59)	0.48
Omit studies of sample size <50	-	-	-	10/3206	1.07 (-1.26, 3.41)	12.86	9/2787	-0.89 (-1.56, -0.23)	0.99	-	-	-
Omit studies with low quality	-	-	-	-	-	-	-	-	-	-		
Omit studies with extreme outliers	-	-	-	11/3042	**2.43 (0.63, 4.23)**	8.01	12/2677	-0.24 (-0.43, -0.05)	0.06	8/7882	0.75 (0.56, 1.00)	0.08
Different statistical estimators												
DL	10/9578	0.71 (0.57, 0.88)	0.09	12/3287	1.18 (-0.94, 3.29)	12.57	13/2922	-0.70 (-1.24, -0.1)	0.89	9/95997	0.90 (0.52, 1.53)	0.48
EB	10/9578	0.72 (0.58, 0.89)	0.06	12/3287	1.16 (-2.02, 4.33)	23.4	13/2922	**-0.71(-1.66, 0.24)**	2.39	9/95997	0.92 (0.53, 1.46)	0.23
REML	10/9578	0.72 (0.58, 0.89)	0.03	12/3287	1.16 (-2.04, 4.35)	23.73	13/2922	**-0.71(-1.66, 0.24)**	2.38	9/95997	0.91 (0.53, 1.55)	0.33

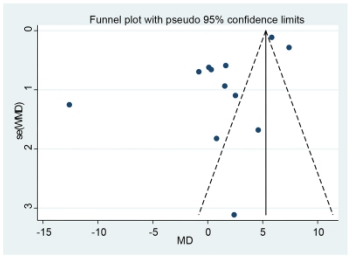
Sensitivity analysis	CV events			SBP			DBP		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD (95%CI)	τ^2
Omit studies of follow-up time <12 months	4/93467	1.03 (0.50, 2.12)	0.45	-	-	-	-	-	-
Omit studies of sample size <50	-	-	-	13/3402	-4.05 (-6.93, -1.17)	22.57	12/3318	-1.45 (-2.34, -0.56)	1.05
Omit studies with low quality	-	-	-	-	-	-	-	-	
Different statistical estimators									
DL	5/93573	0.97 (0.49, 1.93)	0.45	19/3608	-5.34 (-7.93, -2.75)	22.80	15/3524	-2.00(-2.93, -1.06)	1.56
EB	5/93573	1.01 (0.46, 2.22)	0.28	19/3608	-5.81(-9.63, -1.99)	14.19	15/3524	-2.30(-3.61, -0.98)	3.78
REML	5/93573	0.99 (0.41, 2.37)	0.35	19/3608	-5.78(-9.59, -1.97)	52.37	15/3524	-2.27(-3.57, -0.96)	3.41

Note: CI, confidence interval; CV, cardiovascular; DL, DerSimonian-Laird; EB, empirical Bayes; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; No., number of trials; NOS, Newcastle-Ottawa Scale; REML, restricted maximum likelihood; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

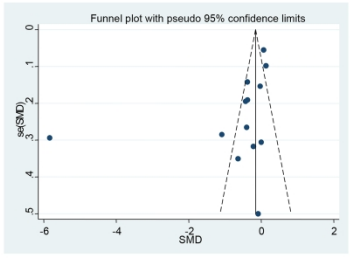
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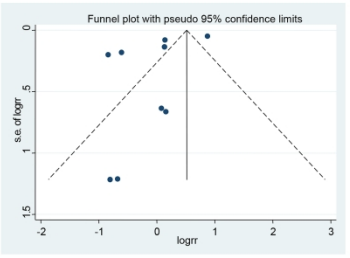
Funnel plot for publication bias on renal composite outcomes events.



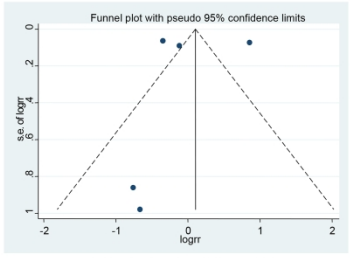
Funnel plot for publication bias on rate of change eGFR.



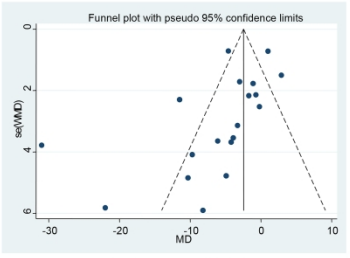
Funnel plot for publication bias on change in proteinuria or albuminuria.



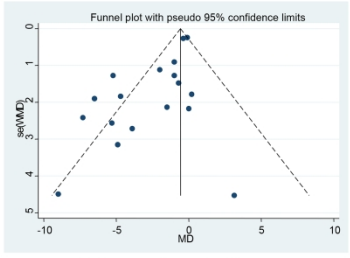
Funnel plot for publication bias on **all-cause mortality events**.



Funnel plot for publication bias on **CV events.**



Funnel plot for publication bias SBP.



Funnel plot for publication bias on **DBP.**

Supplementary file 13 Funnel plot for publication bias on outcomes. CV, cardiovascular; DBP, diastolic blood pressure; MD, mean difference; RR, relative risk; SMD, standard mean difference; SBP, systolic blood pressure.

Supplementary file 14 Table S7 The risk of publication bias of outcomes.

	N	Egger's test(*P*-value)	Bgge's test(*P*-value)
Renal composite outcomes events	10	0.47	0.28
The rate of change in eGFR	12	0.01	0.95
Proteinuria or albuminuria	13	0.10	0.30
All-cause mortality	9	0.09	1.00
CV events	5	0.91	0.46
BP	19	＜0.001	0.005

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5-6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-7

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8-12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15-16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Effects of Low-salt Diet Intervention on Chronic Kidney Disease Outcomes: A Systematic Review and ~~m~~Meta-analysis

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ABSTRACT

Objective The benefits of low-salt diet intervention for CKD patients are still controversial. We performed a systematic review and meta-analysis to investigate the effects of low-salt diet on major clinical outcomes.

Design Systematic review and meta-analysis.

Data sources MEDLINE by Ovid, EMBASE, and the Cochrane Library database.

Eligibility criteria for selecting studies We included RCTs and cohort studies that evaluated the effects of low-salt diet on the renal composite outcomes (more than ~~25% or~~ 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD), decline rate in eGFR, changes in proteinuria, all-cause mortality events, cardiovascular (CV) events, change in SBP and DBP.

Data extraction and synthesis Two independent reviewers extracted data, and assessed quality. RR with 95% CI were used for dichotomous data. Difference in means (MD) or standardized mean differences (SMD) with 95% CI were used to pool continuous data. We used Cochrane Collaboration risk-of-bias tool to assess quality of RCTs, and NOS to assess quality of cohort studies.

Results We identified 9948 potentially records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. Thirty-three studies with 101077

participants were included. Low-salt diet produced a 28% reduction in renal composite outcomes events (RR: 0.72; 95%CI: 0.58-0.89) compared with control groups. But no significant effects were observed on change of proteinuria (SMD: -0.71; 95% CI: -1.66 to 0.24), decline rate of eGFR (MD: 1.16; 95% CI: -2.02 to 4.33), the risk of all-cause mortality (RR: 0.92; 95% CI: 0.58-1.46), and CV events (RR: 1.01; 95% CI: 0.46-2.22).

Conclusion Low-salt diet seems to reduce the risk of renal composite outcomes events in CKD patients. However, no compelling evidence indicated that low-salt diet would reduce rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. More definitive studies are needed.

PROSPERO registration number CRD42017072395.

Strengths and limitations of this study

This study is the first meta-analysis to evaluate the effect of low-salt diet intervention on hard clinical outcomes, such as renal composite outcomes, all-cause mortality and CV events for CKD patients.

We included cohort studies and RCTs ~~No~~without ~~study design or~~ language restriction, and screened reference lists from included articles. ~~t~~Therefore allowing inclusion of the maximum number of studies in this review.

The existence of significant heterogeneity might limit the interpretation and clinical application of the results.

Confounding factors in the observational studies cannot be neglected, which might impact our results.

INTRODUCTION

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Chronic kidney disease (CKD) has become a global public health burden. Dietary interventions that aim to delay the progression of CKD and decrease complications have a pivotal role in the management of CKD. Restriction of dietary sodium intake can augment the antiproteinuric effects of angiotensin converting enzymes inhibitors/angiotensin receptor blockers (ACEIs/ARBs) [1] and decrease blood pressure (BP) [2]. In the 2020 KDIGO guidelines for glomerulonephritis, dietary restriction of sodium to < 2 g/d is a primary tenet for controlling BP and edema and improving urinary protein excretion independently of medications. However, there remains an open question of whether these effects can translate into a worthwhile absolute risk reduction in patients with CKD, including kidney failure events, cardiovascular (CV) events and all-cause mortality. In the Chronic Renal Insufficiency Cohort (CRIC) Study, recruiting patients eGFR between 20 and 70 ml/min/1.73 m^2 depending on age, compared with the lowest quartile of urinary sodium excretion (<116.8 mmol/24 h), high urinary sodium excretion (>194.6 mmol/24 h) was associated with a 54% risk increase in end-stage renal disease (ESRD) or halving of estimated glomerular filtration rate (eGFR) and with a 43% risk increase in all-cause mortality [3]. Contrasting results have been observed in other cohort studies, in which sodium intake does not appear to impact the progression of CKD to ESRD and a decline in eGFR [4,5]. Furthermore, a low salt intake of <6 g/day was associated with an increased risk of all-cause mortality and CV mortality in hemodialysis patients, resulting in an L-shaped association curve [6]. According to data from the NHANE II study, compared with those whose sodium intake exceed 2.3g daily, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans [7]. Randomized controlled trials (RCTs) evidence that discussed the effect of a low-salt diet in CKD patients on kidney failure events, CV events and

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all-cause mortality are controversial. It is difficult to base clinical decisions on findings from these studies, given their small sample sizes, highly selected participants, short-term follow-up and lack of assessment of hard clinical outcomes.

Thus, the association between dietary salt restriction and clinical outcomes in patients with CKD remains unclear. We performed this systematic review to summarize all eligible clinical studies data and evaluate the effects of low-salt diet intervention on kidney disease outcomes, all-cause mortality and CV events in CKD patients.

MATERIALS AND METHODS

Data sources and search strategy

We performed this systematic review according to a pre-specified protocol registered at the International Prospective Register of Systematic Reviews (CRD42017072395) (Supplementary file 1) [8], and reporting was done in line with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines [9]. A comprehensive search was conducted using the following database: MEDLINE by Ovid (from 1946 to July 2021), EMBASE (from 1966 to July 2021), and Cochrane Central Register of Controlled Trials (no date restriction), with relevant keywords and medical subject headings that included all spellings of “CKD”, “RCT”, “Cohort Studies”, “Sodium Chloride”, “Sodium”, “Sodium-Restricted” (see Supplementary file 2 for full search terms). Studies were considered without any language restriction. To ensure a comprehensive literature search, we also screened reference lists from included articles. The ClinicalTrials.gov website was also searched for studies the ongoing, but unpublished in this field.

Study selection and outcome estimation

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We included data from RCTs and cohort studies in which low-salt diet was given to adults with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) compared with usual therapy or different salt intake. Salt reduction was defined as that recommended by guidelines sodium <2.3 g/day (<100 mmol of sodium, <6g of salt), or as defined by the authors of each study.

Pre-defined outcomes that contained analyzable data were listed as follows: First, renal composite outcomes events, defined as a more than 50% decline in eGFR from baseline during follow-up [10], doubling of serum creatinine or ESRD. Second, the rate of change in eGFR per year. Third, changes in urinary protein or urinary albumin during follow-up, including urinary protein excretion, urinary albumin excretion and urinary albumin/creatinine ratio. Fourth, all-cause mortality events. Fifth, CV events, defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, CV disease and CV death. Fifth, change in systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Data extraction and quality assessment

Two independent reviewers (H.S. and X.S.) extracted data, and assessed quality, according to the pre-specified protocol. Disagreements were resolved through a third reviewer (L.W.). Data of all eligible studies was extracted into a spreadsheet. The data sought included characteristics of studies (study type, randomization method, follow-up time, withdrawals/dropouts), baseline patient characteristics (age, gender, baseline eGFR, baseline BP), intake of salt, and outcome events.

We used Cochrane Collaboration risk-of-bias tool [11] to assess all potential sources of bias for the included RCTs. Trails were assessed as being at low, other, some concerns, or high, and the overall risk of bias generally corresponds to the worst risk of bias in any of the domains. However, if

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a study was judged to have some concerns about risk of bias for multiple domains, it might be judged as at high risk of bias overall. We used the Newcastle-Ottawa Scale (NOS) to assess quality of cohort studies. The NOS contains eight items, categorized into three dimensions, including selection of cohorts, comparability of cohorts, and assessment of outcome[12]. A star system is used to allow a semi-quantitative assessment of study quality, such that the highest quality studies are awarded a maximum of one star for each item with the exception of the item related to comparability that allows the assignment of two stars. The NOS ranges between zero up to nine stars.

Data synthesis and statistical analyses

When dichotomous outcome data from individual trials were analyzed, relative risks (RRs) measure and its 95% confidence intervals (CIs) were used. If individual study RR was unavailable in the original article, RR and 95% CI were calculated from event numbers extracted from each study before data pooling. In calculating the RR values, we used the total number of patients randomized in each group as the corresponding denominator. Continuous outcome data from individual trials were analyzed using the difference in means (MD) with 95% CI to pool eGFR, and the standardized mean differences (SMD) with 95% CI were used to pool the proteinuria or albuminuria data. When continuous outcome data were analyzed, the difference in mean change between baseline and end of treatment value were used. If data for change between baseline and end of treatment were not available in the studies, we calculated it using correlations that were estimated from other included studies that had a similar follow-up period and reported in considerable detail according to the imputed formulation and its related interpretations in Cochrane Handbook [13].

In consideration of poor stability of the Der Simonian-Laird procedure for small numbers of studies [14], we used the empirical Bayes procedure [15] to estimate all outcomes. We also used Der

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Simonian-Laird random effects model [16] and restricted maximum likelihood [17] to assess the summary effects as a part of the sensitivity analyses. Considering the inevitable heterogeneity between studies, subgroup analyses, meta-regression, and sensitivity analyses were performed. Subgroup analyses were performed according to pre-specified protocol, including study type, baseline eGFR, and baseline BP, and comparator of control group. In addition, we performed sensitivity analyses by using different random-effects estimation methods as above mentioned, excluding studies with sample size <50, studies with follow-up time <12 months, studies with low quality (high risk of overall bias for RCTs, NOS <5★ for cohort studies), and studies with extreme outliers. Heterogeneity between studies was evaluated by using I^2 statistics or τ^2 statistics. The I^2 value of 25%, 50%, 75% respectively represent low, moderate, and high heterogeneity. Publication bias was assessed using Funnel plot, Egger's test and Begg's test, and P-value < 0.05 indicated obvious publication bias. Stata version 15.0 (StataCorp LP, College Station, TX, USA) was used for statistical analyses , and a two-sided P-value < 0.05 was considered statistically significant.

Patient and Public Involvement statement

No patients and the public were involved in the development of the research question, selection of the outcome measures, design and implementation of the study, or interpretation of the results.

RESULTS

Overview of included trials

We identified 9948 potentially relevant records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. As shown in figure 1, Supplementary file 3 and Supplementary file 4, a total of 33 eligible studies [3-6, 18-48] including

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13 RCTs [18,20,25-27,31,32,35-40,45] and 19 cohort studies [3-6,19,21-24,28,30,33,34,41-44,46-48] reported in 35 publications with 101077 participants, were included in this review. The median follow-up duration was 6 months (interquartile range 3 to 21). Participants were enrolled at an average age of 56 years, and male participants accounted for 60% of the total participants. The average eGFR of the included participants was 47.45 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and twenty-one trials compared different levels of sodium intake.

Figure 1 PRISMA flow chart for the included studies.

The risk of bias varied substantially across the RCTs. The results from the Cochrane Collaboration risk-of-bias tool are shown in Supplementary file 5. In all the RCTs, bias arising from the randomization process was assessed as low risk in 61.54% of the trails; risk of bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result were low risk in all the trails. In terms of overall bias, 61.54% of the trails were assessed as low risk, and 38.46% of the trails were assessed as having some concerns.

According to the NOS checklist, all included cohort studies were generally high quality and awarded 6★- 8★ (Supplementary file 6).

Effect of low-salt diet intervention on renal outcomes

One RCT and 9 observational studies including 9578 patients with 2329 events compared the low-salt diet with the control in preventing renal composite outcomes events. As shown in figure 2, compared with the control, low-salt diet intervention produced a 34% reduction in the risk of renal composite outcomes events (RR: 0.72; 95% CI: 0.58, 0.89), with substantial heterogeneity (I^2

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=87.2%, P <0.001). Subgroup analyses and meta-regression were conducted to discuss the source of heterogeneity. No significant difference between RCTs and observational studies was found, while the effect sizes were greater in studies that enrolled patients aged < 58 years (P = 0.03) (table 1). On univariate meta-regression, no clear relationship between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcomes events was observed (Supplementary file 7, Supplementary file 8).

Figure 2 Forest plot for renal composite outcomes events. Renal composite outcomes events was defined as a more than ~~25% or~~ 50% decline in eGFR from baseline during follow-up, doubling of serum creatinine or ESRD. CI, confidence interval; Cr, creatinine; Na, sodium; RR, relative risk.

Table 1 Subgroup Analysis of Renal Composite Outcomes Events.

Subgroup	No. of Trials	n	RR (95% CI)	*P* for RR	I^2	P for Heterogeneity Test[a]
Study type						
RCT	1	418	0.64 (0.43, 0.98)	0.04	-	0.76
Cohort study	9	9160	0.72 (0.57, 0.90)	0.005	88.6%	
Baseline eGFR (mL/min/1.73m²)						
60-89	1	171	0.14 (0.02, 1.14)	0.07	-	0.48
30-59	7	8869	0.71 (0.55, 0.92)	0.009	90.8%	
15-29	2	538	0.77 (0.63, 0.94)	0.01	0.0%	
Mean age (years)						
<58	5	5691	0.58 (0.49, 0.70)	<0.001	56.4%	**0.03**
≥58	5	3887	0.85 (0.69, 1.05)	0.14	70.1%	
Baseline BP (mmHg)						
<140/90	6	5806	0.62 (0.52, 0.73)	<0.001	58.5%	0.32
SBP≥140 or DBP≥90	2	1646	0.70 (0.44, 1.11)	0.13	74.6%	
Comparator of control group						
Usual care or usual diet	2	589	0.43 (0.11, 1.64)	0.21	50.2%	0.12
Different levels of sodium intake	8	8989	0.73 (0.58, 0.92)	0.008	89.8%	

Note: a *P* value calculated by meta regression. BP,blood pressure; CI, confidence interval; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk.

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Five RCTs and 7 observational studies with a total of 3287 participants reported data regarding the effects of a low-salt diet on the rate of change in eGFR. Low-salt diet therapy did not produce an apparent beneficial effect on the rate of change in eGFR (MD:1.16; 95% CI: -2.02 to 4.33). There was evidence of significant heterogeneity across the included studies (I^2= 98.0%, P< 0.001) (figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to prespecified characteristics (Supplementary file 9).

Figure 3 Forest plot for rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SD, standard deviation.

Data for the effects of low-salt diet intervention on proteinuria or albuminuria were available in 13 studies including 2922 participants. The SMD for change in proteinuria or albuminuria was not statistically significant at -0.71 (95% CI: -1.66 to 0.24) compared with that in the control groups, with significant heterogeneity (I^2 = 97.3%, P=0.009) (figure 4). No statistical heterogeneity was found in any subgroup analyses for the effects of a low-salt diet on proteinuria or albuminuria (Supplementary file 9).

Figure 4 Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; Cr, creatinine; Na, sodium; SD, standard deviation; SMD, standard mean difference.

Effect of low-salt diet intervention on all-cause mortality and CV events

Data for the effects of low-salt diet intervention on all-cause mortality and CV events were available for 9 studies (two RCTs and seven observational studies) and 5 studies (one RCT and four observational studies), respectively. There were no clear effects of a low-salt diet on the risk of

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all-cause mortality (RR: 0.92; 95% CI: 0.53-1.46; 95997 participants and 4455 events) or CV events (RR: 1.01; 95% CI: 0.46-2.22; 93573 participants and 1989 events) compared with those in the control groups, with significant heterogeneity across the included studies (I^2 = 98.0%, 97.6% for all-cause mortality and CV events respectively, and *P* values for heterogeneity both less than 0.001) (figure 5). More significant benefits were found in studies that included patients with an eGFR of 30-59 mL/min/1.73 m^2 (*P* for subgroup heterogeneity = 0.02) in the subgroup analysis of CV events (Supplementary file 9).

Figure 5 Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death. CI, confidence interval; Cr, creatinine; CV, cardiovascular; Na, sodium; N, number of trials; RR, relative risk.

Effect of low-salt diet intervention on BP

Twenty studies reported the effect of a low-salt diet on BP, including 19 studies (3608 individuals) for SBP and 18 studies (3608 participants) for DBP. The MD changes in SBP and DBP were both statistically significant at -5.81 mmHg (95% CI: -9.63 to -1.99) and -2.3 mmHg (95% CI, -3.61 to -0.98), respectively (Supplementary file 10, Supplementary file 11). Significant heterogeneity across the included studies was found (I^2 = 87.1%, 68.2% for SBP and DBP respectively, and *P* values for heterogeneity both less than 0.05). Subgroup analyses showed that the association between the change in SBP or DBP and a low-salt diet was modified by the baseline BP of patients. Patients with baseline SBP >140 mmHg or DBP > 90 mmHg had an average reduction in SBP of 10 mmHg and in DBP of 5 mmHg (Supplementary file 9).

Sensitivity analyses and publication bias

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All sensitivity analyses results were robust (Supplementary file 1~~1~~2). The most notable exception was that the effect of low-salt diet therapy on the rate of change in eGFR became significant when studies with extreme outliers were omitted, and proteinuria or albuminuria became significant when the DL method was used to pool data. The risk of publication bias was found to be significant for the rate of change in eGFR, proteinuria or albuminuria, all-cause mortality and cardiovascular events, and BP (Supplementary file 1~~2~~3, Supplementary file 14).

DISCUSSION

Behavioral modifications can make an important contribution to delaying CKD progression to ESRD, and dietary salt restriction is one of the main concerns. Our meta-analysis showed that compared with control groups, low-salt diet intervention seems to reduced renal composite outcomes events and lowered BP (SBP and DBP) in CKD patients. No significant effects were observed regarding the decline rate of eGFR, change in proteinuria, risk of all-cause mortality or CV events. These results remained broadly consistent across major subgroups and sensitivity analyses. However, the existence of significant heterogeneity might limit the interpretation and clinical application of these results.

The previous review of 16 studies in 2006 addressing salt intake and kidney disease set out to establish whether variations in dietary sodium consumption influence kidney outcomes in patients with CKD [49]. The review suggested that increased salt intake was associated with worsening albuminuria and an increased likelihood of a reduction in GFR. In 2015 [50] and 2018 [51], the two meta-analyses of RCTs both suggested that moderate dietary salt restriction significantly reduced BP and proteinuria. However, it remains unanswered whether these benefits may translate to clinically

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significant reductions in renal composite outcomes events, all-cause mortality events and CV events. Our study, including 33 studies (13 RCTs and 20 observational studies), more than 90,000 patients, and 2329 renal composite outcomes events, first concluded that compared with the control groups, a low-salt diet produced a significant 28% reduction in renal composite outcomes events despite marked heterogeneity. Furthermore, the significant effect of a low-salt diet on BP (decreases in SBP and DBP of 5.8 mmHg and 2.3 mmHg, respectively) was confirmed again in our study and was recognized as an important target to slow the progression of CKD towards ESRD. The benefit of proteinuria from low-salt diet intervention became nonsignificant in our study when the restricted maximum likelihood or empirical Bayes methods were used to pool data. The standard DerSimonian-Laird procedure that was applied in previous meta-analyses can be very unstable with small numbers of studies [52,53].

Of note, statistical heterogeneity was found in all outcomes, and clinical heterogeneity was also an inevitable issue. Low-salt diets are not palatable to many [54], therefore, it is difficult to ensure sustained adherence to a low-sodium diet over long periods of follow-up. Sodium intake is quantified in distinct ways between studies (for example, 24 h urine collection, spot urine collection or dietary questionnaire). Given the use of diuretic may influence the urine sodium excretion, and eGFR, and it would influence the accuracy of quantified salt intake. We extracted data about use of diuretic. Ten studies have the data about diuretic use in the study we included, but only about the percentages of patients using diuretics in the intervention and control group. The percentage of patients using diuretics was no difference in most studies. There were no data about type and dose of diuretics, or the associations with the eGFR. These inherent problems were reflected in the paradoxically opposite findings of various studies, especially observational studies with more confounding factors. Only

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three studies (1 RCT and 2 observational studies) simultaneously reported the outcomes of eGFR and renal composite outcomes events. Furthermore, the different follow-up periods with 52 months of renal composite outcomes events and 12 months of eGFR should be noted. The inconsistent results of eGFR and renal composite outcomes events may partly be explained by the heterogeneity mentioned above. It should be assumed that heterogeneity limited the interpretation and generalizability of our results.

In CKD, BP is usually sodium sensitive, and RAAS blockers are considered the first-line therapy for hypertension and proteinuria. It has been known for over 2 decades that the effects of RAAS blockers are blunted by high sodium intake [55-57]. Thus, the significant effect of dietary sodium restriction on lowering BP and supporting the function of RAAS blockers was speculated to play important roles in reducing the risk of kidney failure events. Interestingly, a BP-independent effect of dietary sodium on the kidney is substantiated by data in healthy volunteers, in which dietary sodium restriction reduces albuminuria within the normal range, without a detectable effect on BP [58]. Similarly, proteinuria reduction by sodium restriction remained significant after adjustment for the decrease in BP [57]. These results suggested an independent renoprotective effect of low-sodium diets, both as a single measure and in combination with RAAS blockers.

In our study, there was no convincing evidence that a low-salt diet was associated with a lower incidence of all-cause mortality and CV events. Limited data on all-cause mortality (nine studies with 4455 events) and CV events (five studies with 1989 events) for the meta-analysis might introduce a risk of false-negative results because of low statistical power. More significant benefits of CV events were found in studies that included patients with eGFR 30-59 mL/min/1.73 m^2 in the subgroup analysis, while the effects were not significant in patients with eGFR less than 15

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mL/min/1.73 m^2. The significant subgroup heterogeneity (P =0.02) could be attributable to more CV risk factors in patients with ESRD than in those with eGFR 30-59 mL/min/1.73 m^2. The benefits of CV events from a low-salt diet were not enough to outweigh the increase in CV events caused by more risk factors (such as hyperuricemia, hyperphosphatemia, and vascular calcification). It is a hint at least for clinicians that low-salt diet education needs to be emphasized in the early stage of CKD.

Observational data from various studies showing a J-curve between sodium intake and renal and CV outcomes have raised concern regarding the safety of rigorous sodium restriction [59-61]. It is uncertain whether the presence of CKD modifies this association. In our meta-regression analysis, no clear relationships between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in kidney failure events were observed. A regimen of ACEIs and rigorous sodium restriction reduced blood pressure, proteinuria, and glomerular damage but unexpectedly aggravated tubule-interstitial damage in experimental renal disease [62]. This could provide a potential explanation for the association of very low sodium intake with worse renal outcome. Both high or extremely low dietary salt intake might be inappropriate for CKD patients. KDIGO recommends a daily sodium intake of < 2.0 g/d in patients with CKD without a clear lower limit. Taken together, these results should be interpreted, and the proper levels of dietary salt intake in CKD patients should be discussed. The current study might not be the final answer for the question of low-sodium diets and progression of kidney disease. Interventional studies are needed to ascertain the optimum level of dietary sodium intake for renal protection in patients with different CKD stages.

Our study has several potential limitations. First, as has been noted, the existence of statistical heterogeneity and inevitable clinical heterogeneity might raise critical concerns regarding validity, although we tried to resolve these concerns through subgroup analysis and sensitivity analysis.

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Heterogeneity in the study population as included in the umbrella term "CKD patients" remains a major concern for meaningful interpretation of the results. Because of the stages of CKD patients in the studies we included varies a lot, some are CKD 1-5, some are CKD 1-3, some are CKD 3-4, some are CKD 2-4, some are CKD 4-5, some are CKD 3-4. So it is difficult to stratify our analysis our outcomes by stage of CKD. We can only perform subgroup analyses according to the baseline eGFR of CKD patients. Second, cohort studies are prone to selection bias that is usually not found in RCT. Therefore, pooling the results from RCT and cohort studies together will bias the results of meta-analysis. So we performed subgroup analysis according to study design. There was no statistical heterogeneity according to study design for the outcomes. What is noteworthy is that low-salt diet therapy did not produce an beneficial effect on the rate of change in eGFR in RCTs (MD:-0.75; 95% CI: -7.58 to 6.08), but produce beneficial effect on the rate of change in eGFR in cohort studies (MD:2.48; 95% CI: 0.32 to 4.64). Third, many of the included studies with low quality had small sample sizes and a low incidence of events. Consequently, the uncertainty in the analyses was increased, resulting in wide CIs of effect measures. Fourth~~Third~~, confounding factors in the observational studies cannot be neglected, which might impact our results. Therefore, the findings of this study should be viewed as hypothesis-generating and need to be confirmed by further studies.

CONCLUSION

In conclusion, our meta-analyses suggested that a low-salt diet may reduce the risk of renal composite outcomes events in patients with CKD. However, there was no compelling evidence indicating that a low-salt diet could reduce the rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. The optimal dietary salt intake for patients with different CKD stages remains unanswered. Further well-designed RCTs targeting patients with different CKDs are

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required to assess and integrate these results.

Acknowledgments We thank American Journal Experts Corporation for helping in proofreading and editing the English of the final manuscript.

Contributors All authors take responsibility for the integrity of the data and the accuracy of data analysis. Study concept and design: L.W., X.S., H.S. and C.L.. Extraction, analysis and interpretation of data: H.S. and X.S.. Drafting of the manuscript: H.S. and L.W.. Critical revision of the manuscript: L.W. and X.S. Statistical analysis: H.S., C.L. and W.G.. Technical support: L.W., X.S. and W.G.. All authors have read and agreed to the submission to this journal of the manuscript.

Funding Support was provided by the National Natural Science Foundation of China (no. 82000655) and the Science and Technology Project of Shanxi Province (no. 201801D121218).The funders had no role in decision to publish and preparation of the manuscript.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data used to support the findings of this study are included within the article and supplementary information.

Ethics statement Our research include published studies, and does not require ethical committee approval.

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Effects of Low-salt Diet on Chronic Kidney Disease Outcomes: A Systematic Review and Meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050843.R3
Article Type:	Original research
Date Submitted by the Author:	15-Nov-2021
Complete List of Authors:	Shi, Honghong; Shanxi Medical University Second Affiliated Hospital, Renal Division Su, Xiaole; Shanxi Medical University Second Affiliated Hospital Li, Chunfang; Shanxi Cardiovascular Hospital Guo, Wenjuan; Shanxi Medical University Second Affiliated Hospital Wang, Lihua; Shanxi Medical University Second Affiliated Hospital
Primary Subject Heading:	Renal medicine
Secondary Subject Heading:	Health services research
Keywords:	Nephrology < INTERNAL MEDICINE, Chronic renal failure < NEPHROLOGY, HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effects of Low-salt Diet on Chronic Kidney Disease Outcomes: A Systematic Review and Meta-analysis

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ABSTRACT

Objective The benefits of low-salt diet for CKD patients are still controversial. We performed a systematic review and meta-analysis to investigate the effects of low-salt diet on major clinical outcomes.

Design Systematic review and meta-analysis.

Data sources MEDLINE by Ovid, EMBASE, and the Cochrane Library database.

Eligibility criteria for selecting studies We included RCTs and cohort studies that evaluated the effects of low-salt diet on the renal composite outcomes (more than 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD), decline rate of eGFR, changes in proteinuria, all-cause mortality events, cardiovascular (CV) events, change in SBP and DBP.

Data extraction and synthesis Two independent reviewers extracted data, and assessed quality. RRs with 95% CIs were used for dichotomous data. Difference in means (MDs) or standardized mean differences (SMDs) with 95% CIs were used to pool continuous data. We used Cochrane Collaboration risk-of-bias tool to assess quality of RCTs, and NOS to assess quality of cohort studies.

Results We identified 9948 potentially records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. Thirty-three studies with 101077 participants were included. Low-salt diet produced a 28% reduction in renal composite outcomes events (RR: 0.72; 95%CI: 0.58-0.89) compared with control groups. But no significant effects were observed on change of proteinuria (SMD: -0.71; 95% CI: -1.66 to 0.24), decline rate of eGFR (MD: 1.16; 95% CI: -2.02 to 4.33), the risk of all-cause mortality (RR: 0.92; 95% CI: 0.58-1.46), and CV events (RR: 1.01; 95% CI: 0.46-2.22).

Conclusion Low-salt diet seems to reduce the risk of renal composite outcomes events in CKD patients. However, no compelling evidence indicated that low-salt diet would reduce rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. More definitive studies are needed.

PROSPERO registration number CRD42017072395.

Strengths and limitations of this study

This study is the first meta-analysis to evaluate the effect of low-salt diet on hard clinical outcomes, such as renal composite outcomes, all-cause mortality and CV events for CKD patients.

We included cohort studies and RCTs without language restriction, and screened reference lists from included articles, allowing inclusion of the maximum number of studies in this review.

The existence of significant heterogeneity might limit the interpretation and clinical application of the results.

Confounding factors in the observational studies cannot be neglected, which might impact our results.

INTRODUCTION

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Chronic kidney disease (CKD) has become a global public health burden. Dietary interventions that aim to delay the progression of CKD and decrease complications have a pivotal role in the management of CKD. Restriction of dietary sodium intake can augment the antiproteinuric effects of angiotensin converting enzymes inhibitors/angiotensin receptor blockers (ACEIs/ARBs) [1] and decrease blood pressure (BP) [2]. In the 2020 KDIGO guidelines for glomerulonephritis, dietary restriction of sodium to < 2 g/d is a primary tenet for controlling BP and edema and improving urinary protein excretion independently of medications. However, there remains an open question of whether these effects can translate into a worthwhile absolute risk reduction in patients with CKD, including kidney failure events, cardiovascular (CV) events and all-cause mortality. In the Chronic Renal Insufficiency Cohort (CRIC) Study, recruiting patients with estimated glomerular filtration rate (eGFR) between 20 and 70 ml/min/1.73 m^2 depending on age, compared with the lowest quartile of urinary sodium excretion (<116.8 mmol/24 h), high urinary sodium excretion (>194.6 mmol/24 h) was associated with a 54% risk increase in end-stage renal disease (ESRD) or halving ofeGFR and with a 43% risk increase in all-cause mortality [3]. However, results in other cohort studies suggested that sodium intake does not appear to impact the progression of CKD to ESRD and a decline in eGFR [4,5]. Furthermore, a low salt intake of <6 g/day was associated with an increased risk of all-cause mortality and CV mortality in hemodialysis patients, resulting in an L-shaped association curve [6]. According to data from the NHANES II study, compared with those whose sodium intake exceed 2.3g daily, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans [7]. Randomized controlled trials (RCTs) evidence that discussed the effects of a low-salt diet in CKD patients on kidney failure events, CV events and

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all-cause mortality are controversial. It is difficult to base clinical decisions on findings from these studies, given their small sample sizes, highly selected participants, short-term follow-up and lack of assessment of hard clinical outcomes.

Thus, the association between dietary salt restriction and clinical outcomes in patients with CKD remains unclear. We performed this systematic review to summarize all eligible clinical studies data and evaluate the effects of low-salt diet on kidney disease outcomes, all-cause mortality and CV events in CKD patients.

MATERIALS AND METHODS

Data sources and search strategy

We performed this systematic review according to a pre-specified protocol registered at the International Prospective Register of Systematic Reviews (CRD42017072395) (Supplementary file 1) [8], and reporting was done in line with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines [9]. A comprehensive search was conducted using the following database: MEDLINE by Ovid (from 1946 to July 2021), EMBASE (from 1966 to July 2021), and Cochrane Central Register of Controlled Trials (no date restriction), with relevant keywords and medical subject headings that included all spellings of “CKD”, “RCT”, “Cohort Studies”, “Sodium Chloride”, “Sodium”, “Sodium-Restricted” (see Supplementary file 2 for full search terms). Studies were considered without any language restriction. To ensure a comprehensive literature search, we also screened reference lists from included articles. The ClinicalTrials.gov website was also searched for the studies ongoing, but unpublished in this field.

Study selection and outcome estimation

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We included data from RCTs and cohort studies in which low-salt diet was given to adults with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) compared with usual therapy or different salt intake. Salt reduction was defined as that recommended by guidelines sodium <2.3 g/day (<100 mmol of sodium, <6g of salt), or as defined by the authors of each study.

Pre-defined outcomes that contained analyzable data were listed as follows: First, renal composite outcomes events, defined as a more than 50% decline in eGFR from baseline during follow-up [10], doubling of serum creatinine or ESRD. Second, the rate of change in eGFR per year. Third, changes in urinary protein or urinary albumin during follow-up, including urinary protein excretion, urinary albumin excretion and urinary albumin/creatinine ratio. Fourth, all-cause mortality events. Fifth, CV events, defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, CV disease and CV death. Fifth, change in systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Data extraction and quality assessment

Two independent reviewers (H.S. and X.S.) extracted data and assessed theri quality according to the pre-specified protocol. Disagreements were resolved through a third reviewer (L.W.). Data from all eligible studies were extracted into a spreadsheet. The data sought included characteristics of studies (study type, randomization method, follow-up time, withdrawals/dropouts), baseline patient characteristics (age, gender, baseline eGFR, baseline BP), intake of salt, and outcome events.

We used Cochrane Collaboration risk-of-bias tool [11] to assess all potential sources of bias for the included RCTs. Trails were assessed as being at low, other, some concerns, or high, and the overall risk of bias generally corresponds to the worst risk of bias in any of the domains. However, if

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a study was judged to have some concerns about risk of bias for multiple domains, it might be judged as at high risk of bias overall. We used the Newcastle-Ottawa Scale (NOS) to assess quality of cohort studies. The NOS contains eight items, categorized into three dimensions, including selection of cohorts, comparability of cohorts, and assessment of outcome[12]. A star system is used to allow a semi-quantitative assessment of study quality, such that the highest quality studies are awarded a maximum of one star for each item with the exception of the item related to comparability that allows the assignment of two stars. The NOS ranges between zero up to nine stars.

Data synthesis and statistical analyses

When dichotomous outcome data from individual trials were analyzed, relative risks (RRs) measure and its 95% confidence intervals (CIs) were used. If the RR for individual study was unavailable in the original article, the RR and 95% CI were calculated from event numbers extracted from each study before data pooling. In calculating the RR values, we used the total number of patients randomized in each group as the corresponding denominator. Continuous outcome data from individual trials were analyzed using the difference in means (MDs) with 95% CIs to pool eGFR, and the standardized mean differences (SMDs) with 95% CIs were used to pool the proteinuria or albuminuria data. When continuous outcome data were analyzed, the difference in the mean change between baseline and end-of-treatment value was used. If data on changes between baseline and end-of-treatment values were not available in the studies, we calculated it using correlations estimated from other included studies that had a similar follow-up period and reported their results in considerable detail according to the imputed formulation and its related interpretations in the Cochrane Handbook [13].

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In consideration of the poor stability of the Der Simonian-Laird procedure for small numbers of studies [14], we used the empirical Bayes procedure [15] to estimate all outcomes. We also used Der Simonian-Laird random effects model [16] and restricted maximum likelihood [17] to assess the summary effects as a part of the sensitivity analyses. Considering the inevitable heterogeneity among studies, subgroup analyses, meta-regression, and sensitivity analyses were performed. Subgroup analyses were performed according to a pre-specified protocol, including study type, baseline eGFR, and baseline BP, and comparator of control group. In addition, we performed sensitivity analyses using different random-effects estimation methods as above mentioned, excluding studies with sample size <50, studies with follow-up time <12 months, studies with low quality (high risk of overall bias for RCTs, NOS <5★ for cohort studies), and studies with extreme outliers. Heterogeneity among studies was evaluated using I^2 or τ^2 statistic. The I^2 value of 25%, 50%, 75% respectively represent low, moderate, and high heterogeneity. Publication bias was assessed using Funnel plot, Egger's test and Begg's test, and P-value < 0.05 indicated obvious publication bias. Stata version 15.0 (StataCorp LP, College Station, TX, USA) was used for statistical analyses , and a two-sided P-value < 0.05 was considered statistically significant.

Patient and Public Involvement statement

No patients and the public were involved in the development of the research question, selection of the outcome measures, design and implementation of the study, or interpretation of the results.

RESULTS

Overview of included trials

We identified 9948 potentially relevant records. After removal of duplicates, we screened the

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titles and abstracts, and 230 publications were reviewed in full text. As shown in figure 1, Supplementary file 3 and Supplementary file 4, a total of 33 eligible studies [3-6, 18-48] including 13 RCTs [18,20,25-27,31,32,35-40,45] and 19 cohort studies [3-6,19,21-24,28,30,33,34,41-44,46-48] reported in 35 publications with 101077 participants, were included in this review. The median follow-up duration was 6 months (interquartile range 3 to 21). Participants were enrolled at an average age of 56 years, and male participants accounted for 60% of the total participants. The average eGFR of the included participants was 47.45 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and twenty-one trials compared different levels of sodium intake.

Figure 1 PRISMA flow chart for the included studies.

The risk of bias varied substantially across the RCTs. The results from the Cochrane Collaboration risk-of-bias tool were shown in Supplementary file 5. In all the RCTs, bias arising from the randomization process was assessed as low risk in 61.54% of the trails, and all studies had a low risk of bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result were low risk in all the trails. In terms of overall bias, 61.54% of the trails were assessed as low risk, and 38.46% of the trails were assessed as having some concerns.

According to the NOS checklist, all included cohort studies were generally high quality and awarded 6★- 8★ (Supplementary file 6).

Effects of low-salt diet on renal outcomes

One RCT and 9 observational studies including 9578 patients with 2329 events compared the low-salt diet with the control in preventing renal composite outcomes events. As shown in figure 2, compared with the control, low-salt diet produced a 34% reduction in the risk of renal composite

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outcomes events (RR: 0.72; 95% CI: 0.58, 0.89), with substantial heterogeneity (I^2 =87.2%, P <0.001). Subgroup analyses and meta-regression were conducted to discuss the source of heterogeneity. No significant difference between RCTs and observational studies was found, while the effect sizes were greater in studies that enrolled patients aged < 58 years (P = 0.03) (table 1). On univariate meta-regression, no clear relationship between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcomes events was observed (Supplementary file 7, Supplementary file 8).

Figure 2 Forest plot for renal composite outcomes events. Renal composite outcomes events was defined as a more than 50% decline in eGFR from baseline during follow-up, doubling of serum creatinine or ESRD. CI, confidence interval; Cr, creatinine; Na, sodium; RR, relative risk.

Table 1 Subgroup Analysis of Renal Composite Outcomes Events.

Subgroup	No. of Trials	n	RR (95% CI)	*P* for RR	I^2	P for Heterogeneity Test[a]
Study type						
RCT	1	418	0.64 (0.43, 0.98)	0.04	-	0.76
Cohort study	9	9160	0.72 (0.57, 0.90)	0.005	88.6%	
Baseline eGFR (mL/min/1.73m²)						
60-89	1	171	0.14 (0.02, 1.14)	0.07	-	0.48
30-59	7	8869	0.71 (0.55, 0.92)	0.009	90.8%	
15-29	2	538	0.77 (0.63, 0.94)	0.01	0.0%	
Mean age (years)						
<58	5	5691	0.58 (0.49, 0.70)	<0.001	56.4%	**0.03**
≥58	5	3887	0.85 (0.69, 1.05)	0.14	70.1%	
Baseline BP (mmHg)						
<140/90	6	5806	0.62 (0.52, 0.73)	<0.001	58.5%	0.32
SBP≥140 or DBP≥90	2	1646	0.70 (0.44, 1.11)	0.13	74.6%	
Comparator of control group						
Usual care or usual diet	2	589	0.43 (0.11, 1.64)	0.21	50.2%	0.12
Different levels of sodium intake	8	8989	0.73 (0.58, 0.92)	0.008	89.8%	

Note: a *P* value calculated by meta regression. BP,blood pressure; CI, confidence interval; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk.

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Five RCTs and 7 observational studies with a total of 3287 participants reported data regarding the effects of a low-salt diet on the rate of change in eGFR. Low-salt diet therapy did not produce an apparent beneficial effect on the rate of change in eGFR (MD:1.16; 95% CI: -2.02 to 4.33). There was evidence of significant heterogeneity across the included studies (I^2= 98.0%, P< 0.001) (figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to pre-specified characteristics (Supplementary file 9).

Figure 3 Forest plot for rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SD, standard deviation.

Data for the effects of low-salt diet on proteinuria or albuminuria were available in 13 studies, including 2922 participants. The SMD for change in proteinuria or albuminuria was not statistically significant at -0.71 (95% CI: -1.66 to 0.24) compared with that in the control groups, with significant heterogeneity (I^2 = 97.3%, P=0.009) (figure 4). No statistical heterogeneity was found in any subgroup analyses for the effects of a low-salt diet on proteinuria or albuminuria (Supplementary file 9).

Figure 4 Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; Cr, creatinine; Na, sodium; SD, standard deviation; SMD, standard mean difference.

Effects of low-salt diet on all-cause mortality and CV events

Data for the effects of low-salt diet on all-cause mortality and CV events were available for 9 studies (two RCTs and seven observational studies) and 5 studies (one RCT and four observational studies), respectively. There were no clear effects of a low-salt diet on the risk of all-cause mortality

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(RR: 0.92; 95% CI: 0.53-1.46; 95997 participants and 4455 events) or CV events (RR: 1.01; 95% CI: 0.46-2.22; 93573 participants and 1989 events) compared with those in the control groups, with significant heterogeneity across the included studies (I^2 = 98.0%, 97.6% for all-cause mortality and CV events respectively, and *P* values for heterogeneity both less than 0.001) (figure 5). More significant benefits were found in studies that included patients with an eGFR of 30-59 mL/min/1.73 m^2 (*P* for subgroup heterogeneity = 0.02) in the subgroup analyses of CV events (Supplementary file 9).

Figure 5 Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death. CI, confidence interval; Cr, creatinine; CV, cardiovascular; Na, sodium; N, number of trials; RR, relative risk.

Effects of low-salt diet on BP

Twenty studies reported the effects of a low-salt diet on BP, including 19 studies (3608 individuals) for SBP and 18 studies (3608 participants) for DBP. The MD changes in SBP and DBP were both statistically significant at -5.81 mmHg (95% CI: -9.63 to -1.99) and -2.3 mmHg (95% CI, -3.61 to -0.98), respectively (Supplementary file 10, Supplementary file 11). Significant heterogeneity across the included studies was found (I^2 = 87.1%, 68.2% for SBP and DBP respectively, and *P* values for heterogeneity both less than 0.05). Subgroup analyses showed that the association between the change in SBP or DBP and a low-salt diet was modified by the baseline BP of patients. Patients with baseline SBP >140 mmHg or DBP > 90 mmHg had an average reduction in SBP of 10 mmHg and in DBP of 5 mmHg (Supplementary file 9).

Sensitivity analyses and publication bias

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All sensitivity analyses results were robust (Supplementary file 12). The most notable exception was that the effects of low-salt diet therapy on the rate of change in eGFR became significant when studies with extreme outliers were omitted, and proteinuria or albuminuria became significant when the DL method was used to pool data. The risk of publication bias was found to be significant for the rate of change in eGFR, proteinuria or albuminuria, all-cause mortality and cardiovascular events, and BP (Supplementary file 13, Supplementary file 14).

DISCUSSION

Behavioral modifications can make an important contribution to delaying CKD progression to ESRD, and dietary salt restriction is one of the main concerns. Our meta-analysis showed that compared with control groups, low-salt diet seems reduced renal composite outcomes events and lowered BP (SBP and DBP) in CKD patients. No significant effects were observed regarding the decline rate of eGFR, change in proteinuria, risk of all-cause mortality and CV events. These results remained broadly consistent across major subgroups and sensitivity analyses. However, the existence of significant heterogeneity might limit the interpretation and clinical application of these results.

The previous review of 16 studies in 2006 addressing salt intake and kidney disease set out to establish whether variations in dietary sodium consumption influence kidney outcomes in patients with CKD [49]. The review suggested that increased salt intake was associated with worsening albuminuria and an increased likelihood of a reduction in GFR. In 2015 [50] and 2018 [51], the two meta-analyses of RCTs both suggested that moderate dietary salt restriction significantly reduced BP and proteinuria. However, it remains unanswered whether these benefits may translate to clinically significant reductions in renal composite outcomes events, all-cause mortality events and CV events.

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Our study, including 33 studies (13 RCTs and 20 observational studies), more than 90,000 patients, and 2329 renal composite outcomes events, first concluded that compared with the control groups, a low-salt diet produced a significant 28% reduction in renal composite outcomes events despite marked heterogeneity. Furthermore, the significant effects of a low-salt diet on BP (decreases in SBP and DBP of 5.8 mmHg and 2.3 mmHg, respectively) was confirmed again in our study and was recognized as an important target to slow the progression of CKD towards ESRD. The benefit of proteinuria from low-salt diet became nonsignificant in our study when the restricted maximum likelihood or empirical Bayes methods were used to pool data. The standard DerSimonian-Laird procedure that was applied in previous meta-analyses can be very unstable with small numbers of studies [52,53].

Of note, statistical heterogeneity was found in all outcomes, and clinical heterogeneity was also an inevitable issue. Low-salt diets are not palatable to many patients[54]. Therefore, it is difficult to ensure sustained adherence to a low-sodium diet over long periods of follow-up. Sodium intake is quantified in distinct ways among studies (for example, 24 h urine collection, spot urine collection, or dietary questionnaire). Given the use of diuretic may influence the urine sodium excretion and eGFR, and it would influence the accuracy of quantified salt intake. We extracted data about use of diuretic. Ten studies have the data about diuretic use in the study we included, but only about the percentages of patients using diuretics in the intervention and control group. The percentage of patients using diuretics was no difference in most studies. There were no data about type and dose of diuretics, or the associations with the eGFR. These inherent problems were reflected in the paradoxically opposite findings of various studies, especially observational studies with more confounding factors. Only three studies (1 RCT and 2 observational studies) simultaneously reported

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the outcomes of eGFR and renal composite outcomes events. Furthermore, the different follow-up periods with 52 months of renal composite outcomes events and 12 months of eGFR should be noted. The inconsistent results of eGFR and renal composite outcomes events may partly be explained by the heterogeneity mentioned above. It should be assumed that heterogeneity limited the interpretation and generalizability of our results.

In CKD patients, BP is usually sodium sensitive, and RAAS blockers are considered the first-line therapy for hypertension and proteinuria. It has been known for over 2 decades that the effects of RAAS blockers are blunted by high sodium intake [55-57]. Thus, the significant effect of dietary sodium restriction on lowering BP and supporting the function of RAAS blockers was speculated to play important roles in reducing the risk of kidney failure events. Interestingly, a BP-independent effect of dietary sodium on the kidney is substantiated by data in healthy volunteers, in which dietary sodium restriction reduces albuminuria within the normal range, without a detectable effect on BP [58]. Similarly, proteinuria reduction by sodium restriction remained significant after adjustment for the decrease in BP [57]. These results suggested an independent renoprotective effect of low-sodium diets, both as a single measure and in combination with RAAS blockers.

In our study, there was no convincing evidence that a low-salt diet was associated with a lower incidence of all-cause mortality and CV events. Limited data on all-cause mortality (nine studies with 4455 events) and CV events (five studies with 1989 events) for the meta-analysis might introduce a risk of false-negative results because of low statistical power. More significant benefits of CV events were found in studies that included patients with eGFR 30-59 mL/min/1.73 m^2 in the subgroup analysis, while the effects were not significant in patients with eGFR less than 15

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mL/min/1.73 m^2. The significant subgroup heterogeneity (P =0.02) could be attributable to more CV risk factors in patients with ESRD than in those with eGFR 30-59 mL/min/1.73 m^2. The benefits of CV events from a low-salt diet were not enough to outweigh the increase in CV events caused by more risk factors (such as hyperuricemia, hyperphosphatemia, and vascular calcification). It is a hint at least for clinicians that low-salt diet education needs to be emphasized in the early stage of CKD.

Observational data from various studies showing a J-curve between sodium intake and renal and CV outcomes have raised concern regarding the safety of rigorous sodium restriction [59-61]. It is uncertain whether the presence of CKD modifies this association. In our meta-regression analysis, no clear relationships between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcomes events were observed. A regimen of ACEIs and rigorous sodium restriction reduced blood pressure, proteinuria, and glomerular damage but unexpectedly aggravated tubule-interstitial damage in experimental renal disease [62]. This could provide a potential explanation for the association of very low sodium intake with worse renal outcome. Both high or extremely low dietary salt intake might be inappropriate for CKD patients. KDIGO recommends a daily sodium intake of < 2.0 g/d in patients with CKD without a clear lower limit. Taken together, these results should be interpreted, and the proper levels of dietary salt intake in CKD patients should be discussed. The current study might not be the final answer to the question of low-sodium diets and progression of kidney disease. Interventional studies are needed to ascertain the optimum level of dietary sodium intake for renal protection in patients with different CKD stages.

Our study has several potential limitations. First, as has been noted, the existence of statistical heterogeneity and inevitable clinical heterogeneity might raise critical concerns regarding validity,

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although we tried to resolve these concerns through subgroup analysis and sensitivity analysis. Heterogeneity in the study population as included in the umbrella term "CKD patients" remains a major concern for meaningful interpretation of the results. Because of the stages of CKD patients in the studies we included varies a lot, some are CKD 1-5, some are CKD 1-3, some are CKD 3-4, some are CKD 2-4, some are CKD 4-5, some are CKD 3-4. So it is difficult to stratify our analysis of outcomes by stage of CKD. We can only perform subgroup analyses according to the baseline eGFR of CKD patients. Second, cohort studies are prone to selection bias that is usually not found in RCTs. Therefore, pooling the results from RCTs and cohort studies together will bias the results of meta-analysis. So we performed subgroup analysis according to study design. There was no statistical heterogeneity according to study design for the outcomes. What is noteworthy is that low-salt diets did not produce beneficial effects on the rate of change in eGFR in RCTs (MD:-0.75; 95% CI: -7.58 to 6.08), but produced beneficial effects on the rate of change in eGFR in cohort studies (MD:2.48; 95% CI: 0.32 to 4.64). Third, many of the included studies with low quality had small sample sizes and a low incidence of events. Consequently, the uncertainty in the analyses was increased, resulting in wide CIs of effect measures. Fourth, confounding factors in the observational studies cannot be neglected, which might impact our results. Therefore, the findings of this study should be viewed as hypothesis-generating and need to be confirmed by further studies.

CONCLUSION

In conclusion, our meta-analyses suggested that low-salt diet may reduce the risk of renal composite outcomes events in patients with CKD. However, there was no compelling evidence indicating that low-salt diet could reduce the rate of eGFR decline, proteinuria, incidence of all-cause mortality and CV events. The optimal dietary salt intake for patients with different CKD stages

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remains unanswered. Further well-designed RCTs targeting patients with different CKDs are required to assess and integrate these results.

Acknowledgments We thank American Journal Experts Corporation for helping in proofreading and editing the English of the final manuscript.

Contributors All authors take responsibility for the integrity of the data and the accuracy of data analysis. Study concept and design: L.W., X.S., H.S. and C.L.. Extraction, analysis and interpretation of data: H.S. and X.S.. Drafting of the manuscript: H.S. and L.W.. Critical revision of the manuscript: L.W. and X.S. Statistical analysis: H.S., C.L. and W.G.. Technical support: L.W., X.S. and W.G.. All authors have read and agreed to the submission to this journal of the manuscript.

Funding Support was provided by the National Natural Science Foundation of China (no. 82000655) and the Science and Technology Project of Shanxi Province (no. 201801D121218).The funders had no role in decision to publish and preparation of the manuscript.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data used to support the findings of this study are included within the article and supplementary information.

Ethics statement Our research include published studies, and does not require ethical committee approval.

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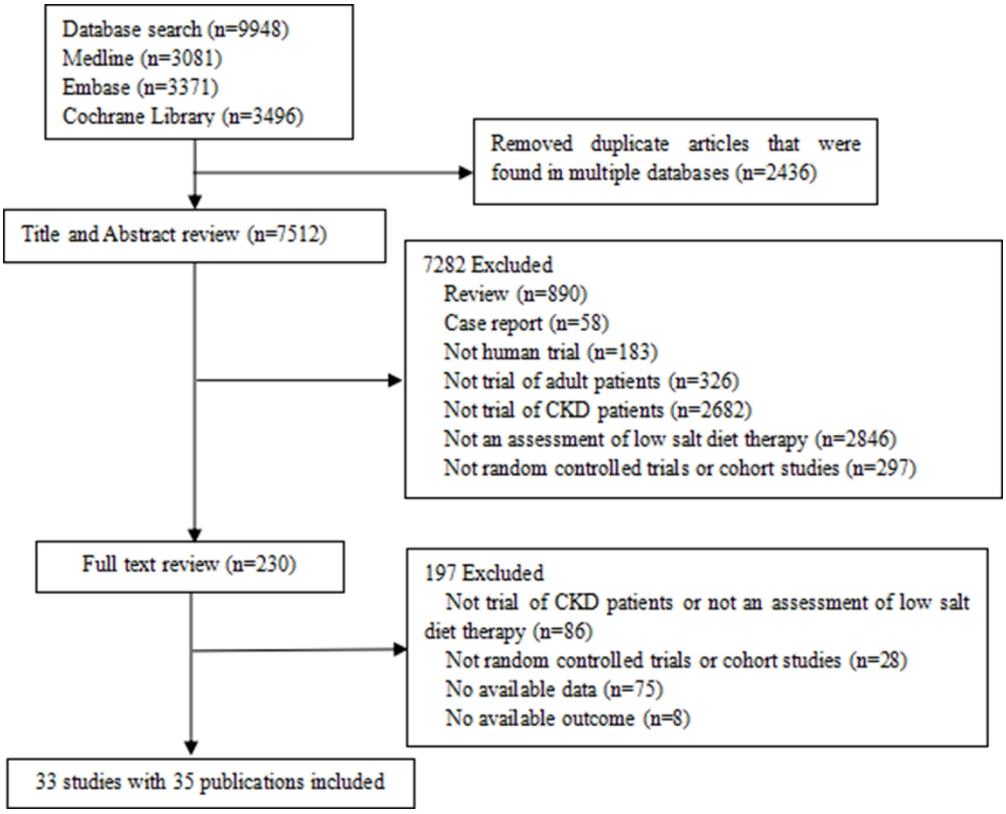
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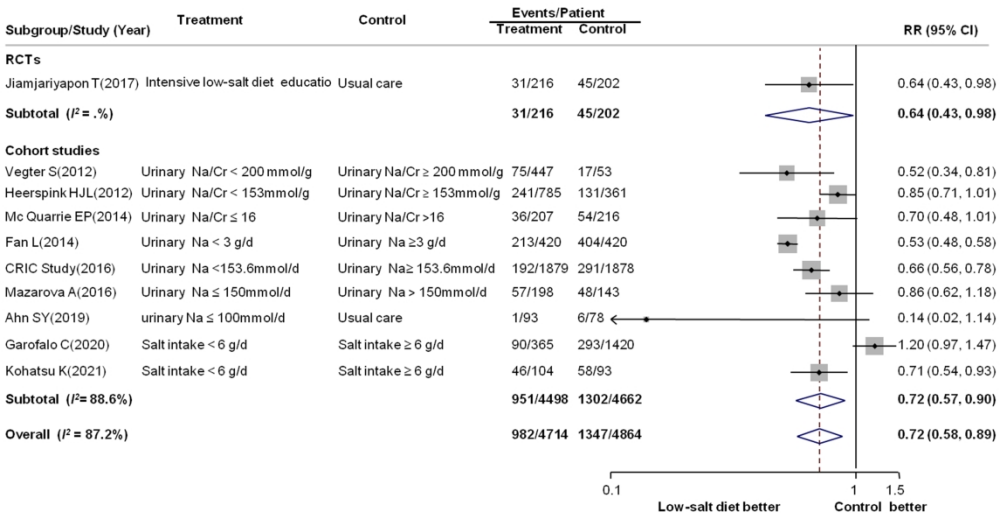
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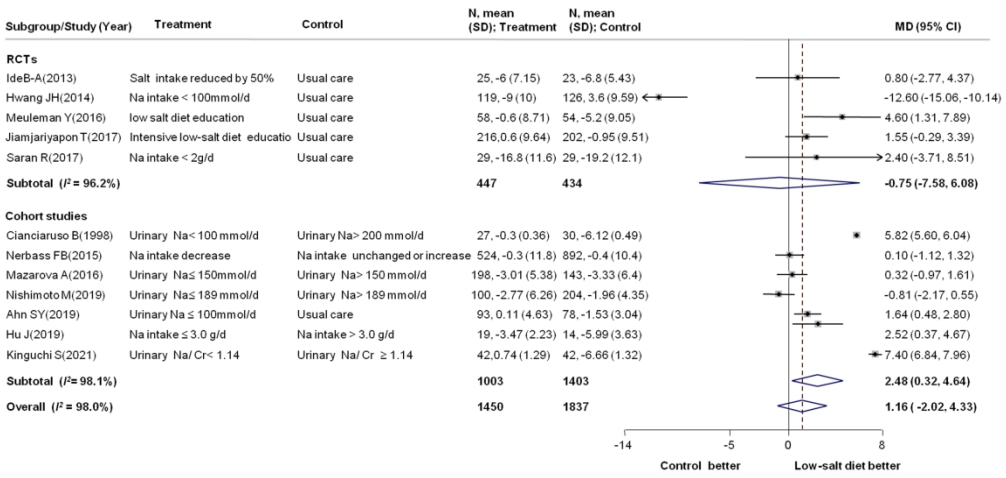
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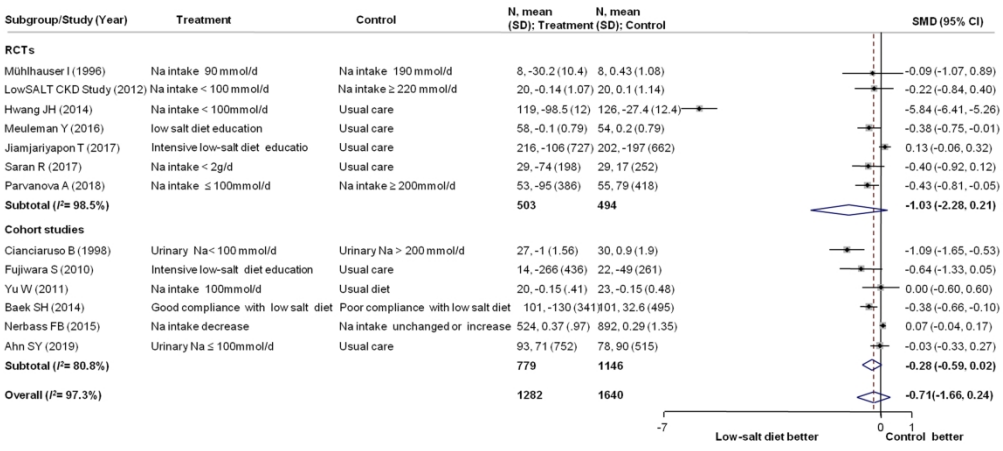
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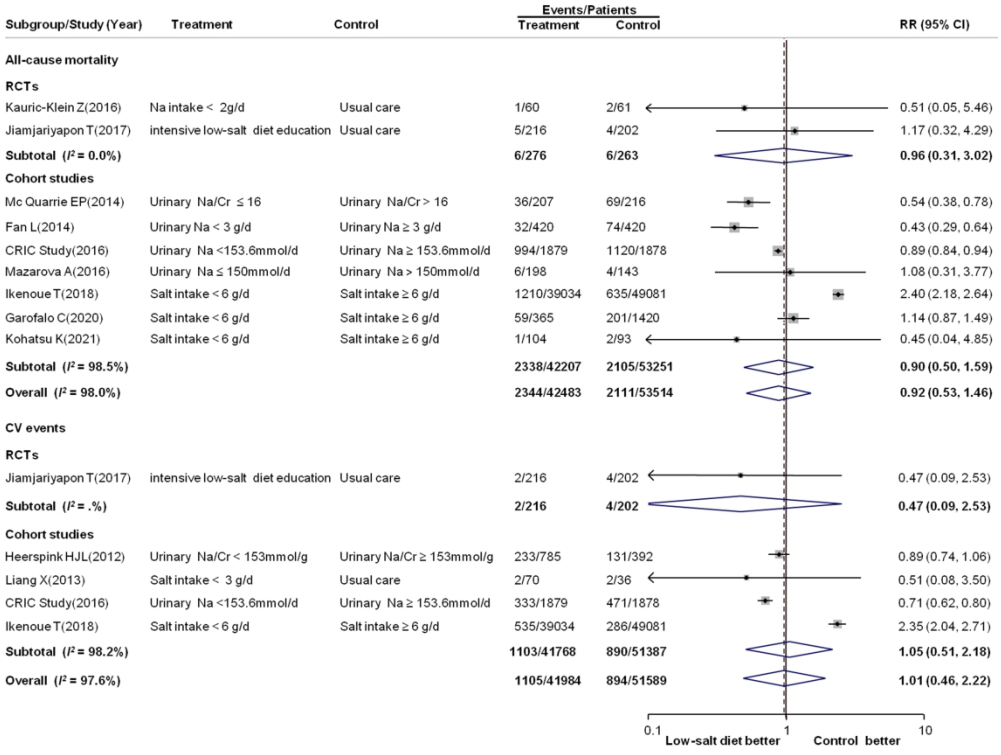
264x136mm (300 x 300 DPI)



304x142mm (300 x 300 DPI)



314x139mm (300 x 300 DPI)



281x213mm (300 x 300 DPI)

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Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang

Citation

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang. Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease. PROSPERO 2017 CRD42017072395 Available from: http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42017072395

Review question

This meta-analysis aims to evaluate the effects of different levels of dietary sodium intake on renal outcome in patients with chronic kidney disease (CKD).

Searches

All relevant randomized controlled trails (RCTs) and cohort studies will be identified and searched in following electronic databases without language restriction: Ovid MEDLINE SP (from 1946 to July 2017), EMBASE (from 1966 to July 2017), and the Cochrane Central Register of Controlled Trails(no date restriction). A search in www.clinicaltrials.gov will be used to identify the ongoing but unpublished studies about this field. We combined this strategy with a manual search of reference lists from identified trials and review articles.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/72395_STRATEGY_20170618.pdf

Types of study to be included

We will include RCTs and cohort studies that compared different levels of sodium intake for adult patients with CKD as well as studies in broader populations in which outcome data for subgroups with CKD could be disaggregated.

Condition or domain being studied

Chronic kidney disease

Participants/population

Inclusion criteria: Adults (no less than 18 years old) with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) at all disease stages will included without restrictions on gender or race. Data from subgroups of participants with CKD within studies with broader inclusion criteria (e.g. people from the general population, people with diabetes, people with cardiovascular disease) were also included.
Exclusion criteria: Studies which do not include any people with chronic kidney disease.

Intervention(s), exposure(s)

Low-sodium diet

Comparator(s)/control

No intervention or different levels of dietary sodium intake

Context

None

Primary outcome(s)

If the data allow:

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1) Kidney failure events: It will include more than 25% or 50% decrease in estimated glomerular filtration rate (eGFR), doubling of serum creatinine, end stage renal disease (ESRD);
2) Rate of change in the eGFR: The difference from baseline in eGFR divided by the number of years between creatinine measurements (ml/min/1.73m2 per year);
3) Change in proteinuria or albuminuria, including urinary protein excretion (UPE), urinary albumin excretion (UAE), protein to creatinine (PCR) and albumin to creatinine ratio (ACR);
4) Major cardiovascular events: It will include a composite including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death.

Secondary outcome(s)

1) 24-hour urinary sodium excretion;
2) Change in blood pressure;
3) All-cause mortality;
4) Change in markers of fluid overload (brain natriuretic peptide, N-terminal pro-brain natriuretic peptide, weight, bio-impedance analysis);
5) Adverse events: hypotensive episodes, undesirable change in blood lipids (low density lipoprotein, high-density lipoprotein);
6) Hospitalization.

Data extraction (selection and coding)

Data extraction will be carried out independently by two reviewers using standard data extraction forms. Disagreement will be resolved by consensus or by discussion with a third author. When detailed information that was needed for the analysis was not available, the original authors will be contacted to obtain the missing information through e-mail. Where multiple publications of one study exists, data will be extracted and grouped together as a single study with the most complete data.
The extracted data will include: study characteristics (study design, method of randomization, and withdrawals / dropouts); baseline patient characteristics (age, sex, race, country, BMI, sodium intake and its measurement, 24-hour urinary sodium excretion, mean proteinuria or albuminuria, eGFR, serum creatinine concentrations, blood lipids, blood pressure, diabetic status, cardiovascular events, smoking status, glycosylated hemoglobin, hemoglobin, antihypertensive drugs, antidiabetic drugs, different kidney disease diagnosis); change in eGFR; follow-up duration; outcomes: as listed in Primary outcomes and Secondary outcomes. When the required quantitative data is only provided in published figures, we will use g3 data software (http://www.frantz.fi/software/g3data.php) to extract exact numbers from figures.

Risk of bias (quality) assessment

For the RCTs, we will assess risk of bias for sequence generation, allocation concealment, blinding, selective reporting, incomplete outcome data and other sources of bias, and determined overall risk of bias based on predefined rules, utilizing the Cochrane Collaboration risk of bias tool. (reference: Higgins JP, Altman DG, et al. Cochrane Statistical Methods G. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials. BMJ 2011;343:d5928.). Furthermore, the quality of RCT study will be quantified with Jadad scale. (reference: Jadad AR, Moore RA, Carroll D, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Control Clin Trials 1996;17:1– 12.). The quality of cohort study will be independently evaluated by each investigator using Newcastle-Ottawa quality assessment scale. (reference: Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010; 25:603–5.).

Strategy for data synthesis

The results of the data extraction and quality assessment for each study will be presented in structured tables. For dichotomous outcomes, relative risks (RR) and 95% Cis will be calculated from the numbers of events and participants at risk for events. For continuous outcomes, the mean difference (MD) will be used, or the standardized mean difference (SMD) if different scales were used. Data will be pooled using the random-effects model but the fixed-effect model will also be used to ensure robustness of the model chosen and susceptibility to outliers.
Heterogeneity will be analyzed using Cochrane's Q and the I-squared statistic. We will explore potential

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heterogeneity by subgroup analysis and meta-regression.
Sensitivity analyses will be conducted to check the robustness of the effect of sodium-restricted treatment by excluding of trials with small sample size and low Jadad scores.
A two-sided p-value less than 0.05 was considered statistically significant and statistical analyses were performed using RevMan version 5.3 and STATA version 12.0.

Analysis of subgroups or subsets

If the data allow, subgroup analysis will be conducted by study design, follow-up duration, measures of sodium intake, different levels of sodium intake, difference in sodium intake between two groups, and baseline measures, including age, sex, BMI, eGFR, 24-hour urinary sodium excretion,mean serum creatinine, blood pressure, mean proteinuria or albuminuria, stage of CKD, presence of comorbidities (hypertension, diabetes and CVD), different kidney disease diagnosis (e.g. GN or Kidney Transplant Recipients) and whether RAAS blocker or other antihypertensive drug treatments.

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Anticipated or actual start date

15 July 2017

Anticipated completion date

01 February 2018

Funding sources/sponsors

None

Conflicts of interest

None known

Language

English

Country

China

Stage of review

Review_Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Acids, Noncarboxylic; Humans; Kidney; Renal Insufficiency, Chronic; Sodium; Sodium Chloride, Dietary; Sodium, Dietary

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Date of registration in PROSPERO
18 July 2017

Date of publication of this version
18 July 2017

Details of any existing review of the same topic by the same authors
None

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions
18 July 2017

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Supplementary file 2 Search Strategy

Ovid MEDLINE

1. exp Sodium Chloride/ or sodium chloride.mp.

2. exp Sodium, Dietary/

3. exp Diet, Sodium-Restricted/

4. ((sodium or salt) adj5 (low or high or alter* or reduce* or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)).mp.

5. (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free).mp.

6. 1 or 2 or 3 or 4 or 5

7. exp Kidney/ or kidney.mp.

8. exp Nephrosis/ or nephrosis*.mp.

9. (renal or nephropath*).mp.

10. (chronic kidney or chronic renal).mp.

11. exp Kidney Diseases/

12. ((kidney or renal) adj (disease* or failure)).mp.

13. exp Kidney Function Tests/ or kidney function test*.mp.

14. (CKF or CKD or CRF or CRD).mp.

15. (end-stage renal or end-stage kidney or endstage renal or endstage kidney).mp.

16. (ESRF or ESKF or ESRD or ESKD).mp.

17. exp Uremia/ or ur?emi*.mp.

18. exp Renal Replacement Therapy/ or renal replacement therap*.mp.

19. exp Renal Insufficiency/ or renal insufficien*.mp.

20. exp Dialysis/ or dialysis.mp.

21. exp Renal Dialysis/ or renal dialys*.mp.

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22. ("pre dialysis" or pre-dialysis or predialysis).mp.

23. (h*emodialysis or h*emofiltration or h*emodiafiltration).mp.

24. exp Peritoneal Dialysis/ or peritoneal dialysis.mp.

25. (PD or CAPD or CCPD or APD or IPD).mp.

26. exp Kidney Transplantation/

27. ((kidney or renal) adj (transplant* or graft* or allograft*)).mp.

28. 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27

29. randomized controlled trial.pt.

30. controlled clinical trial.pt.

31. randomized.ab.

32. placebo.ab.

33. randomly.ab.

34. clinical trials as topic.sh.

35. trial.ti.

36. 29 or 30 or 31 or 32 or 33 or34 or 35

37. (animals not (humans and animals)).sh.

38. 36 not 37

39. exp Cohort Studies/

40. exp Longitudinal Studies/

41. exp Follow-Up Studies/

42. exp Prospective Studies/

43. exp Retrospective Studies/

44. cohort.ti,ab.

45. longitudinal.ti,ab.

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46. prospective.ti,ab.

47. retrospective.ti,ab.

48.38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47

49. 6 and 28 and 48

Embase

#1. 'sodium chloride'/exp OR 'sodium chloride'

#2. 'sodium intake'/exp

#3. 'salt intake'/exp

#4. 'sodium restriction'/exp

#5. 'high sodium intake'/exp

#6. 'high salt diet'/exp

#7. 'high sodium' OR 'low sodium' OR 'sodium alter*' OR 'sodium reduce*' OR 'sodium reduction' OR 'sodium restrict*' OR 'sodium intake*' OR 'sodium diet*' OR 'sodium food' OR 'sodium increas*' OR 'sodium decreas*' OR 'sodium change*' OR 'sodium changing' OR 'sodium free'

#8. 'high salt' OR 'low salt' OR 'salt alter*' OR 'salt reduce*' OR 'salt reduction' OR 'salt restrict*' OR 'salt intake*' OR 'salt diet*' OR 'salt food' OR 'salt increas*' OR 'salt decreas*' OR 'salt change*' OR 'salt changing' OR 'salt free'

#9. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8

#10. 'kidney'/exp OR 'nephrosis'/exp

#11. kidney OR renal OR nephropath* OR nephrosis

#12. 'chronic kidney' OR 'chronic renal'

#13. 'kidney disease'/exp OR 'kidney failure'/exp

#14. 'kidney disease*' OR 'renal disease*' OR 'kidney failure' OR 'renal failure'

#15. 'kidney function test'/exp OR 'kidney function test*'

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#16. ckf OR ckd OR crf OR crd

#17. 'end-stage kidney' OR 'end-stage renal' OR 'endstage kidney' OR 'endstage renal'

#18. esrf OR eskf OR esrd OR eskd

#19. 'uremia'/exp OR ur?emi*

#20. 'renal replacement therapy'/exp OR 'renal replacement therap*'

#21. 'renal insufficien*'

#22. 'dialysis'/exp OR dialys*

#23. 'pre dialysis' OR predialysis

#24. 'hemodialysis'/exp OR 'hemofiltration'/exp OR 'hemodiafiltration'/exp

#25. hemodialysis OR haemodialysis OR hemofiltration OR haemofiltration OR hemodiafiltration OR haemodiafiltration

#26. 'peritoneal dialysis'/exp OR 'peritoneal dialysis'

#27. pd OR capd OR ccpd OR apd OR ipd

#28. 'kidney transplantation'/exp OR 'kidney graft'/exp OR 'kidney allograft'/exp

#29. 'kidney transplant*' OR 'renal transplant*' OR 'kidney graft*' OR 'renal graft*' OR 'kidney allograft*' OR 'renal allograft*'

#30. #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29

#31. 'randomized controlled trial'/exp OR 'randomized controlled trial' OR 'controlled clinical trial'/exp OR 'controlled clinical trial' OR random*OR 'clinical trials'/exp OR 'clinical trials' OR trial:ti

#32. 'human'

#33. animal$ OR nonhuman

#34. #32 AND #33

#35. #33 NOT #34

#36. #31 NOT #35

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#37.'cohort analysis'/exp OR 'longitudinal study'/exp OR 'prospective study'/exp OR 'retrospective study'/exp

#38. (cohort OR concurrent OR incidence OR longitudinal OR followup OR 'follow up' OR prospective OR retrospective) NEXT/1 (analys* OR design* OR evaluation* OR research OR stud* OR survey* OR trial*)

#39. #36 OR #37 OR #38

#40. #9 AND #30 AND #39

CENTRAL

#1 MeSH descriptor: [Sodium Chloride] explode all trees

#2 "sodium chloride":ti,ab,kw (Word variations have been searched)

#3 MeSH descriptor: [Sodium, Dietary] explode all trees

#4 MeSH descriptor: [Diet, Sodium-Restricted] explode all trees

#5 ((sodium or salt) near/5 (low or high or alter* or reduce* or reducing or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)):ti,ab,kw

#6 (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free):ti,ab,kw

#7 (#1 or #2 or #3 or #4 or #5 or #6)

#8 MeSH descriptor: [Kidney] explode all trees

#9 MeSH descriptor: [Nephrosis] explode all trees

#10 (kidney or renal or nephropath* or nephrosis):ti,ab,kw

#11 (chronic next kidney or chronic next renal):ti,ab,kw

#12 MeSH descriptor: [Kidney Diseases] explode all trees

#13 (kidney or renal) next (disease* or failure):ti,ab,kw

#14 MeSH descriptor: [Kidney Function Tests] explode all trees

#15 “kidney function test*”:ti,ab,kw

#16 (CKF or CKD or CRF or CRD):ti,ab,kw

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#17 ("end-stage kidney" or "end-stage renal" or "endstage kidney" or "endstage renal"):ti,ab,kw

#18 (ESRF or ESKF or ESRD or ESKD):ti,ab,kw

#19 MeSH descriptor: [Uremia] explode all trees

#20 ur?emi*:ti,ab,kw

#21 MeSH descriptor: [Renal Replacement Therapy] explode all trees

#22 "renal replacement therap*":ti,ab,kw

#23 MeSH descriptor: [Renal Insufficiency] explode all trees

#24 "renal insufficien*":ti,ab,kw

#25 MeSH descriptor: [Dialysis] explode all trees

#26 dialys*:ti,ab,kw

#27 MeSH descriptor: [Renal Dialysis] explode all trees

#28 "renal dialys*":ti,ab,kw

#29 "pre dialysis" or pre-dialysis or predialysis:ti,ab,kw

#30 (h*emodialysis or h*emofiltration or h*emodiafiltration):ti,ab,kw

#31 MeSH descriptor: [Peritoneal Dialysis] explode all trees

#32 "peritoneal dialys*":ti,ab,kw

#33 (PD or CAPD or CCPD or APD or IPD):ti,ab,kw

#34 MeSH descriptor: [Kidney Transplantation] explode all trees

#35 (kidney or renal) near (transplant* or graft* or allograft*):ti,ab,kw

#36 (#8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35)

#37 (#7 and #36)

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Supplementary file 3 Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up, months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9[a]	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Sodium intake 80-100 mmol/d	24-h urinary sodium	Usual care	1.39c	143/88	NA
Fujiwara S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPAKUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5[b]	144/95	Non- industry

Heerspink HJL[24] (2012)	Type 2 diabetic nephropathy	1177	65	59	Cohort study	30	Renal composite outcomes, CV events	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 153mmol/g	44	155/84	Industry and non- industry
LowSALT CKD Study[25,26] (2012)	Hypertensive patients with stage 3–4 CKD	20	75	69	Randomised crossover trial	1	UACR, SBP, DBP	Sodium intake < 100 mmol/d	Sodium intake < 100 mmol/d	24-h urinary sodium	Sodium intake ≥ 220 mmol/d	32	151/82	Non- industry
de Brito-Ashurst I[27] (2013)	eGFR < 60 mL/min and mean BP > 130/80 mm Hg	48	58	58	Randomised parallel-group controlled trial	6	Change in eGFR	Intensive low-salt diet education	Traditional meals had salt reduced by 50%	24-h urinary sodium	Usual care	41.5	153/85	Non- industry
Liang X[28] (2013)	PD	106	54	33	Cohort study	1	CV events, SBP, DBP	Health education to restrict water-sodium intake strictly	Salt intake not more than 3 g/d	The 3-day diet diary was collected once a month by the special nurse. Compared with the Food Composition, the daily salt intake were estimated	Usual care	NA	149/88	NA
Mc Quarrie EP[29] (2014)	Patients with CKD and had at least one 24-hour urinary sodium measurement	423	50	52	Cohort study	102	Renal composite outcomes, all- cause mortality	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine	Urinary sodium/creatinine > 16	48	139/79	Non- industry
Fan L[4] (2014)	GFR 13–55 mL/min/1.73m^2	840	60	52	Cohort study	72	Renal composite outcomes, all- cause mortality	Urinary sodium < 3 g/d	Urinary sodium < 3 g/d	24-h urinary sodium	Urinary sodium ≥ 3 g/d	32.5	132/81	Non- industry
Baek SH[30] (2014)	Hypertensive patients with albuminuria	202	51	50	Cohort study	2	UAE, SBP, DBP	Good compliance with an low salt diet	Good compliance with an low salt diet (24-hurinary sodium reduction ratio < 50%)	24-hurinary sodium	Poor compliance with an low salt diet	77.1[c]	121/73	Industry and non- industry

Hwang JH[31] (2014)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	245	50	50	Randomised parallel-group controlled trial	2	eGFR, UAE, SBP, DBP	Sodium intake < 100mmol/d	Sodium intake < 100mmol/d	24-hurinary sodium	Usual care	67.3	131/79	Industry
Rodrigues Telini LS[32] (2014)	HD	39	38	58	Randomised parallel-group controlled trial	4	SBP, DBP	2 g of sodium reduction in their habitual diet	2 g of sodium reduction in their habitual diet	Salt intake was based on a 72-halimentary registry	Usual diet	NA	146/86	Non- industry
Nerbass FB[33] (2015)	eGFR 59–30 ml/min/1·73 m^2	1416	39	73	Cohort study	12	eGFR, UACR, SBP, DBP	Sodium intake decrease	Sodium intake decrease	Estimated 24-hurinary sodium	Sodium intake unchanged or increase	52.7	134/73	Industry and non- industry
CRIC Study[3,34] (2016)	eGFR20-70mL/min/1.73 m^2	3757	56	58	Cohort study	78	Renal composite outcomes, all- cause mortality, CV events	Urinary sodium < 153.6mmol/d	Urinary sodium < 153.6mmol/d	24-hurinary sodium	Urinary sodium ≥153.7 mmol/d	44.5	128/71	Non- industry
Mazarova A[5] (2016)	eGFR < 30 mL/min/1.73 m^2	341	62	64	Cohort study	18	Change in eGFR, renal failure, all-cause mortality	Urinary sodium ≤ 150mmol/d	Urinary sodium ≤ 150mmol/d	24-hurinary sodium	Urinary sodium> 150 mmol/d	17.1	NA	No funding supported
Kauric-Klein Z[35] (2017)	HD	121	50	60	Randomised parallel-group controlled trial	3	All- cause mortality, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	Sodium intake checklist developed by Millar and Beard	Usual care	NA	165/87	Non- industry
Meuleman Y[36] (2017)	eGFR ≥ 20mL/min/1.73m^2 and UPE > 0.2 g/L	112	82	55	Randomised parallel-group controlled trial	6	eGFR, UPE, SBP, DBP	Usual care and low salt diet education	low salt diet education	24-hurinary sodium	Usual care	49.7	129/76	Non- industry

Jiamjariyapon T[37] (2017)	Stage 3-4 CKD with diabetes or hypertension	418	27	62	Cluster randomized controlled trial	24	Renal composite outcomes, all-cause mortality, eGFR, CV events, UACR, SBP, DBP	Usual care and intensive low-salt diet education	Intensive low-salt diet educatio (Sodium intake less than 2,400 mg/d)	24-hurinary sodium	Usual care	41.5	125/76	Non- industry
Saran R[38] (2017)	eGFR 15–60 ml/min/1.73 m^2	58	60	61	Randomized crossover trial	1	eGFR, UACR, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	24-hurinary sodium	Usual care	38.7	130.3/NA	Non- industry
Huang B[39] (2017)	HD	90	43	55	Cluster randomized controlled trial	6	SBP, DBP	Self-management support, restrict salt intake	Self-management support, restrict salt intake	Dietary salt intake was measured using slightly modified balance formulas: (1) Sodium intake (mol)=[Na+1(mmol/L)×ECW(L)－Na+ 2(mmol/L)×(ECW– IDWG)(L)]/1000; (2) Salt intake (g/day)=sodium intake(mol)×58.4 (g/mol)/2(day)	Usual care	NA	154/86	Non- industry
Ikenoue T[6] (2018)	HD	88115	60	65	Cohort study	12	All-cause mortality, CV events	Salt intake < 6 g/d	Salt intake < 6 g/d	Intra-dialytic weight loss was used to calculate salt intake	Salt intake ≥ 6 g/d	NA	NA	No funding supported
Parvanova A[40] (2018)	Patients with type 2 diabetes and macroalbuminuria	108	89	64	Randomised parallel-group controlled trial	3	UACR	Sodium intake ≤ 100mmol/d	Sodium intake ≤ 100mmol/d	24-hurinary sodium	Sodium intake ≥ 200mmol/d	87.36	146/80	Industry
Martinez MG[41] (2019)	Pre-dialysis CKD with UPE > 500 mg/d	84	52	56	Cohort study	6	Change in SBP	Urinary sodium reduced	Urinary sodium reduced	24-hurinary sodium	Urinary sodium increased	49[b]	142/NA	NA
Nishimoto M[42] (2019)	UACR 30–599 mg/g and eGFR ≥ 50 ml/min/1.73 m^2	304	70	59	Cohort study	12	Change in eGFR, SBP, and DBP	Urinary sodium ≤ 189 mmol/d	Urinary sodium ≤ 189 mmol/d	24-hurinary sodium	Urinary sodium > 189 mmol/d	64.93	140/83	Non- industry

Ahn SY[43] (2019)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	171	49	50	Cohort study	12	Rate of change in eGFR, renal composite outcomes, change in UAE, SBP, DBP	Intensive low-salt diet education	Intensive low-salt diet education to achieve urinary sodium ≤ 100mmol/d	24-hurinary sodium	Usual care	64.2	124/75	Non- industry
Hu J[44] (2019)	PD	33	58	42	Cohort study	12	Residual renal Ccr	Sodium intake ≤ 3.0 g/d	Sodium intake ≤ 3.0 g/d	Daily sodium intake of the patients was calculated based on the 3-day dietary record	Sodium intake > 3.0 g/d	8.3	146/95	Non- industry
Humalda JK[45] (2020)	CKD stages 1 to 4 or kidney transplants recipients, and eGFR was ≥ 25 mL/min/ 1.73 m2	99	84	57	RCT	6	SBP, DBP	Self-management to restrict salt intake	Self-management to restrict salt intake	A detailed interactive food diary	Usual care	55	139/83	Non- industry
Garofalo C[46] (2020)	CKD stages 1–5	1785	58	64	Cohort study	49	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	37	NA	NA
Kinguchi S[47] (2021)	Diabetic kidney disease patients with HbA1c 7.0–10.0%, eGFR ≥ 45 ml/min/1.73 m2, and UACR ≥ 30 mg/g	84	76	65	Cohort study	6	UACR	urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios	urinary sodium-to-creatinine ratios ≥ 1.14	67	142/79	Non- industry
Kohatsu K[48] (2021)	CKD stages 3–5	197	76	71	Cohort study	17	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	24.2	133/74	Non- industry

Note: Ccr, clearance of creatinine; CKD, chronic kidney disease; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; ESRD, end stage renal disease; ECW, extracellular water; HD, hemodialysis; IDWG, interdialytic weight gain; NA, no available; PD, peritoneal dialysis; SBP, systolic blood pressure; UACR, urinary albumin to creatinine ratio; UAE, urine albumin excretion; UPE, urinary protein excretion; the units for eGFR: mL/min/1.73m². [a] The units for eGFR: mL/min, [b] Only Ccr obtained, the units for Ccr: mL/min; [c] Only serum creatinine obtained, the units for serum creatinine: μmol/L.

Supplementary file 4 Table S2 Summary of characteristics of included trials and patients.

Outcomes	No. of Studies/ No. of Patients	Sample Size	Follow-up, months	Age, years	Male (%)	Lost follow-up rate
Renal composite outcomes	10/9578	462 (305-1329)	43 (23-74)	58 (52-64)	59.2%	3.2%
Rate of change in eGFR	12/3287	142 (57-332)	12 (6-17)	55(48-61)	56.6%	7.5%
Proteinuria	13/2922	108 (40-224)	3 (1-12)	50 (43-66)	57.2%	8.8%
All-cause death	9/95997	423 (231-2771)	24 (15-75)	62 (55-65)	60.6%	0.03%
CV events	5/93573	1177 (262-45936)	24 (6.5-54)	59(45-63)	60.8%	0.02%
SBP	19/3608	103 (38-213)	3 (1-6)	55 (41-60)	57.1%	7.5%
DBP	18/3524	36 (20-90)	12 (2-54)	55 (40-60)	57.3%	7.7%

Note:Values are given as median (quartile). CV, cardiovascular; eGFR, estimated glomerular filtration rate.

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Supplementary file 5 Table S3 Assessment of risk of bias for RCTs.

Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
1. Bias arising from the randomization process																
1.1 Was the allocation sequence random?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias arising from the randomisation process?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
2. Risk of bias due to deviations from intended interventions																
2.1 Were participants aware of their assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	Other	High	High	High	High	Low	High	High
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	High	High	High	High	High	High	High	High
2.3 If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?	N/PN	Y/PY	NA/NI		Low		Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.4 If Y/PY/NI to 2.3: Were these deviations likely to have affected the outcome?	N/PN	Y/PY	NA/NI													
2.5 If Y/PY to 2.4: Were these deviations from intended intervention balanced between groups?	Y/PY	N/PN	NA/NI													

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomised?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to deviations from intended interventions?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
3. Bias due to missing outcome data																
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	High	High	High	High	Low	High	High	Low
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?	Y/PY	N/PN	NA	Low	Low				Low	Low	Low	Low		Low	Low	
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	N/PN	Y/PY	NA/NI													
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to missing outcome data?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
4. Bias in measurement of the outcome																
4.1 Was the method of measuring the outcome inappropriate?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?	N/PN	Y/PY	NI	Low	High	Low	High	High	Low	High	High	High	Low	High	Low	High
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI		Low		Low	Low		Low	Low	Low		Low		Low
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias in measurement of the outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
5. Bias in selection of the reported result																
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:																
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
5.3 ... multiple eligible analyses of the data?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction bias due to selection of the reported results?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
Overall bias																
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the overall predicted direction of bias for this outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null

Note: Y, yes; PY, probably yes; PN, probably no; N, no; NA, not applicable; NI, no information; RCT, randomized controlled trial.

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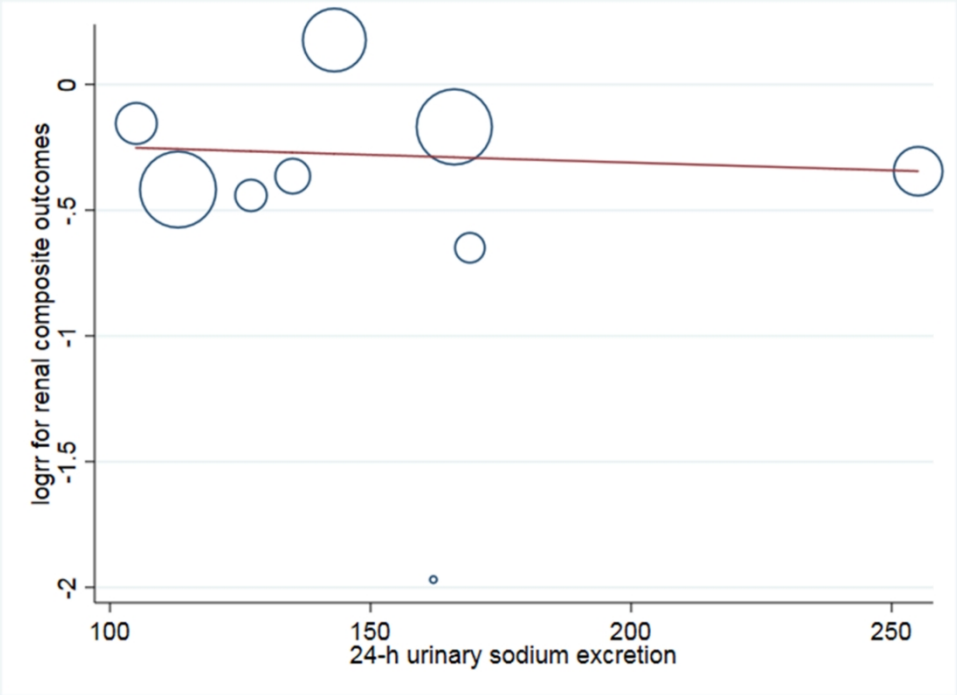
Supplementary file 6 Table S4 Quality assessment for cohort studies.

Study (Year)	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Scores
Cianciaruso B[19] 1998	★	★	★	★	—	★	★	★	7★
Fujiwara S[21] 2010	★	★	★	★	★	★	—	★	7★
Yu W[22] 2011	★	★	★	★	★	★	—	★	7★
Vegter S[23] 2012	★	★	★	★	—	★	★	★	7★
Heerspink HJL[24] 2012	★	★	★	★	★	★	—	★	7★
Liang X[28] 2013	★	★	★	★	★	★	—	★	7★
Mc Quarrie EP[29] 2014	★	★	★	★	★	★	★	★	8★
Fan L[3] 2014	★	★	★	★	—	★	★	—	6★
Baek SH[30] 2014	★	★	★	★	—	★	—	★	6★
Nerbass FB[33] 2015	★	★	★	★	★	★	—	★	7★
CRIC Study[3,34] 2016	★	★	★	★	★	★	★	★	8★
Mazarova A[5] 2016	★	★	★	★	★	★	—	★	7★
Ikenoue T[6] 2018	★	★	★	★	★	★	—	★	7★
Martinez MG[41] 2019	★	★	★	★	★	★	★	★	8★

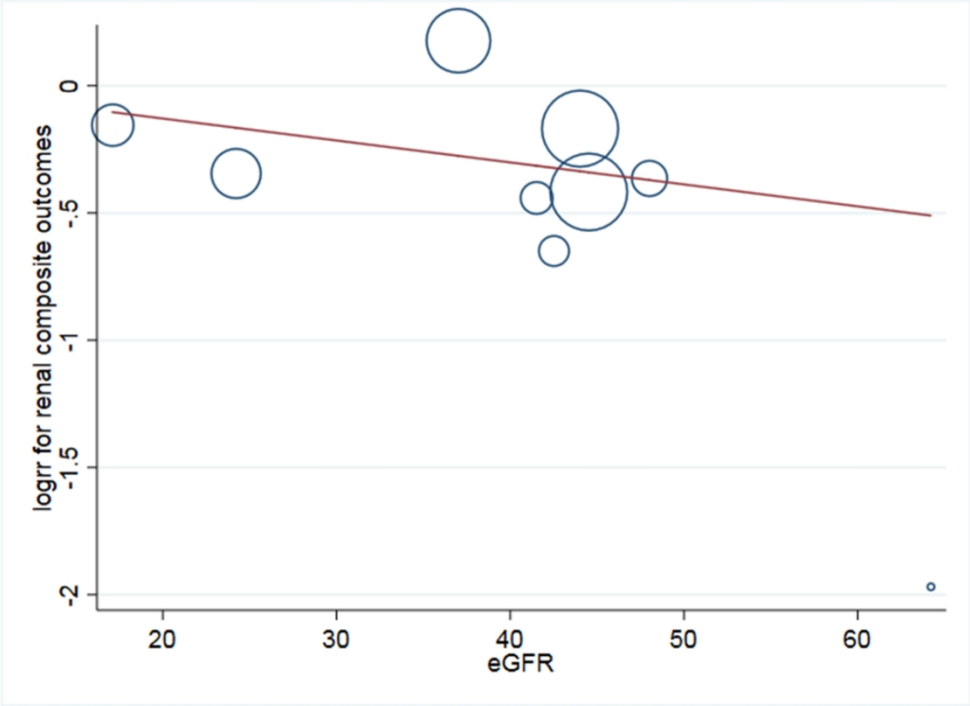
Nishimoto M[42] 2019	★	★	★	★	★	★	—	★	7★
Ahn SY[43] 2019	★	★	★	★	★	★	—	★	7★
Hu J[44] 2019	★	★	★	★	★	★	—	★	7★
Garofalo C[46] 2020	★	★	★	★	★	★	★	★	8★
Kinguchi S[47] 2021	★	★	★	★	★	★	—	★	7★
Kohatsu K[48] 2021	★	★	★	★	★	★	—	★	7★

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability. —: no score; It is considered that the queue follow-up is appropriate when the follow-up rate is more than 80%.

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Supplementary file 7



Supplementary file 8

Supplementary file 9 Table S5 Subgroup analysis of outcomes except renal failure events.

Subgroup	No. of trials	n	RR/MD/SMD (95%CI)	*P* for RR/MD/SMD	I^2	*P* for heterogeneity test[a]
Change in eGFR（mL/min/1.73m²）						
Study type						
RCT	5	881	-0.75 (-7.58, 6.08)	0.83	96.2%	0.28
Cohort studies	7	2406	2.48 (0.32, 4.64)	0.02	98.1%	
Baseline eGFR (mL/min/1.73m²)						
60-89	4	804	-1.00 (-7.41, 5.42)	0.76	99.2%	0.48
30-59	6	2109	2.58 (-0.44, 5.61)	0.10	95.4%	
<15	2	374	1.24 (-0.88, 3.37)	0.25	66.2%	
Mean age (years)						
<58	5	106	0.44 (-4.40, 5.27)	0.86	98.5%	0.70
≥58	6	2669	1.68 (-1.83, 5.18)	0.35	97.8%	
Baseline BP (mmHg)						
<140/90	6	2420	-0.52 (-4.36, 3.32)	0.32	97.8%	0.41
SBP≥140 或 DBP≥90	4	469	2.54 (-2.47, 7.55)	0.55	70.1%	
Comparator of control group						
Usual care or usual diet	6	1052	-0.38 (-5.14, 4.39)	0.88	95.8%	0.85
Different levels of sodium intake	6	2235	2.62 (0.31, 4.94)	0.03	98.2%	
Proteinuria or Albuminuria						
Study type						
RCT	7	997	-1.03 (-2.28, 0.21)	0.11	98.4%	0.45
Cohort studies	6	1925	-0.28 (-0.59, 0.02)	0.07	80.8%	
Baseline eGFR (mL/min/1.73m²)						
60-89	7	821	-1.06 (-2.32, 0.20)	0.10	98.2%	0.42
30-59	6	1201	-0.23 (-0.50, 0.04)	0.10	79.6%	
Mean age (years)						
<58	7	846	-1.12 (-2.35, 0.12)	0.08	98.2%	0.34
≥58	6	2076	-0.13 (-0.34, 0.09)	0.24	64.2%	
Baseline BP (mmHg)						
<140/90	8	200	-0.82 (-1.55, -0.10)	0.01	98.3%	0.65
SBP≥140 或 DBP≥90	4	2665	-0.39 (-0.68, -0.11)	0.03	0.0%	
Comparator of control group						
Usual care or usual diet	7	1083	-1.01 (-2.18, 0.15)	0.09	98.4%	0.12
Different levels of sodium intake	6	1839	-0.34 (-0.69, 0.00)	0.05	81.7%	
All-cause mortality						
Study type						
RCT	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.99
Cohort studies	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
Baseline eGFR (mL/min/1.73m²)						
30-59	4	5438	0.65 (0.41, 1.02)	0.06	85.2%	0.008
15-29	2	2126	1.14 (0.88, 1.48)	0.33	0.0%	
<15	3	88433	1.34 (0.41, 4.39)	0.62	43.4%	

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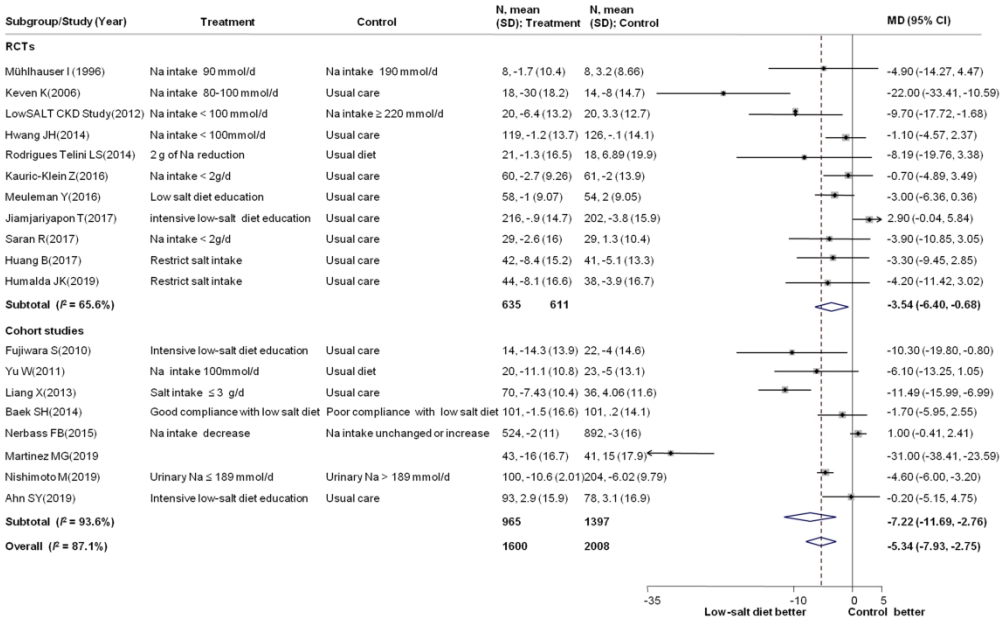
Mean age (years)						
<58	3	704	0.61 (0.37, 1.00)	0.05	90.1%	0.08
≥58	6	90977	1.34 (0.76, 2.37)	0.32	84.2%	
Baseline BP (mmHg)						
<140/90	5	5438	0.64 (0.41, 0.99)	0.05	80.6%	0.86
SBP≥140 或 DBP≥90	1	121	0.51 (0.05, 5.46)	0.58	-	
Comparator of control group						
Usual care or usual diet	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.90
Different levels of sodium intake	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
CV events						
Study type						
RCT	1	418	0.47 (0.09, 2.53)	0.88	-	0.51
Cohort studies	4	93155	1.05 (0.51, 2.18)	0.89	98.2%	
Baseline eGFR (mL/min/1.73m^2)						
30-59	3	5352	0.78 (0.63, 0.95)	0.02	57.1%	**0.02**
<15	2	88221	1.50 (0.39, 5.85)	0.56	58.3%	
Mean age (years)						
<58	2	3863	0.71 (0.62, 0.80)	<0.001	0.0%	0.40
≥58	3	89710	1.20 (0.49, 2.91)	0.69	97.3%	
Baseline BP (mmHg)						
<140/90	2	4175	0.71 (0.62, 0.80)	<0.001	0.0%	0.62
SBP≥140 或 DBP≥90	2	1283	0.88 (0.74, 1.05)	0.17	0.0%	
Comparator of control group						
Usual care or usual diet	2	524	0.49 (0.14, 1.73)	0.27	0.0%	0.52
Different levels of sodium intake	3	93049	1.14 (0.53, 2.46)	0.74	98.8%	
SBP（mmHg）						
Study type						
RCT	11	1246	-3.54 (-6.40, -0.68)	0.02	65.6%	0.43
Cohort studies	8	2362	-7.22 (-11.69, -2.76)	0.002	93.6%	
Baseline eGFR (mL/min/1.73m^2)						
60-89	7	1017	-3.30 (-5.26, -1.34)	0.001	31.7%	0.64
30-59	7	2210	-6.10 (-11.62, -0.59)	0.03	92.8%	
< 15	4	349	-5.66 (-11.62, 0.30)	0.06	76.0%	
Mean age (years)						
< 58	12	1215	-7.45 (-11.76, -3.14)	0.001	85.6%	0.25
≥ 58	7	2393	-2.39 (-5.65, 0.87)	0.15	87.3%	
Baseline BP (mmHg)						
< 140/90	10	2786	-0.97 (-2.75, 0.81)	0.28	46.9%	**0.02**
SBP ≥ 140 or DBP ≥ 90	9	822	-10.24 (-15.37, -5.12)	< 0.001	88.4%	
Comparator of control group						
Usual care or usual diet	13	1546	-4.28 (-7.20, -1.36)	0.004	74.3%	0.05
Different levels of sodium intake	6	2062	-7.55 (-12.83, -2.28)	0.005	94.6%	
DBP（mmHg）						
Study type						
RCT	7	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.85
Cohort studies	11	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

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Baseline eGFR (mL/min/1.73m^2)						
30-59	7	1017	-1.74 (-3.40, -0.08)	0.04	56.7%	0.13
15-29	6	2126	-0.53 (-1.25, 0.20)	0.16	11.9%	
< 15	4	349	-4.94 (-7.08, -2.80)	<0.001	23.2%	
Mean age (years)						
< 58	11	1131	-2.78 (-4.35, -1.22)	<0.001	50.8%	0.34
≥ 58	7	2393	-0.86 (-1.77, 0.04)	0.06	62.2%	
Baseline BP (mmHg)						
< 140/90	9	2747	-0.94 (-1.90, 0.02)	0.056	5.6%	**0.04**
SBP ≥ 140 或 DBP ≥ 90	9	777	-4.05 (-6.61, -1.49)	0.002	0.2%	
Comparator of control group						
Usual care or usual diet	11	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.12
Different levels of sodium intake	7	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

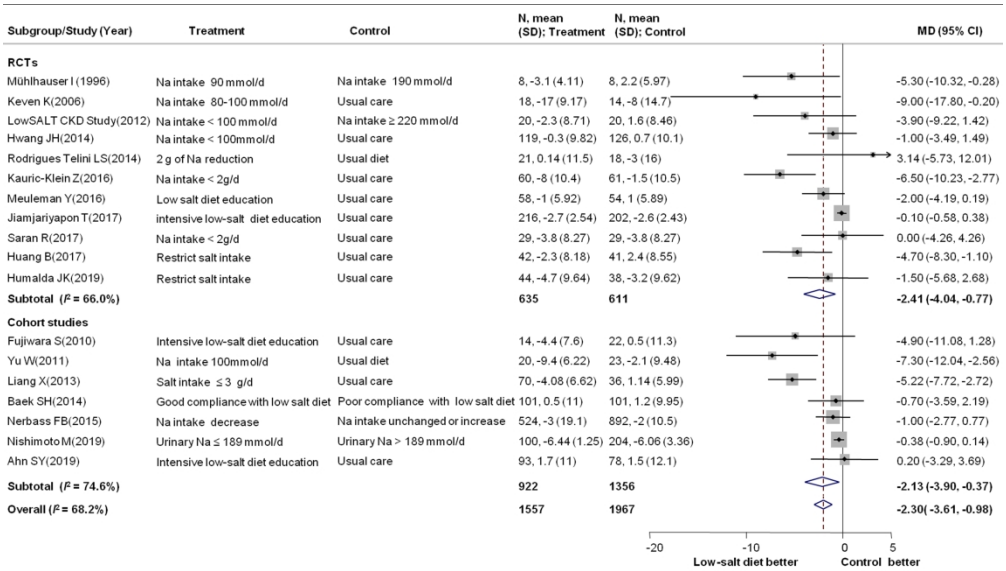
Note: [a] *P* value calculated by meta regression. BP, blood pressure; CI, confidence interval; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

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Supplementary file 10 Forest plot for SBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SBP, systolic blood pressure; SD, standard deviation.

317x210mm (300 x 300 DPI)



Supplementary file 11 Forest plot for DBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; DBP, diastolic blood pressure; SD, standard deviation.

309x186mm (300 x 300 DPI)

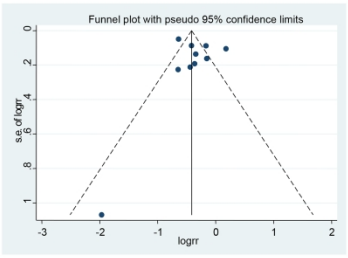
Supplementary file 12 Table S6 Sensitivity analysis of outcomes.

Sensitivity analysis	Renal composite outcomes			The rate of change in eGFR			Proteinuria or albuminuria			All-cause mortaity		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD/SMD (95%CI)	τ^2	No./n	RR (95% CI)	τ^2
Omit studies of follow-up time <12 months	-	-	-	7/2740	1.60 (-1.10, 4.31)	12.78	-	-	-	8/95876	0.92 (0.53, 1.59)	0.48
Omit studies of sample size <50	-	-	-	10/3206	1.07 (-1.26, 3.41)	12.86	9/2787	-0.89 (-1.56, -0.23)	0.99	-	-	-
Omit studies with low quality	-	-	-	-	-	-	-	-	-	-		
Omit studies with extreme outliers	-	-	-	11/3042	**2.43 (0.63, 4.23)**	8.01	12/2677	-0.24 (-0.43, -0.05)	0.06	8/7882	0.75 (0.56, 1.00)	0.08
Different statistical estimators												
DL	10/9578	0.71 (0.57, 0.88)	0.09	12/3287	1.18 (-0.94, 3.29)	12.57	13/2922	-0.70 (-1.24, -0.1)	0.89	9/95997	0.90 (0.52, 1.53)	0.48
EB	10/9578	0.72 (0.58, 0.89)	0.06	12/3287	1.16 (-2.02, 4.33)	23.4	13/2922	**-0.71(-1.66, 0.24)**	2.39	9/95997	0.92 (0.53, 1.46)	0.23
REML	10/9578	0.72 (0.58, 0.89)	0.03	12/3287	1.16 (-2.04, 4.35)	23.73	13/2922	**-0.71(-1.66, 0.24)**	2.38	9/95997	0.91 (0.53, 1.55)	0.33

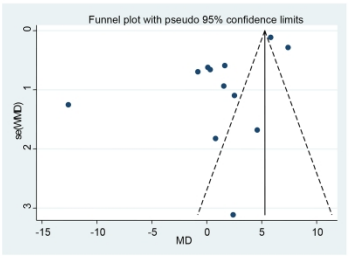
Sensitivity analysis	CV events			SBP			DBP		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD (95%CI)	τ^2
Omit studies of follow-up time <12 months	4/93467	1.03 (0.50, 2.12)	0.45	-	-	-	-	-	-
Omit studies of sample size <50	-	-	-	13/3402	-4.05 (-6.93, -1.17)	22.57	12/3318	-1.45 (-2.34, -0.56)	1.05
Omit studies with low quality	-	-	-	-	-	-	-	-	
Different statistical estimators									
DL	5/93573	0.97 (0.49, 1.93)	0.45	19/3608	-5.34 (-7.93, -2.75)	22.80	15/3524	-2.00(-2.93, -1.06)	1.56
EB	5/93573	1.01 (0.46, 2.22)	0.28	19/3608	-5.81(-9.63, -1.99)	14.19	15/3524	-2.30(-3.61, -0.98)	3.78
REML	5/93573	0.99 (0.41, 2.37)	0.35	19/3608	-5.78(-9.59, -1.97)	52.37	15/3524	-2.27(-3.57, -0.96)	3.41

Note: CI, confidence interval; CV, cardiovascular; DL, DerSimonian-Laird; EB, empirical Bayes; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; No., number of trials; NOS, Newcastle-Ottawa Scale; REML, restricted maximum likelihood; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

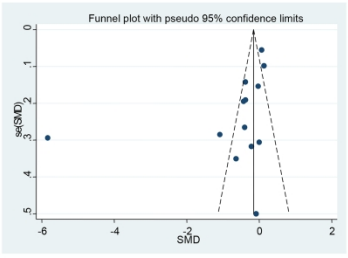
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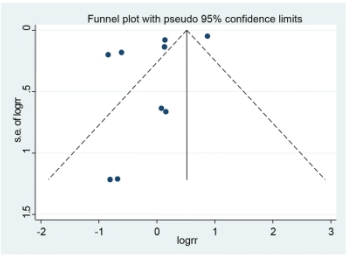
Funnel plot for publication bias on renal composite outcomes events.



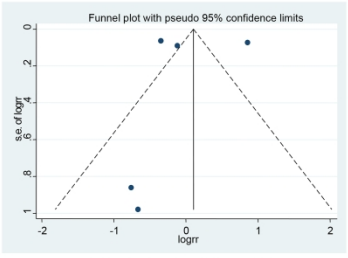
Funnel plot for publication bias on rate of change eGFR.



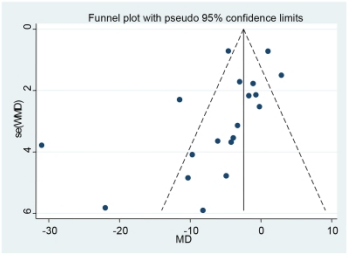
Funnel plot for publication bias on change in proteinuria or albuminuria.



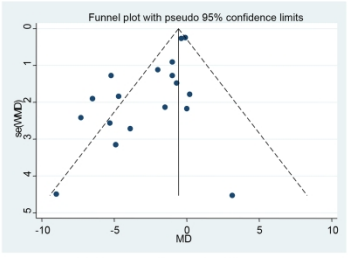
Funnel plot for publication bias on **all-cause mortality events**.



Funnel plot for publication bias on **CV events.**



Funnel plot for publication bias SBP.



Funnel plot for publication bias on **DBP.**

Supplementary file 13 Funnel plot for publication bias on outcomes. CV, cardiovascular; DBP, diastolic blood pressure; MD, mean difference; RR, relative risk; SMD, standard mean difference; SBP, systolic blood pressure.

Supplementary file 14 Table S7 The risk of publication bias of outcomes.

	N	Egger's test(*P*-value)	Bgge's test(*P*-value)
Renal composite outcomes events	10	0.47	0.28
The rate of change in eGFR	12	0.01	0.95
Proteinuria or albuminuria	13	0.10	0.30
All-cause mortality	9	0.09	1.00
CV events	5	0.91	0.46
BP	19	＜0.001	0.005

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5-6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-7

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8-12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15-16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Effect of a Low-Salt Diet on Chronic Kidney Disease Outcomes: A Systematic Review and Meta-Analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050843.R4
Article Type:	Original research
Date Submitted by the Author:	24-Nov-2021
Complete List of Authors:	Shi, Honghong; Shanxi Medical University Second Affiliated Hospital, Renal Division Su, Xiaole; Shanxi Medical University Second Affiliated Hospital Li, Chunfang; Shanxi Cardiovascular Hospital Guo, Wenjuan; Shanxi Medical University Second Affiliated Hospital Wang, Lihua; Shanxi Medical University Second Affiliated Hospital
Primary Subject Heading:	Renal medicine
Secondary Subject Heading:	Health services research
Keywords:	Nephrology < INTERNAL MEDICINE, Chronic renal failure < NEPHROLOGY, HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Effect of a Low-Salt Diet on Chronic Kidney Disease Outcomes: A Systematic Review and Meta-Analysis

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ABSTRACT

Objective The benefits of a low-salt diet for CKD patients are controversial. We conducted a systematic review and meta-analysis of the effect of a low-salt diet on major clinical outcomes.

Design Systematic review and meta-analysis.

Data sources MEDLINE by Ovid, EMBASE, and the Cochrane Library databases.

Eligibility criteria for selecting studies We included RCTs and cohort studies that assessed the effect of a low-salt diet on the renal composite outcomes (more than 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD), rate of eGFR decline, change in proteinuria, all-cause mortality events, cardiovascular (CV) events, and changes in SBP and DBP.

Data extraction and synthesis Two independent researchers extracted data and evaluated their quality. RRs with 95% CIs were used for dichotomous data. Differences in means (MDs) or standardised mean differences (SMDs) with 95% CIs were used to pool continuous data. We used the Cochrane Collaboration risk-of-bias tool to evaluate the quality of RCTs, and NOS to evaluate the quality of cohort studies.

Results We found 9948 potential research records. After removing duplicates, we reviewed the titles and abstracts, and screened the full text of 230 publications. Thirty-three studies with 101,077 participants were included. A low-salt diet produced a 28% reduction in renal

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composite outcome events (RR: 0.72; 95%CI: 0.58-0.89). No significant effects were found in terms of changes in proteinuria (SMD: -0.71; 95% CI: -1.66 to 0.24), rate of eGFR (decline MD: 1.16; 95% CI: -2.02 to 4.33), risk of all-cause mortality (RR: 0.92; 95% CI: 0.58-1.46), and CV events (RR: 1.01; 95% CI: 0.46-2.22).

Conclusion A low-salt diet seems to reduce the risk for renal composite outcome events in CKD patients. However, no compelling evidence indicated that such a diet would reduce the eGFR decline rate, proteinuria, incidence of all-cause mortality, and CV events. Further, more definitive studies are needed.

PROSPERO registration number: CRD42017072395.

Strengths and limitations of this study

►This is the first meta-analysis of the effect of a low-salt diet on hard clinical outcomes, such as renal composite outcomes, all-cause mortality, and CV events, in CKD patients.

►We included cohort studies and RCTs without language restriction, and screened reference lists from included articles, maximising the number of studies included in our review.

►The existence of significant heterogeneity may restrict the explanation and clinical application of the results.

►Confounding factors in the observational studies cannot be ignored, which might affect the results of the meta-analysis.

INTRODUCTION

Chronic kidney disease (CKD) has become an important global public health burden. Dietary interventions aimed at delaying the CKD progression and reducing complications play a crucial role in its management. Dietary sodium restriction can augment the antiproteinuric effects of angiotensin-converting enzyme inhibitors/angiotensin receptor blockers

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(ACEIs/ARBs) [1] and decrease blood pressure (BP) [2]. In the 2020 KDIGO guidelines for glomerulonephritis, reducing dietary sodium intake to < 2 g/day is a primary tenet for controlling BP and oedema and improving urinary protein excretion independently of medications. However, it is unclear whether these effects translate into a significant total risk decrease in CKD patients, including kidney failure events, cardiovascular (CV) events, and all-cause mortality. In the Chronic Renal Insufficiency Cohort (CRIC) Study, which recruited patients with an estimated glomerular filtration rate (eGFR)20 – 70 mL/min/1.73 m^2 depending on age, in contrast to the lowest quartile of urinary sodium excretion (< 116.8 mmol/24 h), high urinary sodium excretion (> 194.6 mmol/24 h) was related to a 54% risk increase in end-stage renal disease (ESRD) or 50% decrease in eGFR and with a 43% risk increase in all-cause mortality [3]. However, other cohort studies have suggested that sodium intake does not affect the progression of CKD to ESRD or eGFR decline [4,5]. Furthermore, in one study, restricting salt intake to < 6 g/day was related to an increased risk of all-cause mortality and CV mortality in haemodialysis patients, resulting in an L-shaped association curve [6]. In the NHANES II study, which recruited 7154 community elderly in the United States, restriction of sodium intake to < 2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality compared with individuals whose sodium intake > 2.3 g daily more than 13.7 years of follow-up period [7]. Evidence from randomised controlled trials (RCTs) of the effect of a low-salt diet in CKD patients on kidney failure events, CV events, and all-cause mortality is controversial. It is hard to make clinical strategy based on results from these studies, considering their small sizes, highly selected patients, short follow-up period, and short of assessment of hard clinical outcomes. Thus, the effect of dietary salt reduction on clinical endpoints in CKD patients remains unclear. We performed this systematic review of relevant

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clinical studies and evaluated the effect of a low-salt diet on renal outcomes, all-cause mortality, and CV events in CKD patients.

MATERIALS AND METHODS

Data sources and retrieval strategy

We performed this systematic review following a pre-specified protocol registered at the International Prospective Register of Systematic Reviews (CRD42017072395) (Supplementary file 1) [8], and reporting was performed in accordance with Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines [9]. A comprehensive search was conducted using the following database: MEDLINE by Ovid (1946 to July 2021), EMBASE (1966 to July 2021), and Cochrane Central Register of Controlled Trials (no date limitation), with related keywords and medical subject headings that included all spellings of "CKD", "RCT", "Cohort Studies", "Sodium Chloride", "Sodium", and "Sodium-Restricted" (see Supplementary file 2 for full search terms). Studies were considered without any language restriction. To ensure a overall literature search, we also screened reference lists from included studies. The ClinicalTrials.gov website was also sarched for ongoing, but unpublished studies in this field.

Study screening and outcome evaluation

We included data from RCTs and cohort studies in which a low-salt diet was given to adults with CKD (as defined by the KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) compared with usual therapy or different salt intake. Salt reduction was defined as that recommended by guidelines: sodium < 2.3 g/day (< 100 mmol sodium, < 6 g salt), or as defined by the authors of the study.

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Pre-defined outcomes that contained analysable data were as follows: renal composite outcome events, defined as a more than50% decline in eGFR from baseline during follow-up [10], doubling of serum creatinine, or ESRD; the rate of change in eGFR per year; changes in urinary protein or urinary albumin during follow-up, including urinary protein excretion, urinary albumin excretion, and urinary albumin/creatinine ratio; all-cause mortality events; CV events, defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularisation, CV disease, and CV death; and changes in systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Data extraction and quality estimation

Two independent reviewers (H.S. and X.S.) extracted data and assessed their quality according to the pre-specified protocol. Disagreements were resolved by a third reviewer (L.W.). Data from all eligible studies were extracted into a spreadsheet. The data sought included characteristics of studies (study type, randomisation method, follow-up time, withdrawals/dropouts), baseline patient traits (age, sex, baseline eGFR, and baseline BP), intake of salt intake, and outcome events.

We used Cochrane Collaboration risk-of-bias tool [11] to assess all potential sources of bias for the included RCTs. Trials were assessed as being at low, some concerns, or high risk, and the overall risk of bias generally corresponded to the worst risk of bias in any of the domains. However, if a study was judged to have some concerns about risk of bias for multiple domains, it might be judged as being at high risk for bias overall. We used the Newcastle-Ottawa Scale (NOS) checklist to assess the quality of cohort studies. The NOS contains eight items, categorised into three dimensions: selection of cohorts, comparability of cohorts, and assessment of outcome [12]. A star system is used for semi-quantitative assessment of study

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quality, such that the highest-quality studies are awarded a maximum of one star for each item, except the item related to comparability, which can be assigned two stars. The NOS ranges from zero to nine stars.

Data synthesis and analyses

When dichotomous outcome data from individual trials were analysed, relative risks (RRs) and 95% confidence intervals (CIs) were used. If the RR and 95% CI for an individual study was unavailable in the original article, we calculated them using event numbers extracted from each study before data pooling. In calculating the RR values, we used the total number of patients randomised in each group as the corresponding denominator. Continuous outcome data from individual trials were analysed using mean differences (MDs) with 95% CIs to pool eGFR, and standardised mean differences (SMDs) with 95% CIs were used to pool the proteinuria or albuminuria data. When continuous outcome data were analysed, the change in MD between baseline and end-of-treatment was used. If these data were not available in the studies, we calculated using correlations estimated from other included studies that had a similar follow-up and reported their results in considerable details according to the imputed formulation and its related interpretations in the Cochrane Handbook [13].

Given the poor stability of the Der Simonian-Laird procedure for a small number of studies [14], we used the empirical Bayes procedure [15] to estimate all outcomes. We also used Der Simonian-Laird random effects model [16] and restricted maximum likelihood approach[17] to assess the summary effects as part of the sensitivity analyses. Considering the inevitable heterogeneity among studies, subgroup, meta-regression, and sensitivity analyses were performed. Subgroup analyses were performed according to a pre-specified protocol, including study type, baseline eGFR, baseline BP, and comparator of control group. In addition,

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we performed sensitivity analyses excluding studies with a sample size < 50, studies with a follow-up < 12 months, studies of low quality (high risk for overall bias in RCTs, NOS < 5 stars for cohort studies), and studies with extreme outliers. Heterogeneity among studies were evaluated using the I^2 or τ^2 statistic. I^2 values of 25%, 50%, and 75%, respectively, represent low, moderate, and high heterogeneity. Publication bias was assessed using a Funnel plot, Egger's test, or Begg's test, and a P-value < 0.05 indicated obvious publication bias. Stata version 15.0 (StataCorp LP, College Station, TX) was used for statistical analyses, and a two-sided P-value < 0.05 was considered indicative of significance.

Patient and public involvement statement

No patient or member of the public was involved in the development of the research question, selection of the outcome measures, design and implementation of the study, or interpretation of the results.

RESULTS

Overview of included trials

We identified 9948 potential relevant records. After removing duplicates, we screened the titles and abstracts, and the full text of 230 publications were reviewed. As shown in Figure 1, Supplementary file 3, and Supplementary file 4, a total of 33 eligible studies [3-6, 18-48] including 13 RCTs [18,20,25-27,31,32,35-40,45] and 19 cohort studies [3-6,19,21-24,28,30,33,34,41-44,46-48] reported in 35 publications with 101077 participants, were included in our review. The median follow-up was 6 months (interquartile range 3 to 21 months). Participants were enrolled at an average age of 56 years, and male participants accounted for 60% of the total. The average eGFR of the participants was 47.45 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and 21 compared different levels

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of sodium intake.

Figure 1. PRISMA flow chart of the included studies.

The risk of bias varied substantially across the RCTs. The results from the Cochrane Collaboration risk-of-bias tool are shown in Supplementary file 5. Bias arising from the randomisation process was considered to be low in 61.54% of the RCTs, and all studies had a low risk of bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported results. In terms of overall bias, 61.54% of the trials were considered to have low risk, and 38.46% had some concern of bias.

According to the NOS checklist, all included cohort studies were generally high quality and awarded 6–8 stars (Supplementary file 6).

Effect of a low-salt diet on renal outcomes

One RCT and nine observational studies including 9578 patients with 2329 events compared a low-salt diet with controls in terms of preventing renal composite outcome events. As shown in Figure 2, compared to controls, a low-salt diet produced a 34% reduction in the risk for renal composite outcome events (RR: 0.72; 95% CI: 0.58, 0.89), with substantial heterogeneity ($I^2 = 87.2\%$, $P < 0.001$). Subgroup analyses and meta-regression were conducted to assess the source of heterogeneity. No significant difference between RCTs and observational studies was found, whereas the effect sizes were greater in studies that enrolled patients aged < 58 years ($P = 0.03$) (Table 1). In univariate meta-regression, no clear relationship between 24 h sodium in urine ($P = 0.80$) or baseline eGFR ($P = 0.51$) and the reduction in renal composite outcome events was observed (Supplementary file 7, Supplementary file 8).

Figure 2. Forest plot for renal composite outcome events. Renal composite outcome events

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were defined as a more than 50% decline in eGFR from baseline during follow-up or a doubling of serum levels of creatinine or ESRD. CI, confidence interval; Cr, creatinine; Na, sodium; RR, relative risk

Table 1. Results of subgroup analysis of renal composite outcome events

Subgroup	No. of Trials	n	RR (95% CI)	*P* for RR	I^2	P for Heterogeneity Test[a]
Study type						
RCT	1	418	0.64 (0.43, 0.98)	0.04	-	0.76
Cohort study	9	9160	0.72 (0.57, 0.90)	0.005	88.6%	
Baseline eGFR (mL/min/1.73m²)						
60–89	1	171	0.14 (0.02, 1.14)	0.07	-	0.48
30–59	7	8869	0.71 (0.55, 0.92)	0.009	90.8%	
15–29	2	538	0.77 (0.63, 0.94)	0.01	0.0%	
Mean age (years)						
<58	5	5691	0.58 (0.49, 0.70)	<0.001	56.4%	**0.03**
≥58	5	3887	0.85 (0.69, 1.05)	0.14	70.1%	
Baseline BP (mmHg)						
<140/90	6	5806	0.62 (0.52, 0.73)	<0.001	58.5%	0.32
SBP≥140 or DBP≥90	2	1646	0.70 (0.44, 1.11)	0.13	74.6%	
Comparator of control group						
Usual care or usual diet	2	589	0.43 (0.11, 1.64)	0.21	50.2%	0.12
Different levels of sodium intake	8	8989	0.73 (0.58, 0.92)	0.008	89.8%	

[a]*P*-value calculated by meta regression. BP, blood pressure; CI, confidence interval; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk

Five RCTs and seven observational studies with a total of 3287 individuals reported data regarding the effect of a low-salt diet on the rate of change in eGFR. The diet did not have a beneficial effect on the rate of change in eGFR (MD: 1.16; 95% CI: -2.02 to 4.33). There was evidence of significant heterogeneity across the included studies (I^2 = 98.0%, P < 0.001) (Figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to pre-specified characteristics (Supplementary file 9).

Figure 3. Forest plot of the rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SD, standard deviation

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Data on the effect of a low-salt diet on proteinuria or albuminuria were available in 13 studies, with 2922 participants. The SMD for change in proteinuria or albuminuria was not statistically significant at -0.71 (95% CI: -1.66 to 0.24) compared to controls, with significant heterogeneity (I^2 = 97.3%, P=0.009) (Figure 4). No statistical heterogeneity was found in subgroup analyses of the effect of a low-salt diet on proteinuria or albuminuria (Supplementary file 9).

Figure 4. Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; Cr, creatinine; Na, sodium; SD, standard deviation; SMD, standard mean difference

Effect of a low-salt diet on all-cause mortality and CV events

Data on the effect of low-salt diet on all-cause mortality and CV events were available for nine studies (two RCTs and seven observational studies) and five studies (one RCT and four observational studies), respectively. There was no effect of a low-salt diet on the risk of all-cause mortality (RR: 0.92; 95% CI: 0.53-1.46; 95,997 participants and 4455 events) or CV events (RR: 1.01; 95% CI: 0.46-2.22; 93,573 participants and 1989 events) compared to controls, with significant heterogeneity across the included studies (I^2 = 98.0%, 97.6% for all-cause mortality and CV events, respectively, both P-values for heterogeneity < 0.001) (Figure 5). A greater benefit was found in studies that included patients with an eGFR of 30–59 mL/min/1.73 m^2 (P for subgroup heterogeneity = 0.02) in the subgroup analyses of CV events (Supplementary file 9).

Figure 5. Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite of fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularisation, cardiovascular disease, and cardiovascular death. CI, confidence interval; Cr, creatinine; CV, cardiovascular; Na, sodium; N, number of trials; RR, relative risk

Effect of a low-salt diet on BP

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Twenty studies reported the effect of a low-salt diet on BP, including 19 studies (3608 participants) for SBP and 18 studies (3608 participants) for DBP. The MD changes in SBP and DBP were statistically significant at -5.81 mmHg (95% CI: -9.63 to -1.99) and -2.3 mmHg (95% CI, -3.61 to -0.98), respectively (Supplementary files 10 and 11). Significant heterogeneity across the included studies was found (I^2 = 87.1% and 68.2% for SBP and DBP, respectively; both *P*-values for heterogeneity < 0.05). Subgroup analyses showed that the association between the change in SBP or DBP and a low-salt diet was modified by the baseline BP. Patients with a baseline SBP > 140 mmHg or DBP > 90 mmHg had an average reduction in SBP and DBP of 10 and 5 mmHg, respectively (Supplementary file 9).

Sensitivity analyses and publication bias

The results of all sensitivity analyses were robust (Supplementary file 12). The most notable exception was that the effect of a low-salt diet on the rate of change in eGFR became significant when studies with extreme outliers were omitted, and proteinuria or albuminuria became significant when the DL method was used to pool data. The risk of publication bias was significant for the rate of change in eGFR, proteinuria or albuminuria, all-cause mortality, CV events, and BP (Supplementary files 13 and 14).

DISCUSSION

Behavioural modifications can delay CKD progression to ESRD, and dietary salt restriction is one of the main concerns. Our meta-analysis showed that compared with the control groups, a low-salt diet reduced renal composite outcome events and lowered BP (SBP and DBP) in CKD patients. No significant effects were observed on the eGFR decline rate, change in proteinuria, risk of all-cause mortality, and CV events. The results were concordant across major subgroups and sensitivity analyses. However, the significant heterogeneity among the

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studies may restrict their explanation and clinical practice.

A prior review of 16 studies on salt consumption and kidney disease investigated whether variation in dietary sodium intake affect kidney outcomes in CKD patients [49]. Increased salt consumption was related to increased albuminuria and the possibility of a decrease in GFR. In 2015 [50] and 2018 [51], two meta-analyses of RCTs suggested that moderate dietary salt limitation obviously reduced BP and proteinuria. However, it is unclear whether these benefits convert to clinically obvious reductions in renal composite outcome events, all-cause mortality events, and CV events. Our meta-analysis of 33 studies (13 RCTs and 20 observational studies), more than 90,000 patients, and 2329 renal composite outcome events showed that compared with the control groups, a low-salt diet produced a significant 28% reduction in renal composite outcome events despite marked heterogeneity. Furthermore, the significant effect of a low-salt diet on BP (decreases in SBP and DBP of 5.8 and 2.3 mmHg, respectively) was confirmed and is an important target for slowing the progression of CKD to ESRD. The benefit of proteinuria from low-salt diet became nonsignificant when the restricted maximum likelihood or empirical Bayes method was used to pool data. The standard DerSimonian-Laird procedure, which was applied in previous meta-analyses, can be unstable with small numbers of studies [52,53].

Statistical heterogeneity was found in all outcomes, and clinical heterogeneity was an inevitable issue. Low-salt diets are not palatable for some patients [54]. Therefore, it is hard to confirm persistent compliance with a salt restrictions during long-term follow-up time. Sodium intake was quantified using different methods (such as, 24 h urine collection, spot urine collection, or dietary questionnaire). Because diuretics influence urine sodium excretion and eGFR, and it would influence the accuracy of quantified salt intake. Ten studies had data on diuretic use, but included only the percentages of patients using diuretics in the intervention

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and control groups. The percentage of patients using diuretics was similar in most of the studies. There were no data on the type and dose of diuretics, or their associations with the eGFR. These inherent problems were reflected in paradoxically opposite findings, especially of observational studies with several confounding factors. Only three studies (one RCT and two observational studies) simultaneously reported the outcomes of eGFR and renal composite outcome events. Furthermore, the different follow-up periods (52 months of renal composite outcome events and 12 months of eGFR) should be noted. The inconsistent results of eGFR and renal composite outcome events may be explained in part by the heterogeneity, which likely limited the interpretation and generalisability of our results.

In CKD patients, BP is typically sodium sensitive, and RAAS blockers are considered the first-line therapy for hypertension and proteinuria. The function of RAAS blockers are weakened by high salt intake [55–57]. Therefore, the significant effect of dietary sodium restriction on lowering BP and supporting the function of RAAS blockers was speculated to play a very important part in reducing the risk of renal failure. Interestingly, a BP-independent effect of dietary sodium on the kidney is substantial by data in healthy volunteers, in which dietary sodium restriction reduces albuminuria to within the normal range, without a detectable effect on BP [58]. Similarly, proteinuria reduction by sodium restriction remained significant after adjustment for the decrease in BP [57]. These results suggest an independent renoprotective function of a low-salt diet.

There was no convincing evidence that a low-salt diet was related to a lower incidence of all-cause mortality and CV events. Limited data on all-cause mortality (nine studies with 4455 events) and CV events (five studies with 1989 events) in the meta-analysis might introduce a risk of false-negative results because of low statistical power. A greater benefit for CV events

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was found in studies that included patients with an eGFR 30–59 mL/min/1.73 m^2 in subgroup analyses, whereas the effects were not significant in patients with an eGFR < 15 mL/min/1.73 m^2. The substantial subgroup heterogeneity (P = 0.02) could be attributable to more CV risk factors in patients with ESRD than in those with an eGFR 30–59 mL/min/1.73 m^2. The benefit of a low-salt diet for CV events did not outweigh the increase in CV events caused by more risk factors (such as hyperuricemia, hyperphosphatemia, and vascular calcification). Therefore, education on the importance of a low-salt diet should be emphasised in early-stage CKD patients.

Observational data showing a J-curve between sodium intake and renal and CV outcomes have raised concern regarding the safety of rigorous sodium restriction [59–61]. It is uncertain whether the presence of CKD modifies this association. In our meta-regression analysis, no clear relationship between 24 h sodium in urine (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcome events was observed. A combination of ACEIs and strict sodium control decreased blood pressure, proteinuria, and glomerular lesion but exacerbated tubule-interstitial injury in experimental renal disease [62]. This could explain the reason of worse renal outcomes by very low salt intake. High or extremely low dietary salt intake might be inappropriate for CKD patients. KDIGO recommends a daily sodium intake of < 2.0 g/day in patients with CKD without a clear lower limit. Therefore, an appropriate dietary salt intake should be discussed with CKD patients. Further studies are necessary to identify the optimal dietary salt intake for renoprotection in CKD patients .

This study had several potential limitations. First, the existence of statistical and clinical heterogeneity may raise concerns regarding validity, although we attempted to resolve these concerns by performing subgroup and sensitivity analyses. Heterogeneity in study populations

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hampers meaningful interpretation of the results. Because the CKD stages varied markedly, it is difficult to stratify the outcomes analysis by CKD stage. Instead, we performed subgroup analyses according to baseline eGFR. Second, unlike RCTs, cohort studies are prone to selection bias. Therefore, pooling the results from RCTs and cohort studies will bias the results of the meta-analysis. We performed a subgroup analysis according to study design. There was no statistical heterogeneity in the outcomes according to study design. However, a low-salt diet did not exert a beneficial effect on the rate of change in eGFR in RCTs (MD: -0.75; 95% CI: -7.58 to 6.08) but did in eGFR in cohort studies (MD: 2.48; 95% CI: 0.32 to 4.64). Third, many of the included low-quality studies had small sample sizes and low incidences of events. Consequently, the uncertainty in the analyses was increased, resulting in wide CIs of effect measures. Fourth, confounding factors in the observational studies cannot be neglected, because they might affect the results. Therefore, the findings of this study should be viewed as hypothesis-generating and need to be confirmed by further studies.

CONCLUSION

This meta-analysis suggests that a low-salt diet may reduce the risk of renal composite outcome events in patients with CKD. However, there was no compelling evidence that a low-salt diet reduces the rate of eGFR decline, proteinuria, incidence of all-cause mortality, and CV events. The optimal dietary salt intake for patients with different CKD stages is unclear. Further well-designed RCTs targeting patients with different CKDs are required.

Acknowledgments We thank Textcheck for proofreading and editing the English of the manuscript.

Contributors All authors take responsibility for the integrity of the data and the accuracy of

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the data analysis. Study concept and design: L.W., X.S., H.S., and C.L. Extraction, analysis, and interpretation of data: H.S. and X.S. Drafting of the manuscript: H.S. and L.W. Critical revision of the manuscript: L.W. and X.S. Statistical analysis: H.S., C.L., and W.G. Technical support: L.W., X.S., and W.G. All authors have read and agreed to submission of the manuscript to this journal.

Funding Support was provided by the National Natural Science Foundation of China (no. 82000655) and the Science and Technology Project of Shanxi Province (no. 201801D121218). The funders had no role in the decision to publish or in preparation of the manuscript.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data used to support the findings of this study are included in the article and the Supplementary information.

Ethics statement This meta-analysis evaluated published studies, and so does not require approval by an ethics committee

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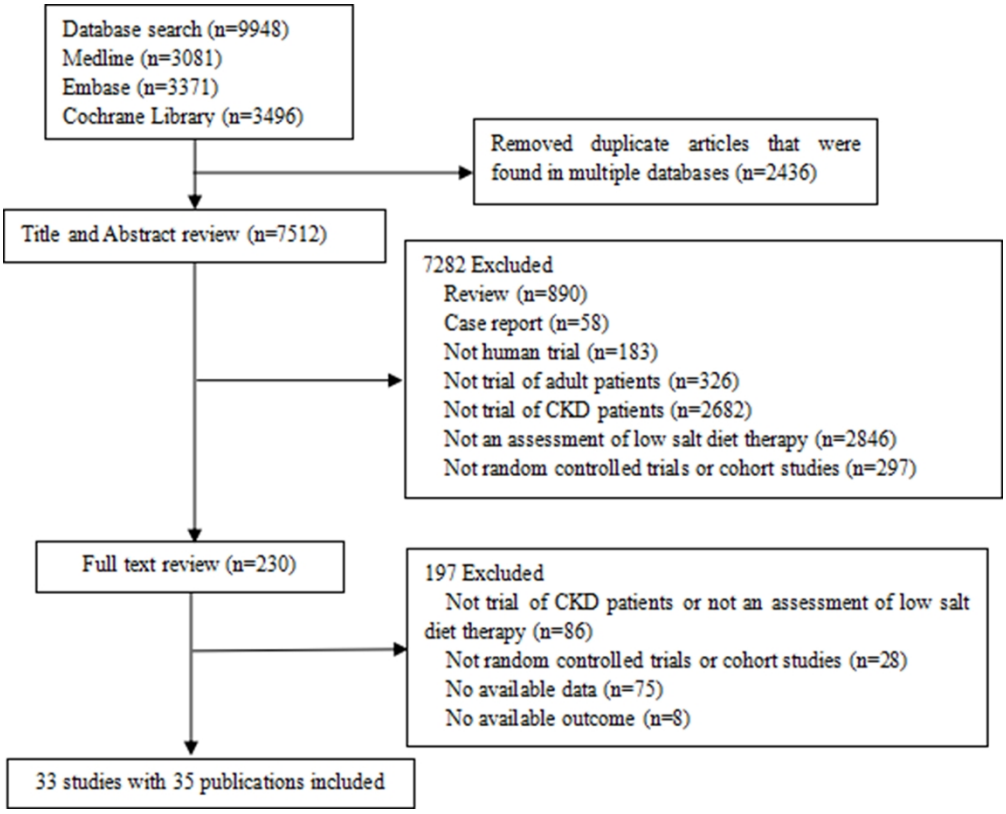
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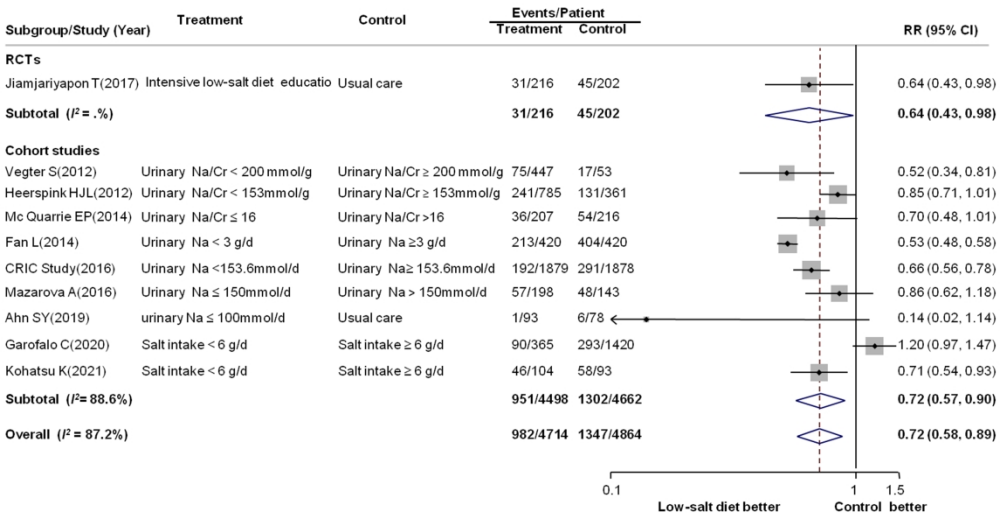
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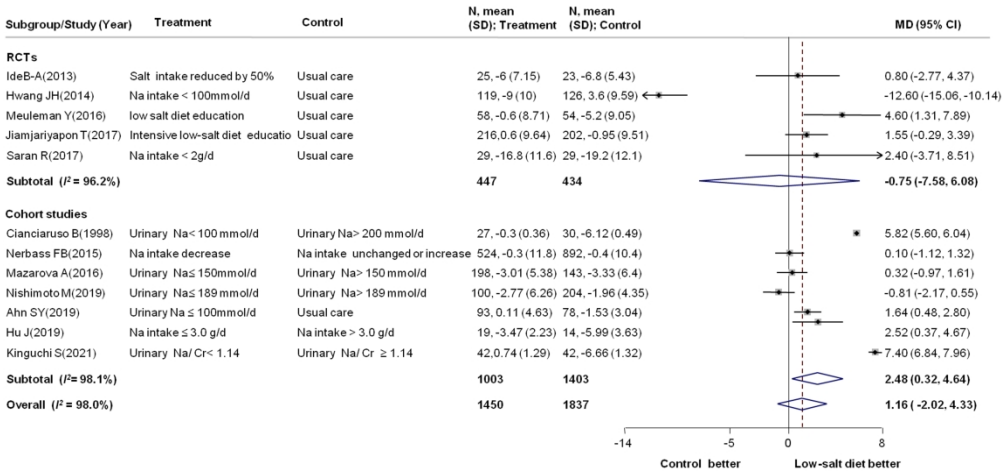
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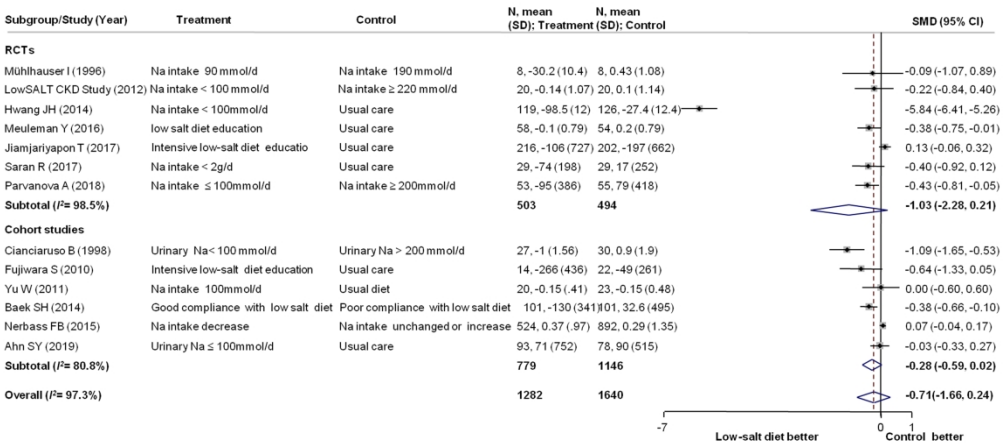
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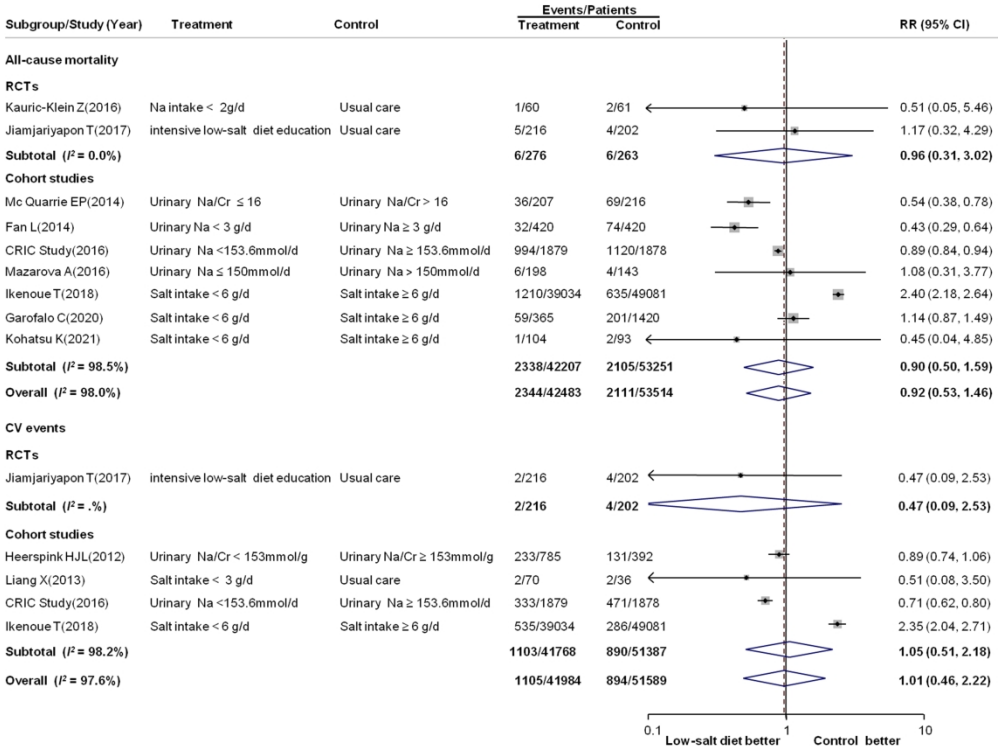
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Database search (n=9948)
Medline (n=3081)
Embase (n=3371)
Cochrane Library (n=3496)
Removed duplicate articles that were found in multiple databases (n=2436)
Title and Abstract review (n=7512)
7282 Excluded
Review (n=890)
Case report (n=58)
Not human trial (n=183)
Not trial of adult patients (n=326)
Not trial of CKD patients (n=2682)
Not an assessment of low salt diet therapy (n=2846)
Not random controlled trials or cohort studies (n=297)
Full text review (n=230)
197 Excluded
Not trial of CKD patients or not an assessment of low salt diet therapy (n=86)
Not random controlled trials or cohort studies (n=28)
No available data (n=75)
No available outcome (n=8)
33 studies with 35 publications included


Subgroup/Study (Year)
Treatment
Control
Events/Patient
Treatment
Control
RR (95% CI)
RCTs
Jiamjariyapon T(2017) Intensive low-salt diet educatio Usual care 31/216 45/202 0.64 (0.43, 0.98)
Subtotal (I² = .%) 31/216 45/202 0.64 (0.43, 0.98)
Cohort studies
Vegter S(2012) Urinary Na/Cr < 200 mmol/g Urinary Na/Cr ≥ 200 mmol/g 75/447 17/53 0.52 (0.34, 0.81)
Heerspink HJL(2012) Urinary Na/Cr < 153mmol/g Urinary Na/Cr ≥ 153mmol/g 241/785 131/361 0.85 (0.71, 1.01)
Mc Quarrie EP(2014) Urinary Na/Cr ≤ 16 Urinary Na/Cr >16 36/207 54/216 0.70 (0.48, 1.01)
Fan L(2014) Urinary Na < 3 g/d Urinary Na ≥3 g/d 213/420 404/420 0.53 (0.48, 0.58)
CRIC Study(2016) Urinary Na <153.6mmol/d Urinary Na≥ 153.6mmol/d 192/1879 291/1878 0.66 (0.56, 0.78)
Mazarova A(2016) Urinary Na ≤ 150mmol/d Urinary Na > 150mmol/d 57/198 48/143 0.86 (0.62, 1.18)
Ahn SY(2019) urinary Na ≤ 100mmol/d Usual care 1/93 6/78 0.14 (0.02, 1.14)
Garofalo C(2020) Salt intake < 6 g/d Salt intake ≥ 6 g/d 90/365 293/1420 1.20 (0.97, 1.47)
Kohatsu K(2021) Salt intake < 6 g/d Salt intake ≥ 6 g/d 46/104 58/93 0.71 (0.54, 0.93)
Subtotal (I²= 88.6%) 951/4498 1302/4662 0.72 (0.57, 0.90)
Overall (I² = 87.2%) 982/4714 1347/4864 0.72 (0.58, 0.89)
0.1
1
1.5
Low-salt diet better
Control better


Subgroup/Study (Year)
Treatment
Control
N, mean (SD); Treatment
N, mean (SD); Control
MD (95% CI)
RCTs
IdeB-A(2013) Salt intake reduced by 50% Usual care 25, -6 (7.15) 23, -6.8 (5.43) 0.80 (-2.77, 4.37)
Hwang JH(2014) Na intake < 100mmol/d Usual care 119, -9 (10) 126, 3.6 (9.59) -12.60 (-15.06, -10.14)
Meuleman Y(2016) low salt diet education Usual care 58, -0.6 (8.71) 54, -5.2 (9.05) 4.60 (1.31, 7.89)
Jiamjariyapon T(2017) Intensive low-salt diet educatio Usual care 216, 0.6 (9.64) 202, -0.95 (9.51) 1.55 (-0.29, 3.39)
Saran R(2017) Na intake < 2g/d Usual care 29, -16.8 (11.6) 29, -19.2 (12.1) 2.40 (-3.71, 8.51)
Subtotal (I² = 96.2%) 447 434 -0.75 (-7.58, 6.08)
Cohort studies
Cianciaruso B(1998) Urinary Na< 100 mmol/d Urinary Na> 200 mmol/d 27, -0.3 (0.36) 30, -6.12 (0.49) 5.82 (5.60, 6.04)
Nerbass FB(2015) Na intake decrease Na intake unchanged or increase 524, -0.3 (11.8) 892, -0.4 (10.4) 0.10 (-1.12, 1.32)
Mazarova A(2016) Urinary Na≤ 150mmol/d Urinary Na> 150 mmol/d 198, -3.01 (5.38) 143, -3.33 (6.4) 0.32 (-0.97, 1.61)
Nishimoto M(2019) Urinary Na≤ 189 mmol/d Urinary Na> 189 mmol/d 100, -2.77 (6.26) 204, -1.96 (4.35) -0.81 (-2.17, 0.55)
Ahn SY(2019) Urinary Na ≤ 100mmol/d Usual care 93, 0.11 (4.63) 78, -1.53 (3.04) 1.64 (0.48, 2.80)
Hu J(2019) Na intake ≤ 3.0 g/d Na intake > 3.0 g/d 19, -3.47 (2.23) 14, -5.99 (3.63) 2.52 (0.37, 4.67)
Kinguchi S(2021) Urinary Na/ Cr< 1.14 Urinary Na/ Cr ≥ 1.14 42, 0.74 (1.29) 42, -6.66 (1.32) 7.40 (6.84, 7.96)
Subtotal (I²= 98.1%) 1003 1403 2.48 (0.32, 4.64)
Overall (I² = 98.0%) 1450 1837 1.16 (-2.02, 4.33)
-14
-5
0
8
Control better
Low-salt diet better


Subgroup/Study (Year)
Treatment
Control
N. mean (SD); Treatment
N. mean (SD); Control
SMD (95% CI)
RCTs
Mühlhauser I (1996) Na intake 90 mmol/d Na intake 190 mmol/d 8, -30.2 (10.4) 8, 0.43 (1.08) -0.09 (-1.07, 0.89)
LowSALT CKD Study (2012) Na intake < 100 mmol/d Na intake ≥ 220 mmol/d 20, -0.14 (1.07) 20, 0.1 (1.14) -0.22 (-0.84, 0.40)
Hwang JH (2014) Na intake < 100mmol/d Usual care 119, -98.5 (12) 126, -27.4 (12.4) -5.84 (-6.41, -5.26)
Meuleman Y (2016) low salt diet education Usual care 58, -0.1 (0.79) 54, 0.2 (0.79) -0.38 (-0.75, -0.01)
Jiamjariyapon T (2017) Intensive low-salt diet educatio Usual care 216, -106 (727) 202, -197 (662) 0.13 (-0.06, 0.32)
Saran R (2017) Na intake < 2g/d Usual care 29, -74 (198) 29, 17 (252) -0.40 (-0.92, 0.12)
Parvanova A (2018) Na intake ≤ 100mmol/d Na intake ≥ 200mmol/d 53, -95 (386) 55, 79 (418) -0.43 (-0.81, -0.05)
Subtotal (I^2= 98.5%) 503 494 -1.03 (-2.28, 0.21)
Cohort studies
Cianciaruso B (1998) Urinary Na< 100 mmol/d Urinary Na > 200 mmol/d 27, -1 (1.56) 30, 0.9 (1.9) -1.09 (-1.65, -0.53)
Fujiwara S (2010) Intensive low-salt diet education Usual care 14, -266 (436) 22, -49 (261) -0.64 (-1.33, 0.05)
Yu W (2011) Na intake 100mmol/d Usual diet 20, -0.15 (.41) 23, -0.15 (0.48) 0.00 (-0.60, 0.60)
Baek SH (2014) Good compliance with low salt diet Poor compliance with low salt diet 101, -130 (341) 101, 32.6 (495) -0.38 (-0.66, -0.10)
Nerbass FB (2015) Na intake decrease Na intake unchanged or increase 524, 0.37 (.97) 892, 0.29 (1.35) 0.07 (-0.04, 0.17)
Ahn SY (2019) Urinary Na ≤ 100mmol/d Usual care 93, 71 (752) 78, 90 (515) -0.03 (-0.33, 0.27)
Subtotal (I^2= 80.8%) 779 1146 -0.28 (-0.59, 0.02)
Overall (I^2= 97.3%) 1282 1640 -0.71(-1.66, 0.24)
-7
0
1
Low-salt diet better
Control better


Subgroup/Study (Year) Treatment Control Events/Patients Treatment Control RR (95% CI)
All-cause mortality
RCTs
Kauric-Klein Z(2016) Na intake < 2g/d Usual care 1/60 2/61 0.51 (0.05, 5.46)
Jiamjariyapon T(2017) intensive low-salt diet education Usual care 5/216 4/202 1.17 (0.32, 4.29)
Subtotal (I² = 0.0%) 6/276 6/263 0.96 (0.31, 3.02)
Cohort studies
Mc Quarrie EP(2014) Urinary Na/Cr ≤ 16 Urinary Na/Cr > 16 36/207 69/216 0.54 (0.38, 0.78)
Fan L(2014) Urinary Na < 3 g/d Urinary Na ≥ 3 g/d 32/420 74/420 0.43 (0.29, 0.64)
CRIC Study(2016) Urinary Na <153.6mmol/d Urinary Na ≥ 153.6mmol/d 994/1879 1120/1878 0.89 (0.84, 0.94)
Mazarova A(2016) Urinary Na ≤ 150mmol/d Urinary Na > 150mmol/d 6/198 4/143 1.08 (0.31, 3.77)
Ikenoue T(2018) Salt intake < 6 g/d Salt intake ≥ 6 g/d 1210/39034 635/49081 2.40 (2.18, 2.64)
Garofalo C(2020) Salt intake < 6 g/d Salt intake ≥ 6 g/d 59/365 201/1420 1.14 (0.87, 1.49)
Kohatsu K(2021) Salt intake < 6 g/d Salt intake ≥ 6 g/d 1/104 2/93 0.45 (0.04, 4.85)
Subtotal (I² = 98.5%) 2338/42207 2105/53251 0.90 (0.50, 1.59)
Overall (I² = 98.0%) 2344/42483 2111/53514 0.92 (0.53, 1.46)
CV events
RCTs
Jiamjariyapon T(2017) intensive low-salt diet education Usual care 2/216 4/202 0.47 (0.09, 2.53)
Subtotal (I² = .%) 2/216 4/202 0.47 (0.09, 2.53)
Cohort studies
Heerspink HJL(2012) Urinary Na/Cr < 153mmol/g Urinary Na/Cr ≥ 153mmol/g 233/785 131/392 0.89 (0.74, 1.06)
Liang X(2013) Salt intake < 3 g/d Usual care 2/70 2/36 0.51 (0.08, 3.50)
CRIC Study(2016) Urinary Na <153.6mmol/d Urinary Na ≥ 153.6mmol/d 333/1879 471/1878 0.71 (0.62, 0.80)
Ikenoue T(2018) Salt intake < 6 g/d Salt intake ≥ 6 g/d 535/39034 286/49081 2.35 (2.04, 2.71)
Subtotal (I² = 98.2%) 1103/41768 890/51387 1.05 (0.51, 2.18)
Overall (I² = 97.6%) 1105/41984 894/51589 1.01 (0.46, 2.22)
0.1
1
10
Low-salt diet better
Control better

PROSPERO
International prospective register of systematic reviews

NHS National Institute for Health Research

Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang

Citation

Chunfang Li, Lihua Wang, Xiaole Su, Xuejiao Wang. Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease. PROSPERO 2017 CRD42017072395 Available from: http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42017072395

Review question

This meta-analysis aims to evaluate the effects of different levels of dietary sodium intake on renal outcome in patients with chronic kidney disease (CKD).

Searches

All relevant randomized controlled trails (RCTs) and cohort studies will be identified and searched in following electronic databases without language restriction: Ovid MEDLINE SP (from 1946 to July 2017), EMBASE (from 1966 to July 2017), and the Cochrane Central Register of Controlled Trails(no date restriction). A search in www.clinicaltrials.gov will be used to identify the ongoing but unpublished studies about this field. We combined this strategy with a manual search of reference lists from identified trials and review articles.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/72395_STRATEGY_20170618.pdf

Types of study to be included

We will include RCTs and cohort studies that compared different levels of sodium intake for adult patients with CKD as well as studies in broader populations in which outcome data for subgroups with CKD could be disaggregated.

Condition or domain being studied

Chronic kidney disease

Participants/population

Inclusion criteria: Adults (no less than 18 years old) with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) at all disease stages will included without restrictions on gender or race. Data from subgroups of participants with CKD within studies with broader inclusion criteria (e.g. people from the general population, people with diabetes, people with cardiovascular disease) were also included.
Exclusion criteria: Studies which do not include any people with chronic kidney disease.

Intervention(s), exposure(s)

Low-sodium diet

Comparator(s)/control

No intervention or different levels of dietary sodium intake

Context

None

Primary outcome(s)

If the data allow:

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1) Kidney failure events: It will include more than 25% or 50% decrease in estimated glomerular filtration rate (eGFR), doubling of serum creatinine, end stage renal disease (ESRD);
2) Rate of change in the eGFR: The difference from baseline in eGFR divided by the number of years between creatinine measurements (ml/min/1.73m2 per year);
3) Change in proteinuria or albuminuria, including urinary protein excretion (UPE), urinary albumin excretion (UAE), protein to creatinine (PCR) and albumin to creatinine ratio (ACR);
4) Major cardiovascular events: It will include a composite including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death.

Secondary outcome(s)

1) 24-hour urinary sodium excretion;
2) Change in blood pressure;
3) All-cause mortality;
4) Change in markers of fluid overload (brain natriuretic peptide, N-terminal pro-brain natriuretic peptide, weight, bio-impedance analysis);
5) Adverse events: hypotensive episodes, undesirable change in blood lipids (low density lipoprotein, high-density lipoprotein);
6) Hospitalization.

Data extraction (selection and coding)

Data extraction will be carried out independently by two reviewers using standard data extraction forms. Disagreement will be resolved by consensus or by discussion with a third author. When detailed information that was needed for the analysis was not available, the original authors will be contacted to obtain the missing information through e-mail. Where multiple publications of one study exists, data will be extracted and grouped together as a single study with the most complete data.
The extracted data will include: study characteristics (study design, method of randomization, and withdrawals / dropouts); baseline patient characteristics (age, sex, race, country, BMI, sodium intake and its measurement, 24-hour urinary sodium excretion, mean proteinuria or albuminuria, eGFR, serum creatinine concentrations, blood lipids, blood pressure, diabetic status, cardiovascular events, smoking status, glycosylated hemoglobin, hemoglobin, antihypertensive drugs, antidiabetic drugs, different kidney disease diagnosis); change in eGFR; follow-up duration; outcomes: as listed in Primary outcomes and Secondary outcomes. When the required quantitative data is only provided in published figures, we will use g3 data software (http://www.frantz.fi/software/g3data.php) to extract exact numbers from figures.

Risk of bias (quality) assessment

For the RCTs, we will assess risk of bias for sequence generation, allocation concealment, blinding, selective reporting, incomplete outcome data and other sources of bias, and determined overall risk of bias based on predefined rules, utilizing the Cochrane Collaboration risk of bias tool. (reference: Higgins JP, Altman DG, et al. Cochrane Statistical Methods G. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials. BMJ 2011;343:d5928.). Furthermore, the quality of RCT study will be quantified with Jadad scale. (reference: Jadad AR, Moore RA, Carroll D, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Control Clin Trials 1996;17:1– 12.). The quality of cohort study will be independently evaluated by each investigator using Newcastle-Ottawa quality assessment scale. (reference: Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010; 25:603–5.).

Strategy for data synthesis

The results of the data extraction and quality assessment for each study will be presented in structured tables. For dichotomous outcomes, relative risks (RR) and 95% Cis will be calculated from the numbers of events and participants at risk for events. For continuous outcomes, the mean difference (MD) will be used, or the standardized mean difference (SMD) if different scales were used. Data will be pooled using the random-effects model but the fixed-effect model will also be used to ensure robustness of the model chosen and susceptibility to outliers.
Heterogeneity will be analyzed using Cochrane's Q and the I-squared statistic. We will explore potential

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heterogeneity by subgroup analysis and meta-regression.
Sensitivity analyses will be conducted to check the robustness of the effect of sodium-restricted treatment by excluding of trials with small sample size and low Jadad scores.
A two-sided p-value less than 0.05 was considered statistically significant and statistical analyses were performed using RevMan version 5.3 and STATA version 12.0.

Analysis of subgroups or subsets

If the data allow, subgroup analysis will be conducted by study design, follow-up duration, measures of sodium intake, different levels of sodium intake, difference in sodium intake between two groups, and baseline measures, including age, sex, BMI, eGFR, 24-hour urinary sodium excretion,mean serum creatinine, blood pressure, mean proteinuria or albuminuria, stage of CKD, presence of comorbidities (hypertension, diabetes and CVD), different kidney disease diagnosis (e.g. GN or Kidney Transplant Recipients) and whether RAAS blocker or other antihypertensive drug treatments.

Contact details for further information

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Anticipated or actual start date

15 July 2017

Anticipated completion date

01 February 2018

Funding sources/sponsors

None

Conflicts of interest

None known

Language

English

Country

China

Stage of review

Review_Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Acids, Noncarboxylic; Humans; Kidney; Renal Insufficiency, Chronic; Sodium; Sodium Chloride, Dietary; Sodium, Dietary

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NHS National Institute for Health Research

Date of registration in PROSPERO
18 July 2017

Date of publication of this version
18 July 2017

Details of any existing review of the same topic by the same authors
None

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions
18 July 2017

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Supplementary file 2 Search Strategy

Ovid MEDLINE

1. exp Sodium Chloride/ or sodium chloride.mp.

2. exp Sodium, Dietary/

3. exp Diet, Sodium-Restricted/

4. ((sodium or salt) adj5 (low or high or alter* or reduce* or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)).mp.

5. (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free).mp.

6. 1 or 2 or 3 or 4 or 5

7. exp Kidney/ or kidney.mp.

8. exp Nephrosis/ or nephrosis*.mp.

9. (renal or nephropath*).mp.

10. (chronic kidney or chronic renal).mp.

11. exp Kidney Diseases/

12. ((kidney or renal) adj (disease* or failure)).mp.

13. exp Kidney Function Tests/ or kidney function test*.mp.

14. (CKF or CKD or CRF or CRD).mp.

15. (end-stage renal or end-stage kidney or endstage renal or endstage kidney).mp.

16. (ESRF or ESKF or ESRD or ESKD).mp.

17. exp Uremia/ or ur?emi*.mp.

18. exp Renal Replacement Therapy/ or renal replacement therap*.mp.

19. exp Renal Insufficiency/ or renal insufficien*.mp.

20. exp Dialysis/ or dialysis.mp.

21. exp Renal Dialysis/ or renal dialys*.mp.

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22. ("pre dialysis" or pre-dialysis or predialysis).mp.

23. (h*emodialysis or h*emofiltration or h*emodiafiltration).mp.

24. exp Peritoneal Dialysis/ or peritoneal dialysis.mp.

25. (PD or CAPD or CCPD or APD or IPD).mp.

26. exp Kidney Transplantation/

27. ((kidney or renal) adj (transplant* or graft* or allograft*)).mp.

28. 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27

29. randomized controlled trial.pt.

30. controlled clinical trial.pt.

31. randomized.ab.

32. placebo.ab.

33. randomly.ab.

34. clinical trials as topic.sh.

35. trial.ti.

36. 29 or 30 or 31 or 32 or 33 or34 or 35

37. (animals not (humans and animals)).sh.

38. 36 not 37

39. exp Cohort Studies/

40. exp Longitudinal Studies/

41. exp Follow-Up Studies/

42. exp Prospective Studies/

43. exp Retrospective Studies/

44. cohort.ti,ab.

45. longitudinal.ti,ab.

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46. prospective.ti,ab.

47. retrospective.ti,ab.

48.38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47

49. 6 and 28 and 48

Embase

#1. 'sodium chloride'/exp OR 'sodium chloride'

#2. 'sodium intake'/exp

#3. 'salt intake'/exp

#4. 'sodium restriction'/exp

#5. 'high sodium intake'/exp

#6. 'high salt diet'/exp

#7. 'high sodium' OR 'low sodium' OR 'sodium alter*' OR 'sodium reduce*' OR 'sodium reduction' OR 'sodium restrict*' OR 'sodium intake*' OR 'sodium diet*' OR 'sodium food' OR 'sodium increas*' OR 'sodium decreas*' OR 'sodium change*' OR 'sodium changing' OR 'sodium free'

#8. 'high salt' OR 'low salt' OR 'salt alter*' OR 'salt reduce*' OR 'salt reduction' OR 'salt restrict*' OR 'salt intake*' OR 'salt diet*' OR 'salt food' OR 'salt increas*' OR 'salt decreas*' OR 'salt change*' OR 'salt changing' OR 'salt free'

#9. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8

#10. 'kidney'/exp OR 'nephrosis'/exp

#11. kidney OR renal OR nephropath* OR nephrosis

#12. 'chronic kidney' OR 'chronic renal'

#13. 'kidney disease'/exp OR 'kidney failure'/exp

#14. 'kidney disease*' OR 'renal disease*' OR 'kidney failure' OR 'renal failure'

#15. 'kidney function test'/exp OR 'kidney function test*'

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#16. ckf OR ckd OR crf OR crd

#17. 'end-stage kidney' OR 'end-stage renal' OR 'endstage kidney' OR 'endstage renal'

#18. esrf OR eskf OR esrd OR eskd

#19. 'uremia'/exp OR ur?emi*

#20. 'renal replacement therapy'/exp OR 'renal replacement therap*'

#21. 'renal insufficien*'

#22. 'dialysis'/exp OR dialys*

#23. 'pre dialysis' OR predialysis

#24. 'hemodialysis'/exp OR 'hemofiltration'/exp OR 'hemodiafiltration'/exp

#25. hemodialysis OR haemodialysis OR hemofiltration OR haemofiltration OR hemodiafiltration OR haemodiafiltration

#26. 'peritoneal dialysis'/exp OR 'peritoneal dialysis'

#27. pd OR capd OR ccpd OR apd OR ipd

#28. 'kidney transplantation'/exp OR 'kidney graft'/exp OR 'kidney allograft'/exp

#29. 'kidney transplant*' OR 'renal transplant*' OR 'kidney graft*' OR 'renal graft*' OR 'kidney allograft*' OR 'renal allograft*'

#30. #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29

#31. 'randomized controlled trial'/exp OR 'randomized controlled trial' OR 'controlled clinical trial'/exp OR 'controlled clinical trial' OR random*OR 'clinical trials'/exp OR 'clinical trials' OR trial:ti

#32. 'human'

#33. animal$ OR nonhuman

#34. #32 AND #33

#35. #33 NOT #34

#36. #31 NOT #35

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#37.'cohort analysis'/exp OR 'longitudinal study'/exp OR 'prospective study'/exp OR 'retrospective study'/exp

#38. (cohort OR concurrent OR incidence OR longitudinal OR followup OR 'follow up' OR prospective OR retrospective) NEXT/1 (analys* OR design* OR evaluation* OR research OR stud* OR survey* OR trial*)

#39. #36 OR #37 OR #38

#40. #9 AND #30 AND #39

CENTRAL

#1 MeSH descriptor: [Sodium Chloride] explode all trees

#2 "sodium chloride":ti,ab,kw (Word variations have been searched)

#3 MeSH descriptor: [Sodium, Dietary] explode all trees

#4 MeSH descriptor: [Diet, Sodium-Restricted] explode all trees

#5 ((sodium or salt) near/5 (low or high or alter* or reduce* or reducing or reduction or restrict* or intake* or diet* or food or increas* or decreas* or change* or changing)):ti,ab,kw

#6 (sodium-restricted or salt-restricted or low-sodium or low-salt or salt-free or sodium-free):ti,ab,kw

#7 (#1 or #2 or #3 or #4 or #5 or #6)

#8 MeSH descriptor: [Kidney] explode all trees

#9 MeSH descriptor: [Nephrosis] explode all trees

#10 (kidney or renal or nephropath* or nephrosis):ti,ab,kw

#11 (chronic next kidney or chronic next renal):ti,ab,kw

#12 MeSH descriptor: [Kidney Diseases] explode all trees

#13 (kidney or renal) next (disease* or failure):ti,ab,kw

#14 MeSH descriptor: [Kidney Function Tests] explode all trees

#15 “kidney function test*”:ti,ab,kw

#16 (CKF or CKD or CRF or CRD):ti,ab,kw

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#17 ("end-stage kidney" or "end-stage renal" or "endstage kidney" or "endstage renal"):ti,ab,kw

#18 (ESRF or ESKF or ESRD or ESKD):ti,ab,kw

#19 MeSH descriptor: [Uremia] explode all trees

#20 ur?emi*:ti,ab,kw

#21 MeSH descriptor: [Renal Replacement Therapy] explode all trees

#22 "renal replacement therap*":ti,ab,kw

#23 MeSH descriptor: [Renal Insufficiency] explode all trees

#24 "renal insufficien*":ti,ab,kw

#25 MeSH descriptor: [Dialysis] explode all trees

#26 dialys*:ti,ab,kw

#27 MeSH descriptor: [Renal Dialysis] explode all trees

#28 "renal dialys*":ti,ab,kw

#29 "pre dialysis" or pre-dialysis or predialysis:ti,ab,kw

#30 (h*emodialysis or h*emofiltration or h*emodiafiltration):ti,ab,kw

#31 MeSH descriptor: [Peritoneal Dialysis] explode all trees

#32 "peritoneal dialys*":ti,ab,kw

#33 (PD or CAPD or CCPD or APD or IPD):ti,ab,kw

#34 MeSH descriptor: [Kidney Transplantation] explode all trees

#35 (kidney or renal) near (transplant* or graft* or allograft*):ti,ab,kw

#36 (#8 or #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16 or #17 or #18 or #19 or #20 or #21 or #22 or #23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35)

#37 (#7 and #36)

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Supplementary file 3 Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up , months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9[a]	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Sodium intake 80-100 mmol/d	24-h urinary sodium	Usual care	1.39c	143/88	NA
Fujiwara S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPAKUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5[b]	144/95	Non- industry

Heerspink HJL[24] (2012)	Type 2 diabetic nephropathy	1177	65	59	Cohort study	30	Renal composite outcomes, CV events	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine< 153mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 153mmol/g	44	155/84	Industry and non- industry
LowSALT CKD Study[25,26] (2012)	Hypertensive patients with stage 3–4 CKD	20	75	69	Randomised crossover trial	1	UACR, SBP, DBP	Sodium intake < 100 mmol/d	Sodium intake < 100 mmol/d	24-h urinary sodium	Sodium intake ≥ 220 mmol/d	32	151/82	Non- industry
de Brito-Ashurst I[27] (2013)	eGFR < 60 mL/min and mean BP > 130/80 mm Hg	48	58	58	Randomised parallel-group controlled trial	6	Change in eGFR	Intensive low-salt diet education	Traditional meals had salt reduced by 50%	24-h urinary sodium	Usual care	41.5	153/85	Non- industry
Liang X[28] (2013)	PD	106	54	33	Cohort study	1	CV events, SBP, DBP	Health education to restrict water-sodium intake strictly	Salt intake not more than 3 g/d	The 3-day diet diary was collected once a month by the special nurse. Compared with the Food Composition, the daily salt intake were estimated	Usual care	NA	149/88	NA
Mc Quarrie EP[29] (2014)	Patients with CKD and had at least one 24-hour urinary sodium measurement	423	50	52	Cohort study	102	Renal composite outcomes, all- cause mortality	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine ≤ 16	Urinary sodium/creatinine	Urinary sodium/creatinine > 16	48	139/79	Non- industry
Fan L[4] (2014)	GFR 13–55 mL/min/1.73m^2	840	60	52	Cohort study	72	Renal composite outcomes, all- cause mortality	Urinary sodium < 3 g/d	Urinary sodium < 3 g/d	24-h urinary sodium	Urinary sodium ≥ 3 g/d	32.5	132/81	Non- industry
Baek SH[30] (2014)	Hypertensive patients with albuminuria	202	51	50	Cohort study	2	UAE, SBP, DBP	Good compliance with an low salt diet	Good compliance with an low salt diet (24-hurinary sodium reduction ratio < 50%)	24-hurinary sodium	Poor compliance with an low salt diet	77.1[c]	121/73	Industry and non- industry

Hwang JH[31] (2014)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	245	50	50	Randomised parallel-group controlled trial	2	eGFR, UAE, SBP, DBP	Sodium intake < 100mmol/d	Sodium intake < 100mmol/d	24-hurinary sodium	Usual care	67.3	131/79	Industry
Rodrigues Telini LS[32] (2014)	HD	39	38	58	Randomised parallel-group controlled trial	4	SBP, DBP	2 g of sodium reduction in their habitual diet	2 g of sodium reduction in their habitual diet	Salt intake was based on a 72-halimentary registry	Usual diet	NA	146/86	Non- industry
Nerbass FB[33] (2015)	eGFR 59–30 ml/min/1·73 m^2	1416	39	73	Cohort study	12	eGFR, UACR, SBP, DBP	Sodium intake decrease	Sodium intake decrease	Estimated 24-hurinary sodium	Sodium intake unchanged or increase	52.7	134/73	Industry and non- industry
CRIC Study[3,34] (2016)	eGFR20-70mL/min/1.73 m^2	3757	56	58	Cohort study	78	Renal composite outcomes, all- cause mortality, CV events	Urinary sodium < 153.6mmol/d	Urinary sodium < 153.6mmol/d	24-hurinary sodium	Urinary sodium ≥153.7 mmol/d	44.5	128/71	Non- industry
Mazarova A[5] (2016)	eGFR < 30 mL/min/1.73 m^2	341	62	64	Cohort study	18	Change in eGFR, renal failure, all-cause mortality	Urinary sodium ≤ 150mmol/d	Urinary sodium ≤ 150mmol/d	24-hurinary sodium	Urinary sodium> 150 mmol/d	17.1	NA	No funding supported
Kauric-Klein Z[35] (2017)	HD	121	50	60	Randomised parallel-group controlled trial	3	All- cause mortality, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	Sodium intake checklist developed by Millar and Beard	Usual care	NA	165/87	Non- industry
Meuleman Y[36] (2017)	eGFR ≥ 20mL/min/1.73m^2 and UPE > 0.2 g/L	112	82	55	Randomised parallel-group controlled trial	6	eGFR, UPE, SBP, DBP	Usual care and low salt diet education	low salt diet education	24-hurinary sodium	Usual care	49.7	129/76	Non- industry

Jiamjariyapon T[37] (2017)	Stage 3-4 CKD with diabetes or hypertension	418	27	62	Cluster randomized controlled trial	24	Renal composite outcomes, all-cause mortality, eGFR, CV events, UACR, SBP, DBP	Usual care and intensive low-salt diet education	Intensive low-salt diet educatio (Sodium intake less than 2,400 mg/d)	24-hurinary sodium	Usual care	41.5	125/76	Non- industry
Saran R[38] (2017)	eGFR 15–60 ml/min/1.73 m^2	58	60	61	Randomized crossover trial	1	eGFR, UACR, SBP, DBP	Sodium intake < 2g/d	Sodium intake < 2g/d	24-hurinary sodium	Usual care	38.7	130.3/NA	Non- industry
Huang B[39] (2017)	HD	90	43	55	Cluster randomized controlled trial	6	SBP, DBP	Self-management support, restrict salt intake	Self-management support, restrict salt intake	Dietary salt intake was measured using slightly modified balance formulas: (1) Sodium intake (mol)=[Na+1(mmol/L)×ECW(L)－Na+ 2(mmol/L)×(ECW– IDWG)(L)]/1000; (2) Salt intake (g/day)=sodium intake(mol)×58.4 (g/mol)/2(day)	Usual care	NA	154/86	Non- industry
Ikenoue T[6] (2018)	HD	88115	60	65	Cohort study	12	All-cause mortality, CV events	Salt intake < 6 g/d	Salt intake < 6 g/d	Intra-dialytic weight loss was used to calculate salt intake	Salt intake ≥ 6 g/d	NA	NA	No funding supported
Parvanova A[40] (2018)	Patients with type 2 diabetes and macroalbuminuria	108	89	64	Randomised parallel-group controlled trial	3	UACR	Sodium intake ≤ 100mmol/d	Sodium intake ≤ 100mmol/d	24-hurinary sodium	Sodium intake ≥ 200mmol/d	87.36	146/80	Industry
Martinez MG[41] (2019)	Pre-dialysis CKD with UPE > 500 mg/d	84	52	56	Cohort study	6	Change in SBP	Urinary sodium reduced	Urinary sodium reduced	24-hurinary sodium	Urinary sodium increased	49[b]	142/NA	NA
Nishimoto M[42] (2019)	UACR 30–599 mg/g and eGFR ≥ 50 ml/min/1.73 m^2	304	70	59	Cohort study	12	Change in eGFR, SBP, and DBP	Urinary sodium ≤ 189 mmol/d	Urinary sodium ≤ 189 mmol/d	24-hurinary sodium	Urinary sodium > 189 mmol/d	64.93	140/83	Non- industry

Ahn SY[43] (2019)	eGFR ≥ 30 mL/min/1.73m2 and UACR ≥ 30 mg/g	171	49	50	Cohort study	12	Rate of change in eGFR, renal composite outcomes, change in UAE, SBP, DBP	Intensive low-salt diet education	Intensive low-salt diet education to achieve urinary sodium ≤ 100mmol/d	24-hurinary sodium	Usual care	64.2	124/75	Non- industry
Hu J[44] (2019)	PD	33	58	42	Cohort study	12	Residual renal Ccr	Sodium intake ≤ 3.0 g/d	Sodium intake ≤ 3.0 g/d	Daily sodium intake of the patients was calculated based on the 3-day dietary record	Sodium intake > 3.0 g/d	8.3	146/95	Non- industry
Humalda JK[45] (2020)	CKD stages 1 to 4 or kidney transplants recipients, and eGFR was ≥ 25 mL/min/ 1.73 m2	99	84	57	RCT	6	SBP, DBP	Self-management to restrict salt intake	Self-management to restrict salt intake	A detailed interactive food diary	Usual care	55	139/83	Non- industry
Garofalo C[46] (2020)	CKD stages 1–5	1785	58	64	Cohort study	49	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	37	NA	NA
Kinguchi S[47] (2021)	Diabetic kidney disease patients with HbA1c 7.0–10.0%, eGFR ≥ 45 ml/min/1.73 m2, and UACR ≥ 30 mg/g	84	76	65	Cohort study	6	UACR	urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios < 1.14	Urinary sodium-to-creatinine ratios	urinary sodium-to-creatinine ratios ≥ 1.14	67	142/79	Non- industry
Kohatsu K[48] (2021)	CKD stages 3–5	197	76	71	Cohort study	17	Renal composite outcomes, all-cause mortality	Salt intake < 6 g/d	Salt intake < 6 g/d	24-hurinary sodium	Salt intake ≥ 6 g/d	24.2	133/74	Non- industry

Note: Ccr, clearance of creatinine; CKD, chronic kidney disease; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; ESRD, end stage renal disease; ECW, extracellular water; HD, hemodialysis; IDWG, interdialytic weight gain; NA, no available; PD, peritoneal dialysis; SBP, systolic blood pressure; UACR, urinary albumin to creatinine ratio; UAE, urine albumin excretion; UPE, urinary protein excretion; the units for eGFR: mL/min/1.73m². [a] The units for eGFR: mL/min, [b] Only Ccr obtained, the units for Ccr: mL/min; [c] Only serum creatinine obtained, the units for serum creatinine: μmol/L.

Supplementary file 4 Table S2 Summary of characteristics of included trials and patients.

Outcomes	No. of Studies/ No. of Patients	Sample Size	Follow-up, months	Age, years	Male (%)	Lost follow-up rate
Renal composite outcomes	10/9578	462 (305-1329)	43 (23-74)	58 (52-64)	59.2%	3.2%
Rate of change in eGFR	12/3287	142 (57-332)	12 (6-17)	55(48-61)	56.6%	7.5%
Proteinuria	13/2922	108 (40-224)	3 (1-12)	50 (43-66)	57.2%	8.8%
All-cause death	9/95997	423 (231-2771)	24 (15-75)	62 (55-65)	60.6%	0.03%
CV events	5/93573	1177 (262-45936)	24 (6.5-54)	59(45-63)	60.8%	0.02%
SBP	19/3608	103 (38-213)	3 (1-6)	55 (41-60)	57.1%	7.5%
DBP	18/3524	36 (20-90)	12 (2-54)	55 (40-60)	57.3%	7.7%

Note:Values are given as median (quartile). CV, cardiovascular; eGFR, estimated glomerular filtration rate.

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Supplementary file 5 Table S3 Assessment of risk of bias for RCTs.

Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
1. Bias arising from the randomization process																
1.1 Was the allocation sequence random?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	Other	Other	Low	Other	Low	Low	Low	Low
1.3 Did baseline differences between intervention groups suggest a problem with the randomisation process?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias arising from the randomisation process?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
2. Risk of bias due to deviations from intended interventions																
2.1 Were participants aware of their assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	Other	High	High	High	High	Low	High	High
2.2 Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?	N/PN	Y/PY	NI	Low	High	Low	High	High	High	High	High	High	High	High	High	High
2.3 If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?	N/PN	Y/PY	NA/NI		Low		Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.4 If Y/PY/NI to 2.3: Were these deviations likely to have affected the outcome?	N/PN	Y/PY	NA/NI													
2.5 If Y/PY to 2.4: Were these deviations from intended intervention balanced between groups?	Y/PY	N/PN	NA/NI													

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomised?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to deviations from intended interventions?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
3. Bias due to missing outcome data																
3.1 Were data for this outcome available for all, or nearly all, participants randomised?	Y/PY	N/PN	NI	Other	Other	Low	Low	Low	High	High	High	High	Low	High	High	Low
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?	Y/PY	N/PN	NA	Low	Low				Low	Low	Low	Low		Low	Low	
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?	N/PN	Y/PY	NA/NI													
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias due to missing outcome data?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
4. Bias in measurement of the outcome																
4.1 Was the method of measuring the outcome inappropriate?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?	N/PN	Y/PY	NI	Low	High	Low	High	High	Low	High	High	High	Low	High	Low	High
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI		Low		Low	Low		Low	Low	Low		Low		Low
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?	N/PN	Y/PY	NA/NI													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction of bias in measurement of the outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
5. Bias in selection of the reported result																
5.1 Were the data that produced this result analysed in accordance with a prespecified analysis plan that was finalised before unblinded outcome data were available for analysis?	Y/PY	N/PN	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Is the numerical result being assessed likely to have been selected, on the basis of the results, from:																
5.2 ... multiple eligible outcome measurements (eg, scales, definitions, time points) within the outcome domain?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
5.3 ... multiple eligible analyses of the data?	N/PN	Y/PY	NI	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low

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Bias domain and signaling question	Response options			Mühlhauser I 1996	Keven K 2006	Low SALT CKD Study 2012	de Brito-Ashurst I 2016	Hwang JH 2014	Rodrigues Telini LS 2014	Kauric-KleinZ 2017	Meuleman Y 2017	Jiamjariyapon T 2017	Saran R 2017	Huang B 2017	Parvanova A 2018	Humalda JK 2020
	Low	High	Other													
Risk-of-bias judgment (low/high/some concerns)				**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**	**Low**
Optional: What is the predicted direction bias due to selection of the reported results?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null
Overall bias																
Risk-of-bias judgment (low/high/some concerns)				**Some concerns**	**Some concerns**	**Low**	**Low**	**Low**	**Some concerns**	**Some concerns**	**Low**	**Some concerns**	**Low**	**Low**	**Low**	**Low**
Optional: What is the overall predicted direction of bias for this outcome?				Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null	Towards null

Note: Y, yes; PY, probably yes; PN, probably no; N, no; NA, not applicable; NI, no information; RCT, randomized controlled trial.

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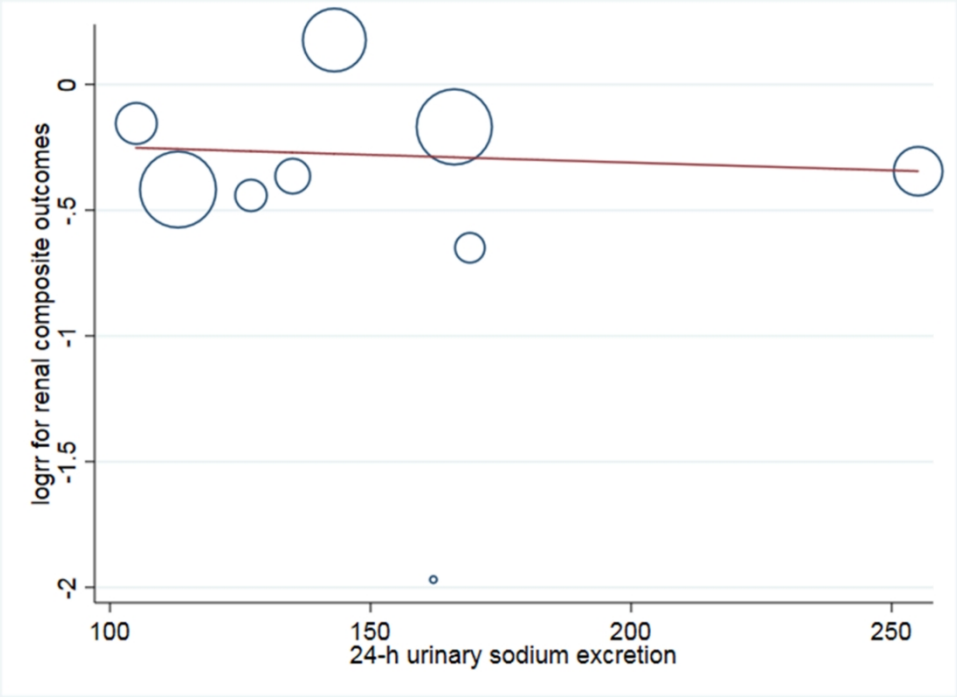
Supplementary file 6 Table S4 Quality assessment for cohort studies.

Study (Year)	Representativeness of the exposed cohort	Selection of the non exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Scores
Cianciaruso B[19] 1998	★	★	★	★	—	★	★	★	7★
Fujiwara S[21] 2010	★	★	★	★	★	★	—	★	7★
Yu W[22] 2011	★	★	★	★	★	★	—	★	7★
Vegter S[23] 2012	★	★	★	★	—	★	★	★	7★
Heerspink HJL[24] 2012	★	★	★	★	★	★	—	★	7★
Liang X[28] 2013	★	★	★	★	★	★	—	★	7★
Mc Quarrie EP[29] 2014	★	★	★	★	★	★	★	★	8★
Fan L[3] 2014	★	★	★	★	—	★	★	—	6★
Baek SH[30] 2014	★	★	★	★	—	★	—	★	6★
Nerbass FB[33] 2015	★	★	★	★	★	★	—	★	7★
CRIC Study[3,34] 2016	★	★	★	★	★	★	★	★	8★
Mazarova A[5] 2016	★	★	★	★	★	★	—	★	7★
Ikenoue T[6] 2018	★	★	★	★	★	★	—	★	7★
Martinez MG[41] 2019	★	★	★	★	★	★	★	★	8★

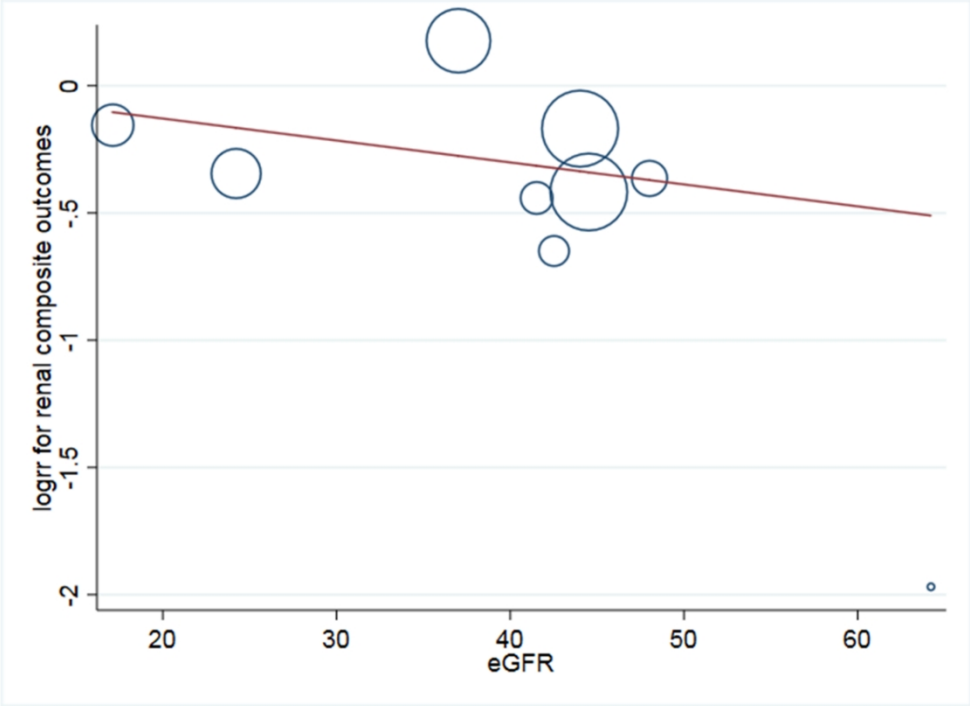
Nishimoto M[42] 2019	★	★	★	★	★	★	—	★	7★
Ahn SY[43] 2019	★	★	★	★	★	★	—	★	7★
Hu J[44] 2019	★	★	★	★	★	★	—	★	7★
Garofalo C[46] 2020	★	★	★	★	★	★	★	★	8★
Kinguchi S[47] 2021	★	★	★	★	★	★	—	★	7★
Kohatsu K[48] 2021	★	★	★	★	★	★	—	★	7★

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability. —: no score; It is considered that the queue follow-up is appropriate when the follow-up rate is more than 80%.

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Supplementary file 7



Supplementary file 8

Supplementary file 9 Table S5 Subgroup analysis of outcomes except renal failure events.

Subgroup	No. of trials	n	RR/MD/SMD (95%CI)	*P* for RR/MD/SMD	I^2	*P* for heterogeneity test[a]
Change in eGFR（mL/min/1.73m²）						
Study type						
RCT	5	881	-0.75 (-7.58, 6.08)	0.83	96.2%	0.28
Cohort studies	7	2406	2.48 (0.32, 4.64)	0.02	98.1%	
Baseline eGFR (mL/min/1.73m²)						
60-89	4	804	-1.00 (-7.41, 5.42)	0.76	99.2%	0.48
30-59	6	2109	2.58 (-0.44, 5.61)	0.10	95.4%	
<15	2	374	1.24 (-0.88, 3.37)	0.25	66.2%	
Mean age (years)						
<58	5	106	0.44 (-4.40, 5.27)	0.86	98.5%	0.70
≥58	6	2669	1.68 (-1.83, 5.18)	0.35	97.8%	
Baseline BP (mmHg)						
<140/90	6	2420	-0.52 (-4.36, 3.32)	0.32	97.8%	0.41
SBP≥140 或 DBP≥90	4	469	2.54 (-2.47, 7.55)	0.55	70.1%	
Comparator of control group						
Usual care or usual diet	6	1052	-0.38 (-5.14, 4.39)	0.88	95.8%	0.85
Different levels of sodium intake	6	2235	2.62 (0.31, 4.94)	0.03	98.2%	
Proteinuria or Albuminuria						
Study type						
RCT	7	997	-1.03 (-2.28, 0.21)	0.11	98.4%	0.45
Cohort studies	6	1925	-0.28 (-0.59, 0.02)	0.07	80.8%	
Baseline eGFR (mL/min/1.73m²)						
60-89	7	821	-1.06 (-2.32, 0.20)	0.10	98.2%	0.42
30-59	6	1201	-0.23 (-0.50, 0.04)	0.10	79.6%	
Mean age (years)						
<58	7	846	-1.12 (-2.35, 0.12)	0.08	98.2%	0.34
≥58	6	2076	-0.13 (-0.34, 0.09)	0.24	64.2%	
Baseline BP (mmHg)						
<140/90	8	200	-0.82 (-1.55, -0.10)	0.01	98.3%	0.65
SBP≥140 或 DBP≥90	4	2665	-0.39 (-0.68, -0.11)	0.03	0.0%	
Comparator of control group						
Usual care or usual diet	7	1083	-1.01 (-2.18, 0.15)	0.09	98.4%	0.12
Different levels of sodium intake	6	1839	-0.34 (-0.69, 0.00)	0.05	81.7%	
All-cause mortality						
Study type						
RCT	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.99
Cohort studies	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
Baseline eGFR (mL/min/1.73m²)						
30-59	4	5438	0.65 (0.41, 1.02)	0.06	85.2%	0.008
15-29	2	2126	1.14 (0.88, 1.48)	0.33	0.0%	
<15	3	88433	1.34 (0.41, 4.39)	0.62	43.4%	

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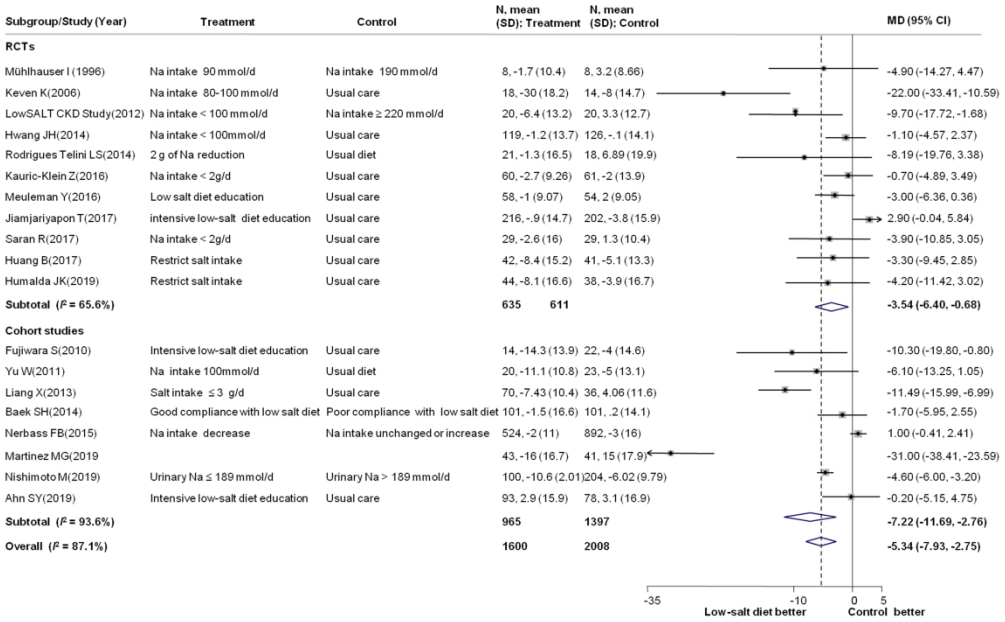
Mean age (years)						
<58	3	704	0.61 (0.37, 1.00)	0.05	90.1%	0.08
≥58	6	90977	1.34 (0.76, 2.37)	0.32	84.2%	
Baseline BP (mmHg)						
<140/90	5	5438	0.64 (0.41, 0.99)	0.05	80.6%	0.86
SBP≥140 或 DBP≥90	1	121	0.51 (0.05, 5.46)	0.58	-	
Comparator of control group						
Usual care or usual diet	2	539	0.96 (0.31, 3.02)	0.95	0.0%	0.90
Different levels of sodium intake	7	95458	0.90 (0.50, 1.59)	0.71	98.5%	
CV events						
Study type						
RCT	1	418	0.47 (0.09, 2.53)	0.88	-	0.51
Cohort studies	4	93155	1.05 (0.51, 2.18)	0.89	98.2%	
Baseline eGFR (mL/min/1.73m^2)						
30-59	3	5352	0.78 (0.63, 0.95)	0.02	57.1%	**0.02**
<15	2	88221	1.50 (0.39, 5.85)	0.56	58.3%	
Mean age (years)						
<58	2	3863	0.71 (0.62, 0.80)	<0.001	0.0%	0.40
≥58	3	89710	1.20 (0.49, 2.91)	0.69	97.3%	
Baseline BP (mmHg)						
<140/90	2	4175	0.71 (0.62, 0.80)	<0.001	0.0%	0.62
SBP≥140 或 DBP≥90	2	1283	0.88 (0.74, 1.05)	0.17	0.0%	
Comparator of control group						
Usual care or usual diet	2	524	0.49 (0.14, 1.73)	0.27	0.0%	0.52
Different levels of sodium intake	3	93049	1.14 (0.53, 2.46)	0.74	98.8%	
SBP（mmHg）						
Study type						
RCT	11	1246	-3.54 (-6.40, -0.68)	0.02	65.6%	0.43
Cohort studies	8	2362	-7.22 (-11.69, -2.76)	0.002	93.6%	
Baseline eGFR (mL/min/1.73m^2)						
60-89	7	1017	-3.30 (-5.26, -1.34)	0.001	31.7%	0.64
30-59	7	2210	-6.10 (-11.62, -0.59)	0.03	92.8%	
< 15	4	349	-5.66 (-11.62, 0.30)	0.06	76.0%	
Mean age (years)						
< 58	12	1215	-7.45 (-11.76, -3.14)	0.001	85.6%	0.25
≥ 58	7	2393	-2.39 (-5.65, 0.87)	0.15	87.3%	
Baseline BP (mmHg)						
< 140/90	10	2786	-0.97 (-2.75, 0.81)	0.28	46.9%	**0.02**
SBP ≥ 140 or DBP ≥ 90	9	822	-10.24 (-15.37, -5.12)	< 0.001	88.4%	
Comparator of control group						
Usual care or usual diet	13	1546	-4.28 (-7.20, -1.36)	0.004	74.3%	0.05
Different levels of sodium intake	6	2062	-7.55 (-12.83, -2.28)	0.005	94.6%	
DBP（mmHg）						
Study type						
RCT	7	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.85
Cohort studies	11	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

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Baseline eGFR (mL/min/1.73m^2)						
30-59	7	1017	-1.74 (-3.40, -0.08)	0.04	56.7%	0.13
15-29	6	2126	-0.53 (-1.25, 0.20)	0.16	11.9%	
< 15	4	349	-4.94 (-7.08, -2.80)	<0.001	23.2%	
Mean age (years)						
< 58	11	1131	-2.78 (-4.35, -1.22)	<0.001	50.8%	0.34
≥ 58	7	2393	-0.86 (-1.77, 0.04)	0.06	62.2%	
Baseline BP (mmHg)						
< 140/90	9	2747	-0.94 (-1.90, 0.02)	0.056	5.6%	**0.04**
SBP ≥ 140 或 DBP ≥ 90	9	777	-4.05 (-6.61, -1.49)	0.002	0.2%	
Comparator of control group						
Usual care or usual diet	11	1246	-2.41 (-4.04, -0.77)	0.004	66.0%	0.12
Different levels of sodium intake	7	2278	-2.13 (-3.90, -0.37)	0.02	74.6%	

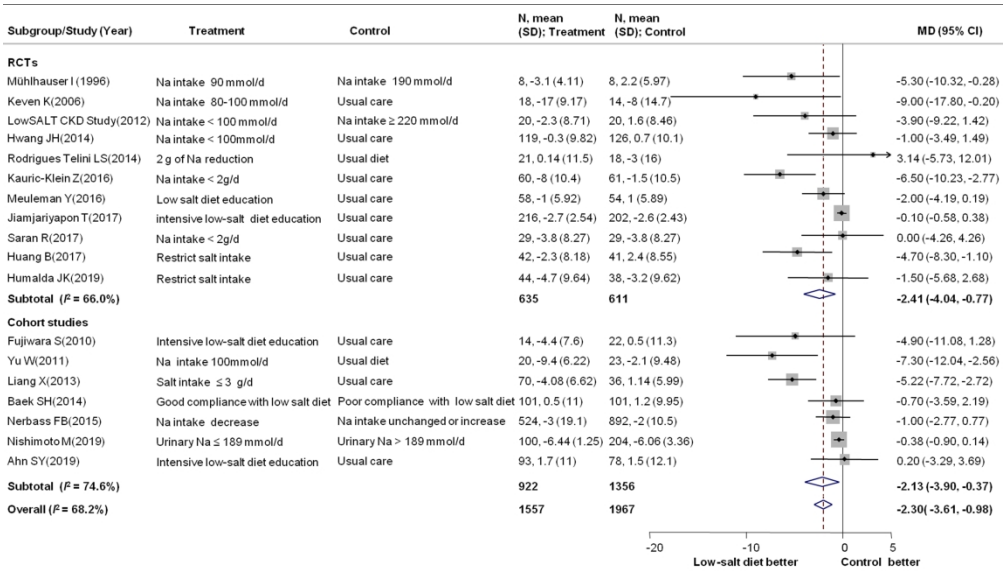
Note: [a] *P* value calculated by meta regression. BP, blood pressure; CI, confidence interval; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; RCT, randomised parallel-group controlled trial; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

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Supplementary file 10 Forest plot for SBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SBP, systolic blood pressure; SD, standard deviation.

317x210mm (300 x 300 DPI)



Supplementary file 11 Forest plot for DBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; DBP, diastolic blood pressure; SD, standard deviation.

309x186mm (300 x 300 DPI)

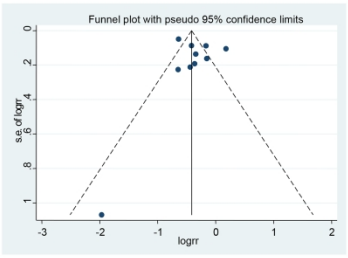
Supplementary file 12 Table S6 Sensitivity analysis of outcomes.

Sensitivity analysis	Renal composite outcomes			The rate of change in eGFR			Proteinuria or albuminuria			All-cause mortaity		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD/SMD (95%CI)	τ^2	No./n	RR (95% CI)	τ^2
Omit studies of follow-up time <12 months	-	-	-	7/2740	1.60 (-1.10, 4.31)	12.78	-	-	-	8/95876	0.92 (0.53, 1.59)	0.48
Omit studies of sample size <50	-	-	-	10/3206	1.07 (-1.26, 3.41)	12.86	9/2787	-0.89 (-1.56, -0.23)	0.99	-	-	-
Omit studies with low quality	-	-	-	-	-	-	-	-	-	-		
Omit studies with extreme outliers	-	-	-	11/3042	**2.43 (0.63, 4.23)**	8.01	12/2677	-0.24 (-0.43, -0.05)	0.06	8/7882	0.75 (0.56, 1.00)	0.08
Different statistical estimators												
DL	10/9578	0.71 (0.57, 0.88)	0.09	12/3287	1.18 (-0.94, 3.29)	12.57	13/2922	-0.70 (-1.24, -0.1)	0.89	9/95997	0.90 (0.52, 1.53)	0.48
EB	10/9578	0.72 (0.58, 0.89)	0.06	12/3287	1.16 (-2.02, 4.33)	23.4	13/2922	**-0.71(-1.66, 0.24)**	2.39	9/95997	0.92 (0.53, 1.46)	0.23
REML	10/9578	0.72 (0.58, 0.89)	0.03	12/3287	1.16 (-2.04, 4.35)	23.73	13/2922	**-0.71(-1.66, 0.24)**	2.38	9/95997	0.91 (0.53, 1.55)	0.33

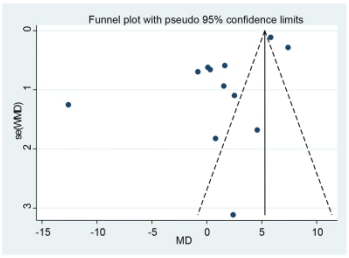
Sensitivity analysis	CV events			SBP			DBP		
	No./n	RR (95% CI)	τ^2	No./n	MD (95%CI)	τ^2	No./n	MD (95%CI)	τ^2
Omit studies of follow-up time <12 months	4/93467	1.03 (0.50, 2.12)	0.45	-	-	-	-	-	-
Omit studies of sample size <50	-	-	-	13/3402	-4.05 (-6.93, -1.17)	22.57	12/3318	-1.45 (-2.34, -0.56)	1.05
Omit studies with low quality	-	-	-	-	-	-	-	-	
Different statistical estimators									
DL	5/93573	0.97 (0.49, 1.93)	0.45	19/3608	-5.34 (-7.93, -2.75)	22.80	15/3524	-2.00(-2.93, -1.06)	1.56
EB	5/93573	1.01 (0.46, 2.22)	0.28	19/3608	-5.81(-9.63, -1.99)	14.19	15/3524	-2.30(-3.61, -0.98)	3.78
REML	5/93573	0.99 (0.41, 2.37)	0.35	19/3608	-5.78(-9.59, -1.97)	52.37	15/3524	-2.27(-3.57, -0.96)	3.41

Note: CI, confidence interval; CV, cardiovascular; DL, DerSimonian-Laird; EB, empirical Bayes; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; MD, mean difference; MAP, mean arterial pressure; n, number of patients; No., number of trials; NOS, Newcastle-Ottawa Scale; REML, restricted maximum likelihood; RR, relative risk; SMD, standardized mean difference; SBP, systolic blood pressure.

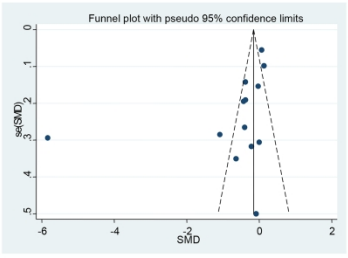
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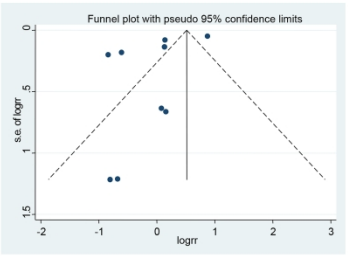
Funnel plot for publication bias on renal composite outcomes events.



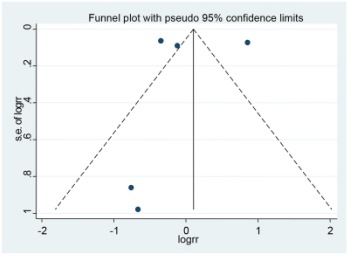
Funnel plot for publication bias on rate of change eGFR.



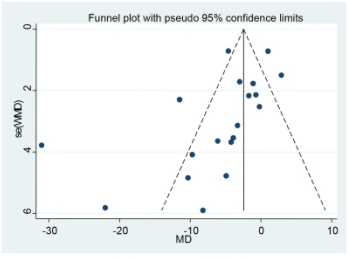
Funnel plot for publication bias on change in proteinuria or albuminuria.



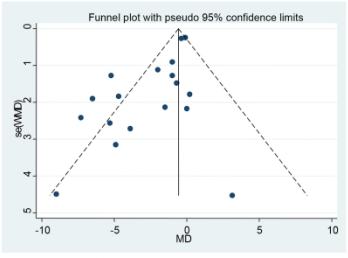
Funnel plot for publication bias on **all-cause mortality events**.



Funnel plot for publication bias on **CV events.**



Funnel plot for publication bias SBP.



Funnel plot for publication bias on **DBP.**

Supplementary file 13 Funnel plot for publication bias on outcomes. CV, cardiovascular; DBP, diastolic blood pressure; MD, mean difference; RR, relative risk; SMD, standard mean difference; SBP, systolic blood pressure.

Supplementary file 14 Table S7 The risk of publication bias of outcomes.

	N	Egger's test(*P*-value)	Bgge's test(*P*-value)
Renal composite outcomes events	10	0.47	0.28
The rate of change in eGFR	12	0.01	0.95
Proteinuria or albuminuria	13	0.10	0.30
All-cause mortality	9	0.09	1.00
CV events	5	0.91	0.46
BP	19	＜0.001	0.005

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5-6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6-7

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8-12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15-16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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